BASIC SCIENCE — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 109 chapters (1 from TO notes) –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

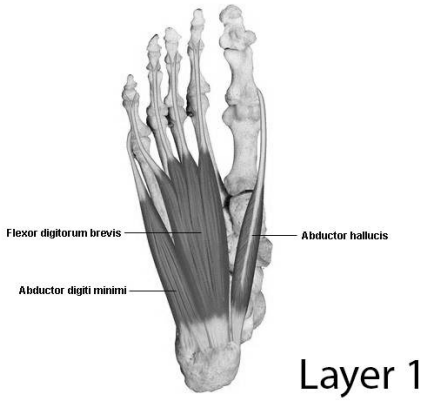
ABDUCTOR DIGITI MINIMI

– Basic Science · 1 / 108 –

1 Basics

Anatomy – foot ADM (Orthobullets)

	Abductor digiti minimi (foot)
Origin	Calcaneal tuberosity
Insertion	Base of 5th toe
Action	Abducts small toe
Innervation	**Lateral plantar nerve**
Arterial supply	Lateral plantar artery



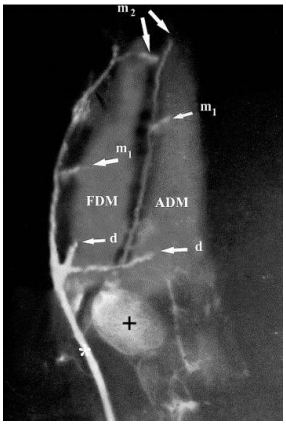
Abductor digiti minimi of the foot

Blood supply – hand ADM

Hand ADM hypothenar · abducts 5th digit · **deep branch of ulnar nerve (C8, T1)**

Main blood supply ✓ **ulnar artery** (deep branch) · 1 major + 2 constant minor pedicles · major pedicle leaves ulnar artery just distal to **pisiform**

Flap use ADM can be used as a local flap in crushed hands



Ulnar artery supplying ADM and FDM (hypothenar excised)

– – Active space – –

X vs Y

Artery	Supplies
Ulnar artery	✓ **Hypothenar muscles (ADM, FDM)**
Radial artery	Thenar muscles
Posterior interosseous artery	Superficial + deep extensor compartments of forearm
Anterior interosseous artery	Forearm flexor compartment
Palmar metacarpal artery	Palmar interossei

	Hand ADM	Foot ADM
Nerve	Deep branch **ulnar**	**Lateral plantar**
Artery	**Ulnar**	Lateral plantar

How it's asked

"Main blood supply to abductor digiti minimi (hand)?"	→ **ulnar artery** ✗ Wrong options: PIA · AIA · palmar metacarpal · radial
"Nerve to abductor digiti minimi of the foot?"	→ **lateral plantar nerve**

ADDUCTOR LONGUS

– Basic Science · 2 / 108 –

1 Basics

Anatomy

	Adductor longus
Origin	Anterior body of pubis, just lateral to symphysis
Insertion	**Middle third of linea aspera** · between adductor magnus/brevis (medial) and vastus medialis origin (lateral)
Action	Adducts + flexes thigh · helps laterally rotate hip
Innervation	**Anterior division of obturator nerve (L2, L3, L4)**
Arterial supply	Obturator artery + medial circumflex femoral artery



Adductor longus

Cross-sectional anatomy – proximal thigh

Adductor longus lies **just deep to the superficial femoral artery**

Adductor brevis between longus and magnus · **deep to the deep femoral vessels**

Adductor magnus largest, most posterior adductor

Sartorius superficial · **covers the superficial femoral vessels**

Gracilis medial + superficial

– – Active space – –



Proximal thigh cross-section — identify the arrowed muscle

4 Diagnosis & Classification

L3 radiculopathy vs femoral neuropathy

Key test	**hip adduction** (obturator nerve) · weak in **L2/L3 radiculopathy**, spared in **femoral neuropathy (L2–4)**
Femoral-innervated	quadriceps · iliacus · sartorius · psoas ✗ Wrong options: distinguishing

X vs Y

Muscle	Nerve	Distinguishes L3 root from femoral nerve?
Adductor longus	**Obturator (ant. division)**	Yes
Quadriceps	Femoral	No
Iliacus 174	Femoral	No
Sartorius 342	Femoral	No
Psoas 297	Femoral (listed as such in the OB key)	No

How it's asked

"L3 radiculopathy best differentiated from femoral neuropathy by testing?"	→ **adductor longus** (obturator) ✗ Wrong options: quadriceps · iliacus · sartorius · psoas
"Muscle just deep to the superficial femoral artery on proximal-thigh cross-section?"	→ **adductor longus** ✗ Wrong options: magnus (large, posterior) · brevis (deep to profunda) · sartorius (covers SFA) · gracilis (medial)

– – Active space – –

AMYOTROPHIC LATERAL SCLEROSIS (ALS)

– Basic Science · 3 / 108 –

1 Basics

Pathology

Definition	**Lou Gehrig's disease** · progressive motor neuron disorder · **degeneration of anterior horn cells** · commonest degenerative disease of the motor neuron system
LMN component	loss of motor neurons in **anterior horns** + motor cranial nerve nuclei
UMN component	loss of myelinated fibres in **corticospinal + corticobulbar** tracts
Hallmark	**abnormal motor conduction, normal sensory conduction**

Genes & lab markers

Familial	**~5%** · some map to **superoxide dismutase** gene on **chromosome 21** · cause otherwise unknown
Labs	**no lab test confirms ALS** · SPEP/immunoelectrophoresis only exclude differentials

3 Presentation

Symptoms	**painless weakness** of one limb → spreads · fasciculations · cramps · dysarthria/dysphagia · ↓ head control · breathing difficulty · urinary frequency/incontinence (late) · **sensation normal**
Neck ptosis	neck drop from neck-extensor weakness

UMN signs	LMN signs
Spasticity	Atrophy
Hyperreflexia	Weakness
(+) Hoffmann	**Fasciculations**
(+) Babinski	Clumsiness
Spastic dysarthria	↓ reflexes, ↓ tone

4 Diagnosis & Classification

Diagnostic criteria

Clinical	**UMN + LMN signs in the same limb** · no pain, no sensory symptoms · ± cranial-nerve findings → highly indicative
Period of observation	documents progressive loss of UMN + LMN function
Often misdiagnosed	cervical myelopathy / radiculopathy

– – Active space – –

Electrodiagnostics — definitive test

Definitive diagnosis ✓ **electrodiagnostic studies (NCS + EMG)** · also exclude peripheral mimics

EMG/NCS finding in ALS	
Motor	↓ **CMAP amplitude** widespread · slowed motor CV
Denervation	**Fibrillations + positive sharp waves** + ↓ recruitment in **≥3 extremities**
Spontaneous	Fibrillation + **fasciculation** potentials
Reinnervation	Present (**denervation + reinnervation**)
Sensory	**Normal SNAPs**

Not definitive genetic testing (only some familial types) · MRI brain/cord (rules out structural mimics) · muscle biopsy (not required) · SPEP

5 Management

No cure supportive · maintain function + comfort

Riluzole blocks **tetrodotoxin-sensitive Na^+ channels** in damaged neurons · delays ventilator dependence · prolongs life **2–3 months**

X vs Y

	ALS	Peripheral compressive neuropathy (CTS/cubital)
UMN signs (Hoffmann, Babinski, hyperreflexia)	**Present**	Absent
Sensory symptoms	**None**	Present
SNAPs	**Normal**	Abnormal
Distribution	Spreads, multiple limbs	Single nerve territory

How it's asked

"Most useful study for definitive diagnosis of ALS?"	→ **electrodiagnostic studies** ✗ Wrong options: genetic testing · MRI · muscle biopsy · SPEP
"Hallmark EMG/NCS pattern?"	→ **abnormal motor, normal sensory conduction** · denervation + reinnervation
"Hand weakness + wasting + hyperreflexia/Hoffmann, normal sensation?"	→ **ALS** ✗ Wrong options: carpal/cubital tunnel
"Riluzole mechanism?"	→ **blocks tetrodotoxin-sensitive sodium channels** · +2–3 months survival

ANTERIOR INTEROSSEOUS NERVE

– Basic Science · 4 / 108 –

1 Basics

Innervated muscles

AIN muscles	**FPL · FDP index + long · pronator quadratus**
Motor only	no cutaneous sensation



AIN branching from the median nerve

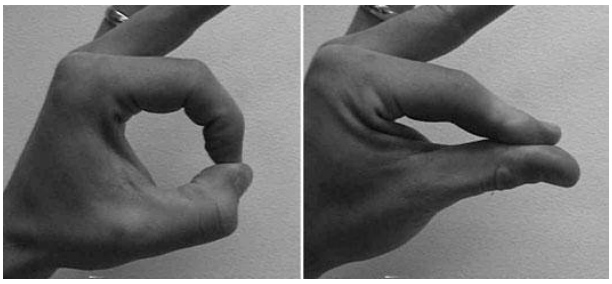
Gantzer muscle (TO)

Gantzer muscle	✓ **accessory head of FPL** (or of FDP) · anatomical variant in up to **70% (TO)**
Origin	deep surface of FDS, coronoid process or medial epicondyle **(TO)**
Insertion	FPL or FDP **(TO)**
Innervation	exclusively **AIN (TO)**
Traps	✗ Wrong options: accessory popliteus · accessory biceps femoris · accessory rectus femoris **(TO)**

3 Presentation

Supracondylar humerus fracture (extension type)	**AIN = commonest nerve injury** in children
Muscle affected	✓ **pronator quadratus** (most common electrophysiological abnormality) · also FPL, FDP index/long
Often misdiagnosed	as tendon injury (only 3 of 13 correct initially in Seror's series)

– – Active space – –



Inability to make an "OK" sign — AIN lesion

Book test (TO, FNB 2024)

"Book test assesses?" answer key not provided by source · options: adductor pollicis, opponens pollicis, FPL, dorsal interossei · module files it under **FPL** (AIN muscle) — read the options

X vs Y

Muscle	Nerve	Lost in AIN palsy?
Pronator quadratus	**AIN**	Yes
FPL	AIN	Yes
FDP index/long	AIN	Yes
FDP ring/small	Ulnar	No
FCR	Median (main trunk)	No
FCU	Ulnar	No
EDC	PIN	No

How it's asked

"Extension supracondylar fracture, 4-yr-old, nerve deficit — muscle most likely affected?"	→ **pronator quadratus** (AIN) ✗ Wrong options: EDC · FCR · FDP ring/small · FCU
"True about Gantzer muscle (TO)?"	→ **accessory head of FPL** · AIN-innervated ✗ Wrong options: popliteus · biceps femoris · rectus femoris
"Book test (FNB, TO)?"	→ answer key not provided · module = **FPL**

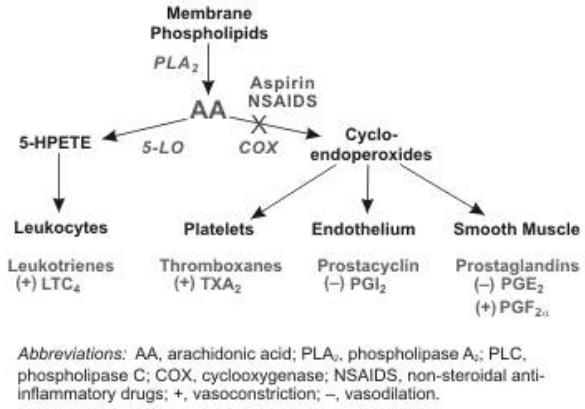
ANTI-INFLAMMATORY MEDICATIONS

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1 Basics

NSAID mechanism (COX-1, COX-2)

NSAID effects	anti-inflammatory · antipyretic · analgesic · **antiplatelet**
Arachidonic acid	directly metabolised by ✓ **cyclooxygenase** → prostaglandins, prostacyclin, **thromboxane**
5-lipoxygenase	arachidonic acid → **leukotrienes** ✗ Wrong options: "1-lipoxygenase"
HMG-CoA reductase	mevalonate / cholesterol pathway (statins) · not arachidonic acid
Steroids	inhibit ✓ **phospholipase A2** → no arachidonic acid from membrane phospholipids · earliest step of cascade



Arachidonic acid metabolism – steroids block PLA2, NSAIDs block COX

Enzyme	Role
COX-1	Constitutive · **gastric mucosal protection (PGE2)** · renal blood flow · **platelet TXA2**
COX-2	Inducible · **inflammation, pain, fever** · no platelet effect · **needed for endochondral ossification**

Drug classes

Drug	Binding	Note
Aspirin	**Irreversible** · acetylates **serine 530** of COX	Blocks **TXA2** → antiplatelet · effect >1 wk (platelet life)
Ibuprofen	**Reversible**, competitive · non-selective COX-1 + COX-2	GI ulcer, ↓ TX + ↓ PG
Indomethacin, naproxen, diclofenac, meloxicam	**Reversible** COX-1	Indomethacin = **HO prophylaxis** · **highest GI toxicity**
Ketorolac (Toradol)	COX-1 selective (OB)	–
Celecoxib, rofecoxib	**COX-2 selective**	Spare platelets + gastric mucosa · **cardiac toxicity**

– – Active space – –

Aspirin – dose-dependent action

Dose	Effect
Low 75–81 mg/day	Irreversibly acetylates COX-1 → ↓ platelet TXA2 → **antithrombotic**
Intermediate 650 mg–4 g/day	Inhibits COX-1 + COX-2 → ↓ PG → **analgesic, antipyretic**

5 Management

Indications	pain · **heterotopic ossification prophylaxis**
Contraindications	severe renal disease · gastric ulcers
COX-2 selective	anti-inflammatory without COX-1 loss (gastric mucosa, renal blood flow, platelets) · usable **perioperatively** (no platelet effect) · **no more effective for OA** than non-selective
Perioperative NSAID (OB)	dose- and time-dependent effect on bone healing → **lowest effective dose, ≤7 days postop**

Corticosteroids

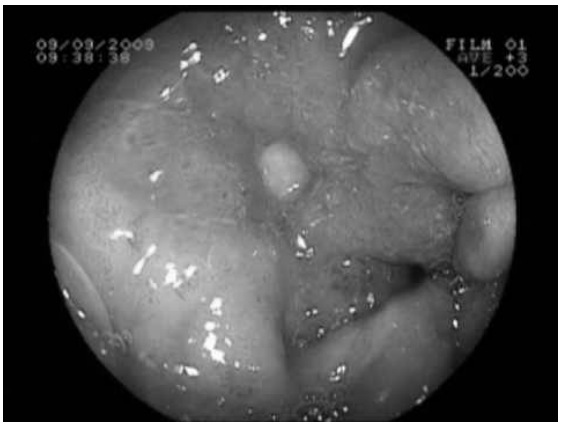
Systemic GC bone loss	adverse skeletal effects **within 3 months** · at doses **as low as 2.5–5.0 mg/day** prednisone-equivalent · dose-dependent
High fracture risk on long-term oral GC	✓ give **bisphosphonate** (antiresorptive) + calcium + vitamin D
After stopping GC	fracture risk normalises unless previously high-dose
Epidural steroid injection	✓ **↓ lumbar BMD** (esp. postmenopausal women) · knee/other large-joint injections do not ↓ lumbar BMD
Other GC effects	transient **hyperglycaemia** (caution in diabetics) · interfere with RANKL pathway

6 Complications

NSAID side effects & fracture healing

Most strongly associated	✓ **renal impairment** (all NSAIDs, incl. COX-2) ✗ Wrong options: hepatic · QTc · seizures · haematuria
GI	dyspepsia · ulcer perforation/bleeding/obstruction **2–4%** of chronic users · elderly **4–5×** risk of death from GI haemorrhage
GI mechanism	**non-specific COX-1 + COX-2 inhibition → ↓ thromboxane + ↓ prostaglandin** → uncoupled mitochondrial oxidative phosphorylation, ↑ permeability, **disruption of intestinal phospholipid bilayer**
GI risk factors	**concurrent anticoagulant (most important)** · age >60 · prior GI disorder
Other	**↑ bleeding time** · **↑ creatinine** · **tinnitus**/sensorineural hearing loss (reversible) · cardiac toxicity · NOT ↑ IL-6 (NSAIDs ↓ IL-6)
Indomethacin for HO	most likely side effect = ✓ **GI ulceration**

– – Active space – –



Peptic ulcer from NSAID use

Fracture healing

Mechanism	↓ COX-2 → **↓ PGE2** → ↓ mesenchymal stem cell → osteoblast/ chondrocyte differentiation (PGE2 acts via RANKL/OPG balance)
Effect	✓ **↓ endochondral ossification** → ↓ callus → delayed union/nonunion
Which healing	**secondary (callus) > primary**
Risk ↑	prolonged use · large doses · **adults** · no effect shown in **children**
COX-2	thought **necessary for normal endochondral ossification** (animal data) · not "definitively shown in human studies"

Intra-articular corticosteroid injection complications

Described	local flare · fat atrophy · skin pigmentation (depigmentation) · facial flushing · injection-site pain, bleeding
Not described	**apoptosis of myocytes**

X vs Y

	Steroids	**NSAIDs**
Enzyme inhibited	**Phospholipase A2**	**COX-1/COX-2**
Level in cascade	Earliest (no arachidonic acid)	Arachidonic acid → prostanoids
Leukotrienes	↓	Unaffected (5-LOX intact)

	COX-1	**COX-2**
Expression	Constitutive	Inducible (inflammation)
Function	Gastric protection, platelets, renal flow	Pain, fever, inflammation, **endochondral bone**
Selective drug	Ketorolac (OB)	**Celecoxib**
Inhibiting it causes	GI ulcer, bleeding	↓ fracture healing, cardiac events

– – Active space – –

How it's asked

"Arachidonic acid is directly metabolised by?"	→ **cyclooxygenase** ✗ Wrong options: 1-lipoxygenase · HMG-CoA reductase · carbonic anhydrase · thromboxane synthetase
"All bind COX-1 reversibly EXCEPT?"	→ **aspirin** (irreversible) ✗ Wrong options: meloxicam, diclofenac, indomethacin, naproxen
"Mechanism of aspirin?"	→ **binding of COX-1 and COX-2** ✗ Wrong options: vitamin K carboxylation · Xa via antithrombin · direct thrombin inhibition
"Chronic ibuprofen → GI ulcer mechanism?"	→ **non-specific COX-1 + COX-2 inhibition → ↓ thromboxane + ↓ PG → phospholipid bilayer disruption**
"Indomethacin for HO prophylaxis — likely side effect?"	→ **GI ulceration**
"Side effect most strongly associated with NSAIDs?"	→ **renal impairment**
"NOT a side effect of prolonged NSAIDs?"	→ **↑ circulating IL-6** ✗ Wrong options: ulcer · ↑ creatinine · ↑ bleeding time · tinnitus
"True about COX-2?"	→ **necessary for normal endochondral ossification** ✗ Wrong options: constitutive form · more gastric irritation · proven in humans
"True about NSAIDs in femoral nailing?"	→ **prolonged use ↓ endochondral ossification** ✗ Wrong options: inhibits PLA2 · COX-1 selectivity ↓ GI effects · adults lower risk than children · primary > secondary healing
"How do NSAIDs impair bone healing?"	→ **↓ PGE2 and COX-mediated osteoblastic MSC differentiation** ✗ Wrong options: ↑ osteoclastogenesis · ↑ COX-2
"Corticosteroids inhibit?"	→ **phospholipase A2** ✗ Wrong options: lipoxygenase · COX-1 · TXA2 · prostacyclin
"NOT a complication of corticosteroid injection?"	→ **apoptosis of myocytes**
"Most accurate about corticosteroid complications?"	→ **epidural steroid injection ↓ lumbar BMD** ✗ Wrong options: no effect <1 yr · bisphosphonate contraindicated · 2.5–5 mg safe · knee injection ↓ lumbar BMD

– – Active space – –

ANTIBIOTIC CLASSIFICATION & MECHANISM

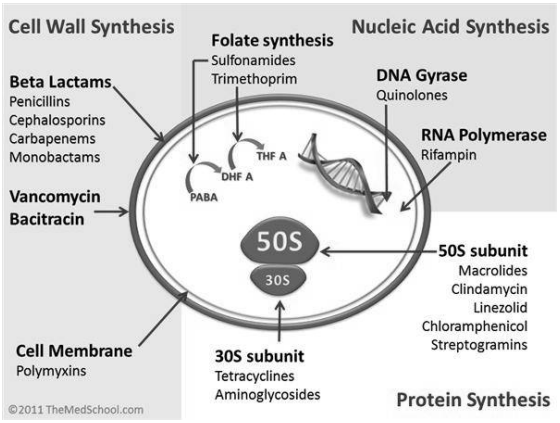
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1 Basics

Mechanism of action by class

Five basic mechanisms

cell-wall synthesis · membrane permeability · ribosomal (protein) inhibition · DNA metabolism · antimetabolite (folate)



Antibiotic mechanisms of action

– – Active space – –

Target	Drugs	Cidal / static
Cell wall (transpeptidase / PBP)	**Penicillins, cephalosporins**, carbapenems (imipenem), aztreonam, β-lactamase inhibitors	Bactericidal
Cell wall – D-Ala-D-Ala	**Vancomycin** (binds D-Ala-D-Ala terminal → blocks transpeptidation, peptidoglycan cross-linking)	Bactericidal
Membrane permeability	Bacitracin · polymyxin	–
Membrane depolarisation	**Daptomycin**	Bactericidal
Fungal membrane sterols	**Amphotericin, nystatin** – bind sterols → disrupt membrane	–
30S ribosome	**Aminoglycosides** (gentamicin, tobramycin – **irreversible**) · **tetracyclines** (doxycycline – block tRNA binding)	AG cidal · tetracycline static
50S ribosome	**Macrolides** (erythromycin – reversible, blocks aminoacyl translocation) · **clindamycin** · chloramphenicol (peptidyl transferase) · streptogramins	Static
50S 23S rRNA – initiation complex	**Linezolid** (oxazolidinone) – prevents 30S–50S (70S) initiation complex	Static vs staph/ enterococci · cidal vs strep
DNA gyrase (type II topoisomerase)	**Fluoroquinolones** (cipro, levo, moxi)	Bactericidal
DNA (metabolic by-products)	Metronidazole	Bactericidal
RNA polymerase (DNA-dependent)	**Rifampin**	Bactericidal
Mycolic acid	Isoniazid	–
Folic acid (PABA mimic)	**TMP/SMX**, sulfonamides	Static

Classification & indications

– – Active space – –

Class	Drugs	Drug of choice / coverage	Toxicity
Natural penicillin	Penicillin G, V	**Strep pyogenes (A), Strep agalactiae (B), C. perfringens**	Hypersensitivity, haemolytic anaemia
Aminopenicillins	Ampicillin, amoxicillin	+ **E. coli**, E. faecalis	As above
Penicillinase-resistant	Methicillin, **nafcillin, oxacillin**, cloxacillin	+ PCNase-producing **S. aureus**	+ **Interstitial nephritis**
Antipseudomonal penicillins	Piperacillin, ticarcillin	+ **Pseudomonas**	–
1st-gen cephalosporin	**Cefazolin**, cephalexin	**S. aureus, S. epidermidis**, some GN (E. coli, Klebsiella)	Allergy · **Coombs + anaemia (3%)**
2nd-gen	Cefoxitin, cefaclor, cefuroxime	+ GN	Disulfiram reaction
3rd-gen (4th = cefepime)	Ceftriaxone, ceftazidime	+ GN, **Pseudomonas**	Disulfiram reaction
Vancomycin	–	**MRSA**, PCN/ceph allergy, S. epidermidis	**Red man syndrome**, nephro-, ototoxicity
Carbapenems	Imipenem (+ cilastatin), meropenem	**Broadest spectrum** (✗ Wrong options: MRSA, Mycoplasma)	–
Aztreonam	–	GN rods, aerobes, hospital-acquired	–

– – Active space – –

Class (non-β-lactam)	Drugs	Drug of choice / coverage	Toxicity
Aminoglycosides	Gentamicin, tobramycin, amikacin	**Aerobic GN**, Pseudomonas · with cephalosporin in **open fractures**	**Nephrotoxicity, ototoxicity**
Tetracyclines	Doxycycline, minocycline	Rickettsia, Mycoplasma, **Lyme**	**Tooth/bone staining** · avoid <12 yr
Macrolides	Erythromycin, azithro	Strep, H. influenzae, Mycoplasma	**Warfarin interaction (P450)**
Chloramphenicol	–	Meningitis, brain abscess	Aplastic anaemia, grey baby
Clindamycin	–	B. fragilis, S. aureus · **excellent bone penetration**	**Pseudomembranous colitis**
Linezolid	–	Resistant G+ (**MRSA, VRE**)	**Serotonin syndrome** (MAOI)
Streptogramins	Quinupristin/ dalfopristin	VRE	–
Fluoroquinolones	Cipro (+ Pseudomonas), levo, moxi (+ anaerobes)	Equivalent IV/PO · osteomyelitis, PJI	Phototoxicity · **Achilles rupture** · **↓ fracture healing**
Metronidazole	–	**Anaerobes**	Seizures, cerebellar, disulfiram
Rifampin	–	**Staph** (intracellular), **TB**	Orange body fluids · hepatotoxic (with INH)
TMP/SMX	–	UTI, Proteus	Thrombocytopenia · avoid 3rd trimester

Splenectomy

Required pneumococcal · **Hib** · meningococcal C · influenza vaccines · **lifelong oral penicillin V or erythromycin**

2 Mechanism

– – Active space – –

Resistance (mecA, biofilm, efflux, D-zone test)

Three basic mechanisms (OB) **avoidance** · decreased susceptibility · inactivation

Mechanism	Confers resistance to	Example
β-lactamase (hydrolysis)	Penicillin	E. coli, S. epidermidis, Pseudomonas, Klebsiella → add clavulanate (Augmentin)
Genetic mutation – mecA (on SCCmec) → **PBP2a** (low β-lactam affinity)	**Methicillin, oxacillin, nafcillin**, cephalosporins	**MRSA** · SCCmec type IV = **CA-MRSA** (less multidrug-resistant)
Altered cell-wall permeability	Tetracyclines, quinolones, trimethoprim, β-lactams	–
Biofilm (glycocalyx) = **avoidance**	Adherence to implants, resists phagocytosis	**S. epidermidis**, Salmonella
Active efflux pumps	**Erythromycin, tetracycline**	msrA gene (Staph)
Altered peptidoglycan (D-Ala-D-Ala)	**Vancomycin**	**VRE**
Ribosome alteration – erm gene methylates 23S rRNA	Inducible **MLS** (macrolide–lincosamide–streptogramin)	**D-zone test**
Altered DNA gyrase	Quinolones	–
Altered RNA polymerase	Rifampin	–
Altered membrane-binding protein	β-lactams	–

Tetracycline resistance (3) **increased drug efflux** · ribosome protection · tetracycline modification

Biofilm biology forms when bacterial mass sufficient → **quorum sensing** → stationary/reduced growth phase · **high-shear** environments stimulate it

MRSA options (don't bind PBPs) – **vancomycin, TMP/SMX, clindamycin, linezolid**

4 Diagnosis & Classification

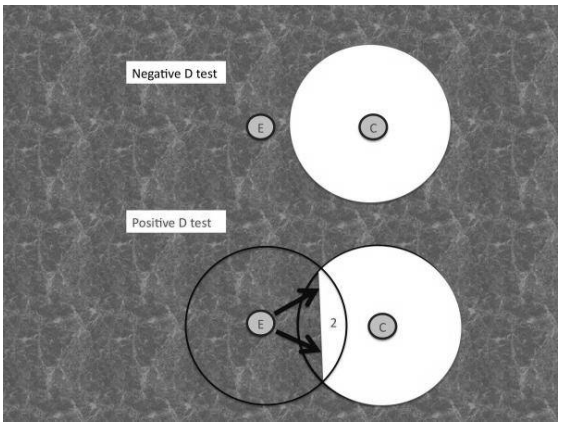
D-zone test

Indication **erythromycin-resistant MRSA** (or β-haemolytic strep) **before starting clindamycin**

Basis plasmid alters **50S** binding site for both clindamycin + erythromycin → **inducible clindamycin resistance**

Method erythromycin disc next to clindamycin disc → **D-shaped zone of inhibition = positive → don't use clindamycin**

– – Active space – –



D-zone test – flattening next to the erythromycin disc

Test	Use
D-zone	✓ **Inducible clindamycin resistance**
Etest	MIC
CAMP	Group B strep
Ziehl–Neelsen	Acid-fast (mycobacteria)
Oxidase	Cytochrome c oxidase +/–
Phenylalanine deaminase	Proteus
Citrate / decarboxylase	Enterobacteriaceae
β-lactamase	Penicillin resistance

5 Management

Clinical use (rifampin, antibiotic cement, dental prophylaxis)

Rifampin	**lipophilic → high cell penetration** → effective vs **intracellular phagocytised S. aureus** (in macrophages) · always combined (synergy)
Vancomycin indications	serious β-lactam allergy · MRSA/MRSE infection · prosthesis prophylaxis where **MRSA/MRSE rate is high**
Vancomycin coverage	G+ only · no GN bacilli (too large) · time-dependent killing
Antibiotic cement (PMMA)	max **5% of cement weight** without ↓ mechanical strength → **2 g per 40 g** · commercial = **1 g / 40 g** · add as **powder, never liquid**
Antibiotic nail for septic tibial nonunion	PMMA + **vancomycin (G+) + tobramycin (GN)**
Dental prophylaxis after TJA	only **high-risk**: immunocompromised (transplant, AIDS, inflammatory arthritis, chemo) · **prior PJI** · **uncontrolled diabetes**
Regimen	**cephalexin / amoxicillin / cephradine 2 g PO within 1 h** ✗ Wrong options: routine IV

6 Complications

Adverse effects & contraindications

Quinolones	**inhibit early fracture healing** (toxic to **chondrocytes** → immature callus, more cartilage) · **Achilles tendinitis/rupture (RR 3.7)**
Tetracycline	✓ **contraindicated in children** (stains teeth + bone) ✗ Wrong options: penicillin, cephalexin, erythromycin, ampicillin
Linezolid	reversible non-selective **MAOI → serotonin syndrome** (esp. with SSRIs): headache, diaphoresis, tachycardia, hypertension, tremor, **clonus**, unreactive pupils → stop drug + **IV benzodiazepines** · severe → serotonin antagonist
Vancomycin	flushing = **red man syndrome**

X vs Y

	Aminoglycoside (gentamicin/ tobramycin)	Tetracycline (doxycycline)	Macrolide (erythromycin)	Clindamycin	Linezolid
Subunit	**30S**	**30S**	**50S**	**50S**	**50S (23S rRNA)**
Binding	**Irreversible**	Blocks tRNA	Reversible	Translocation / peptidyl transferase	Blocks **initiation complex**
Action	Cidal	Static	Static	Static (cidal high dose)	Variable

	Vancomycin	Tobramycin
Target	Cell wall (D-Ala-D-Ala)	30S ribosome (irreversible)
Spectrum	**Gram-positive**	**Gram-negative**

How it's asked

"Mechanism of vancomycin?"	→ **inhibition of cell-wall synthesis** ✗ Wrong options: permeability · ribosome · DNA · antimetabolite
"All inhibit ribosomes EXCEPT?"	→ **vancomycin** ✗ Wrong options: gentamicin, tobramycin, erythromycin, linezolid
"All interfere with cell-wall synthesis EXCEPT?"	→ **gentamicin** ✗ Wrong options: cefazolin, penicillin G, vancomycin, imipenem
"Family that inhibits DNA gyrase?"	→ **quinolones** ✗ Wrong options: penicillins, aminoglycosides, macrolides, sulfonamides

– – Active space – –

"Binds the 30S subunit?"	→ **aminoglycosides** ✗ Wrong options: macrolides (50S) · rifampin (RNA polymerase)
"Linezolid (oxazolidinone) targets?"	→ **ribosomal activity** ✗ Wrong options: DNA-dependent RNA polymerase (rifampin) · membrane (daptomycin)
"Cephalosporins block transpeptidase → effect?"	→ **inhibit cell-wall synthesis**
"Fluoroquinolones inhibit?"	→ **DNA gyrase** ✗ Wrong options: RNA gyrase · DNA polymerase · telomerase
"Amphotericin acts by?"	→ **binding sterols, disrupting the cell membrane**
"Tobramycin in an antibiotic nail — mechanism?"	→ **irreversibly binds 30S** ✗ Wrong options: reversible · 50S · D-Ala-D-Ala = vancomycin
"Correct class–mechanism pairing?"	→ **fluoroquinolones – DNA gyrase**
"Prevents 30S–50S initiation complex?"	→ **linezolid** ✗ Wrong options: gentamicin · doxycycline · chloramphenicol · erythromycin
"Binds the 50S subunit?"	→ **clindamycin** ✗ Wrong options: cefazolin, cipro, doxycycline, vancomycin
"Binds D-Ala-D-Ala to block transpeptidation?"	→ **vancomycin** ✗ Wrong options: daptomycin (membrane depolarisation)
"Species producing mecA?"	→ **MRSA**
"How does S. epidermidis adhere and resist phagocytosis?"	→ **biofilm**
"MRSA resistance to penicillinase-stable antibiotics via?"	→ **genetic mutation (mecA → PBP2a)** ✗ Wrong options: efflux · biofilm · permeability · peptidoglycan
"Tetracycline resistance: ribosome protection, modification and?"	→ **increased drug efflux**
"Biofilm resistance is an example of?"	→ **avoidance**
"Erythromycin-resistant MRSA, want clindamycin — test?"	→ **D-zone test** ✗ Wrong options: oxidase, CAMP, Ziehl–Neelsen, Etest
"MSSA abscess → recurrence now methicillin-resistant — mechanism?"	→ **mecA gene mutation**
"Antibiotic toxic to chondrocytes, inhibits early fracture healing?"	→ **quinolones**
"Contraindicated in children?"	→ **tetracycline**

"Linezolid → clonus, tachycardia, hypertension — next step?"	→ **IV benzodiazepines (serotonin syndrome)**
"Why rifampin kills intracellular staph?"	→ **high cell penetration** (lipophilic)
"Max vancomycin in 40 g PMMA?"	→ **2 g** (5% by weight)
"TKA 8 mo ago, A1c 9.0, gingival dental procedure — prophylaxis?"	→ **cephalexin 2 g PO within 1 h** ✗ Wrong options: IV daptomycin · none · delay · repeat A1c

– – Active space – –

– – Active space – –

ANTICOAGULATION MEDICATIONS

– Basic Science · 7 / 108 –

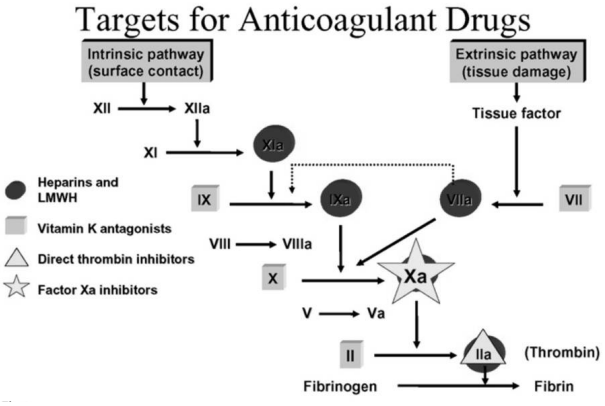
1 Basics

Coagulation & Virchow's triad

Cascade series of reactions → **fibrin** → platelet activation → clot · imbalance → thromboembolism / DVT

Virchow's triad **venous stasis** · **endothelial damage** · **hypercoagulable state**

Factor Ia = **fibrin** · broken down by **plasmin** (from plasminogen)



Targets for anticoagulant drugs

Overview of anticoagulants

Agent	Mechanism	Metabolism	Reversal	Advantage / risk
Compression stocking	Mechanical	–	–	No bleeding · compliance
Aspirin	**Irreversible COX** acetylation in **platelets** → ↓ TXA2, PG	Renal	None (acts at platelet)	Convenient
Unfractionated heparin	**Activates ATIII** → inhibits **IIa, III, Xa**	**Hepatic**	**Protamine**	Reversible · **highest bleeding + HIT**
LMWH (enoxaparin, dalteparin)	ATIII → principally **Xa** (also IIa)	**Renal**	Protamine (**partial** – anti-Xa not normalised)	Fixed dose, no monitoring · t½ **4–5 h**
Fondaparinux (Arixtra)	**Indirect Xa** via ATIII · synthetic pentasaccharide · no IIa effect	**Renal**	–	No monitoring · ↓ DVT vs enoxaparin (hip fx, TKA) · **highest bleeding** · avoid with epidurals · t½ **17–21 h**
Warfarin	**Vitamin K 2,3-epoxide reductase (VKORC1)** → ↓ γ-carboxylation of **II, VII, IX, X, C, S**	**Hepatic**	**Vit K** (≤3 days) · **FFP** (immediate)	**Most effective** · hard to reverse/dose
Dextran	Dilutional · coats RBC/ platelets/endothelium (↑ electronegativity) · ↓ vWF/VIII-Ag	–	–	Fluid overload
Rivaroxaban, apixaban, edoxaban	**Direct Xa** ("-xa-ban")	Liver (major) + kidney, gut	Andexanet alfa (see below)	**Oral** · peak **2–4 h**
Dabigatran (Pradaxa)	**Direct thrombin (IIa)**, reversible	**Renal**	**Idarucizumab** (FDA 2015)	GI upset, bleeding
Hirudin (leech), bivalirudin	Direct thrombin	–	–	Hirudin = most potent thrombin inhibitor
Clopidogrel	**ADP receptor** on platelets	–	–	–

5 Management

Warfarin mechanism & pharmacology

Mechanism ✓ **inhibits vitamin K-dependent clotting factors** · blocks reduction of vit K epoxide → active vit K (hydroquinone) · **inhibits carboxylation**

Factors **II (prothrombin), VII (first affected), IX, X** + proteins **C and S** · not ATIII, VIII, Xa directly

Alcoholic / liver disease caution due to **factor VII** (narrow margin)

Pharmacokinetics oral, rapid GI absorption · **affected by liver function + diet** · plasma **t½ 36–42 h** · therapeutic in **72–96 h** · INR not affected for **2–3 days**

– – Active space – –

Target INR	**2–3** for orthopaedic patients ✗ Wrong options: 2.5–3.5
Elective surgery	stop **3–5 days** before
Drug interactions	rifampin · phenobarbital · diuretics · cholestyramine

Heparin, LMWH & fondaparinux

Activates ATIII	✓ **heparin** ✗ Wrong options: warfarin, aspirin, rivaroxaban, dabigatran
Heparin inhibits IIa (thrombin)	✓ heparin ✗ Wrong options: fondaparinux (Xa only) · rivaroxaban (Xa only) · TXA
Enoxaparin	primarily **Xa** (antithrombin complex irreversibly inactivates Xa) · protamine less effective than for UFH
Indirect Xa inhibitor	✓ **fondaparinux** ✗ Wrong options: rivaroxaban, apixaban, edoxaban (direct) · dabigatran (direct IIa)
Protamine reverses	✓ **LMWH** (partially), UFH ✗ Wrong options: aspirin, clopidogrel (no medical reversal) · warfarin (vit K)
Highest bleeding + thrombocytopenia	✓ **unfractionated heparin** (**HIT** – antibodies activate platelets → intravascular clot) · rare with LMWH ✗ Wrong options: coumadin

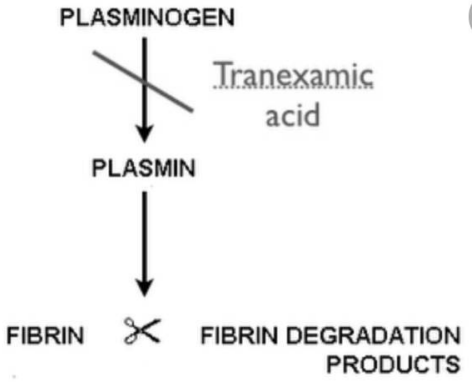
Factor Xa inhibitors

Direct Xa	✓ **rivaroxaban (Xarelto)**, apixaban, edoxaban · antithrombin-independent ✗ Wrong options: fondaparinux (indirect)
Class facts	**oral** · **max effect 2–4 h** · predictable → **no routine monitoring** · **low** food/drug interaction · excretion liver (major) + kidneys + intestine
Half-life	rivaroxaban **8 h** · apixaban **12 h** · urgent surgery delayed until a half-life has elapsed
Reversal	Orthobullets: **no current antidote, andexanet alfa under investigation** · OB: **FDA-approved coagulation factor Xa (recombinant) inactivated-zhzo** for life-threatening bleeding – read the options

Tranexamic acid

Mechanism	synthetic **lysine** analogue · **competitively inhibits plasminogen activation** (binds lysine-binding site) · non-competitive plasmin inhibitor at high dose · ↓ **active plasmin** → ↓ fibrinolysis
Potency	**8–10×** ε-aminocaproic acid

– – Active space – –



TXA blocks plasminogen → plasmin

Route	Fact
IV	10–20 mg/kg bolus ± repeat every 3 h ×1–4 or infusion 1–10 mg/kg/h
Topical	**As effective as IV** · sprayed on wound at closure · **no detectable serum level**
Oral	**As effective as IV** · 1.95 g **2 h preop** · **cheaper**

Kinetics <5% metabolised · joint-fluid t½ **3 h** · in tissues up to **17 h**

Safety **no ↑ thromboembolic events** (no route difference in DVT) · avoid with **cardiac stents, prior VTE/ischaemic stroke**

Jehovah's Witness THA TXA ↓ active plasmin
✗ Wrong options: epoetin alfa (↑ Hb preop) · dexamethasone (pain) · celecoxib

Antiplatelet effects & supplements

Aspirin's target cell ✓ **platelets** (irreversibly blocks TXA2)

Supplement	Effect
Ginkgo (ginkgolide B displaces PAF)	↑ bleeding – **platelets**
Ginseng (panaxynol)	↑ bleeding – platelets · hypoglycaemia
Garlic (aged extract – ↑ cyclic nucleotides, ↓ fibrinogen binding)	↑ bleeding – platelets
Omega-3 fish oil	↑ INR (platelets + vit K factors)
St John's wort	↓ INR (↑ warfarin catabolism) · CNS reuptake inhibition · **not a platelet effect**
Coenzyme Q10 · green tea	↓ INR (green tea = direct warfarin antagonist)
Ephedra	Cardiovascular instability
Kava, valerian	↑ sedative effect of anaesthetics

VTE prophylaxis guidelines

CRISTAL (9,711 THA/TKA) symptomatic VTE **aspirin 3.45% vs LMWH 1.8%** → aspirin **failed non-inferiority** (1% margin) · **no difference in mortality**, joint readmission, major bleeding

ACCP mechanical + pharmacological · minimum **10–14 days**, ideally up to **35 days** · LMWH preferred, aspirin acceptable

– – Active space – –

Aspirin dose — **81 mg BID** = efficacy of 325 mg BID with **less GI** · <4 weeks → ↑ mortality + bleeding

No prophylaxis — DVT >**50%** after TKA → mechanical alone not recommended

Weight-based LMWH — = therapeutic dose → ↑ bleeding/wound drainage (avoid without confirmed VTE)

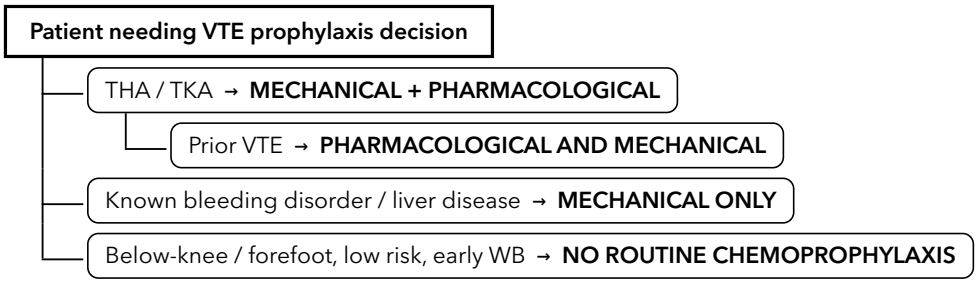
AAOS VTED guideline (elective THA/TKA)	
Postop duplex screening	**Not recommended**
Only risk factor clearly supported for VTE	**Previous VTE**
Only bleeding risk factors clearly supported	**Known bleeding disorder, active liver disease**
Antiplatelet agents	**Stop preop**
All patients	Pharmacological and/or mechanical
Previous VTE	Pharmacological **AND** mechanical
Bleeding disorder / liver disease	**Mechanical**
Anaesthesia	**Neuraxial** (spinal/epidural)
Mobilisation	Early
IVC filter	Not clearly supported
Aspirin	✓ **Included** as a prevention option

Operative criteria – when chemoprophylaxis is not mandatory

Below-knee injury, no risk factors — e.g. healthy 37-yr-old, isolated rotational ankle ORIF → **not absolutely indicated** (DVT rate = placebo)

Elective forefoot/ midfoot surgery, low risk, immediate WB — ✓ **no routine chemical prophylaxis** (ICM-VTE Foot & Ankle) · NWB = independent risk, mitigated by **>50% load-bearing**

Risk factors that mandate it — older age · obesity · prior DVT · **malignancy** · OCP · smoking · non-ambulatory · SCI / axial injury · hip fracture · THA



X vs Y

	Direct	**Indirect (via ATIII)**
Xa	**Rivaroxaban, apixaban, edoxaban**	**Fondaparinux** (Xa only), LMWH (mainly Xa)
IIa	**Dabigatran**, bivalirudin, hirudin	**UFH** (IIa + Xa)

	Protamine	**Vitamin K / FFP**	**Idarucizumab**	**Andexanet alfa**
Reverses	UFH, LMWH (partial)	Warfarin	Dabigatran	Xa inhibitors

How it's asked

– – Active space – –

"Mechanism of warfarin?"	→ **inhibits vitamin K-dependent clotting factors** ✗ Wrong options: platelets · COX · ↓ ATIII
"Inhibits carboxylation of clotting factors?"	→ **warfarin** ✗ Wrong options: enoxaparin, dalteparin, heparin, hirudin
"Prevents γ-carboxylation of X, IX and prothrombin?"	→ **warfarin** ✗ Wrong options: clopidogrel (ADP receptor)
"Alcoholic, venogram DVT – warfarin caution due to factor?"	→ **VII**
"Specific target of warfarin?"	→ **vitamin K 2,3-epoxide reductase** ✗ Wrong options: ATIII · protein C · VIII · Xa
"Plasma half-life of warfarin?"	→ **36–42 h**
"DVT agent affected by liver function and diet?"	→ **warfarin** ✗ Wrong options: LMWH, fondaparinux (renal)
"Protamine reverses?"	→ **LMWH**
"Highest risk of bleeding and thrombocytopenia?"	→ **unfractionated heparin** (HIT)
"Activates antithrombin III?"	→ **heparin**
"Inhibits the circled factor IIa?"	→ **heparin** ✗ Wrong options: fondaparinux · rivaroxaban (Xa only)
"Where does enoxaparin act?"	→ **factor Xa**
"Indirect inhibitor of factor X?"	→ **fondaparinux**
"Direct factor Xa inhibitor?"	→ **rivaroxaban** ✗ Wrong options: fondaparinux (indirect) · dextran · coumadin
"True of Xarelto class?"	→ **maximal effect 2–4 h** ✗ Wrong options: food interactions · IV/SC only · monitoring · mainly renal
"Competitive inhibition of plasminogen activation?"	→ **tranexamic acid**
"Inhibits activation of the enzyme that breaks down factor Ia?"	→ **TXA**
"TXA in TKA – true?"	→ **similar blood-loss reduction IV vs oral** ✗ Wrong options: IV better · topical better · ↑ DVT
"Jehovah's Witness THA – drug ↓ active plasmin?"	→ **TXA**

– – Active space – –

"Supplement affecting clotting via platelets?"	→ **ginkgo** ✗ Wrong options: vitamin D, ephedra, St John's wort, selenium
"↑ bleeding by direct platelet effect EXCEPT?"	→ **St John's wort**
"Aspirin's primary cellular target (image)?"	→ **platelets**
"LMWH vs aspirin after THA/TKA — true?"	→ **no difference in mortality** ✗ Wrong options: aspirin non-inferior · aspirin fewer DVT · LMWH more bleeding/readmission
"AAOS VTED recommendations EXCEPT?"	→ **"aspirin not approved in low-risk patients"** (false — aspirin is included)
"Uncomplicated TKA, BMI 39 — optimal prophylaxis?"	→ **aspirin 81 mg BID for 6 weeks** ✗ Wrong options: 325 mg × 2 wk · weight-based LMWH · INR 2.5–3.5 · SCDs only
"DVT prophylaxis NOT absolutely indicated?"	→ **healthy 37-yr-old, isolated ankle ORIF**
"Healthy 49-yr-old, elective bunion surgery, immediate heel WB — VTE strategy?"	→ **no routine chemical prophylaxis**

ARTICULAR CARTILAGE

– Basic Science · 8 / 108 –

1 Basics

Forms of cartilage

Form	Example
Hyaline / articular	Joint surfaces
Fibroelastic	Meniscus
Fibrocartilage	Tendon/ligament insertion into bone
Elastic	Trachea
Physeal	Growth plate

Composition & function (aggrecan, water, type II collagen)

Function ↓ friction · distributes load · **stress-shielding of the solid matrix** (high water, incompressible water, PG–collagen organisation) · **anisotropic**
✗ Wrong options: isotropic

% by weight **water > collagen > proteoglycan > non-collagenous protein > cells**

Component	Amount	Role
Water	**65–80%** of wet weight · **80% superficial, 65% deep**	Load-dependent deformation · ↓ with age · **↑ in OA**
Collagen	10–20% · **type II = 90–95%** (minor V, VI, IX, X, XI)	**Tensile strength** · framework · **resists swelling** from aggrecan osmotic pressure
Proteoglycans	10–15%	**Compressive strength** · attract water · fluid/electrolyte balance
Chondrocytes	**~2% of volume** (OB) · "<10%" / "up to 5%" (TO)	Make collagen, PG, enzymes · only cell in matrix

Type I collagen gives tensile strength **false (TO)** · it is **type II**

Type X collagen **calcified cartilage** (mineralisation) · ↑ in **early OA** at subchondral interface (strain >2.5% induces Col X mRNA)

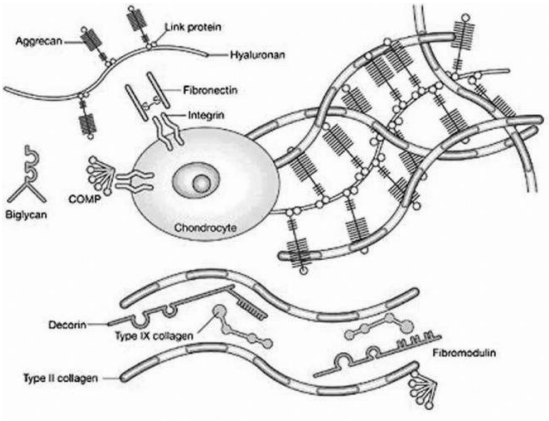
Type IV collagen **basal lamina**
✗ Wrong options: calcified cartilage

Proteoglycans

Aggrecan **most responsible for hydrophilic behaviour** · GAGs (**chondroitin sulfate, keratan sulfate**) on a core protein → aggrecan → **link protein** binds it to **hyaluronic acid → proteoglycan aggregate**

Negative charge sulfated GAG chains → osmotic swelling pressure · holds **50× its weight** in water · swelling restrained ~20% by type II collagen · MW ~1,000,000
✗ Wrong options: positive

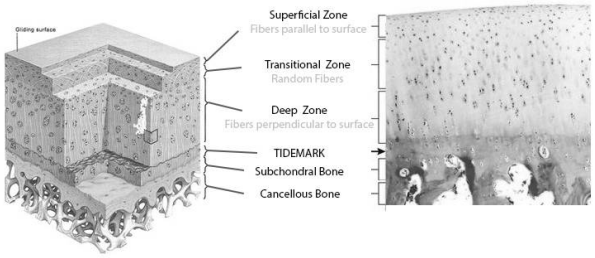
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Aggrecan aggregate on hyaluronic acid

Molecule	Role
Aggrecan	✓ **Hydrophilicity, swelling pressure**
Decorin, biglycan, fibromodulin	Small PGs · bind type II collagen · matrix stabilisation (decorin organises collagen)
Versican	Cell-surface PG
Matrilin-1	Cartilage tissue organisation
Type IX collagen	Stability · links matrix molecules to type II
COMP	Regulates chondrocyte proliferation
Osteocalcin	Most prevalent non-collagenous protein of **bone**

Structure & zones



Zones of articular cartilage

Zone	Collagen	Chondrocytes	Content
Lamina splendens	Cell-free, random flat fibril bundles	None	Surface layer
Superficial (tangential, zone 1)	**Parallel** to surface (resists shear)	**Flattened**, parallel to tidemark	**Highest collagen, lowest PG**, high water · thinnest · **only zone with progenitor cells**
Transitional / intermediate (zone 2)	Oblique / **random**, larger fibres	Round, active	**Thickest** · more PG
Deep / radial / basal (zone 3)	**Perpendicular**, **crosses tidemark**, largest fibrils	**Round, in columns**	**Highest PG, lowest water** (lowest collagen)
Tidemark	Basophilic line	–	Boundary **uncalcified / calcified** cartilage
Calcified (zone 4)	Type X	Small, metabolically inactive	Anchors to subchondral bone

Tidemark (TO) lies between **radial (deep) and calcified zone** · part of zone 4
✗ Wrong options: calcified/subchondral · superficial/transitional

Tidemark (Orthobullets) separates true articular cartilage from remnant of **cartilage anlage** · divides nutritional sources · found only in joints, most prominent in adults

Water vs depth (TO) as PG ↑ from superficial → deep, **water ↓**

Nourishment, growth factors

Avascular fed by **synovial fluid** (surface) + **subchondral bone** (base) · ATP by **glycolysis**

Immature cartilage has stem cells · mature does not

Growth factor	Effect
TGF-β1	**Most investigated** · ↑ PG + ECM synthesis · ↓ IL-1 + MMP catabolism · **synovial proliferation + fibrosis** · osteophytes
b-FGF (FGF-2)	↑ chondrocyte DNA synthesis, PG · but ↑ inflammation, osteophytes · no defect healing
IGF-1	↑ DNA + matrix synthesis (adult) · ↓ synovial inflammation · **additive with TGF-β** · ↓ in OA
PDGF	Laceration healing (from PRP data) · **no adverse or positive effect in vivo**
IL-1	Catabolic · ↑ **MMP**, COX-2, nitric oxide synthase
Glucosamine	Anabolic · prevents IL-1β-induced ↓ PG synthesis

– – Active space – –

2 Mechanism

Response to loading & mechanotransduction

Stimulates matrix synthesis	Suppresses synthesis, ↑ chondrolysis
Cyclic stress **1–5 MPa**	Excess **>5 MPa**
Moderate frequency **0.1–1 Hz**	**Static** <0.01 Hz
Low rate <1000 MPa/s	High rate >1000 MPa/s

Mechanosensors — **primary cilia** (chondrocytes + osteoblasts) · transduction via **integrins**

Marathon runner, cartilage 2× thick — ✓ **chondrocyte modulation via mechanotransduction**

Moderate running — ↑ thickness · ✓ **↑ GAG** production (T1 gadolinium MRI) · lateral femoral GAG > medial

Strenuous loading — thinning + PG loss

Immobilisation / NWB — thinning, softening, ↓ PG · **only in the off-loaded limb**
✗ Wrong options: contralateral hypertrophy

Lubrication (lubricin, boundary, fluid film)

Type	Mechanism
Elastohydrodynamic	**Main** during dynamic function · elastic deformation + thin fluid film
Boundary	Non-deformable surfaces, lubricant partially separates · **lubricin** (superficial zone protein) + HA
Boosted	Fluid trapped in pools between contact regions
Hydrodynamic	Fluid separates sliding surfaces
Weeping	Fluid exuded under load → hydrostatic separation
Squeeze film	Fluid slowly squeezed out from between surfaces

Lubricin — mucinous glycoprotein · binds **hyaluronic acid → boundary** layer
✗ Wrong options: vitronectin (doesn't bind HA)

Fluid-film in MoM THA — ↑ by **larger head**, slight **radial clearance** (most important), low roughness · **complete congruence prevents fluid film**

Wear mechanisms — adhesion · abrasion · transfer · fatigue · third body

4 Diagnosis & Classification

Aging vs osteoarthritis changes

	Aging	Osteoarthritis
Water	↓	↑ (→ ↑ permeability, ↓ strength)
Modulus / stiffness	↑ (less elastic, brittle)	↓ (more elastic) · ↓ tensile + shear + compressive modulus
Chondrocytes	Fewer but larger (hypocellular)	Clusters / clones (stage 2)
GAG ratio	↑ **keratan sulfate : chondroitin** (KS ↑ until ~30 yr, CS ↓; C4S → C6S)	↑ **chondroitin 4-sulfate : keratan sulfate**
Proteoglycans	↓ synthesis, ↓ size · ↑ decorin	↑ **degradation** · unbound from HA · **aggrecan ↑ early, ↓ late**
Collagen……101	↑ **cross-linking** (passive glycation)	**Disorganised** (↑ collagenase)
AGEs	↑ (→ ↑ stiffness, brittleness, fatigue failure)	AGE accumulation thought to lead to knee/ankle OA

Common to both ✓ **↓ proteoglycan content**

OA collagen ↓ collagen content (TO, FNB 2016) · "relative ↑ collagen concentration from PG loss" (TO second key) – read the options

AGEs non-enzymatic glycation (glucose, fructose, ribose + lysine/arginine) · **pentosidine** · cross-link type II collagen · ↓ chondrocyte response to TGF-β, IGF-1, BMP-7

OA stages

Stage	Event
1 (early)	Matrix disruption → **↑ water** (hallmark), ↓ PG aggregation, ↓ aggrecan → ↓ stiffness
2	Chondrocytes detect damage → mediators → **↑ proliferation (clusters/clones)** + ↑ matrix synthesis **and** ↑ catabolism
3	↓ proliferative + anabolic response → visible cartilage loss → clinical OA

Initiators inflammation · overload · ↓ matrix production

Microscopy fibrillation of superficial + transitional zones → **tidemark breached by vessels** → subchondral remodelling

Late / terminal water ↓ with degradation

Molecular Wnt-induced signal protein 1 ↑ proteases · ↓ response to TGF-α, IGF-1 · ROS, senescence

- - Active space - -

5 Management

Healing

Deep laceration (through tidemark into subchondral bone)	haematoma → marrow MSCs → **fibrocartilage (type I collagen)** · poor wear, ↓ stiffness
Superficial laceration (above tidemark)	chondrocyte proliferation but **no healing** (avascular)

6 Complications

Post-traumatic OA & inflammatory mediators

Initiating event	✓ **chondrocyte apoptosis** along superficial matrix cracks
Direct mediator of chondrocyte death	✓ **caspases** (mainly **caspase-3**) · aspartate-specific cysteine proteases
Cytokines after intra-articular fracture	**IL-1β, TNF-α, nitric oxide, MMPs, aggrecans, DAMPs**
Neutrophil recruitment	✓ **IL-8** (with IL-6) · IL-6 → MMPs, hypertrophy (STAT3/ERK) ✗ Wrong options: IL-10 anti-inflammatory · RANKL osteoclasts

Intervertebral disc degeneration

DDD	✓ **↑ MMP-1**, **↓ aggrecan/PG**, **↓ water** → ↓ osmotic pressure → disc height loss · **↓ T2 signal** ("dark disc")

X vs Y

	Superficial zone	**Deep zone**
Collagen orientation	**Parallel**	**Perpendicular**
Chondrocytes	**Flat**	**Round, columns**
Collagen 101	**Highest**	Lowest
Proteoglycan	**Lowest**	**Highest**
Water	Highest (80%)	**Lowest** (65%)

	Aging	**OA**
Water	↓	↑
Stiffness	↑	↓
KS : CS	↑	↓

How it's asked

"Tidemark lies between (TO)?"	→ **radial and calcified zone**

"TRUE about articular cartilage?"	→ **stress-shielding of solid matrix** ✗ Wrong options: isotropic · water deep · type IV in calcified cartilage
"Highest PG, lowest water?"	→ **deep zone**
"Zone containing progenitor cells?"	→ **superficial (A)** ✗ Wrong options: lamina splendens · tidemark
"Superficial zone appearance?"	→ **flattened chondrocytes + collagen both parallel to tidemark**
"Majority by weight of ECM?"	→ **water**
"Superficial-zone collagen orientation?"	→ **parallel, low PG content**
"Chondrocyte % volume?"	→ **2%**
"Arrowed superficial zone?"	→ **high collagen, low PG**
"All true EXCEPT (TO)?"	→ **"type I collagen provides tensile strength"**
"Contributes most to water attraction?"	→ **aggrecan** ✗ Wrong options: biglycan, decorin, fibromodulin, osteocalcin
"True statement?"	→ **type II collagen resists swelling from osmotic pressure** ✗ Wrong options: collagen hydrophilic · water 25% · aggrecans form GAGs · positive charge
"Main biologic effect of aggrecan?"	→ **hydrophilic behaviour**
"Swelling pressure mainly due to water with?"	→ **aggrecan**
"MoM feature NOT improving fluid-film lubrication?"	→ **complete congruence**
"Lubricin provides?"	→ **boundary lubrication**
"Binds HA to form boundary layer?"	→ **lubricin**
"Marathon runner, thick cartilage?"	→ **mechanotransduction**
"Non-weightbearing after ankle ORIF — knees at 2 months?"	→ **thinning ipsilateral, no change contralateral**
"Moderate running effect?"	→ **↑ GAG production**
"OA changes EXCEPT (TO)?"	→ **decreased water content**
"In OA but not aging (TO)?"	→ **↑ water + ↑ PG degradation**
"Common to aging and OA?"	→ **↓ proteoglycan content**
"Best describes OA degeneration?"	→ **initiated by inflammation, overload or ↓ matrix production** ✗ Wrong options: stage 2 ↓ catabolism · repair improves with age

– – Active space – –

"Change with age?"	**→ PG synthesis ↓** ✗ Wrong options: cells ↑ · CS ↑ · KS ↓ · water ↑
"Arthritic cartilage characterised by?"	**→ PG loss** ✗ Wrong options: ↑ moduli · ↓ water
"Collagen ↑ in early OA?"	**→ type X**
"First stage of OA?"	**→ ↓ PG/aggrecan + ↑ water**
"↑ water is the hallmark of which stage?"	**→ early**
"AGEs characteristic of?"	**→ aging cartilage → ↑ stiffness**
"Most common change in arthritic cartilage?"	**→ ↑ water content** ✗ Wrong options: ↑ KS:CS · ↑ stiffness · ↑ cross-linking
"↑ early, ↓ late in OA?"	**→ aggrecan** ✗ Wrong options: water, MMP, CS:KS (all uniformly ↑) · modulus (uniformly ↓)
"Bone-on-bone OA – cellular environment?"	**→ ↓ aggrecan, ↑ water, ↑ permeability**
"Growth factor: most studied, ↑ ECM, synovial fibrosis?"	**→ TGF-β1**
"IL-1 increases?"	**→ MMP**
"Initiating cellular event in PTOA?"	**→ chondrocyte apoptosis**
"Cytokines after intra-articular fracture → PTOA?"	**→ IL-1β, TNF-α, NO, MMPs, aggrecans, DAMPs**
"Mediator of chondrocyte death after chondral injury?"	**→ caspases**
"Recruits neutrophils after intra-articular fracture?"	**→ IL-8**
"Degenerative disc composition?"	**→ ↑ MMP-1, ↓ water, ↓ PG**
"Why is L5/S1 darker than L3/4 on T2?"	**→ ↓ aggrecan, ↓ water**

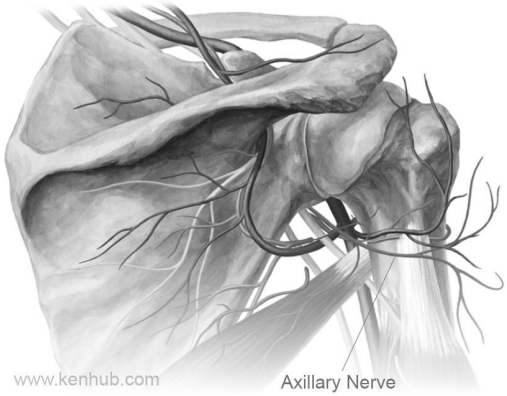
AXILLARY NERVE

– Basic Science · 9 / 108 –

1 Basics

Course

Origin **posterior cord** (with radial nerve) → just **inferior to subscapularis** → winds posteriorly through **quadrangular space** with **posterior circumflex humeral artery**



Axillary nerve, posterior view

Motor & sensory distribution

Branch	Supplies
Anterior branch	Winds round **surgical neck** → **anterior (half) deltoid** + overlying skin
Posterior branch	**Teres minor** + **posterior deltoid** · skin over lower ⅔ of posterior deltoid (**superior lateral cutaneous nerve of arm**)
Collateral branch	Long head of triceps

Axillary nerve muscles ✓ **deltoid + teres minor**
✗ Wrong options: teres major (**lower subscapular n.**) · supraspinatus (**suprascapular n.**) · latissimus (thoracodorsal)

3 Presentation

Lateral deltoid numbness + weak Hornblower's (teres minor, ER in abduction) ✓ **posterior branch of axillary nerve** (e.g. after greater-tuberosity fracture)

Deltoid palsy test weakness of **abduction with the arm internally rotated** (supraspinatus rotated anteriorly, contributes less)

Posterior cord injury would cause wider deficit (radial nerve too)

– – Active space – –

6 Complications

Iatrogenic	large **anterior shoulder exposure** (e.g. **Latarjet**) · uncommon but debilitating
Treatment	**Leechavengvongs radial → axillary nerve transfer**: a triceps branch of the radial nerve (medial, lateral or long head) → axillary nerve, as proximal as possible on axillary nerve, maximal excursion of donor

X vs Y

Muscle	Nerve
Deltoid, **teres minor**	**Axillary**
Teres major	**Lower subscapular**
Supraspinatus, infraspinatus	Suprascapular
Coracobrachialis, biceps, brachialis (medial)	Musculocutaneous

How it's asked

"Healed GT fracture, lateral-deltoid numbness, weak Hornblower's – structure injured?"	→ **posterior branch of axillary nerve** ✗ Wrong options: anterior branch · posterior cord · suprascapular · musculocutaneous
"Axillary lesion weakens deltoid and?"	→ **teres minor** ✗ Wrong options: teres major · latissimus
"Post-Latarjet weak abduction in IR, radial-to-axillary transfer – injured nerve supplies?"	→ **deltoid + teres minor**

BICEPS BRACHII

– Basic Science · 10 / 108 –

1 Basics

Anatomy & innervation

	Biceps brachii
Origin	**Short head:** tip of **coracoid** · **Long head: supraglenoid tubercle**
Insertion	**Radial tuberosity** + forearm fascia via **bicipital aponeurosis**
Action	**Supinates** + flexes forearm
Innervation	**Musculocutaneous nerve (C5, C6)**
Arterial supply	Muscular branches of brachial artery



Biceps brachii

Long head in the rotator interval

Rotator interval superior **supraspinatus** · inferior **subscapularis** · medial **glenoid**

Sagittal MRI LHB seen in cross-section in the interval between subscapularis + supraspinatus → from superior labrum laterally → turns inferiorly into **bicipital groove**

MGHL on sagittal MRI runs obliquely anteroinferiorly from superior labrum, **deep (posterior) to subscapularis**

– – Active space – –



Sagittal MRI — arrow on the long head of biceps in the rotator interval

Arthroscopy (posterosuperior portal)	LHB tendon from supraglenoid tubercle → muscle innervated by **musculocutaneous**

How it's asked

"Sagittal shoulder MRI, structure between subscapularis and supraspinatus?"	→ **long head of biceps** ✗ Wrong options: MGHL (oblique, deep to subscap) · CA ligament
"Arthroscopic tendon from supraglenoid tubercle — nerve to its muscle?"	→ **musculocutaneous** ✗ Wrong options: axillary, median, radial, ulnar

BICEPS FEMORIS SHORT HEAD

– Basic Science · 11 / 108 –

1 Basics

Anatomy & innervation

	Biceps femoris short head
Origin	Lateral lip of **linea aspera**, lateral supracondylar ridge, lateral intermuscular septum
Insertion	**Fibular head** (mainly) · also LCL + lateral tibial condyle
Action	Flexes knee · externally rotates tibia (long head also extends hip)
Innervation	**Common peroneal (peroneal division of sciatic) – L5, S1, S2**
Arterial supply	Perforators of profunda femoris, inferior gluteal, superior muscular branches of popliteal



Biceps femoris short head

X vs Y

	Long head	Short head
Nerve	**Tibial division** of sciatic (L5–S3)	**Peroneal division** of sciatic (L5–S2)
Crosses hip	Yes (extends hip)	No

Only hamstring on peroneal division short head of biceps femoris

How it's asked

"Nerve to short head of biceps femoris?" → **peroneal division of sciatic**
✗ Wrong options: tibial division (long head) · femoral · superior gluteal · saphenous

– – Active space – –

BIOABSORBABLE MATERIALS

– Basic Science · 12 / 108 –

1 Basics

Why bioabsorbable

Problems solved migration · growth disturbance · rigidity · radio-opacity · infection · **need for removal**

Indications paediatrics (transphyseal **SR PLGA 80/20** screws → only temporary growth arrest in rabbits) · osteomyelitis (antibiotic-eluting PLA) · **carriers for rhBMP-2/-7** · iliac-crest harvest site

Types

Material	Properties
PGA	**Hydrophilic**, crystalline · Tg **36 °C** · **early degradation + strength loss** · must be heated/cooled intraop
PLA	More **hydrophobic** than PGA
PLLA (L-isomer)	Hydrophobic, crystalline · Tg **57 °C** · **degrades over years → late reactions**
D-isomer PLA	Amorphous, less stable · for copolymers
P(L/D)LA	Adding D → less tightly packed → **less crystalline, faster** than PLLA · e.g. 70/30 (maxillofacial)
PLGA (PLA + PGA)	Low crystallinity
Self-reinforced (SR)	Oriented fibres in matrix (composite) · ↑ **rigidity + strength** · **malleable at room temp** · withstands 4× bending · minimal memory · **gamma sterilisation OK** (✗ Wrong options: non-reinforced (↓ molecular weight))

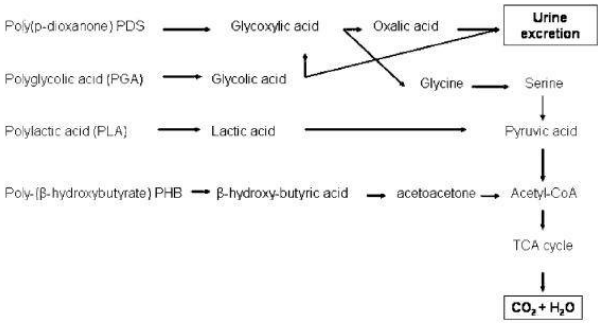
Absorption time

Material	Time
SR PLLA	**>5–6 yr**
PLLA	>5 yr
P(D/L)LA 70/30	2–3 yr
P(D/L)LA 96/4	2 yr
PLGA 80/20	1–2 yr
SR PGA	0.5–1 yr
PDS	2 months
PGA	**1–2 months**

2 Mechanism

Biodegradation & excretion

Main mechanism	**random hydrolysis of ester bonds** → ↓ molecular weight → loss of strength → **CO_2 + H_2O + Krebs (TCA) cycle** products



Poly-hydroxy-acid degradation

PLA excretion	lactic acid → Krebs cycle → ✓ **CO_2 via lungs (respiratory)** ✗ Wrong options: hepatic · renal · GI
Kidney	**PGA and PDS** (poly-p-dioxanone) products may be renally excreted
Enzymes	involved in PLA + PGA degradation
↓ pH	breakdown products lower pH → **accelerate** breakdown
Crystallinity	**amorphous / hydrophilic → faster** · crystalline / hydrophobic → slower
Other variables	composition, MW · fibre orientation (SR vs simple) · monomer concentration · stereoisomerism · porosity, surface area/volume · sterilisation · enzymatic vs hydrolytic

4 Diagnosis & Classification

Histopathology

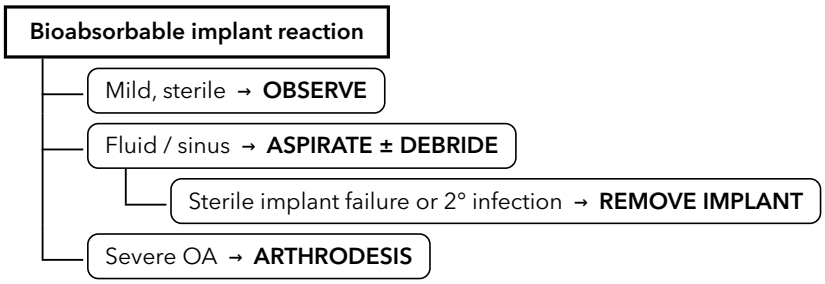
Granulomatous inflammation	T lymphocytes (**CD4 > CD8**) · plasma cells · endothelial cells · **birefringent polymer debris** · thin macrophage layer · **multinucleated giant cells**
Capsule	**inner** 2–3-cell layer (**type III collagen**) · **outer** fibrous layer (**type I collagen**)
Sequence	neutrophils → CD4 T cells → **macrophages last**

6 Complications

Adverse tissue reactions

Incidence	**3% children · up to 60% adults**
Cause	degradation outpaces debris clearance → **sterile fluid** accumulation · fluctuant papules · synovitis · sinus
Timing	late with slow materials (**PLLA ~5 yr**)
Labs	cultures **sterile** unless secondary infection
X-ray	**osteolysis** in up to **60%**

– – Active space – –



X vs Y

	PGA	PLLA
Water affinity	Hydrophilic	Hydrophobic
Tg	36 °C	57 °C
Absorption	**1–2 months**	**>5 yr**
Problem	Early strength loss	Late reactions
Excretion	Kidney possible	Lungs (CO_2)

How it's asked

"PLA implants excreted after absorption by?"	→ **respiratory system (CO_2)** ✗ Wrong options: hepatic · renal (PGA/PDS) · GI
"Bioabsorbable that degrades fastest?"	→ **PGA** (1–2 months) · slowest **SR PLLA**
"Degrade faster — amorphous or crystalline?"	→ **amorphous / hydrophilic**

BISPHOSPHONATES

– Basic Science · 13 / 108 –

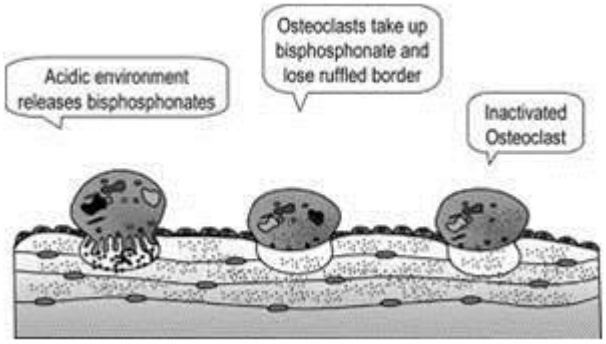
1 Basics

Overview

Class stable **pyrophosphate analogues** · strong affinity for **hydroxyapatite** (calcium) → concentrate in bone → taken up by **osteoclasts** during resorption

Net effect ✓ **inhibit osteoclast bone resorption** · ↓ recruitment + activity, ↑ **apoptosis** · loss of **ruffled border** (cytoskeleton/microtubules) · ↓ bone turnover · bone formed on therapy is **histologically normal**

Excretion ✓ **renal** → avoid in severe renal disease

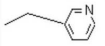
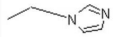


Bisphosphonate disrupts the osteoclast ruffled border

Mechanism (nitrogen vs non-nitrogen containing)

	Nitrogen-containing	Non-nitrogen (simple)
Drugs	**Alendronate, risedronate, pamidronate, zoledronate** (IV yearly)	**Etidronate, clodronate, tiludronate**
Target	**Farnesyl pyrophosphate (diphosphate) synthase** – **mevalonate (cholesterol) pathway**	Metabolised to **non-functional / toxic ATP analogue** (replaces terminal pyrophosphate of ATP)
Result	↓ **protein prenylation** (geranylgeranylation) of **small GTPases** → osteoclast inactivation/ apoptosis	Competes with ATP → targets **mitochondria** → **osteoclast apoptosis**
Potency	**Much more potent**	Earlier, weaker

– – Active space – –

Agent	R_1 side chain	R_2 side chain
Etidronate	-OH	$-CH_3$
Clodronate	-Cl	-Cl
Tiludronate	-H	-S-  -Cl
Pamidronate	-OH	$-CH_2-CH_2-NH_2$
Neridronate	-OH	$-(CH_2)_5-NH_2$
Olpadronate	-OH	$-(CH_2)_2N(CH_3)_2$
Alendronate	-OH	$-(CH_2)_3-NH_2$
Ibandronate	-OH	$-CH_2-CH_2N(CH_3)(CH_2)_4-CH_3$
Risedronate	-OH	
Zoledronate	-OH	

Nitrogen vs non-nitrogen bisphosphonates

Traps ✗ Wrong options: HMG-CoA reductase (statins) · caspase · MMP · reverse transcriptase · ↑ osteoblast activity · ↑ gut Ca · ↓ PTH

5 Management

Clinical uses (OI, fibrous dysplasia, osteolysis, fracture prevention)

Indications osteoporosis · metastatic bone disease · **multiple myeloma** · **Paget's** · **polyostotic fibrous dysplasia** · osteolysis after TJA · early AVN · **osteogenesis imperfecta** · **metastatic hypercalcaemia**

Use	Outcome
Osteoporotic VCF (T-score ≤ −2.5)	1 yr therapy → vertebral fracture risk ↓ **50–60%**
Hip fracture risk (N-BP, e.g. alendronate)	↓ **~50%** (45–55%)
Non-vertebral fragility fracture	↓ risk of multiple fractures
After hip fracture fixation	Resume at **1 week** → **no effect on outcome** (✗ Wrong options: nonunion, cutout, AVN, faster union)
OI – cyclical **IV pamidronate**	↓ bone pain, ↓ fractures, ✓ ↑ **bone density**, ↑ ambulation
McCune-Albright / polyostotic FD pain	✓ **Bisphosphonates** (✗ Wrong options: calcitonin, PTH, vit D, MTX)
Myeloma / metastases (pamidronate, zoledronate)	↓ skeletal events, pain, radiotherapy need · **no survival benefit** · monthly IV most effective
TJA osteolysis	✓ ↓ periprosthetic bone loss (alendronate, 6 yr DXA) · may ↑ BMP-2
Precollapse AVN	Investigational · RCT collapse at 2 yr **6.9% vs 76%** placebo

Contraindications

Severe renal disease renal excretion

After lumbar fusion animal models: ↑ fusion-mass size but **↓ fusion rate**

Invasive dental work complete **before** starting bisphosphonates

6 Complications

Osteonecrosis of the jaw & dental work

ONJ	✓ chronic **alendronate** / bisphosphonates ✗ Wrong options: warfarin, LMWH, diclofenac, ketorolac
Risk factors	**IV** therapy · duration **>36 months** (length of exposure most important) · elderly **myeloma** · **recent dental extraction** · incidence **4–17%**
Teriparatide	used to treat bisphosphonate ONJ



Osteonecrosis of the jaw

Atypical femur fractures

Site	✓ **subtrochanteric / proximal diaphyseal femur** · **lateral cortical thickening ("beaking")** → transverse fracture
ASBMR features	no/minimal trauma · **transverse lateral, oblique medial** · periosteal reaction · behaves as stress fracture
Numbers	**17–29%** of subtroch fractures atypical · **28% bilateral** → ✓ **contralateral films** · risk 3.2–50 / 100,000 person-yr → up to **100** with prolonged use
Prodrome	thigh pain (often misdiagnosed as quadriceps strain)



Lateral cortical beaking, subtrochanteric femur

Incomplete AFF with "dreaded black line"	stop bisphosphonate + **prophylactic IM fixation**

– – Active space – –

Atypical femur fracture — drug management after nailing (TO)

Denosumab AFF	✓ **stop denosumab → start teriparatide (anabolic window) → mandatory short-acting bisphosphonate after the anabolic course (TO)**
Why	denosumab **doesn't embed** in matrix → withdrawal = **rebound osteoclastogenesis**, rapid bone loss, **multiple vertebral fractures** → bisphosphonate "locks in" bone **(TO)**
Traps (TO)	✗ Wrong options: drug holiday alone (rebound) · continue denosumab ± teriparatide (suppressed turnover) · switch to IV zoledronate (keeps turnover paralysed)

AFF nailing technique (TO)

Step	Rule (TO)
Implant	**Full-length cephalomedullary nail** (protects neck) · nail with **smaller radius of curvature (1.0–1.5 m)** vs standard 2.0–3.0 m
Plates	**Contraindicated** — fatigue failure before union
Entry	**Lateralised** (± slightly posterior) trochanteric start
Reaming	✓ **Over-ream ≥1.5–2.0 mm** above nail diameter · (✗ Wrong options: line-to-line (incarceration)) · sharp reamers, slow, flexible shafts (avoid anterior blowout, thermal necrosis)
Nail size	Largest feasible (fatigue life) · never deliberately under-ream/downsize as routine
Extreme bow	Downsize slightly **or** closing-wedge osteotomy at apex
Biology	Pack **reamer dust** into fracture · **no distraction** (→ nonunion) — back-slap or compress

Paediatric effects

Osteopetrosis-like radiographic changes in children	✓ **bisphosphonates** (osteoclast is the common pathway) ✗ Wrong options: RANKL antibody, TNF inhibitor, BMP, fluoroquinolone

X vs Y

Drug	Mechanism
Bisphosphonate	Inhibits osteoclasts
Denosumab	Anti-RANKL antibody · no matrix binding → rebound on withdrawal
Teriparatide	PTH analogue → **activates osteoblasts** (anabolic)
SERM (raloxifene)	Oestrogen action on bone → ↓ vertebral fractures
Corticosteroid	Cause of osteonecrosis

How it's asked

"Non-nitrogen BP (clodronate, etidronate) mechanism for osteoclast apoptosis?"	→ **toxic ATP analogue targeting mitochondria** ✗ Wrong options: farnesyl diphosphate synthase (N-BP)

"Mechanism of non-nitrogen BPs?"	→ **increased osteoclast apoptosis** ✗ Wrong options: ↑ turnover · FPPS
"Binds hydroxyapatite and prevents protein prenylation?"	→ **alendronate** ✗ Wrong options: calcitonin, PTH, raloxifene
"N-BPs target?"	→ **farnesyl diphosphate synthase** ✗ Wrong options: HMG-CoA reductase
"Inhibits prenylation and GTPase formation?"	→ **alendronate** ✗ Wrong options: etidronate
"Paget's patient on non-nitrogen BP – mechanism?"	→ **non-functioning ATP analogue**
"Mechanism of bisphosphonates?"	→ **inhibit osteoclast resorption**
"N-BPs work by?"	→ **GTPase inhibition**
"BP mechanism in OI?"	→ **inhibits osteoclasts**
"Osteonecrosis hip, drug inhibits FPPS in figure?"	→ **nitrogen-containing BP**
"Denosumab 7 yr, nailed incomplete AFF – drug plan (TO)?"	→ **stop denosumab → teriparatide → then short-acting BP**
"Bowed femur AFF, cephalomedullary nail – critical technique (TO)?"	→ **lateralised entry + over-ream 1.5–2.0 mm**
"Long-term BP → stress fracture where?"	→ **subtrochanteric femur** ✗ Wrong options: jaw (ONJ not stress fracture)
"Thigh pain then low-energy transverse subtroch fracture?"	→ **long-term alendronate**
"10 yr alendronate, proximal diaphyseal fatigue fracture – aetiology?"	→ **prolonged bisphosphonates** ✗ Wrong options: HCTZ · metastasis · rickets
"ONJ is a complication of?"	→ **alendronate**
"Finish invasive dental work before starting?"	→ **bisphosphonates** ✗ Wrong options: teriparatide (treats ONJ)
"BP therapy in myeloma/metastases associated with?"	→ **ONJ** ✗ Wrong options: improved survival
"Medication ↓ osteolysis after TJA?"	→ **bisphosphonates**
"OI child on cyclical IV pamidronate – associated with?"	→ **↑ bone density** ✗ Wrong options: no change in pain/ fractures · osteosarcoma
"McCune-Albright, bone pain – next treatment?"	→ **bisphosphonates**
"Restart BP 1 week after hip fracture fixation – effect?"	→ **no effect on outcomes**
"Which hip fracture needs contralateral films?"	→ **atypical (subtrochanteric) femur fracture**
"T-score −2.7 + VCF, FPPS inhibitor – hip fracture risk?"	→ **↓ ~50%**

– – Active space – –

"Osteopetrosis-like changes in children from?"	→ **bisphosphonates**
"Primary excretion of BPs?"	→ **renal**

BONE CELLS

– Basic Science · 14 / 108 –

1 Basics

Lineage

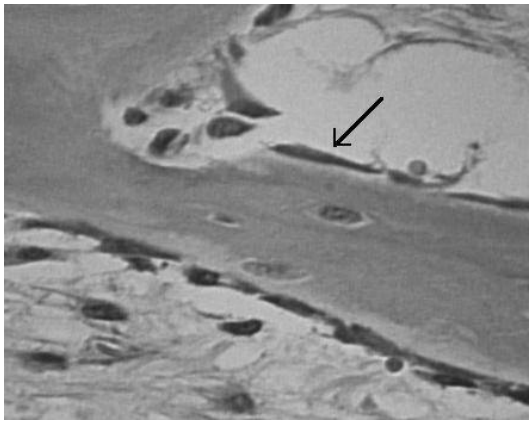
Cell	Origin
Osteoblast, osteocyte, osteoprogenitor, chondroblast	**Mesenchymal** stem cell
Osteoclast (preosteoclast)	**Myeloid haematopoietic** – **monocyte/macrophage** lineage · mononuclear cells **fuse**

MSCs form bone, cartilage, fat, muscle, tendon (fibroblasts), marrow stroma · **not osteoclasts**

Osteoprogenitor fate

Environment	Becomes
Low strain, high O_2	**Bone (osteoblast)**
Intermediate strain, low O_2	**Cartilage**
High strain	Fibrous tissue

Transcription factor	Lineage
Runx2 (Cbfa1), Osx	✓ **Osteoblast**
Sox9	Chondroblast
PPARγ2, C/EBPα	Adipocyte
MyoD	Myoblast
c-Fos	Osteoclast



Osteoprogenitor cell differentiation

– – Active space – –

Osteoblasts (Runx2, osteocalcin, RANKL, PTH)

Structure ↑ ER, Golgi, mitochondria · **cuboidal**, line immature osteoid · most active at bone surface

Products non-mineralised matrix: **type I collagen**, **alkaline phosphatase**, osteonectin, **osteocalcin** (↑ by 1,25-$(OH)_2D$)

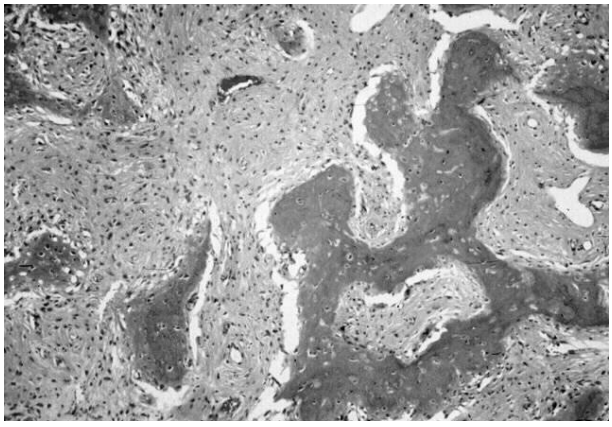
Most specific marker of mature osteoblast ✓ **osteocalcin** (not in immature, proliferating cells)
✗ Wrong options: TGF-β · COL2A1 (chondrocyte) · c-Fos (osteoclast) · IL-1

Differentiation signals **BMP** (MSC → osteoprogenitor) · **Runx2** · **stable β-catenin** (→ intramembranous bone) · PDGF · IGF

Signal on osteoblast	Effect
PTH receptor	↑ ALP, type I collagen · → **RANKL** + M-CSF · **intermittent rhPTH (1-34) → anabolic** · chronic ↑ PTH → resorption
1,25-$(OH)_2D$ receptor	↑ matrix, ALP, **osteocalcin**
Oestrogen	↓ resorption, ↑ formation (inhibits adenylyl cyclase)
Glucocorticoids	↓ collagen + matrix
Prostaglandins	↑ resorption (activate adenylyl cyclase)

Coupling osteoblast **RANKL → RANK** on osteoclast precursor → resorption · osteoblast **OPG** = decoy for RANKL → ↓ osteoclasts

Jagged1–Notch ✓ **PTH** induces **Jagged1** on osteoblasts → Notch on **HSCs** → HSC proliferation (immune/haematopoietic regulation)



Osteoblasts lining osteoid

Osteoclasts (origin, cathepsin K, integrins, RGD)

Appearance **multinucleated giant cell**, monocyte lineage, **ruffled border** · sits in **Howship's lacunae**

Differentiation **M-CSF + RANKL** (progenitors express RANK) · also PU.1, CSF-1

Activators **RANKL** (osteoblasts, tumour cells) · **IL-1** (next to loose TJA implants)

Inhibitors **calcitonin** · IL-10 · OPG

Acid phosphatase ✓ **osteoclast (TRAP) (TO)**
✗ Wrong options: osteoblast · osteocyte

Step	Molecule
Attachment (sealing zone)	**Integrins** – **αVβ3 = vitronectin receptor** · bind **RGD (Arg-Gly-Asp)** sequence on vitronectin, fibronectin, osteopontin · **reversible** binding · anti-αVβ3/RGD block resorption
Polarisation	Ruffled border · functional secretory domain · basolateral membrane
Mineral dissolution	**HCl** into resorption lacuna · ATP proton pumps · H^+ from **carbonic anhydrase II** · Cl^- channels · CA deficiency → no resorption
Organic matrix	**Cathepsin K** (major) · **MMP-9** · lysosomal cysteine proteinases
Removal	Transcytosis to FSD · **TRAP** in vesicles → reactive O_2 species destroy collagen
End	Apoptosis or return to non-resorbing state

Cathepsin K function = ✓ **bone resorption** · mutation → **pycnodysostosis** · inhibitors studied for osteoporosis

Bisphosphonates block ruffled border + acid hydrolase production

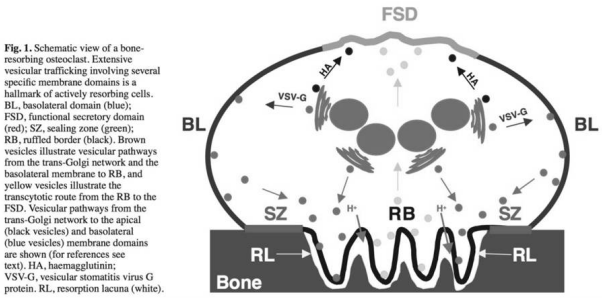


Fig. 1. Schematic view of a bone-resorbing osteoclast. Extensive vesicular trafficking involving several specific membrane domains is a hallmark of actively resorbing cells. BL, basolateral domain (blue); FSD, functional secretory domain (red); SZ, sealing zone (green); RB, ruffled border (black). Brown vesicles illustrate vesicular pathways from the trans-Golgi network and the basolateral membrane to RB, and yellow vesicles illustrate the transcytotic route from the RB to the FSD. Vesicular pathways from the trans-Golgi network to the apical (black vesicles) and basolateral (blue vesicles) membrane domains are shown (for references see text). HA, haemagglutinin; VSV-G, vesicular stomatitis virus G protein. RL, resorption lacuna (white).

Osteoclast zones – sealing zone, ruffled border, resorption lacuna

Disuse ↓ integrin expression → disuse osteoporosis

Osteocytes

Origin **former osteoblasts trapped in their own matrix** · **90%** of cells in mature skeleton

Structure high nucleus:cytoplasm · long processes through **canaliculi → gap junctions** (connexins; cAMP, IP3 pass) · radial around Haversian canal

Function maintain bone matrix · **Ca/P regulation** · no ALP

Signals **stimulated by calcitonin**, **inhibited by PTH**

Structure	Role
Canaliculi	✓ **Osteocyte–osteocyte communication**
Haversian canal	Centre of osteon · vessels + nerves
Volkmann canal	Perpendicular · links Haversian canals + periosteum
Cement line	Boundary of osteon

– – Active space – –

X vs Y

	Osteoblast	Osteoclast	Osteocyte
Origin	Mesenchymal	**Myeloid (monocyte)**	Trapped osteoblast
Marker	**Osteocalcin**, ALP	**TRAP**, cathepsin K, calcitonin receptor	No ALP
PTH	Receptor ✓ (→ RANKL)	**No receptor** (indirect)	Inhibited
Calcitonin	–	Inhibited (receptor ✓)	Stimulated

Function	Molecule / cell
Osteoclast attachment	**Integrin αVβ3 / vitronectin / RGD**
Binds RANKL, prevents RANK activation	**OPG**
Chemotaxis in fracture haematoma	PDGF
Induces type II collagen + PG in callus	TGF-β

How it's asked

"Acid phosphatase contained in (TO)?"	→ **osteoclast**
"Function of cathepsin K?"	→ **bone resorption** ✗ Wrong options: RANK activation · water absorption
"Vitronectin receptor function?"	→ **osteoclast attachment to bone** ✗ Wrong options: RANKL interaction · chemotaxis (PDGF) · type II collagen (TGF-β)
"MSCs can form all EXCEPT?"	→ **osteoclasts**
"Multinucleated giant cells, monocyte lineage, ruffled border?"	→ **osteoclasts**
"True about the cell in Figure A (osteoclast)?"	→ **haematopoietic, macrophage lineage** ✗ Wrong options: mesenchymal (osteoblast) · trapped osteoblast (osteocyte)
"Integrins function?"	→ **osteoclast attachment to bone** ✗ Wrong options: binds RANKL (OPG)
"RGD sequence allows?"	→ **interaction with integrins**
"Enzyme used by mature osteoclasts to resorb bone?"	→ **cathepsin K** ✗ Wrong options: ALP, TIMP, osteonectin, osteopontin
"Lineage and mediator of osteoclast differentiation?"	→ **myeloid / RANKL** ✗ Wrong options: lymphoid · BMP · OPG
"Cell of myeloid origin?"	→ **preosteoclast**

"Primary function of RGD bone protein?"	→ **reversible osteoclast binding to bone**
"True about normal remodelling (osteopetrosis patient)?"	→ **Howship's lacunae form at resorption sites** ✗ Wrong options: PTH acts directly on osteoclasts · OPG ↑ turnover · TRAP from osteocytes · lamellar → woven
"Function of Cbfa1/Runx2?"	→ **osteoblastic differentiation**
"Releases RANKL?"	→ **osteoblasts**
"Most specific marker of mature osteoblasts?"	→ **osteocalcin**
"Intermittent rhPTH targets?"	→ **osteoblasts**
"Transcription factor committing to bone?"	→ **Runx2** ✗ Wrong options: PPARγ2, MyoD, Sox9, C/EBPα
"Activates Jagged1–Notch pathway?"	→ **PTH**
"Osteocyte communication network?"	→ **canaliculi** ✗ Wrong options: Volkmann · cement lines · Haversian
"Spindled cells in mature osteoid connected by canaliculi?"	→ **osteocytes**

– – Active space – –

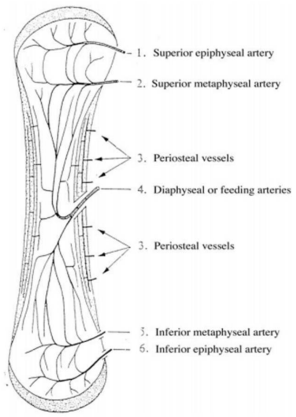
BONE CIRCULATION

– Basic Science · 15 / 108 –

1 Basics

Overview

Share of cardiac output	**5–10%** · intraosseous flow **5–20 mL/min/100 g**
Tenuous supply	**scaphoid** · **talus** · **femoral head** · **odontoid**
Three sources (long bone)	nutrient artery · metaphyseal–epiphyseal · periosteal



Arterial vascularisation of a long bone

Blood supply to physis & diaphysis

System	Pressure	Supplies
Nutrient artery	**High**	Enters via nutrient foramen → medullary canal → ascending/ descending → **inner ⅔ of mature cortex** (Haversian)
Metaphyseal–epiphyseal	–	From periarticular plexus (e.g. geniculates)
Periosteal arterioles	**Low**	**Outer ⅓ of diaphyseal cortex** · linked by Volkmann (perpendicular) + Haversian (parallel)
Perichondrial artery (of LaCroix)	–	✓ **Major source of nutrition of growth plate**
Epiphyseal artery	–	Proliferative zone of physis

Emissary system **venous** only · no "emissary artery"

Direction of flow

– – Active space – –

	Arterial	Venous
Mature bone	**Centrifugal** (inside → out; nutrient high-pressure dominates)	**Centripetal** (cortical capillaries → sinusoids → emissary veins)
Immature bone	**Centripetal** (periosteal dominates)	–

↑ Bone blood flow hypoxia · hypercapnia · sympathectomy

2 Mechanism

After fracture

Phase	Flow
Immediate	**↓ flow** · becomes **centripetal** (nutrient system disrupted, periosteal dominates)
Hours–days	**↑ flow** (regional acceleratory phenomenon) · **peaks 2 wk** · normal by **3–5 months**

Intramedullary nails

Unreamed nail preserves endosteal supply

Reaming devascularises **inner 50–80%** of cortex · delays endosteal revascularisation

Loose-fitting nail spares cortical perfusion, faster reperfusion · **tight-fitting** → compromised, slow

How it's asked

"Disruption interrupts major nutrient supply to growth plate?"	→ **perichondrial artery** ✗ Wrong options: diaphyseal, metaphyseal, synovial
"Supplies outer third of diaphysis?"	→ **periosteal arterioles** ✗ Wrong options: nutrient (inner ⅔) · emissary artery (doesn't exist) · perichondral (physis)
"Direction of arterial flow in mature bone?"	→ **centrifugal** · immature → **centripetal**
"Immediately after fracture, flow becomes?"	→ **↓ and centripetal**

– – Active space – –

BONE GRAFTING

– Basic Science · 16 / 108 –

1 Basics

Overview

Volume	~**1 million** grafting procedures/yr in the US · growth ~12–13%/yr
Indications	fractures, delayed union, **nonunion** · arthrodesis, **spinal fusion** · defects after trauma/tumour

Graft properties (osteogenic, osteoinductive, osteoconductive)

Property	Definition	Example
Osteoconductive	**Scaffold** – permits ingrowth of capillaries, perivascular tissue, osteoprogenitors → **creeping substitution**	Allograft, ceramics, DBM
Osteoinductive	Factors that **recruit + differentiate MSCs** into osteoblasts (bind transmembrane receptors)	**BMP** (TGF-β superfamily), DBM
Osteogenic	**Living bone-forming cells** – MSCs, osteoblasts, osteocytes	**Fresh autograft, bone marrow aspirate** only
Osseointegration	Direct bone–implant contact (Brånemark) · **not a graft property** (TO)	Ca-phosphate-coated stems

Most common / most important property (TO)	✓ **osteoconduction**
Live mesenchymal precursor cells	✓ only **autograft / bone marrow aspirate** ✗ Wrong options: fresh-frozen allograft · rhBMP · DBM · Ca phosphate

Cancellous > cortical for new bone — larger surface area

Material	Conductive	Inductive	Osteogenic
Autograft (ICBG, **RIA**, local cancellous/cortical)	Yes	Yes	Yes
Bone marrow aspirate	No (no structure)	Yes (OB)	✓ **(TO: osteogenic potential)**
DBM	Yes	Yes	No
BMP (rhBMP-2, -7)	No	✓ **only**	No
PRP	No	"**solely osteoinductive**" (Orthobullets/OB) · TO: mitogenic + angiogenic only, **not true osteoinduction**	No
Allograft (freeze-dried, cortical, cancellous, fibular strut)	✓ **only**	No (fresh/fresh-frozen retain some BMP)	No
Ceramics (Ca phosphate, sulfate, carbonate, TCP, HA), polymers	✓ **only**	No	No

Most osteoinductive (of listed) **cancellous autograft**

Diamond concept & graft biology in nonunion (TO)

Giannoudis Diamond (4 pillars) **osteoconductive scaffold** · **osteoinductive mediators** · **osteogenic cells** · **mechanical stability (TO)**

HA/β-TCP + rhBMP-2 + LR-PRP + locking plate deficient in ✓ **osteogenesis** (no viable MSCs) → add **autograft or BMAC (TO)**

rhBMP-2 serine/threonine kinase receptors → **SMAD** → MSC → osteoblast **(TO)**

Antigenicity

Antigens class I + II MHC on graft cells → host **T cells** · **marrow cells most immunogenic** · endothelium · type I collagen (humoral + cell-mediated) · non-collagenous matrix

Hydroxyapatite no immune response

Rejection **cell-mediated**, MHC incompatibility · greater mismatch → less new bone

Ligament allograft antigenicity related to **bone** > soft tissue → **BPTB for ACL = highest** (also meniscal bone plugs)

- - Active space - -

2 Mechanism

Graft healing stages

Stage	Event
1 Inflammation	Necrotic debris → chemotaxis
2 Osteoblast differentiation	From mesenchymal precursors
3 Osteoinduction	Osteoblast + osteoclast stimulation
4 Osteoconduction	Bone forms on scaffold
5 Remodelling	Years

Types of autograft

Type	Facts
Cancellous	Little structure · ↑ osteoconduction · **rapid incorporation by creeping substitution**
Cortical	Slow (remodel existing Haversian canals) · structural · interstitial lamellae preserved
Vascularised	Faster union, cells preserved · **free fibula (peroneal a.)**, free iliac crest (deep circumflex iliac a.), **distal radius 1,2-ICSRA** for scaphoid

Structural allograft biology & outcomes

Incorporation — particulate > structural · massive cortical graft largely **never remodels, stays devascularised**

Insufficiency fracture — **15–30%** (~**25%**) of massive structural allografts

Retrieved bulk allograft (10 yr) — junction heals by **bridging external callus** + persistent callus **perpendicular** to junction · little penetration
✗ Wrong options: osteonal/Haversian remodelling · complete incorporation

Osteoarticular allograft cartilage at 5 yr — **completely acellular** (no donor or host chondrocytes)

Meniscal allograft — success depends most on **accurate sizing (within 5%)**
✗ Wrong options: donor cell viability (repopulated by host) · immunosuppression · cryopreservation

5 Management

Types of allograft & processing

Allograft	Immunogenicity	BMP	Disease transmission	Shelf life
Fresh	**Highest**	Preserved	**Highest**	Hours–days
Fresh-frozen	Less	Preserved (inductive)	Intermediate · **stronger** – better structural	**−20 °C: 1–2 yr** · **−70 °C: up to 5 yr**
Freeze-dried ("croutons")	**Least**	**Depleted** (purely conductive)	**Lowest** (theoretical HIV 1 : 2.6 billion)	**Indefinite** (5 yr–∞)

Shelf life order **freeze-dried > fresh-frozen −70 °C > fresh-frozen −20 °C > fresh**

Processing debride soft tissue → **ethanol** wash (removes cells) → **gamma irradiation** (dose-dependent: kills pathogens but ↓ mechanical properties)

Method	Process
Cryopreservation	✓ **Chemicals (DMSO/glycerol) displace cellular water + controlled-rate freezing** → no ice crystals · keeps some chondrocytes (osteoarticular)
Freeze-drying	Water → alcohol to 5% moisture → vacuum (sublimation) · alters growth-factor proteins
Fresh-frozen	Frozen twice, packaged dry at −80 °C
Fresh	Not frozen · max viability, short shelf life

DBM vs freeze-dried allograft DBM additionally **osteoinductive** (factors bind transmembrane receptors on MSCs) · neither is osteogenic

DBM **acid extraction** of allograft → removes mineral, keeps collagen + non-collagenous proteins (**BMPs, TGF-β, IGF, FGF**, residual Ca) · no structural support · **inter-lot variability**



Demineralised bone matrix

Bone graft substitutes (calcium sulfate, calcium phosphate, HA)

All ceramics osteoconductive void fillers · **scaffold for progenitor cells** · **low shear, tensile, torsional strength** · biocompatible with steel + titanium · used in spinal fusion

– – Active space – –

Resorption (fastest → slowest) **calcium sulfate > tricalcium phosphate > hydroxyapatite**

Substitute	Resorption	Strength	Notes
Calcium sulfate (plaster of Paris, OsteoSet)	**4–12 wk** (5–7 wk; gone by 6 wk) · **passive dissolution** · **faster than bone grows**	≈ cancellous	**Sterile serous wound drainage** (osmotic load) **(TO)** · fails alone in nonunion
β-TCP	**6–18 months** · **cell (osteoclast)-mediated (TO)** · dissolution + fragmentation	≤ cancellous · brittle	Less bone formed than resorbed → used as adjunct
Calcium phosphate cement	~1 yr (1–10 yr)	**Highest compressive – 4–10× cancellous**	Hardens to **dahllite** · **early WB** without subsidence (plateau, calcaneus) · **osseointegration + osteoconduction** (stem coatings) · reduce + fix **before** injecting
Coralline HA (Interpore)	Very slow (decades)	Slightly > cancellous	Coral $CaCO_3$ → HA by hydrothermal exchange · **neovascularisation + appositional bone** (✗ Wrong options: creeping substitution · dissolution)
Calcium carbonate (Biocoral)	–	–	Unaltered coral · conductive only
Collagen matrix + HA coating	Collagen fast, HA slow	< cancellous	–
Aluminium oxide	–	–	Bonds to bone under stress/strain

Level I evidence Ca-phosphate fills defects, allows early rehab, **prevents articular subsidence** (distal radius, tibial plateau)

Ca-phosphate cure (TO) isothermal / mildly exothermic · no PMMA-type thermal necrosis

HO complication of **BMP**, not calcium sulfate **(TO)**

Bone morphogenetic proteins

rhBMP-2, rhBMP-7 FDA-approved, long bones + spine · **solely osteoinductive** · perivascular MSC → osteoblast via **serine-threonine kinase receptors**

Complications under/over-production of bone · inflammation · early bone resorption · near-zero disease transmission (synthetic)

Autograft harvest (iliac crest, RIA)

Autograft **gold standard** · all three properties · **least immunogenic**

Iliac crest commonest site · cortical + cancellous · complications **2–36%**

Anterior vs posterior **anterior = more complications** (pain, gait) · **posterior = more stem cells** (viability same), less pain

Structure at risk	Where
Superior cluneal nerves — commonest posterior complication (buttock numbness)	Cross crest **>8 cm lateral to PSIS → ✓ stay within 8 cm (TO)**
Superior gluteal artery	Retractor/elevator slipping **caudally into sciatic notch** · brisk bleeding (✗ Wrong options: vein · inferior gluteal)
SI joint posterior ligaments	Full-thickness posterior graft → **SI instability/dislocation**
LFCN, ilioinguinal nerve	**Anterior** crest
Deep circumflex iliac artery	Anterior crest
Corona mortis	Anterior intrapelvic approaches
External iliac artery	✓ **Not at risk (TO)**

Other complications haematoma · hernia · infection · fracture · cosmetic defect · chronic pain

Vascularised autograft sources **fibula, ribs**

Reamer–irrigator–aspirator (RIA)

RIA vs ICBG	
Volume	**Higher** (~50–90 mL per femur)
Union	**Comparable (~90%)**
Operative time	**Shorter** · technically easier
Immediate complications	**Fewer**
Blood loss	**Higher → transfusion** (up to 44%)
Structural support	**Less** (fragments ≤2 mm)
Equipment	More expensive
Biology	All three properties · **>95% cell viability** · ≥ ICBG growth factors/MSCs · similar RNA transcription profile
Specific risks	**Femoral shaft fracture** (eccentric reaming) · insertion-site (abductor) pain

RIA false statement (TO) false: "**heat injury from reaming abolishes osteoinductive properties**" · RIA is a viable alternative to cancellous autograft, product of long-bone nailing, osteogenic **(TO)**

– – Active space – –



RIA harvesting from the femoral canal

6 Complications

Allograft disease transmission & processing

Pathogen	Risk (fresh-frozen bone)
HIV	**1 in 1,000,000 – 1,670,000** (1 in 1.6 million despite screening) · last bone transmission 1985
Hepatitis C	1 in 100,000 · last 2000
Hepatitis B	1 in 63,000
HTLV	One case (improper screening)
CJD	**Never** in bone (cornea, dura only)
Bacteria	Up to 12% massive allograft · 0.7% morselised · Clostridium (2001)

AATB required serology
✓ **HIV, HBV, HCV, syphilis** (+ HTLV-I/II)
✗ Wrong options: CMV · hepatitis A

Donor exclusions (TO)
malignancy (except skin BCC) · collagen vascular disease · metabolic bone disease · toxins

Positive preimplantation culture (CoNS), no clinical infection
✓ **observe** · 4.8–13.3% positive, none became infected · routine culture not recommended

Calcium sulfate
↑ **serous wound drainage**

X vs Y

	DBM	Freeze-dried allograft	BMP	BMA
Conductive	Yes	Yes	No	No
Inductive	Yes	No	Yes	Yes
Osteogenic	No	No	No	Yes

	Calcium sulfate	β-TCP	Calcium phosphate cement	HA
Resorption	**Fastest (weeks)**	6–18 mo	~1 yr	Decades
Mechanism	Passive dissolution	Osteoclasts	–	Appositional
Strength	≈ cancellous	≤ cancellous	**Highest**	> cancellous
Signature	**Serous drainage**	Adjunct	**Early WB**	Coral-derived

How it's asked

"Contains live mesenchymal precursor cells?"	→ **autologous iliac crest marrow aspirate**
"Most osteoinductive?"	→ **cancellous autograft**
"Least immune response?"	→ **freeze-dried cancellous chips** ✗ Wrong options: fresh Achilles (highest)
"Alternative to autograft acting by osteoinduction?"	→ **BMP**
"Osteoconductive graft property?"	→ **permits capillary + perivascular ingrowth**
"Osteoinductive + osteoconductive for plateau void?"	→ **DBM** ✗ Wrong options: Ca phosphate · BMP · allograft
"Solely osteoinductive?"	→ **PRP** (OB) · ✗ Wrong options: DBM · freeze-dried · ICBG
"Osteogenic + conductive + inductive adjuvant?"	→ **RIA bone** ✗ Wrong options: BMA (no conduction) · DBM · fibular strut · BMP
"Conductive + inductive, not osteogenic?"	→ **DBM**
"DBM differs from freeze-dried allograft by?"	→ **osteoinductive factors**
"RIA autograft capacities?"	→ **osteogenic, osteoconductive, osteoinductive**
"Ceramic degradation + complication of calcium sulfate (TO)?"	→ **β-TCP osteoclast-mediated 6–18 mo · $CaSO_4$ passive dissolution → sterile serous exudate**
"Fastest-resorbing substitute?"	→ **calcium sulfate**
"Disappears most quickly in vivo (plateau)?"	→ **calcium sulfate**
"Resorbs faster than bone grows?"	→ **calcium sulfate**
"True of both Ca phosphate and sulfate?"	→ **scaffold for progenitor cells** ✗ Wrong options: shear/torsion resistant · contraindicated in fusion
"$CaSO_4$ vs autograft in plateau — expected ↑?"	→ **serous drainage complications**
"Permits earliest WB without subsidence?"	→ **calcium phosphate**

– – Active space – –

"Porous HA incorporation?"	**→ neovascularisation + appositional bone growth**
"Highest early compressive strength?"	**→ calcium phosphate**
"Osseointegration stem coating?"	**→ calcium phosphate**
"HIV risk, fresh-frozen bone?"	**→ 1 in 1,000,000** · "despite screening" **→ 1 in 1.6 million** · "1 in 1.67 million" **→ HIV**
"AATB required serology?"	**→ HBV, HCV, HIV, syphilis**
"BPTB allograft culture + CoNS, well patient?"	**→ observation**
"Cryopreservation achieved by?"	**→ chemicals remove water + controlled-rate freezing**
"Decreasing shelf life?"	**→ freeze-dried > FF −70 °C > FF −20 °C > fresh**
"Highest theoretical disease transmission?"	**→ fresh allograft**
"Donor chondrocytes in osteoarticular allograft at 5 yr?"	**→ none – acellular**
"Structural allograft insufficiency fractures?"	**→ 15–30%**
"Histology at bulk allograft junction, 10 yr?"	**→ bridging external callus + perpendicular callus**
"Highest antigenicity knee allograft?"	**→ BPTB for ACL**
"Most critical for meniscal allograft success?"	**→ accurate graft size**
"NOT true of RIA (TO)?"	**→ "heat injury abolishes osteoinductive properties"**
"Commonest posterior ICBG complication avoided by (TO)?"	**→ staying within 8 cm of PSIS**
"Not at risk in ICBG (TO)?"	**→ external iliac artery**
"Anterior vs posterior iliac crest harvest?"	**→ anterior: ↑ complications**
"Posterior marrow aspiration vs anterior?"	**→ ↑ number of stem cells** ✗ Wrong options: ↑ viability
"Elevator slips caudally, brisk bleeding (posterior crest)?"	**→ superior gluteal artery**
"RIA vs ICBG – true?"	**→ ↑ blood loss needing transfusion** · **greater volume** · equal union · shorter time · fewer immediate complications
"Most common property of a bone graft (TO)?"	**→ osteoconduction** ✗ Wrong options: osseointegration (not a graft property)
"Substitute with osteogenic potential (TO)?"	**→ bone marrow aspirate** ✗ Wrong options: DBM, PRP, BMP
"HA/β-TCP + rhBMP-2 + LR-PRP – missing Diamond pillar (TO)?"	**→ osteogenesis**

BONE GROWTH FACTORS

– Basic Science · 17 / 108 –

1 Basics

Overview of growth factors

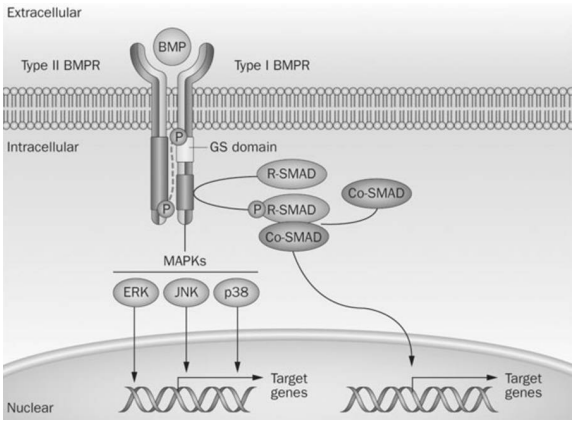
Factor	Source	Receptor	Function
TGF-β	Platelets, bone ECM, cartilage	**Serine-threonine kinase**	Pleiotropic · **undifferentiated MSC proliferation** · type II collagen + PG by MSCs · osteoblast collagen synthesis · in fracture haematoma · coats porous implants
BMP	Osteoprogenitors, osteoblasts, bone ECM	**Serine-threonine kinase → SMADs**	✓ **MSC → chondrocytes + osteoblasts** · skeletal patterning
FGF (FGF-1, -2 most abundant)	Macrophages, MSCs, chondrocytes, osteoblasts	**Tyrosine kinase**	**Mitogenic** · angiogenesis · early fracture healing
IGF-1 (somatomedin C)	Bone matrix, osteoblasts, chondrocytes	**Tyrosine kinase**	**Most abundant GF in bone** · GH–IGF-1 → physeal **proliferation without maturation** (linear growth) · intramembranous defects
IGF-2	–	Tyrosine kinase	**More potent than IGF-1** · type I collagen, cartilage matrix, proliferation, bone
PDGF	Platelets, osteoblasts	**Tyrosine kinase**	Mitogen · **macrophage/inflammatory-cell chemotaxis** to fracture
PPAR-γ	–	**Nuclear receptor** (binds DNA)	**Adipogenic** differentiation
Wnt	–	**LRP5/6** → β-catenin to nucleus	Osteoblast differentiation

Thyroid hormone	nuclear receptor → physeal **proliferation and maturation**
Morphogens vs mitogens (disc)	BMP-2, BMP-7, TGF-β change phenotype (↑ chondrogenic phenotype + matrix) without proliferation · IGF-1, FGF ↑ proliferation

BMP signalling (SMADs, serine-threonine kinase)

Family	TGF-β superfamily (with GDFs, AMH, activin, Nodal) · ~30 members
Target	undifferentiated **perivascular MSCs**
Cascade	BMP → heterodimer of **type I + II serine-threonine kinase receptors** → phosphorylate **SMAD 1, 5, 8** → complex with **SMAD 4** → nucleus → osteoblast genes · 8 SMADs known

– – Active space – –



BMP receptor → SMAD signalling

BMP	Fact
BMP-2, 4, 6, 7	**Osteoinductive** (BMP-2 potent osteogenic + osteoinductive)
BMP-3	**Not osteoinductive** · **antagonises BMP-2** · **most abundant BMP in DBM** · knockout → 2× trabecular bone · ↑ chondrogenic markers (type II collagen, aggrecan)
BMP-1	**Metalloprotease** (protease for collagen I, II, III) · not bone-inducing
BMP-4	Altered signalling → **fibrodysplasia ossificans progressiva**

Lineage factors

Factor	Drives
BMP, **RUNX2**, Wnt/β-catenin, ATF4 (with Runx2)	**Osteogenic**
SOX9	Chondrogenic
PPAR-γ	✓ **Adipogenic** (pre-adipocyte → adipocyte) · inhibited by Wnt5a

Thiazolidinediones — **PPAR-γ agonists** → ↑ marrow adiposity, **↓ osteoblastogenesis** → **↓ BMD** (femoral neck, hip), ↑ fracture risk

GSK-3 inhibition — ↑ β-catenin → ↑ osteoblasts
✗ Wrong options: ↓ BMD

5 Management

rhBMP-2 clinical use

Product	FDA-approved use
rhBMP-2	**Single-level ALIF L2–S1** for DDD with **LT-Cage**
rhBMP-2	**Open tibial shaft fracture + IM nail, within 14 days**
rhBMP-7 (HDE)	Recalcitrant long-bone nonunion where autograft unfeasible + alternatives failed
rhBMP-7 (HDE)	Revision posterolateral lumbar fusion in compromised hosts (osteoporosis, smoking, DM)

Not approved (rhBMP-2)	plates (MIPO tibia) · posterior/posterolateral lumbar · multilevel · **cervical**
PLF in >60 yr	rhBMP-2 **cost-effective** vs ICBG and ↓ revision surgery (off-label)
Contraindications	pregnancy · allergy to **bovine type I collagen** or rhBMP-2 · infection · tumour · **skeletal immaturity**
Concerns	**uncontrolled heterotopic bone** · cost

6 Complications

BMP-4 & FOP

FOP ("stone man")	altered **BMP-4** signalling → massive spontaneous HO · no cure · excision helps mobility/seating

Disease	Gene
FOP	**BMP-4 signalling**
Mazabraud	**GNAS1** (polyostotic FD + intramuscular myxomas)
Ollier	PTHrP / Indian hedgehog
Osteopetrosis	**CLCN7**
Osteopoikilosis	**LEMD3**

X vs Y

Serine-threonine kinase receptor	Tyrosine kinase receptor
TGF-β, BMP	**FGF, IGF-1/2, PDGF**

How it's asked

"All true of BMP-3 EXCEPT?"	→ **"it is osteoinductive"**
"BMP receptor class?"	→ **serine-threonine kinase** ✗ Wrong options: tyrosine kinase, MAPK, aurora, GSK-3
"Phosphorylated directly after BMP binding?"	→ **SMADs** ✗ Wrong options: Myc · β-catenin (Wnt) · adenylate cyclase
"Serine-threonine receptor, MSC metaplasia into bone-forming cells?"	→ **BMP** ✗ Wrong options: IGF-2, PDGF (tyrosine kinase) · OPG · RANKL
"Osteoinduction and osteogenesis?"	→ **BMP-2** ✗ Wrong options: BMP-1 (metalloprotease) · BMP-3 · PPAR-γ · IGF-1
"Promotes MSC → chondrocytes + osteoblasts?"	→ **BMP** ✗ Wrong options: TGF-β (proliferation) · FGF (mitogen)

– – Active space – –

"↑ chondrogenic phenotype of disc cells?"	**→ BMPs**
"FDA-approved rhBMP-2 scenario?"	**→ L5–S1 ALIF for DDD** ✗ Wrong options: plated open tibia · posterolateral · cervical
"rhBMP-2 true EXCEPT?"	**→ "FDA indicated for single-level posterolateral lumbar fusion"**
"Binds LRP5/6?"	**→ Wnt**
"Drives MSCs to adipose?"	**→ PPAR-γ** ✗ Wrong options: BMP, RUNX2, SOX9, Wnt
"Thiazolidinedione effect on BMD?"	**→ activates PPAR-γ → adipocytes ↑, osteoblastogenesis ↓, BMD ↓**
"Altered BMP-4 signalling?"	**→ fibrodysplasia ossificans progressiva**

BONE MATRIX

– Basic Science · 18 / 108 –

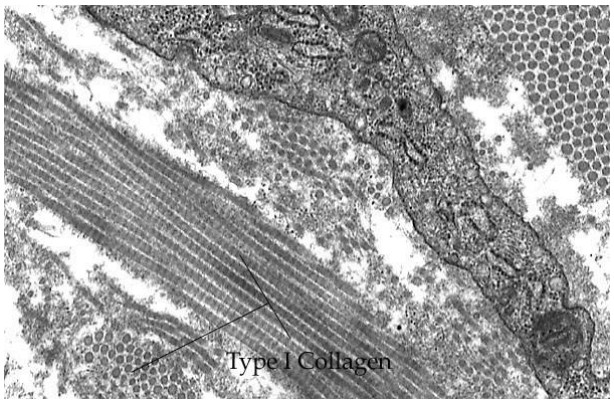
1 Basics

Composition of bone

Component	% dry weight	Contents
Organic	**40%**	Collagen (90% of organic) · proteoglycans · matrix proteins · cytokines/ GFs
Inorganic	**60%**	**Calcium hydroxyapatite $Ca_{10}(PO_4)_6(OH)_2$** · osteocalcium phosphate (**brushite**)

Osteocalcin & matrix components

Component	Role
Type I collagen	**Tensile strength** · triple helix: **two α1 + one α2** chains
Proteoglycans	✓ **Compressive strength** (with HA) · **inhibit mineralisation** · bind growth factors · GAG–protein complexes
Hydroxyapatite	**Compressive strength**
Osteocalcin	**Most abundant non-collagenous protein** (10–20%) · by **mature osteoblasts** · promotes mineralisation · regulates bone density · attracts osteoclasts · ↑ by **$1,25\text{-}(OH)_2D_3$**, ↓ by **PTH** · **marker of bone formation/turnover** (serum, urine)
Osteonectin	**Glycoprotein that binds calcium** · from **platelets + osteoblasts** · regulates Ca / organises mineral
Osteopontin	**Cell-binding** protein
Cytokines / GFs	IL-1, IL-6, IGF, TGF-β, BMPs



Type I collagen triple helix

Osteocalcin clinical — **most specific marker of mature osteoblast** · serum levels track **BMD gain on osteoporosis treatment** and anabolic trials (teriparatide)

– – Active space – –

X vs Y

	Osteocalcin	Osteonectin
Source	**Osteoblasts** (✗ Wrong options: osteoclasts)	Platelets + osteoblasts
Key fact	Most abundant non-collagenous protein	**Ca-binding glycoprotein**

Strength	Provided by
Tensile	**Type I collagen**
Compressive	**Proteoglycans + hydroxyapatite**

How it's asked

"Osteocalcin true EXCEPT?"	→ **"glycoprotein that binds calcium"** (that is osteonectin)
"Osteocalcin true EXCEPT (2)?"	→ **"secreted by osteoclasts"**
"Most responsible for compressive strength?"	→ **proteoglycans** ✗ Wrong options: type I collagen (tensile) · osteocalcin · osteonectin · osteopontin

BONE REMODELING

– Basic Science · 19 / 108 –

1 Basics

Remodelling laws & cortical vs cancellous turnover

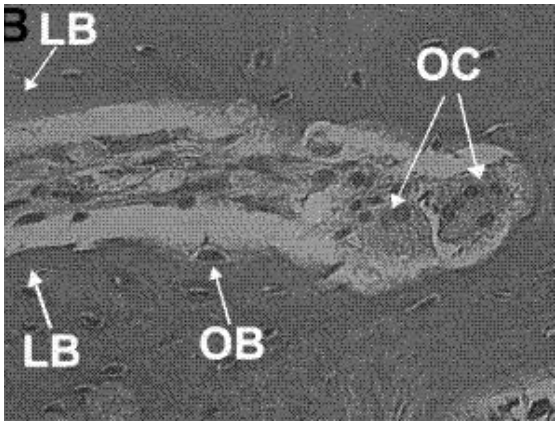
Law	Statement
Wolff's law	Bone **remodels/adapts to the loads (mechanical stress)** placed on it · ↑ load → stronger · ↓ load → osteopenia · osteocytes sense load
Davis's law	**Soft tissues** heal according to how they are mechanically stressed **(TO)**
Hueter–Volkmann	**Compression on physis slows growth**, tension/distraction accelerates (immature bone)
Delpech's law	**Tension on physis speeds growth (TO)**
Piezoelectric	**Compression side electronegative → osteoblasts (formation)** · tension side electropositive → osteoclasts
Maxwell's law	**Not a physeal law (TO)**

Law governing remodelling (TO) ✓ **both Wolff's and Davis's** (bone and soft tissue)

Remodelling mechanism

Basic multicellular unit (BMU, Frost) osteoclasts + osteoblasts + osteocytes in the remodelling cavity

	Cortical bone	Cancellous bone
Mechanism	**Osteoclastic tunnelling – cutting cone** (head = osteoclasts → capillaries → osteoblasts fill = closing cone → cement line)	**Osteoclastic resorption on trabecular surface** + osteoblastic lamellae
Cycle (TO)	**Shorter – median ~120 days**	**~200 days**



Cutting cone – osteoclasts lead, osteoblasts follow

Turnover extremes (TO) **~100 days** in thyrotoxicosis, 1° hyperparathyroidism · **>1,000 days** in myxoedema and **after bisphosphonates** (slower, not "faster")

– – Active space – –

Sclerostin & Wnt pathway

Sclerostin	glycoprotein from **osteocytes** · ✓ **inhibits osteoblastogenesis → ↓ bone formation**
Sclerostin + Dkk-1	block Wnt binding to **LRP5/6** → ✓ inhibit **Wnt/β-catenin** pathway
Wnt on	β-catenin not phosphorylated → accumulates → nucleus → osteoblast genes · also ↑ **OPG**
Knockout / inhibition of sclerostin or Dkk-1	↑ bone mass

Age-related changes

Ageing cortex	**↓ cortical thickness/area**, **↑ medullary canal** volume (bone volume constant) → ↑ fracture risk

2 Mechanism

Calcium, vitamin D & calcitonin

Sunlight	UV converts **7-dehydrocholesterol → cholecalciferol (D3)** in skin → liver 25-OH → kidney 1,25-$(OH)_2D$
↓ Sun exposure	✓ **impaired GI calcium absorption** (no 1,25-D) · liver/kidney hydroxylation intact · PTH ↑ ✗ Wrong options: ↓
Phosphate ↓ urinary Ca	✓ **complexes with Ca in the intestine** (less absorbed) · also ↑ PTH, complexes in bone/soft tissue (↓ filtered load)
Calcitonin	✓ **interferes with osteoclast maturation** (receptor on osteoclasts) · ↓ renal Ca + phosphate reabsorption · chondroprotective (MMP inhibition) · no osteoblast receptor · no major gut Ca effect · used in **Paget's when bisphosphonate-intolerant**

Anabolic therapy — intermittent PTH

Direct anabolic	✓ **intermittent PTH** ✗ Wrong options: bisphosphonate, calcitonin (antiresorptive) · corticosteroid · TNF-α inhibitor
Continuous ↑ PTH	bone resorption
TNF-α	↓ osteoblasts, ↑ osteoclastic resorption
Excess cortisol	↓ gut Ca absorption, ↑ renal Ca loss, ↓ matrix formation, secondary hyperparathyroidism

Bone healing

Rigid compression plating (absolute stability)	✓ **primary healing by Haversian remodelling** (osteonal cutting cones cross fracture) · no visible callus
Relative stability (IM nail)	**secondary healing, callus**, endochondral ossification

5 Management

Hueter–Volkmann principle

Clinical uses — **tension-band (8-) plate guided growth** · scoliosis progression (concave side compressed → slower growth) · **Scheuermann extension brace** (anterior bodies in tension → growth corrects wedging)

X vs Y

	Wolff	Hueter–Volkmann
Governs	**Remodelling** (mature + immature)	**Longitudinal growth** (immature physis)
Stimulus	Intermittent ↑ stress → apposition	Sustained compression → growth suppression

	Primary healing	Secondary healing
Stability	Absolute (compression plate)	Relative (nail, cast)
Mechanism	**Haversian / cutting cones**	**Callus, endochondral**

How it's asked

"Direct anabolic effect on bone?"	→ **intermittent PTH** ✗ Wrong options: bisphosphonate, calcitonin, corticosteroid, TNF inhibitor
"Inhibits osteoblastogenesis to ↓ bone formation?"	→ **sclerostin** ✗ Wrong options: RANKL, IL-1, Wnt, OPG
"Sclerostin + Dkk-1 inhibit?"	→ **Wnt/β-catenin**
"Physeal growth modulation by compression/tension (TO)?"	→ **Hueter–Volkmann**
"Law NOT pertaining to growth plate (TO, FNB 2023)?"	→ **Maxwell's law**
"Scheuermann extension brace corrects wedging by?"	→ **Hueter–Volkmann principle** ✗ Wrong options: Hooke, Kirchhoff, Wolff
"Tension-band plate relies on?"	→ **Hueter–Volkmann** ✗ Wrong options: cutting cones · Wolff · piezoelectric
"Phosphate ↓ urinary Ca by?"	→ **intestinal Ca–phosphate complex**
"↓ Sun exposure harms bone by?"	→ **↓ GI calcium absorption**
"Calcitonin mechanism (Paget's)?"	→ **interferes with osteoclast maturation**
"Rigid compression plating — healing type?"	→ **primary, Haversian remodelling**
"25-yr-old vs 85-yr-old femur?"	→ **thicker cortex, smaller canal**
"Law governing remodelling (TO)?"	→ **both Wolff's and Davis's**

– – Active space – –

"True about remodelling (TO)?"

→ cortical cycle shorter than cancellous
✗ Wrong options: cutting cones in cancellous · bisphosphonates speed remodelling

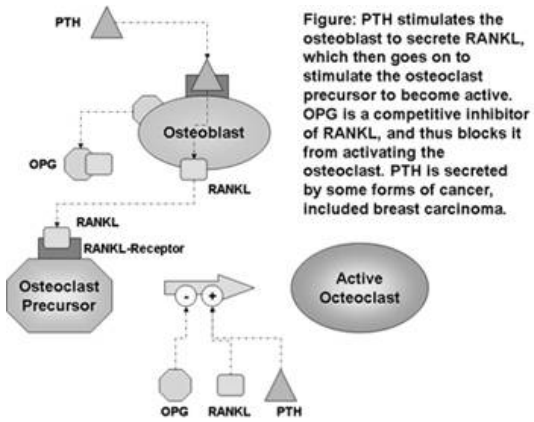
BONE SIGNALING & RANKL

– Basic Science · 20 / 108 –

1 Basics

RANK–RANKL–OPG axis

Central pathway	couples osteoblast and osteoclast activity · resorption stimulated by **RANKL (normal)** · **PTHrP (pathological – metastases)**
RANKL	major source = ✓ **osteoblasts** (+ marrow stromal cells, **activated T cells**, osteocytes) · binds **RANK** on osteoclast precursors + mature osteoclasts · RANKL + **M-CSF** → fusion into multinucleated osteoclasts · secreted in response to **inflammatory stimuli** · TNF superfamily
OPG (osteoprotegerin)	**decoy receptor** from **osteoblasts + stromal cells** · ✓ **binds and sequesters RANKL** → ↓ osteoclast differentiation, fusion, activation
Primary stimulus for osteoclastic resorption	✓ **RANK–RANKL interaction** ✗ Wrong options: PTH–osteoclast direct · OPG–RANKL



RANK, RANKL and OPG

Osteoclast transcription factors	PU.1, NF-κB, AP-1, **NFATc1**, Mitf, Myc, Src · **DC-STAMP** required for fusion

– – Active space – –

Osteoclast activators vs inhibitors

Stimulate resorption	Mechanism
RANKL	Binds RANK
PTH	Receptor on **osteoblasts** → **adenylyl cyclase/cAMP/PKA** → ↑ RANKL + M-CSF, ↓ OPG
IL-1	↑ osteoclast differentiation
1,25-(OH)$_2$D	↑ RANKL expression · terminal osteoclast differentiation
PGE2	Activates adenylyl cyclase
IL-6, MIP-1α	**Myeloma**
TNF-α	↑ resorption (✗ Wrong options: not an inhibitor)
Thyroid hormone, GH	Systemic stimulators

Inhibit resorption	Mechanism
OPG	Sequesters RANKL
Calcitonin	✓ **Binds directly to receptor on osteoclast** → loss of ruffled border, retraction, ↓ number + activity · from thyroid **parafollicular cells** · ↑ by hypercalcaemia, gastrin, pentagastrin · on osteocytes: ↓ apoptosis, ↑ sclerostin
Oestrogen	↓ RANKL · inhibits adenylyl cyclase · also anabolic
TGF-β	↑ OPG
IL-10	Suppresses osteoclasts
Androgens	Systemic
Bisphosphonates 49	Osteoclast apoptosis
Denosumab	**Anti-RANKL IgG2** (OPG-like)

Dexamethasone / corticosteroids ↓ OPG → ↑ osteoclast formation + longevity → osteoporosis

Wnt/β-catenin signalling (TO)

Wnt binds **frizzled + LRP5/6** → β-catenin accumulates → binds **lymphoid enhancer factor (LEF)** → bone-formation genes + **OPG (TO)**

Wnt antagonists **Dkk-1**, soluble frizzled-related proteins (**sFRP**), **sclerostin (SOST) (TO)**

Favour formation (TO)	Favour resorption (TO)	Coupling factors (TO)
OPG, BMPs, Wnt, β-catenin–LEF complex	RANK, RANKL, M-CSF, DC-STAMP, Dkk-1, sFRP, sclerostin	**TGF-β** (released + activated by acid during resorption) · **ephrinB2** (osteoclast) → **EphB4** (osteoblast)

Does NOT inhibit bone formation (TO) ✓ **β-catenin–LEF complex**
✗ Wrong options: sclerostin, Dkk-1, RANKL

Smad signalling

BMP-2 cascade	receptor phosphorylates **regulatory Smad 1, 5, 8** → complex with ✓ **co-Smad 4** (not phosphorylated) → nucleus
Smad 6, 7	**inhibitory** · Smad 2, 3 – TGF-β pathway

Second messengers

Adenylyl cyclase / cAMP	✓ **PTH** (Gαs/PKA in osteocytes) · PGE2
Serine-threonine kinase	BMP, TGF-β (→ SMADs) · **Tyrosine kinase** – IGF-II

Osteoporosis drugs by target

Drug	Target
Calcitonin	**Directly binds osteoclast**
Denosumab	RANKL antibody
Romosozumab	**Sclerostin antibody** → activates Wnt
Teriparatide	PTH analogue → **activates osteoblasts**
Vitamin D	Osteoclast differentiation + osteoid mineralisation

4 Diagnosis & Classification

Genes & lab markers

Alkaline phosphatase	from **bone (osteoblasts) + liver** · bone ↑: Paget's, metastases, osteomalacia · liver ↑: hepatitis, cirrhosis, **liver metastases**, biliary obstruction
↑ ALP, negative bone scan, known cancer	✓ check **ALT/AST** (liver) ✗ Wrong options: CEA · PSA · CT hip · MRI brain
Antibodies to RANKL or sclerostin	↑ BMD

- - Active space - -

6 Complications

Clinical implications

Condition	RANKL-pathway link
Osteoporosis	**Loss of OPG function** → constitutive osteoclast activation · **OPG knockout mice → osteoporosis**
Osteopetrosis	Absent osteoclastic resorption · **RANKL knockout → osteopetrosis-like** · not OPG knockout
Paget's	Altered RANK cytoplasmic binding or **OPG mutations**
Lytic metastases	✓ **Cancer cells directly produce RANKL** · OPG blockade ↓ metastasis (animals) · denosumab ↓ pathological fractures
TJA osteolysis	PE debris → macrophages → cytokines incl. **RANKL** → osteoclasts
Inflammatory arthropathy (RA pannus, gout tophi)	IL-6, IL-8, TNF-α · ↑ RANKL, ↓ OPG

OPG overexpression would help
✓ **osteoporosis** (also Paget's, metastases)
✗ Wrong options: osteopetrosis, rickets, achondroplasia, scurvy

X vs Y

	RANKL	OPG	RANK
Made by	Osteoblasts, T cells, tumour	Osteoblasts, stromal cells	– (receptor)
Located	Ligand	Soluble decoy	Osteoclast precursors
Effect	↑ resorption	↓ resorption	Mediates differentiation
Knockout	**Osteopetrosis-like**	**Osteoporosis**	–

How it's asked

"How OPG affects RANKL-mediated resorption?"	→ **inhibits by binding RANKL directly** ✗ Wrong options: binds RANK
"Major source of RANKL?"	→ **osteoblasts** ✗ Wrong options: osteoclasts · integrins · macrophages
"Lytic breast metastasis in humerus – involved molecule?"	→ **RANK** ✗ Wrong options: OPG, IL-4, TNF-α, Sox9
"True of OPG?"	→ **binds and sequesters RANKL** ✗ Wrong options: from osteoclasts · ↑ resorption · knockout → osteopetrosis
"Why metastases are lytic?"	→ **cancer cells directly produce RANKL** ✗ Wrong options: osteoclasts produce RANKL

– – Active space – –

"Cells producing RANKL?"	→ **osteoblasts**
"Osteoclastogenic factor secreted by osteoblasts after inflammation?"	→ **RANKL**
"Primary stimulus for osteoclastic resorption?"	→ **RANK–RANKL interaction**
"Function of OPG?"	→ **sequesters RANKL** ✗ Wrong options: adenylyl cyclase (oestrogen) · binds osteoclasts (calcitonin) · ↑ RANKL (PTH)
"Drug causing OPG overexpression helps?"	→ **osteoporosis**
"Mediator reducing bone resorption?"	→ **calcitonin** ✗ Wrong options: RANKL, IL-6, PTH, dexamethasone
"Calcitonin mechanism?"	→ **directly binds osteoclast receptor → ↓ activity**
"Osteoporosis drug binding osteoclasts directly?"	→ **calcitonin** ✗ Wrong options: denosumab, romosozumab, teriparatide, vitamin D
"All inhibit osteoclasts EXCEPT?"	→ **TNF-α**
"Uses adenylyl cyclase in osteocytes?"	→ **PTH** ✗ Wrong options: SMADs, IGF-II, BMP-2, TGF-β
"↑ ALP, negative bone scan, metastatic lung cancer — next test?"	→ **ALT/AST**
"Complexes with Smad 1/5/8?"	→ **Smad 4** ✗ Wrong options: Smad 6/7 (inhibitory)
"Does NOT inhibit bone formation (TO)?"	→ **β-catenin–LEF complex**

– – Active space – –

BRACHIAL PLEXUS

– Basic Science · 21 / 108 –

1 Basics

Variants

Type	Roots	Frequency
Standard	C5–T1	**77%**
Prefixed	+ C3, C4	22%
Postfixed	Below T1	1%

Anatomy – "Robert Turner Drinks Cold Beer"

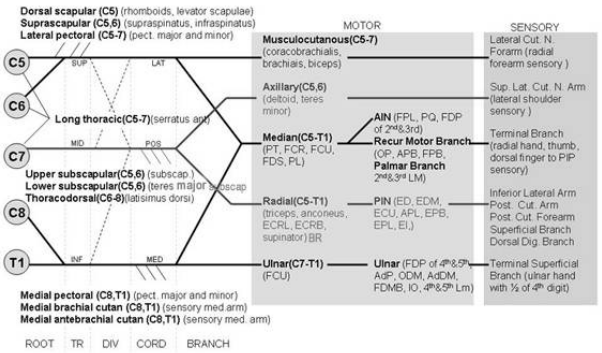
Roots (5) ventral rami **C5–T1** · superior + posterior to subclavian artery

Interscalene course ✓ exits between **scalenus anterior and scalenus medius** (+ first rib) **(TO)**
✗ Wrong options: medius/posterior · medius/levator scapulae

Trunks (3) superior (C5, 6) · middle (C7) · inferior (C8, T1)

Divisions (6) 3 anterior + 3 posterior

Cords (3) posterior (all 3 posterior divisions) · lateral (anterior superior + middle, C5–7) · medial (anterior inferior, C8, T1)



Brachial plexus with branches

Branches by level

Level	Branch	Supplies
Root	**Dorsal scapular (C5)**	Levator scapulae, rhomboids
Root	**Long thoracic (C5–7)**	Serratus anterior
Superior trunk	**Suprascapular (C5, 6)**	Supra- + infraspinatus, AC + GH joints (via suprascapular notch)
Superior trunk	Nerve to subclavius (C5, 6)	Subclavius
Lateral cord	Lateral pectoral (C5–7)	Pectoralis major
Medial cord	Medial pectoral (C8, T1)	Pierces pec minor → pec minor + major
Medial cord	Medial brachial + antebrachial cutaneous	Skin
Posterior cord	Upper subscapular (C5, 6)	Subscapularis 369
Posterior cord	**Lower subscapular (C5, 6)**	Subscapularis + **teres major**
Posterior cord	**Thoracodorsal (C6–8)**	Latissimus dorsi

Lateral cord branches – terminal branches

Cord	Terminal branches
Lateral	✓ **Musculocutaneous + median (lateral root)**
Medial	**Ulnar + median (medial root)**
Posterior	**Axillary + radial**

Musculocutaneous (C5–7) most superficial · pierces **coracobrachialis 1.5–9 cm** from origin → biceps, coracobrachialis, brachialis → **lateral antebrachial cutaneous**

Median lateral + medial roots join **anterior to axillary artery**

How it's asked

"Terminal branches of the lateral cord?"	→ **musculocutaneous + median** ✗ Wrong options: axillary (posterior cord) · ulnar (medial cord)
"Brachial plexus exits between (TO)?"	→ **scalenus anterior and scalenus medius**

– – Active space – –

BRACHIALIS

– Basic Science · 22 / 108 –

1 Basics

Anatomy

	Brachialis
Origin	Distal half of anterior humerus
Insertion	**Coronoid process + ulnar tuberosity**
Action	**Major forearm flexor** – flexes in all positions
Innervation	**Medial (major) part: musculocutaneous (C5, C6)** · **inferolateral part: radial**
Arterial supply	Muscular branches of brachial artery, recurrent radial artery



Brachialis

Anterior approach to humerus **split brachialis** using its dual innervation (internervous plane)

Dual innervation (composite / hybrid muscles)

Muscle	Nerves
Brachialis	**Musculocutaneous + radial**
FDP	Median (AIN) + ulnar
Adductor magnus	Obturator + sciatic (tibial)
Pectineus	Femoral + obturator
FPB	Median + ulnar

Single nerve **gracilis** (obturator) **(TO)** · pronator teres · FDS · coracobrachialis · latissimus dorsi
✗ Wrong options: dual

How it's asked

– – Active space – –

"All have dual nerve supply EXCEPT (TO)?"	→ **gracilis** ✗ Wrong options: FDP, adductor magnus, brachialis
"Muscle with dual innervation?"	→ **brachialis** ✗ Wrong options: pronator teres, FDS, coracobrachialis, latissimus

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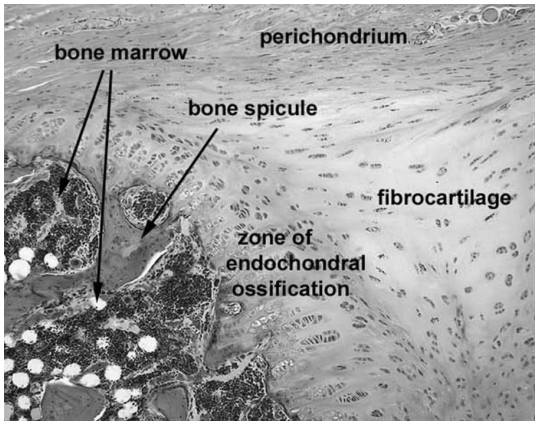
CARTILAGE

– Basic Science · 23 / 108 –

1 Basics

Types of cartilage

Type	Location	Collagen	Function
Hyaline (articular)	Articular surfaces, ribs, nasal septum	**Type II**	↓ friction, distributes load
Fibrocartilage	Tendon/ligament–bone junction, **pubic symphysis**, **annulus fibrosus**, menisci	**Type I** (predominantly)	Compressive strength · **repair tissue** after microfracture, drilling, abrasion arthroplasty
Fibroelastic	Menisci	–	Fewer PGs + glycoproteins than hyaline
Elastic	**Auricle, epiglottis, auditory tube**	**Type II** + elastin network	Tolerates repetitive deformation
Epiphyseal (physeal)	Between epiphysis + metaphysis	–	**Linear growth (endochondral ossification)** · remodelling potential



Fibrocartilage

Physeal zones reserve → proliferative → hypertrophic → primary spongiosa → secondary spongiosa · ECM holds PGs + GFs (BMP-2)

Cartilage biology

Properties **avascular, aneural, alymphatic** · nutrients by **diffusion from synovial fluid** · pH **7.4** · metabolism regulated mechanically

Composition (hyaline) water (65–80%) > collagen > PG > non-collagenous protein > cells · **stress-shielding** of solid matrix

Loading physiological = chondroprotective · **underloading** → thinning, softening, ↓ PG → fibrillation, erosion · overloading → damage (**in vitro only**)

SOX9 & chondrocyte differentiation

SOX9	✓ **"master switch" transcription factor for chondrocytes** · HMG superfamily · binds type II collagen promoter/enhancer · ↑ **type II, IX, XI collagen, aggrecan, link protein** · not a tyrosine kinase receptor

Matrix components (aggrecan, type II & IX collagen)

Type II collagen	✓ **tensile strength + reinforcement of PG gel** · restrains swelling · superficial-zone collagen bears ~**20%** of load · degradation → fibrillation
Aggrecan	negative GAG charge (chondroitin, keratan sulfate) → ↑ osmolality → attracts water → compressive strength
↓ Aggrecan expression	✓ **↓ water content** (and ↓ osmolality, ↓ compressive strength) · no direct AGE/collagen change
Pericellular integrin anchors	fibronectin, **type VI collagen**

Genes & lab markers

OA linkage in women	✓ **type IX collagen** (400-family study · knockout mice → early OA) · also **ADAMTS-5** (aggrecanase)

PTH effect on cartilage

PTH	**chondroprotective** · ✓ **↓ cartilage degradation** · via **Indian hedgehog (IHH)** signalling · synergy with bFGF, IGF
Traps	✗ Wrong options: ↑ MMP-13 (degradation) · ↑ type X collagen (end-stage hypertrophic/OA) · ↓ PG

X vs Y

	Hyaline	Fibrocartilage	Elastic
Collagen .101	**II**	**I**	**II + elastin**
Example	Joint surface	Annulus, symphysis, meniscus, enthesis	Ear, epiglottis

How it's asked

"SOX9 drives differentiation of?"	→ **chondrocytes** ✗ Wrong options: osteoclasts, osteoblasts, fibroblasts
"SOX9 regulates which tissue (histology)?"	→ **cartilage** ✗ Wrong options: bone, fat, muscle, nerve
"Strong genetic linkage with OA in women?"	→ **type IX collagen**
"Downregulate aggrecan genes → effect?"	→ **↓ water content**
"Primary biomechanical function of type II collagen?"	→ **tensile strength, reinforcing PG matrix** ✗ Wrong options: osmotic swelling (PGs)

– – Active space – –

"PTH effect on cartilage?" → **↓ degradation (chondroprotective)**

CLINICAL BILLING

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1 Basics

Definitions

Term	Meaning
CPT	5-digit code for the **procedure/service** → reimbursement · Category I (00100–99499) · Category III = provisional/emerging
RVU	Unit value per CPT code (4 factors)
ICD-10	WHO **diagnosis** classification · **3–7 alphanumeric characters** (1st alpha, 2nd numeric) · >68,000 codes (vs ~13,000 ICD-9)
E&M	Evaluation and management (office/hospital visits)
Modifier	2-digit code for a special situation

ICD-10 & modifiers

ICD-10 describes chronicity · **phase of care** · effects (routine, complication) · **laterality**

7th character ✓ **phase of treatment / encounter**
✗ Wrong options: site · laterality · aetiology · category

7th char	Meaning
A	Initial, closed fracture
B	Initial, open type I/II
C	Initial, open IIIA–C
D	Subsequent, routine healing
G	Subsequent, delayed healing
K	Subsequent, **nonunion**
P	Subsequent, **malunion**
S	**Sequela**
E/F, H/J, M/N, Q/R	Open-fracture equivalents (I/II vs IIIA–C)

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Modifier	Use
-22	Unusually complicated (difficulty, time, effort)
-24	**Unrelated E&M** in the postoperative period
-25	**Significant, separate E&M on the day of a (minor) procedure**
-58	✓ **Staged / planned / more extensive procedure in global period** · **restarts global** · 100% paid (e.g. spacer removal + graft, 2nd-stage PJI)
-59	Distinct procedural service, same encounter (different site, incision, lesion)
-76	Repeat procedure
-78	**Unplanned return to OR** for related condition/complication in global period
-79	**Unrelated** procedure in global period · overlapping globals · 100% paid

5 Management

New vs established patient

New patient — no professional services from the physician **or anyone in the same group and same specialty within 3 years**

Established — seen by physician or same-group same-specialty partner **within 3 years**

Exception — hand + sports surgeons with a **CAQ** may bill patients seen by other group orthopaedists as new

Consultation — requested by another physician for **advice** · request documented · level documented · **written response** to requester · no 3-yr rule · **Medicare no longer recognises consult codes**
✗ Wrong options: transfer of care

Coding error rate — 32.4% of Medicare claims (Shalowitz) · consult errors common

E&M coding & global period

Component-based level — history + examination + MDM · **billed at the lowest of the three**

Time-based — visit mostly counselling/coordination · time documented

Global period — 0, 10 or **90 days** · single payment

In the 90-day global	Not in the global
Preop visits/work, marking, positioning, prep	Diagnostic tests (e.g. POD1 CBC)
Skin-to-skin work	**Immobilisation applied outside OR**
Immobilisation/bracing applied in OR	Visits for separate problems
Inpatient + postop visits (physician **or extender**)	Evaluation by a different group
Closed fracture "restorative treatment" follow-up (code 99024, $0)	**Reoperation for complications** (infection) · **staged procedures**

Fraud — services not provided · not provided by billing physician · not supported by documentation · **intentional or not** · responsibility = **billing physician**

How it's asked

"Which scenario is a new patient (2009 guidelines)?"	→ **arthroscopy by partner 4 years ago** ✗ Wrong options: seen <3 yr by you or partner
"Time to revert to new patient?"	→ **3 years**
"Referred with consult request, partner saw 4 yr ago, history L3/exam L4/MDM L3, letter sent?"	→ **level 3 consultation**
"Cast removal 4 wk after closed reduction of phalanx fracture?"	→ **not billed** (90-day global)
"ICD-10 seventh character?"	→ **phase of treatment**
"Planned second-stage spacer removal + graft at 6 wk — modifier?"	→ **58** ✗ Wrong options: 78 (unplanned) · 79 (unrelated) · 59 · 25

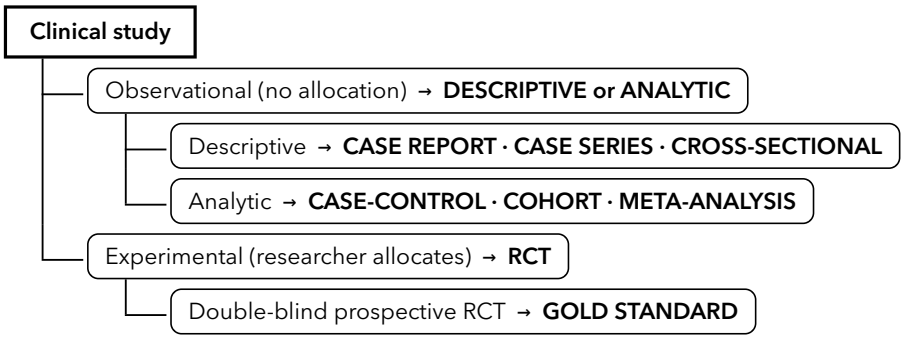
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CLINICAL TRIAL DESIGN

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1 Basics

Observational vs experimental



Internal validity methodological rigour (↓ bias, confounding) · **external validity** – generalisability · homogeneous, tightly controlled populations ↑ internal, ↓ external

Study types (case-control, cohort, parallel, factorial, RCT)

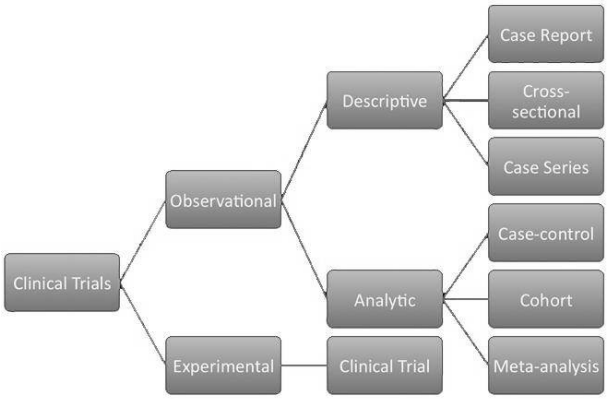
Design	Definition	Level	Measure
RCT	Random assignment to treatment/control, followed prospectively	**I**	– · **best for causality**
Prospective cohort	Groups separated **non-randomly by exposure**, exposure **after** study start	**II**	**Relative risk**, incidence
Retrospective cohort	Exposure **before** study start	**III**	Relative risk
Case-control	Groups by **presence (cases) / absence (controls) of disease** → look back for exposure · always retrospective	**III** (most)	**Odds ratio**
Case series	Retrospective, same injury/treatment, **no control** · generates hypotheses	**IV**	–
Cross-sectional	**Snapshot** at one time, no follow-up (e.g. prevalence)	–	Prevalence
Meta-analysis	Systematic review pooling studies · **↑ power of under-powered studies**	Level of included studies	–
Case report / expert opinion	–	**V**	–

RCT with 70% follow-up downgraded to **Level II** (<80% follow-up)

RCT designs

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Design	Allocation
Parallel	Each group gets one intervention (commonest RCT) · e.g. bisphosphonate vs placebo
Crossover	Each participant gets **both therapies sequentially** in random order · needs adequate **washout**
Factorial	Groups get **A + B, A, B or neither** → tests interaction
Cluster	**Pre-existing groups** (schools, villages) randomised



Study types

2 Mechanism

Randomisation & selection bias

Purpose of computer randomisation
✓ **prevent selection bias** (unbiased placement into groups)
✗ Wrong options: ↑ power · ↑ treatment effect · significance · prevent crossover

Sealed opaque envelopes / allocation concealment
minimise **selection bias** (recruiters can't predict next allocation)

Randomisation also
distributes **confounders** (known + unknown) evenly → ✓ **best reduces confounding** · facilitates blinding · allows probability theory

Confounders
external factors affecting both predictor and outcome · controlled by randomisation, stratification, multivariable regression

Blinding & other biases

Single-blind
patients unaware, surgeon delivers + assesses (experimenter bias remains)

Double-blind
patient + doctor/assessor unaware

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Blinding the outcome assessor ✓ ↓ **reported treatment effect** (unblinded assessors inflate effect — Poolman, 32 RCTs)

Bias	Definition
Selection	Dissimilar groups / non-random enrolment · prevented by randomisation + concealment
Washout period	Crossover — first therapy's effect persists (drug t½ long)
Recall	Subjects' recollection differs by outcome or **after unblinding** · common in retrospective/survey studies
Observer-expectancy	Investigator finds what they expect · prevented by masking
Procedure	Groups treated differently beyond the intervention (e.g. extra PT)
Attrition	Patients don't complete study
Detection	Look harder for outcome in one group
Verification	Test result influences group assignment
Incorporation	Diagnostic test includes features of target outcome
Confirmation	Interpretation skewed by researcher's opinion
Sampling	Homogeneous population → ↓ generalisability
Publication	Unfavourable results unpublished (industry funding)
Hawthorne effect	Behaviour changes when observed

Intention-to-treat analysis

Analysis	Who is analysed
Intention-to-treat	**As randomised**, regardless of treatment received/compliance · preserves randomisation · ↓ non-responder bias
As-treated	By treatment actually received · no longer an RCT · used with high crossover
Per-protocol	**Excludes non-compliant** patients

Bayesian analysis prior probability + new data → posterior probability

Effect size (difference between groups) ÷ SD

5 Management

Evidence-based care & conflict of interest

Evidence-based health care EBM principles applied to **all health professions incl. purchasing + management**

Analysis	Effects expressed as
Cost-benefit	Money
Cost-effectiveness	Health units (cost per event prevented)
Cost-utility	Utilities (cost per **QALY**)
Decision analysis	Explicit quantitative choice under uncertainty

Conflict of interest statement ✓ **discloses all financial interests** of each author

– – Active space – –

X vs Y

	Cohort	Case-control
Grouped by	**Exposure**	**Outcome/disease**
Direction	Forward (or retrospective)	Backward
Measure	**Relative risk**	**Odds ratio**
Level	II (prospective) / III	III

How it's asked

"Purpose of computer random-number generator in Level I trial?"	→ **prevent selection bias** ✗ Wrong options: power · significance · as-treated
"Intern's RCT, attending recommends computer randomisation — why?"	→ **prevent selection bias** ✗ Wrong options: blinding · prevent crossover
"Randomisation in sealed opaque envelopes minimises?"	→ **selection bias** ✗ Wrong options: attrition · NNH
"Reason for computer randomisation (hemi vs screws)?"	→ **unbiased placement into groups** ✗ Wrong options: confirmation bias · heterogeneity
"Closure-technique QI study — best reduces confounding?"	→ **randomisation** ✗ Wrong options: Hawthorne · blinding · power analysis
"Crossover PT vs drug (t½ 2 wk), immediate switch — bias?"	→ **washout period bias**
"Patients unaware, surgeon injects and scores outcomes?"	→ **single-blinded randomised**
"Effect of blinding the outcome assessor?"	→ **↓ reported treatment effect**
"Patient discovers their drug before final evaluation — bias?"	→ **recall bias** ✗ Wrong options: selection · observer-expectancy · procedure
"50 knee ON patients vs 23 without, compare alcohol intake?"	→ **case-control**
"Two groups randomised to bisphosphonate vs placebo?"	→ **parallel design**
"Two interventions applied separately or combined?"	→ **factorial**
"Dislocated vs non-dislocated THAs, look for factors?"	→ **case-control**
"Definition of cohort study?"	→ **groups separated non-randomly by exposure, before or after study start**

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"Randomised prospective drug vs standard for AVN — true?"	→ **best method for causality** ✗ Wrong options: OR (case-control) · RR (cohort) · ↑ power (meta-analysis) · hypothesis-generating (case series)
"RCT with 70% follow-up — level?"	→ **II**
"Athletes choose cleat, followed prospectively through season — level?"	→ **II (prospective cohort)**
"30 non-completers kept in PT group?"	→ **intent-to-treat** ✗ Wrong options: per-protocol
"Evidence use across health professions incl. purchasing?"	→ **evidence-based health care**
"Purpose of conflict-of-interest statement?"	→ **disclose all financial interests**

COLLAGEN

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1 Basics

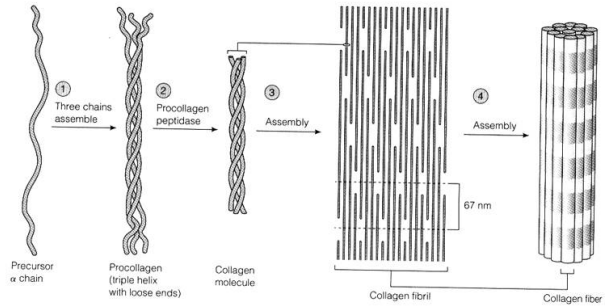
Overview

Most abundant protein in humans	~25% of body protein · >25 types · made by **fibroblasts** · synthesis **inside and outside** the cell
Structure	**triple helix: two α1 + one α2** chains · repeats **Gly-X-hydroxyproline**, **Gly-Pro-X** (glycine every 3rd residue)
Type I genes	**COL1A1** (pro-α1) + **COL1A2** (pro-α2)

Assembly

Collagen synthesis

- Intracellular → α-CHAINS → **PROCOLLAGEN (2 α1 + 1 α2, triple helix)**
- Extracellular → **TROPOCOLLAGEN (enzymatic processing after Golgi)**
  - Lysine/hydroxylysine aldehyde cross-links → FIBRIL → **FIBRE**



Collagen processing

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Collagen types by tissue

Type	Tissue	Disease
I (>90% of body collagen)	**Bone, tendon, ligament, meniscus, annulus fibrosus, skin, healed cartilage/fibrocartilage, scar, nerve** (tensile strength), deep calcified cartilage	**Osteogenesis imperfecta**, Ehlers–Danlos
II (90–95% of cartilage collagen)	**Articular/hyaline cartilage**, **nucleus pulposus**, **elastic cartilage** · very long half-life	–
III	**Skin, blood vessels** · **early tendon/ligament healing** (inflammatory + early regenerative)	**Ehlers–Danlos (vascular, type IV)**, **Dupuytren's**
IV	**Basement membrane**	Goodpasture, Alport
V, VI, IX	Small amounts in articular cartilage · **VI only pericellular** (lacunae, mechanotransduction) · V, VI in disc	–
IX	Cartilage 90	Deletion → **OA in women**, knockout mice · **COL9A1–3 → MED**
VII, VIII	Epithelial basement membrane	–
X	**Deep calcified cartilage** · only **hypertrophic chondrocytes** in endochondral ossification (physis, callus, HO) · ↑ **early OA**	**Schmid metaphyseal chondrodysplasia**
XI	**Adhesive holding cartilage collagen lattice together**	–
XII	Annulus only	–

Disc **annulus = type I** · **nucleus = type II** · type VI both

Bone collagen

Bone weight inorganic **60–70%** · water **5–8%** · organic remainder · collagen = 90% of organic = **20–25% of bone weight** · collagen gives **flexibility**

Multiple epiphyseal dysplasia

Genes **COMP (70%)** · COL9A1, COL9A2, COL9A3

Features fragmented ossific centres · **coxa vara** · **genu valgum** · short stubby metacarpals



MED – coxa vara and fragmented capital epiphyses

2 Mechanism

Tendon & ligament healing

Inflammatory + early regenerative phase	✓ **type III** predominates → **type I** in late regenerative phase

How it's asked

"Type I collagen in peripheral nerve provides?"	→ **tensile strength** ✗ Wrong options: conduction velocity/ strength · compression
"Collagen first produced in inflammatory phase of tendon/ligament healing?"	→ **type III**
"Major cartilage collagen, 90–95%?"	→ **type II**
"Collagen % of bone weight?"	→ **20–25%**
"Predominant collagen of nucleus pulposus?"	→ **type II** (annulus – type I)
"Type II most abundant in all EXCEPT?"	→ **blood vessels** (type III)
"Collagen of hypertrophic chondrocytes, ↑ in early OA, Schmid?"	→ **type X**

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COMPLEX REGIONAL PAIN SYNDROME (CRPS)

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1 Basics

Epidemiology

Synonyms	**RSD** (type I) · **causalgia** (type II)
Incidence	5.5 / 100,000 person-yr (US) · 6–26 / 100,000 (OB) · ↑ with age to 70
Sex	**female 4:1** (Orthobullets/OB) · 2–4× (TO)
Site	**upper limb 60%** (Orthobullets) · **70%** (TO) · type I **88%** vs type II 12% (TO)
Postoperative rate	10–37% (OB)

Risk factors

Listed	trauma with exaggerated response (**commonest cause of poor outcome after foot crush**) · surgery (multiple procedures) · **prolonged immobilisation**, tight dressings/casts (early cast complaints) · **anxiety/ depression** · **ACE inhibitor at time of trauma** · migraine, asthma · **smoking** · **fibromyalgia (↑ up to 2.5×)** · RA, carpal tunnel
Not a risk factor	**male sex**

Pathophysiology

Triad	✓ **aberrant inflammatory response** · **vasomotor dysfunction** · **maladaptive neuroplasticity** → sustained sympathetic activity in a perpetuated reflex arc
Mediators	IL-1β, IL-6, TNF-α, **substance P** · ↑ osteoclasts → regional migratory osteoporosis / **bone marrow oedema**

Genes & lab markers

Genetics	↑ risk in siblings of young-onset cases · **TNF-α and ACE gene polymorphisms**
Labs	WBC, ESR, CRP **normal** (distinguishes from infection)

3 Presentation

Cardinal signs	**pain out of proportion** · **swelling** · **stiffness** · **skin discolouration**
Exam	vasomotor disturbance · trophic skin changes · **hyperhidrosis** · **allodynia, hyperalgesia, hyperpathia** · cool shiny skin · **flamingo gait** (knee) · **equinovarus** (ankle) · not **crepitus**
Not typical	↓ fingertip perfusion (Raynaud) · ptosis/anhidrosis (Horner, preganglionic plexus) · fingertip ulcers (Buerger) · positive Spurling

Natural course (TO) warm, ↑ hair/nail growth, oedema, erythema → later slow hair growth, brittle nails, weakness, **cold limb**

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Lankford & Evans stages

Stage	Onset	Exam	Imaging
Acute	0–3 mo	Burning pain, redness, swelling, warmth, hyperhidrosis	X-ray normal · **bone scan +**
Subacute (dystrophic)	3–12 mo	↑ pain, cyanosis, dry skin, stiffness, skin atrophy	**Subchondral osteopenia**
Chronic (atrophic)	>12 mo	↓ pain, glossy skin, fibrosis, contractures, hair/nail loss	**Extreme osteopenia**

4 Diagnosis & Classification

IASP classification (type I vs II)

	Type I	Type II
Nerve damage	✓ **None identifiable**	**Identifiable nerve injury**
Frequency	**Most common**	Less common
Sympathetic block	Responds	**Minimal / no response**
Distribution	Whole region	Single nerve territory

Not distinguishing autonomic dysfunction · trophic changes · functional impairment · positive bone scan (all in both)

Diagnostic criteria – Budapest

Criterion	Requirement
1	**Continuing pain disproportionate** to inciting event
2 Symptoms	≥1 symptom in **3 of 4** categories (**clinical**) · **4 of 4** (research)
3 Signs	≥1 sign in **≥2** categories at evaluation
4	**No other diagnosis** explains it

Category	Symptoms / signs
Sensory	Hyperaesthesia, **allodynia**, hyperalgesia (pinprick)
Vasomotor	**Temperature asymmetry**, skin colour change/asymmetry
Sudomotor / oedema	**Oedema, sweating change/asymmetry**
Motor / trophic	↓ ROM, weakness, tremor, dystonia, hair/nail/skin changes

Budapest vs IASP IASP sensitivity 1.00, **specificity 0.41** · Budapest sensitivity 0.99, **specificity 0.68–0.79**

Best diagnostic test ✓ **history + physical examination** (clinical) · confirmed by **relief with sympathetic block** · early diagnosis critical

– – Active space – –

Key radiology findings

Modality	Finding
X-ray	Osteopenia (**patella** – commonest knee finding) · soft-tissue swelling · subperiosteal resorption · **joint spaces preserved**
Three-phase bone scan	↑ uptake all phases · **phase III (2–3 h) most sensitive** · **high NPV** (rules out type I) · **most sensitive investigation (TO)** · TO note: high specificity, ↓ sensitivity vs Budapest, not confirmatory
MRI	**Diffuse bone marrow oedema** (warm phenotype)
Thermography	Limb temperature difference **≥1 °C** significant · questionable utility
EMG/NCV	Slowing in known nerve (type II, e.g. median) · not diagnostic for type I · myoclonus only 11–36%



Three-phase bone scan in CRPS

Bone scan phase	Shows
I (2 min)	Arteriogram
II (5–10 min)	Cellulitis, synovitis
III (2–3 h)	**Bone**
IV (24 h)	Osteomyelitis vs cellulitis

Differential

Listed soft-tissue infection · malingering · **clenched fist syndrome** (psychiatric) · neuropathic/chronic pain · **Raynaud** (digital vasospasm) · TOS · arterial insufficiency · erythromelalgia

Clenched fist syndrome **non-organic somatoform** disorder · fixed finger flexion · **macerated foul-smelling palm**, paradoxical stiffness · psychiatric association, poor prognosis · normal imaging/EMG → therapy + psychiatry

5 Management

Vitamin C prophylaxis

Regimen ✓ **vitamin C 500 mg daily × 50 days** · 200 mg if renal impairment

Evidence	**Level I** (Zollinger, 2 RCTs) in **distal radius fractures** · Lancet: CRPS **7% vs 22%** placebo · JBJS: 200 mg 4.2%, **500 mg 1.8%**, 1500 mg 1.7% → 500 mg recommended · later studies conflicting · also ↓ **CRPS type I** after **foot & ankle surgery** (non-diabetics)
Mechanism	↓ lipid peroxidation · scavenges hydroxyl radicals · protects capillary endothelium · ↓ vascular permeability
Other prevention	avoid tight dressings + prolonged immobilisation

Treatment (physiotherapy, blocks, sympathectomy)

CRPS treatment ladder

- First line → **PT (GENTLE / ACTIVE, desensitisation) + DRUGS**
  (avoid casting, immobilisation, aggressive PROM)
- Failed → SYMPATHETIC / NERVE BLOCK → **CHEMICAL SYMPATHECTOMY**
  - Focal LE CRPS failing blocks → **DRG STIMULATION (TO)**
- Refractory → **SURGICAL SYMPATHECTOMY**
  - Type II with known nerve → **DECOMPRESS NERVE (e.g. CTR)**

Early CRPS	✓ **immediate PT: desensitisation + active ROM + multidisciplinary follow-up** ✗ Wrong options: restrict PT · LSB first · bone scan before treating
Modalities	**gentle physiotherapy** · contrast baths · **tactile discrimination training** · **graded motor imagery** · mirror therapy · splinting, elevation, massage · **TENS — positive only in type II** (OB)
Graded motor imagery sequence (TO)	✓ **laterality restoration → explicit motor imagery → mirror therapy**
Drugs	NSAIDs · **α-blockers (phenoxybenzamine, prazosin)** · β-blocker (propranolol) · antidepressants · anticonvulsants · Ca-channel blockers · **gabapentin** · **bisphosphonates** · antiarrhythmics · corticosteroids · calcitonin · vitamin C
Also used (TO)	**ketamine** (NMDA antagonist, topical/IV) · **botulinum toxin A** · **mirror therapy** · naltrexone · plasma exchange · cannabis inhaler · **pramipexole — not used**
Warm phenotype + bone marrow oedema on MRI (<6 mo, TO)	✓ **IV bisphosphonates (neridronate)** first-line pharmacological
Post-TKA CRPS	✓ **α-blockers + PT + tactile discrimination + graded motor imagery** ✗ Wrong options: explore knee · I&D/poly exchange · L-spine MRI
Nerve stimulation	TENS · peripheral nerve · spinal cord stimulator · indicated when symptoms mainly in **one peripheral nerve**

– – Active space – –

Neuromodulation – focal LE CRPS after failed blocks (TO)

✓ **dorsal root ganglion stimulation** (ACCURATE trial: up to 80% pain ↓, fewer postural changes vs tonic SCS) · diffuse whole-limb → SCS

Block / sympathectomy	Target
Stellate ganglion	**Upper limb**
Lumbar sympathetic	**Lower limb**
Agents (block)	Lidocaine, bupivacaine, bretylium, guanethidine
Agents (chemical sympathectomy)	**Phenol, alcohol**

Second line after failed early PT modalities (OB)

✓ **surgical sympathectomy** · best candidates = responded to sympathetic block

✗ Wrong options: cast · boot NWB · exchange nail · SCS

Best success

find + treat an associated nerve problem (type II)

6 Complications

Prognosis

poor response to treatment · **better: upper limb, warm CRPS, children** · late CRPS refractory → permanent disability

X vs Y

	CRPS	Raynaud	Horner (preganglionic plexus)	Clenched fist
Hallmark	Pain out of proportion, allodynia, sweating	Digital vasospasm in cold	Ptosis, miosis, anhidrosis	Fixed clenched fist, psychiatric
Mechanism	Inflammatory + vasomotor	Exaggerated arterial vasoconstriction	Sympathetic chain injury	Somatoform

How it's asked

"Most sensitive investigation for CRPS (TO)?"	→ **bone scan** ✗ Wrong options: CT, MRI, US
"Clinical features of LE CRPS EXCEPT?"	→ **crepitus**
"Commonest X-ray finding in knee CRPS?"	→ **patellar osteopenia**
"Healed DRF, now hyperalgesia, allodynia, hyperhidrosis – pathophysiology?"	→ **aberrant inflammatory and vasomotor response**
"Post-Kidner, allodynia, hyperalgesia, oedema of medial foot?"	→ **RSD**
"All common in CRPS EXCEPT?"	→ **↓ perfusion to fingertips**
"Fibromyalgia patient with DRF – at ↑ risk of?"	→ **hyperalgesia, allodynia, oedema (CRPS)**

"CRPS type I vs II differs by?"	→ **no identifiable peripheral nerve involvement**
"Whole-leg pain, shiny cool cyanotic skin after tibia ORIF – diagnosis + best test?"	→ **CRPS type I; history and examination** ✗ Wrong options: type II bone scan · EMG
"Wrist sprain, allodynia, sweating, relief with sympathetic block?"	→ **CRPS type I** (type II doesn't respond)
"Vitamin C after DRF ORIF prevents?"	→ **pain out of proportion + ↑ sweating (CRPS)**
"Adjuvant improving outcome after DRF ORIF?"	→ **oral vitamin C**
"Level I evidence for vitamin C reducing CRPS in?"	→ **distal radius fractures**
"Vitamin C after F&A surgery ↓?"	→ **CRPS type I**
"Protocol shown to prevent CRPS?"	→ **vitamin C 500 mg × 50 days**
"NOT used in CRPS treatment (TO)?"	→ **pramipexole**
"Graded motor imagery sequence (TO)?"	→ **laterality → explicit imagery → mirror therapy**
"Focal knee CRPS after ACL, failed blocks – neuromodulation (TO)?"	→ **DRG stimulation**
"Post-TKA burning pain, oedema, ↑ hair, marrow oedema – first-line drug (TO)?"	→ **IV bisphosphonate (neridronate)**
"Positive effect in early CRPS?"	→ **gentle physiotherapy** ✗ Wrong options: casting, plyometrics
"Post-TKA burning, sweating, purple leg, aspirate negative?"	→ **α-blockers + PT + tactile discrimination + GMI**
"Failed splinting, contrast baths, TENS – second line?"	→ **surgical sympathectomy**
"Early CRPS after ankle sprain – initial step?"	→ **immediate PT, desensitisation + AROM, multidisciplinary follow-up**
"Crush foot CRPS – risk factors EXCEPT?"	→ **male gender**
"Painful clenched hand, depression, normal studies?"	→ **clenched fist syndrome**

– – Active space – –

DEEP VEIN THROMBOSIS

– Basic Science · 28 / 108 –

1 Basics

Epidemiology

Incidence	80 / 100,000 / yr · LE 1 / 1,000 / yr · 2.5–5% of population · usually **>40 yr** · rare in children
High-risk procedures	spine fracture with paralysis · **TKA (2–3× DVT rate of THA)** · THA · hip fracture · polytrauma
DVT rate ≠ PE or death	AAOS review: DVT rate does **not correlate** with PE/death after THA/TKA
Risk factors	immobilisation · mechanical compression · trauma · **prior DVT** · surgery · polycythaemia vera · thrombocytosis · dehydration · malignancy · CHF · obstructive airway disease
No ↑ risk	**factor V Leiden in THA** (no ↑ DVT or PE vs controls) · **rheumatoid arthritis in TKA** (age/comorbidity-matched)

Pathophysiology

Virchow's triad	hypercoagulability · venous stasis · endothelial injury
Factor V Leiden	factor V resistant to **protein C** inactivation → hypercoagulable (factor V = cofactor for Xa → thrombin)

Anatomy

	Deep veins
Proximal LE	**Popliteal and above** (above trifurcation)
Distal LE – axial	Peroneal, anterior + posterior tibial
Distal LE – muscular	Soleal, gastrocnemius
UE	Radial, ulnar, brachial, axillary, subclavian, IJ, brachiocephalic

2 Mechanism

Timing & intraoperative embolisation

Peak marrow embolisation in cemented THA	✓ **femoral stem insertion** (also acetabular prep, reduction)
Symptomatic VTE after TJA	✓ **1–6 weeks** (2nd–6th week) · events on day of surgery = fat embolism or cardiac · ~10% readmitted within 3 months

3 Presentation

Symptoms	**swelling (70%)** · **calf pain (50%)** · redness · warmth

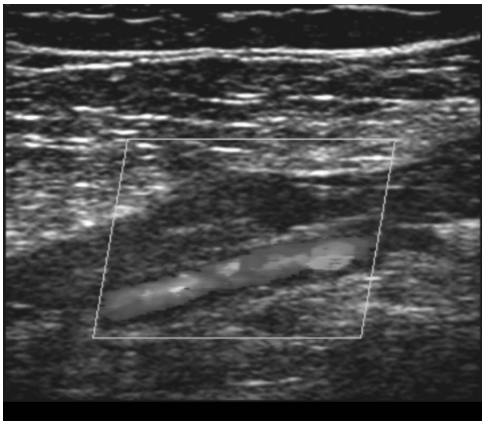
Exam unilateral/bilateral swelling · erythema · dilated veins · tenderness · **Homans' sign not specific** (or sensitive)

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Test	Sens / spec	Role
Complete duplex US	**96% / 97%**	✓ **First line** · non-compressible vein, loss of phasic flow with Valsalva, ↑ superficial flow · operator-dependent · **routine screening not recommended**
Contrast venography	–	**Historical gold standard** · invasive · if US unavailable/inconclusive
CT venography	90% / 95%	Suspected PE or inconclusive US
MR direct thrombus imaging	90% / 90%	Not widely used
Plethysmography (proximal)	75% / 90%	–



Duplex ultrasound – non-compressible vein

D-dimer

High sensitivity, low specificity excludes VTE in **low clinical probability** · >500 ng/mL suggests PE · **useless after injury/surgery** (false positives)

Differential

Mimic	Clue
Cellulitis	Infective labs
Post-thrombotic syndrome	Prior DVT, chronic swelling, stasis ulcers
Ruptured Baker cyst	US: no venous thrombus
Superficial thrombophlebitis	Cord-like superficial vein
CHF, nephrotic, cirrhosis	Chronic bilateral oedema

– – Active space – –

5 Management

Anticoagulation

Agent	Role
DOACs (dabigatran, rivaroxaban, apixaban, edoxaban)	✓ **First line** for DVT + PE (ACCP 2016, ESC)
LMWH (enoxaparin)	**Early postoperative period** · high bleeding risk
Warfarin	**Bridge with LMWH until INR 2–3** · preferred with **mechanical heart valve**

Indication postoperative **above-knee DVT** · below-knee treatment controversial

Scenario	Duration
Isolated provoked DVT	**3 months**
With PE	**>3 months** (longer if large clot, persistent risk, recurrence)
>3 months	Unprovoked · recurrent · active malignancy · paralysis/immobilisation · thrombophilia

Procedures

Intervention	Indication
IVC filter	**DVT identified preop** in high-risk pelvic/LE trauma · contraindication to anticoagulation
Thrombolysis (systemic)	Life/limb-threatening PE or acute DVT · high morbidity
Catheter-directed thrombolysis	Initial treatment **only for threatened limb loss**
Thrombectomy	Anticoagulation ineffective, high-risk thrombus

Risk factors & individualised prophylaxis

Foot & ankle (ICM-VTE)

✓ **patient-specific risk assessment** (age, DM, prior DVT, NWB, obesity) · **NWB = independent risk factor** but no prophylaxis rule based on NWB alone
✗ Wrong options: routine chemoprophylaxis for all NWB · blanket "none for below-knee" · surveillance duplex · postop D-dimer

6 Complications

Post-thrombotic syndrome chronic venous insufficiency · venous HTN · swelling, pain, pigmentation, ulceration, induration · **recurrent DVT 4–8×**

PE 1 in 1,000 / yr · anticoagulation ± intervention

Bleeding anticoagulants, thrombolysis

How it's asked

"Peak marrow embolisation in cemented THA?"	→ **femoral stem insertion**
"When do most symptomatic VTE occur after TJA?"	→ **1–6 weeks**
"Factor V Leiden and THA?"	→ **no increased risk of DVT or PE**
"RA (RF, anti-CCP, MTX) vs OA patient for TKA — VTE prophylaxis?"	→ **equal risk, standard prophylaxis for both**
"Hindfoot fusion, NWB 6 wk, BMI 31, DM, prior DVT — strategy?"	→ **patient-specific risk assessment, consider chemoprophylaxis** ✗ Wrong options: none · routine for all · screening duplex · D-dimer
"First-line imaging for DVT?"	→ **complete duplex ultrasound** · routine screening not recommended
"First-line DVT treatment?"	→ **DOAC** · warfarin if mechanical valve

– – Active space – –

EMBRYOLOGY

– Basic Science · 29 / 108 –

1 Basics

Timeline

Event	Time
Gastrula (3 germ layers)	~**3 weeks (TO)**
Neural plate	✓ **Day 18–19 (TO)**
Limb bud	**4 weeks** (stage 12, **day 26**)
Appendicular skeleton forms	**4–8 weeks**
Chondrification	Humerus day 36 → distal phalanges day 50 (proximal → distal)
Shoulder interzone / hand interzones	Day 36 / day 47
Digital rays in hand paddle	Stage 17, **day 41**
Interdigital apoptosis	Stages 19–22, **days 47–54**
Limb bud on transvaginal US	**8 weeks**

Early stages (TO) zygote → morula (16 cells) → blastula (128) → blastocyst (trophoblast + inner cell mass) → gastrula

Limb bud signalling (AER, ZPA, timing)

Limb bud lateral plate + somatic mesoderm under ectoderm · Shh from notochord regulates formation · growth = **AER ↔ progress-zone mesoderm** interaction (FGF-driven)

All UL bones endochondral except distal tufts of distal phalanges (membranous)

Signalling centre	Order	Molecule	Axis	Defect
Apical ectodermal ridge (AER)	**1st**	**FGF8 (obligatory, earliest)** · FGF4, 9, 17 redundant · forms under **FGF10**	**Proximal → distal** + interdigital necrosis	Removal/insult → **transverse truncation** (congenital transverse absence of forearm), central deficiency (cleft hand), radial club hand
Zone of polarising activity (ZPA) – posterior	**2nd**	**Sonic hedgehog (Shh)** – dose-dependent	**Anteroposterior (radioulnar)**	**Duplication/anterior graft → mirror hand (ulnar dimelia)**
Non-AER dorsal ectoderm (Wnt centre)	**3rd**	**WNT7a → Lmx1b**	**Dorsoventral** (all dorsal features incl. **nails**)	–
Ventral ectoderm	–	**Engrailed-1** (antagonises WNT7a)	Ventral patterning	–

– – Active space – –

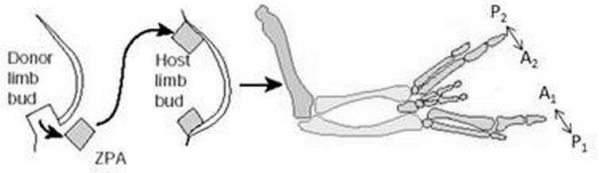
Axis and Signaling Center	Action	Area-Specific Secreted Proteins and Transcription Factors
Dorsoventral WNT pathway	Development of dorsal and ventral structures	Ventral ectoderm expressing en-1 Dorsal ectoderm expressing WNT7a and r-Fng Dorsal mesoderm expressing Lmx1
Anteroposterior ZPA	Development of ulnar and radial structures	Posterior mesoderm (ZPA) expressing Shh. Other specific markers of ZPA include Gli 3, FGF10, Msx1, Nmyc, Slug, higher number Hox genes Anterior mesoderm expressing Gli 3R, lower number Hox genes
Proximodistal apical ectodermal ridge	Outgrowth of the limb from proximal to distal	Apical ectodermal ridge expressing p63, WNT3a, FGF2, FGF4, FGF8, BMP2

en-1, engrailed-1; Gli 3R, Gli 3 repressor; Hox, homeobox; r-Fng, radical fringe.

Limb bud signalling centres

Shh gradient

Little finger	**highest** Shh (posterior/ulnar) · **thumb** — **lowest** Shh (anterior/radial)
↑ Shh posteriorly	**ulnar (postaxial) polydactyly** (dose-dependent) · ↓ Shh posteriorly → loss of ulnar digits
↑ Shh anteriorly	**loss of thumb**
Timing	**posterior elements (ulna, little finger) form early**, anterior (radius, thumb) late → AP disruption loses **radius/thumb**
Shh activates Gremlin	inhibits BMPs that would block AER FGF
Mesoderm markers	anterior **ALX4** · posterior **Hox8**



Mirror hand (ulnar dimelia)

Chondrification & joints

Chondrification factors	**Sox-5, -6, -9** · TGF-β, BMP-2 · FGFs · retinoids · hedgehog · PTHrP · cadherins · WNT5a, WNT7a
Joint formation	repress chondrogenesis: **WNT4, WNT14, GDF5** (CDMP-1)
Finger separation	**Msx2** in interdigital mesenchyme → **BMP4**-mediated apoptosis · Hox-7
Apert (acrocephalosyndactyly)	**FGFR2** mutation · severe craniofacial + mild syndactyly (gain-of-function, FGF2) · mild face + severe syndactyly (binds FGF10)

Somites & axial skeleton (Hox genes)

Somites	paired **paraxial mesoderm** beside notochord · **cranial → caudal** (paraxis gene) · **42 pairs (TO)**

Somite layer	Becomes
Sclerotome	✓ **Axial skeleton — vertebral bodies + annulus fibrosus**
Myotome	Muscle (dorsal epimere → dorsal muscles)
Dermatome	Dermis/skin

Vertebra precursor (TO)	✓ **somites** · each vertebra = **caudal half of one sclerotome + cranial half of the next** (two somite levels) · sclerotome cells engulf notochord in week 4
Metameric shift	spinal nerves, originally mid-sclerotome, exit **between** vertebral bodies

– – Active space – –

Hox (homeobox) genes ✓ **regulate segmentation/identity of axial skeleton** (rostral → caudal) · with Shh, AP patterning

Structure	Origin
Nucleus pulposus	**Notochord**
Annulus fibrosus	**Sclerotome**
Spinal cord	Neural tube
PNS, pia, spinal ganglia, sympathetic trunk	**Neural crest**

Neural tube DV patterning **Shh** ventral (floor plate, notochord) vs **Wnt/β-catenin** dorsal (roof plate)

Vertebral primary ossification centres (3) **centrum** · **neural arch** (posterior elements, pedicles) · costal element

X vs Y

Gene	Function
FGF8 / FGF10	Proximodistal growth
Shh	AP (radioulnar), limb bud formation
WNT7a	Dorsoventral
Hox	Axial segmentation
COL1A1	Collagen cross-linked fibres

Hand anomaly	Mechanism
Mirror hand	**ZPA duplication**
Preaxial polydactyly	Sporadic · Holt-Oram, Fanconi
Postaxial polydactyly	AD in African-Americans
Syndactyly	Failed apoptosis · Poland, Apert
Hypoplastic thumb	VACTERL, Holt-Oram, TAR, Fanconi

How it's asked

"Congenital transverse absence of forearm – vascular insult to?"	→ **apical ectodermal ridge** ✗ Wrong options: limb bud mesoderm · ectoderm · notochord
"Duplication of ZPA causes?"	→ **mirror hand** ✗ Wrong options: preaxial/postaxial polydactyly, syndactyly, hypoplastic thumb
"Transplant along AP axis → mirror hand – site + molecule?"	→ **ZPA, Shh** ✗ Wrong options: AER/FGF8 · WNT7a
"Limb bud timing and US?"	→ **forms 4 wk, transvaginal US 8 wk**

"Sclerotome becomes?"	→ **axial skeleton**
"Function of homeobox genes in axial skeleton?"	→ **regulate segmentation (somatisation)** ✗ Wrong options: limb bud (Shh) · PD (FGF10) · DV (Wnt)
"Neural plate formation by (TO)?"	→ **18 days**
"Embryological precursor of vertebrae (TO)?"	→ **somites** ✗ Wrong options: notochord (NP)

– – Active space – –

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ENDOCHONDRAL BONE FORMATION

– Basic Science · 30 / 108 –

1 Basics

Where it occurs

Three settings — **longitudinal physeal growth** · **embryonic long-bone formation** · **non-rigid (secondary) fracture healing** (cast, brace, **IM nail**)

Principle — **cartilage model** → chondrocytes hypertrophy/calcify → osteoclasts resorb → osteoblasts lay bone **on** the cartilage framework (bone **replaces** cartilage, not converted from cartilage) · deposition on **metaphyseal side** · **type X collagen**

Molecular regulation

Sox-9 — master regulator of chondrogenesis · collagen II, IV, XI, aggrecan

Low β-catenin — ✓ mesenchyme → **cartilage** (↑ SOX9) → collagen II, IX, XI · anlage grows interstitially + appositionally **(TO)**

High β-catenin (Wnt) — chondrocytes stop growing, **hypertrophy** → express **RUNX2 + VEGF-A** → vessels, osteoblasts, osteoclasts → **primary ossification centre (TO)**

Ihh–PTHrP loop — prehypertrophic chondrocytes secrete **Indian hedgehog** → ↑ **PTHrP** → **delays hypertrophic differentiation**

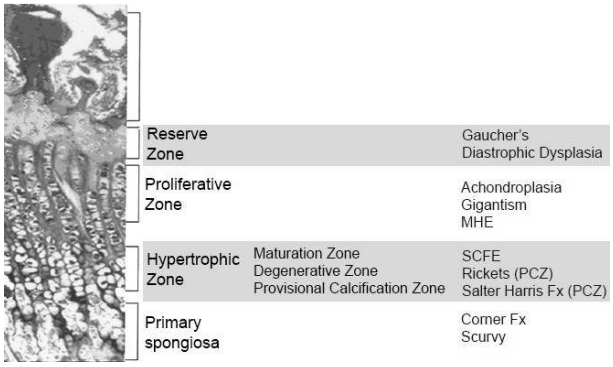
Mutation	Effect on hypertrophic maturation
Deactivating PTHrP receptor	✓ **Accelerated**
Activating Ihh	Delayed (↑ PTHrP)
Activating TGF-β / SMAD-3	Delayed (TGF-β inhibits hypertrophy, type X, ALP)
PGE2	No physeal effect shown

Physeal blood supply

Perichondrial artery — **major nutrition of the physis**

2 Mechanism

Physeal zones & pathology



Physeal zones with associated diseases

Zone	Features	Diseases
Reserve / resting / germinal	Stores **lipids, glycogen, PG aggregates** · **max PG storage (TO)** · **low O_2** · supplies cells, organises plate	**Gaucher's**, diastrophic dysplasia, Kniest, pseudoachondroplasia
Proliferative	Chondrocyte division + **columns (stacking)** · ↑ O_2 inhibits calcification · **highest ECM production (TO)**	**Achondroplasia, gigantism**, MHE
Hypertrophic – maturation, degeneration (cells ↑ 5×), **provisional calcification**	Chondrocyte hypertrophy → death → Ca release → matrix calcifies · **type X collagen** + ALP · Ihh/PTHrP · **weakest zone (TO)**	**SCFE (non-renal)**, **rickets (provisional calcification)**, enchondromas, MPS, Schmid · **physeal fractures (SH I/II) through zone of provisional calcification**
Primary spongiosa (metaphysis)	Vascular invasion, transverse septa resorbed, osteoblasts on cartilage bars → woven bone	**Metaphyseal corner fracture (abuse)**, scurvy
Secondary spongiosa	Internal remodelling (woven → lamellar), funnelisation	**Renal SCFE**

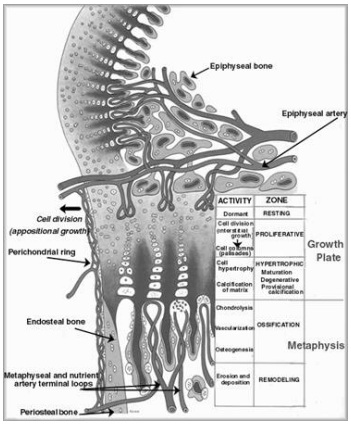
Type X collagen produced by terminal **hypertrophic** chondrocytes · ✓ **abrupt appearance heralds ossification (TO)** · type II still most abundant in hypertrophic zone **(TO)**

Longitudinal growth **chondrocyte hypertrophy** contributes most (cell volume/height) (OB8254) · proliferative zone "responsible for longitudinal growth" (OB3234) – read the options

– – Active space – –

Physis periphery

Structure	Function	Disease
Groove (zone) of Ranvier	Wedge of germinal cells · supplies chondrocytes to periphery · ✓ **appositional (circumferential) growth** · ↑ mechanical strength	**Osteochondroma** · SH VI injury (lawnmower, burns)
Perichondrial ring of LaCroix	Dense fibrous ring · **peripheral stability** · links Ranvier zone to metaphyseal periosteum	–
Perichondrial artery	Physeal nutrition	–



Ring of LaCroix

Growth regulation

Hueter–Volkmann **sustained compression → decelerated longitudinal growth** · tension accelerates

Embryonic long-bone formation

Mesenchymal anlage (6th week in utero)

- Cartilage model → **CHONDROBLASTS (hyaline precursor)**
- Vascular buds invade → **PRIMARY OSSIFICATION CENTRE (~8 wk) + MARROW**
- Ends keep growing → **SECONDARY OSSIFICATION CENTRES (epiphyses, physis)**

Non-rigid fracture healing

Soft callus = cartilage intermediate → proliferation → **hypertrophy** → calcification → bone · IM nail → both endochondral and intramembranous

How it's asked

"Physeal zone with maximum PG storage (TO)?"	→ **resting zone**
"Collagen whose abrupt appearance heralds ossification (TO)?"	→ **type X**
"SH I fractures occur through?"	→ **zone of provisional calcification**
"SH II distal radius – zone?"	→ **zone of provisional calcification**

"Gigantism affects?"	→ **proliferative zone (C)**
"Type X collagen prominent in?"	→ **zone of hypertrophy**
"Accelerated hypertrophic maturation?"	→ **deactivating PTHrP-receptor mutation** ✗ Wrong options: activating TGF-β, SMAD-3, Ihh
"Found during chondrocyte differentiation / anlage formation (TO)?"	→ **low β-catenin** ✗ Wrong options: RUNX2, VEGF-A (hypertrophy/ossification) · RANKL
"Sustained compression on physis?"	→ **decelerated longitudinal growth**
"Groove of Ranvier responsible for?"	→ **appositional growth**
"Process contributing most to longitudinal growth?"	→ **chondrocyte hypertrophy**

– – Active space – –

ETHICS IN ORTHOPAEDIC PRACTICE

– Basic Science · 31 / 108 –

1 Basics

Ethical principles

Pillar	Definition
Beneficence	Duty to do good
Non-maleficence	Duty not to harm
Respect for autonomy	Patient self-determination
Justice	**Fair, equitable, appropriate** treatment (✗ Wrong options: "everyone gets the **same** treatment")
Confidentiality	Patient's right + doctor's duty to protect information

5 Management

Industry relationships & conflicts of interest

AAOS Standards of Professionalism	
Gifts	Market value **<$100** (except textbooks, patient-education material) · dinners **≤$100/person** with useful education
Industry non-CME course	✓ Tuition, travel, modest hospitality acceptable (✗ Wrong options: "must decline non-CME courses")
Faculty at industry meeting	✓ Travel + hotel paid (contribution must match expenses)
CME event	Corporate subsidy **to the sponsor** OK · not OK: direct industry reimbursement to the surgeon, spouse's travel
On-site training (company HQ workshop, new technique/implant)	✓ **Expenses reimbursable**
Disclosure	All industry relationships → **patients, colleagues, institution** · **even without financial compensation**
Royalties	**Disclose full relationship incl. royalties to patient** before surgery · may still use own device
Kick-backs	No payment for using a product ($500/knee, "consultant" after 25 implants)
Stark II	No referral to centres in which physician has a **financial interest**
Consulting	Defined duties + work product · fair pay for time · no thinly veiled payment
Residency grants	**Unrestricted** for CME ✓ · (✗ Wrong options: restricted to industry course)
Personal deals (e.g. rep buys 90% of a boat)	**Reject — unacceptable conflict** (equal share doesn't fix it)

Implant selection ✓ surgeon responsible for knowing the **implant's clinical track record** · (✗ Wrong options: rely on sales rep) · publish negative results too

Research conduct

Industry-funded RCT ✓ paid fair market value, disclose, register (**ClinicalTrials.gov — FDA law for implanted devices**), regulatory binder, publish
✗ Wrong options: **company statistician analysing data** (bias → use independent blinded statistician)

Sunshine Act all industry payments to physicians reported to **CMS**

Publication government-funded studies must publish regardless of outcome · private studies need only report serious adverse events to FDA (publication bias)

– – Active space – –

Advertising

Who regulates	AMA/AAOS **cannot stop** physicians advertising (FTC Act) · **FTC, AAOS, state medical boards, consumer protection agencies** sanction false advertising

Avoid	✓ Acceptable
"**Cure**" (if no true cure)	"**Relatively little pain**"
"**Painless / pain-free**", "**bloodless**"	"**Less invasive / minimally invasive**" (no outcome claims)
"Board certified in joint replacement" (doesn't exist)	"Relatively safe"
"World renowned" · "safe surgery for all"	–
"Return to all sports after TJA"	–

Interpreters & consent

Preferred	✓ **certified (professional) interpreter**, on-site or telephonic (medico-legal)
Family members	not as interpreters (confidentiality) unless patient offers/agrees · **minors** not as interpreters except emergencies
Interpreter summarising (short phrases)	consent invalid → ✓ **repeat consent with qualified interpreter** ✗ Wrong options: another nurse · proceed
Interpretation style	conduit, literal, first person, no omissions

Patient transfer – EMTALA

Requirements	patient **screened** (+ stabilised) at original ED · **benefit of transfer > risk** · **accepting facility knows and formally accepts** · ✓ **exiting facility lacks resources to treat** (OB key)
Not required	insurance discussion · full lab work-up · cost of transport

Team physician obligations

Team-employed physician	✓ **agent of the team → must disclose findings to employer** · record = employment record, **not HIPAA-protected**
Private physician	HIPAA applies · no obligation to disclose to prospective teams without athlete consent

6 Complications

Patient complaints, communication & litigation

Leading cause of litigation	✓ **poor communication** · ↓ litigation with **effective communication**, standard of care, thorough informed consent, documentation
Orthopaedic surgeons rated	highest for **technical skills** · lowest for **empathy**, listening, communication ("high tech, low touch") · doctors interrupt within **23 s**
Most common patient complaint	✓ **lack of empathy**

Formal written complaint about care	written response letter to patient
Formal medical-board complaint	✓ **written response to the board with all relevant records** · governed by **state boards** · (✗ Wrong options: automatic licence suspension · contact patient's attorney) · attorney not required ✗ Wrong options: federal

Physician burnout & suicide

Burnout	loss of enthusiasm, cynicism (depersonalisation), ↓ personal accomplishment · ✓ **45–55%** of US physicians · rising · **1.5–2×** general population · ortho among higher-risk specialties · Maslach Burnout Inventory
Suicide	✓ orthopaedic surgeons = **highest number among surgical subspecialties (28%)** · surgeons **6%** more likely than general population

Disparities & diversity

Cultural competency goal	✓ better patient–physician relationships, **access**, quality of care ✗ Wrong options: quotas · stereotypes · foreign languages · travel abroad
Implicit bias	unconscious prejudice influencing care · ✓ **determinant of healthcare disparity**
Paediatric access	determined by **type of insurance**

How it's asked

"Standards of professionalism EXCEPT?"	→ **"decline industry-sponsored non-CME courses"**
"You receive royalties on a TKA that suits your patient?"	→ **disclose before surgery**
"True of ethical implant management?"	→ **surgeon must know the implant's track record**
"Most appropriate industry relationship?"	→ **travel/hotel for faculty at non-CME industry meeting** ✗ Wrong options: spouse travel · $200 dinner · restricted resident grant
"Rep offers 90% share of a $200,000 boat?"	→ **reject — conflict of interest**
"Acceptable way to learn the Genuflex knee?"	→ **company pays expenses for HQ workshop** ✗ Wrong options: company pays CME tuition · $500 per knee
"Royalty-receiving designer uses product — obligation?"	→ **disclose full relationship incl. royalties**
"Violates standards of professionalism?"	→ **not reporting a conflict because no compensation received**
"Inappropriate in company-funded TKA RCT?"	→ **company statistician analysing data**
"Acceptable newspaper advert?"	→ **"relatively little pain"** ✗ Wrong options: bloodless · board certified in joint replacement · cure

– – Active space – –

"Acceptable advert phrase?"	→ **"less invasive hip and knee replacements"**
"Non-English-speaking mother-in-law of colleague — interpreter?"	→ **certified translator**
"Nurse interpreter gives short phrases, patient offers to sign?"	→ **repeat consent with qualified interpreter**
"True about burnout?"	→ **45–55%**
"True about physician suicide?"	→ **ortho = highest among surgical subspecialties (28%)**
"Formal complaint after infected TKA?"	→ **written response to medical board with records**
"Ortho surgeons rated highest for?"	→ **technical skills**
"Commonest patient complaint about ortho surgeons?"	→ **lack of empathy**
"Associated with ↓ litigation?"	→ **effective communication**
"Benefit of diversity/cultural competency training?"	→ **better relationships, access, quality of care**
"Prisoner with hip OA — important to?"	→ **understand implicit bias**
"EMTALA requires?"	→ **exiting facility lacks resources to treat** ✗ Wrong options: insurance discussion · complete labs
"Incorrectly paired ethical term?"	→ **justice = everyone gets the same treatment**
"Free-agent athlete's private surgeon — disclosure?"	→ **team physician must disclose to own team** · private physician not obligated

EXTENSOR DIGITORUM LONGUS

– Basic Science · 32 / 108 –

1 Basics

Anatomy

	Extensor digitorum longus
Origin	Lateral tibial condyle, proximal ¾ anterior fibula, upper interosseous membrane, fascia cruris, anterior intermuscular septum
Insertion	4 slips after inferior extensor retinaculum → dorsal middle + distal phalanges (extensor expansion)
Action	Extends toes 2–5 · dorsiflexes ankle
Innervation	**Deep peroneal nerve (L4, L5, S1)**
Arterial supply	Anterior tibial artery



Extensor digitorum longus

Deep peroneal nerve function

Supplies	tibialis anterior · **EHL** · EDL · peroneus tertius · EDB · EHB
Course	posterior half of tibial shaft proximally · crosses distal tibia **40–110 mm above the ankle joint**
Anterolateral ankle approach	always exposes **superficial peroneal nerve** (subcutaneous)

– – Active space – –

X vs Y

Deficit	Nerve
Great-toe extension	✓ **Deep peroneal**
Hindfoot eversion (peronei)	Superficial peroneal
Hindfoot inversion (tibialis posterior)	Tibial
Great-toe flexion (FHL)	Tibial
Ankle plantarflexion (gastrosoleus)	Tibial
Posterior distal leg + lateral foot skin	Sural

How it's asked

"Deep peroneal nerve injury causes?"

→ **weak great-toe extension**
✗ Wrong options: eversion (SPN) · inversion/toe flexion/plantarflexion (tibial)

EXTENSOR INDICIS PROPRIUS

– Basic Science · 33 / 108 –

1 Basics

Anatomy

	Extensor indicis proprius
Origin	Posterior ulna + interosseous membrane
Insertion	Extensor expansion of index finger
Action	Extends index finger, assists wrist extension
Innervation	**Posterior interosseous nerve (C7, C8)**
Arterial supply	Posterior interosseous artery



Extensor indicis proprius

Identification in the 4th compartment

Position **deep and ulnar** to the EDC tendons · same (4th) compartment · single tendon, no subcompartment

Distinguishing feature ✓ **most distal muscle belly**

Traps ✗ Wrong options: separate compartment · circular cross-section (both flat) · most radial/superficial · two slips

How it's asked

"Dorsal wrist laceration – which 4th-compartment stump is EIP?"	**→ the one with the most distal muscle belly**

– – Active space – –

FAT EMBOLISM SYNDROME

– Basic Science · 34 / 108 –

1 Basics

Epidemiology

Incidence	**3–4%** isolated long-bone fracture · **10–15%** polytrauma · ~**90%** of long-bone fractures have some fat emboli (subclinical ↓ PaO_2 in almost all)
Mortality	up to **15%**
Causes	long-bone fractures (femur, tibia, pelvis) · **IM nailing** · **hip & knee arthroplasty** · rare: crush, burns, marrow biopsy/transplant, liposuction, CPR, haemoglobinopathy, steroids, pancreatitis, fatty liver **(TO)**
Risk factors (TO)	young age · polytrauma, multiple fractures · **closed** fractures · **conservative** treatment of long-bone fractures · IM nailing: over-zealous nailing, reaming, ↑ reaming speed, ↑ nail–cortex gap

Pathophysiology

Theory	Mechanism
Mechanical	Marrow fat droplets enter venous system (shear of intraosseous vessels, reaming) → obstruct pulmonary vessels
Biochemical	✓ **Lipoprotein lipase → free fatty acids + glycerol** → endothelial toxicity → cytokines → **ARDS-like** picture · ↓ surfactant

Arthroplasty emboli	**fat + marrow** (obstructive, act like PE) vs free fat of fractures (chemical injury)

2 Mechanism

Intraoperative embolisation in arthroplasty

Peak intramedullary pressure	**cement pressurisation + femoral component insertion** (whole canal pressurised) > reaming/broaching
Uncemented hemiarthroplasty stem insertion → hypotension, hypoxia	✓ **intramedullary fat + marrow embolisation**
TKA	IM alignment rods → emboli · ✓ **extramedullary guides ↓ embolisation risk** · overdrilling the femoral canal entry ↓ risk

3 Presentation

Onset	within **24 h** (Orthobullets) · **24–72 h** (OB) · **12–72 h** (TO)
Classic triad	**respiratory distress (first) · cerebral signs (confusion) · petechial rash (last)**

Petechiae	**axillae**, upper chest, shoulders, **conjunctivae**, oral mucosa · appear within 36 h, gone within a week · RBC extravasation from distended capillaries · pathognomonic **(TO)**
Other	tachycardia, tachypnoea, fever (non-specific but near-universal) · jaundice, oliguria **(TO)** · respiratory recovery in ~3 days if no further embolism **(TO)**

4 Diagnosis & Classification

Diagnostic criteria

Major (≥1)	Minor (≥4)
Hypoxaemia (PaO_2 <60 mmHg) + respiratory signs, radiographic changes	**Tachycardia >110**
CNS depression (unrelated to head injury)	**Pyrexia >38.5 °C**
✓ **Petechial rash**	**Retinal** fat/petechiae
Pulmonary oedema (Orthobullets)	Fat in urine/sputum · renal changes · jaundice
–	**Thrombocytopenia** · ↓ Hct/Hb · ↑ ESR

Gurd & Wilson (TO)	**1 major + 4 minor + fat macroglobulinaemia**
Additional	PCO_2 >55 · pH <7.3 · RR >35 · dyspnoea · anxiety

Key radiology findings

ABG (TO)	early hypoxia + **hypocarbia, respiratory alkalosis** → later respiratory acidosis · ↑ A–a shunt within 24–48 h without ARDS
CXR (TO)	initially normal → diffuse fluffy bilateral infiltrates (**"snowstorm"**), bases + periphery
CT angiogram	negative for PE (distinguishes)

5 Management

Prevention

Most important	✓ **early fracture stabilisation (<24 h)** of long bones ✗ Wrong options: delay fixation, TO
Techniques	narrower-shaft reamers · overdrill femoral entry in TKA · **external fixation** in medically unstable patients

Treatment

Supportive (ICU)	O_2 → **CPAP** (↑ PaO_2 without ↑ FiO_2) → **mechanical ventilation with high PEEP** if FiO_2 >60% + CPAP >10 cm needed **(TO)**
Fluids	**albumin** preferred (restores volume, lipophilic) **(TO)** · CVP monitoring (pulmonary HTN, RV failure)
Not recommended (TO)	**heparin** (bleeding in polytrauma) · **high-dose steroids** (insufficient evidence)

– – Active space – –

6 Complications

Bone cement implantation syndrome (TO)

Definition — **hypoxia, hypotension** ± **unexpected loss of consciousness**, arrhythmia, ↑ PVR, cardiac arrest around **cementation, prosthesis insertion, reduction or tourniquet deflation** in cemented arthroplasty **(TO)**

Not a feature — **hypertension (TO)**

Timing (5 stages) — femoral reaming · acetabular/femoral cement · prosthesis insertion · joint reduction · (tourniquet deflation in TKA) **(TO)**

Mechanism — exothermic cement (up to **96 °C**) expands, traps air + marrow under high intramedullary pressure → embolisation **(TO)**

BCIS grade (TO)	Criteria
1	SpO_2 **<94%** or SBP fall **>20%**
2	SpO_2 **<88%** or SBP fall **>40%** or unexpected LOC
3	**Cardiovascular collapse → CPR**

Risk factors (TO) — old age · poor reserve · cardiopulmonary disease · **pulmonary hypertension** · osteoporosis · **bony metastases** · hip fracture (**pathological/intertrochanteric**) · **previously uninstrumented canal** · **long-stem** component

Treatment (TO) — IV fluids, inotropes, vasopressors → **ECMO** if no recovery

X vs Y

	FES	PE (DVT)	BCIS
Timing	12–72 h after fracture/ nail	1–6 wk postop	**Intraop** – cement, stem, reduction, tourniquet release
Rash	**Axillary petechiae**	No	No
CNS	Confusion	Rare	LOC
Test	Clinical · CTA negative	**CTA positive**	Clinical

How it's asked

"Hypotension + hypoxia during uncemented hemi stem insertion?"	→ **intramedullary fat and marrow embolisation** ✗ Wrong options: DVT-PE · MI · fluid deficit
"Benefit of extramedullary guides in TKA?"	→ **↓ embolisation risk**
"Cemented bipolar, death after hypoxia; marrow emboli in lungs – cause?"	→ **femoral component insertion after cement pressurisation** ✗ Wrong options: reaming · undersized broaches
"Emboli during TKA composed of?"	→ **fat and marrow**

"Major criterion for fat embolism (TO)?"	→ **petechial rash** ✗ Wrong options: tachycardia, pyrexia, retinal changes (minor)
"True regarding FES (TO)?"	→ **CPAP and PEEP for hypoxia** ✗ Wrong options: heparin · steroids · delay fixation
"Bilateral femur + humerus fractures, tachycardia, hypoxia, confusion, axillary petechiae, CTA negative?"	→ **fat embolism syndrome**
"Day after reamed tibial nail: ↑ O_2, delirium, chest rash — mechanism?"	→ **lipoprotein lipase activation → free fatty acids** ✗ Wrong options: tissue factor/ collagen (PE) · IgE · plasmin inhibition (TXA) · ↑ surfactant
"TKR — hypotension + dyspnoea after tourniquet deflation (TO)?"	→ **bone cement implantation syndrome** ✗ Wrong options: fat embolism (IM rod insertion)
"NOT a component of BCIS (TO)?"	→ **hypertension**

– – Active space – –

FEMORAL NERVE

– Basic Science · 35 / 108 –

1 Basics

Anatomy

Course	runs **within the fascial sheath of iliopsoas** → vulnerable to iliopsoas haematoma

3 Presentation

Iliopsoas haematoma

Setting	**over-anticoagulation** (warfarin), alcoholism → spontaneous bleed into iliopsoas sheath → compartment-like syndrome
Features	rapid **LLQ abdominal/groin pain → anterior thigh** · **"psoas sign"** (pain on resisted hip flexion) · **quadriceps weakness → knee buckling** · progressive femoral motor loss
Management	rapid evaluation · **release iliopsoas fascia + evacuate haematoma** to prevent permanent palsy

X vs Y

Cause of acute quadriceps weakness	Clue
Iliopsoas haematoma	Anticoagulated · **abdominal pain** · psoas sign
L3–4 posterolateral / L4–5 far-lateral disc	Back/leg pain, **no abdominal pain**
Epidural haematoma	No abdominal pain · cord/cauda signs
Femoral pseudoaneurysm	Palpable mass · rare cause of palsy

How it's asked

"Anticoagulated alcoholic, LLQ pain to anterior thigh, knee buckling, quads 1/5, pain on resisted hip flexion?"

→ **spontaneous bleed into iliopsoas sheath**
✗ Wrong options: disc herniation · epidural haematoma · pseudoaneurysm

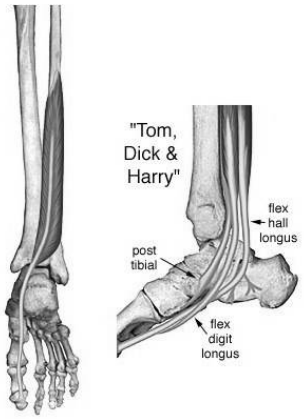
FLEXOR HALLUCIS LONGUS

– Basic Science · 36 / 108 –

1 Basics

Anatomy

	Flexor hallucis longus
Origin	**Inferior ⅔ of posterior fibula** + lower interosseous membrane
Insertion	Plantar base of **distal phalanx of great toe**
Action	Flexes great toe · helps supinate · very weak ankle plantarflexor
Innervation	**Tibial nerve (S2, S3)**
Arterial supply	Muscular branches of peroneal + posterior tibial arteries



Flexor hallucis longus

Origin – leg compartments (TO)

Compartment	Origin rule (TO)
Anterior	All from **fibula** except **tibialis anterior** (tibia)
Lateral	All from fibula (peroneus longus, brevis)
Posterior	✓ **Only FHL from fibula alone** · soleus + tibialis posterior from **both** tibia and fibula · FDL from tibia

– – Active space – –

Posterior muscle (TO)	Origin	Insertion
Gastrocnemius	Femoral condyles	Calcaneus (Achilles)
Plantaris	Lateral supracondylar ridge	Calcaneus
Soleus	Tibia + fibula	Calcaneus (Achilles)
Popliteus.....286	Lateral femoral condyle	Tibial shaft
FDL	**Tibia**	Distal phalanges, lateral 4 toes
FHL	**Fibula**	Distal phalanx, great toe
Tibialis posterior	Tibia + fibula + IOM	Navicular tuberosity

MRI anatomy

Axial ankle MRI — FHL tendon runs on **posterior tibia, just anterior to Achilles** · peroneals behind fibula · tibialis posterior + FDL behind medial malleolus

How it's asked

"Axial ankle MRI, tendon on posterior tibia just anterior to Achilles?"	→ **FHL** ✗ Wrong options: tibialis posterior, FDL (behind medial malleolus) · peroneals (behind fibula)
"Posterior-compartment muscle arising from fibula only (TO)?"	→ **FHL** ✗ Wrong options: tibialis posterior, soleus (both bones) · FDL (tibia)

FLEXOR POLLICIS LONGUS

– Basic Science · 37 / 108 –

1 Basics

Anatomy

	Flexor pollicis longus
Origin	Anterior radius + adjacent interosseous membrane
Insertion	Base of **distal phalanx of thumb**
Action	Flexes thumb phalanges (IP joint)
Innervation	**Anterior interosseous nerve (C8, T1)**
Arterial supply	Anterior interosseous artery



Flexor pollicis longus

6 Complications

AIN injury

AIN branches **posteriorly from median nerve deep to pronator teres** → at risk in forearm fracture fixation

Weak FPL + FDP index after both-bone forearm fixation
✓ **AIN injury**
✗ Wrong options: PIN, radial, ulnar (don't supply) · LABCN (sensory)

How it's asked

"Weak FPL and index FDP after both-bone ORIF?" → **anterior interosseous nerve**

– – Active space – –

FRACTURE HEALING

– Basic Science · 38 / 108 –

1 Basics

Stem-cell sources

Two major sources — **periosteum** + **endosteum** · periosteal response ↑ by motion, ↓ by rigid fixation

Primary vs secondary healing & strain theory

Strain (definition) — ✓ **change in length ÷ original length** (ΔL/L) · stress = force/area

Perren's interfragmentary strain theory — gap tissue differentiation depends on strain · **comminuted fractures tolerate more motion** (motion shared across many gaps) · simple transverse fractures in a nail → higher strain

Strain	Outcome
<2%	**Primary healing** (intramembranous, **Haversian remodelling / cutting cones**)
2–10%	**Secondary healing** (endochondral, callus)
10–100%	Fibrous tissue (or hypertrophic nonunion)
>100%	Nonunion

Tissue tolerance (TO) — **cortical bone tolerates only 2%** · **lamellar bone up to 10%** (TO wording) · granulation tissue tolerates the **greatest** strain

Osteoblasts — stimulated at 0.3–2.8% strain · optimal interfragmentary motion **0.2–1 mm**

Construct	Healing
Compression plate, **lag screw + neutralisation plate**, tension band, medial malleolar screws	**Primary – Haversian remodelling** (absolute stability, anatomic reduction, <2%)
IM nail (incl. CMN), **external fixator**, **bridge plate**, cast/brace/splint	**Secondary – endochondral** (relative stability, 2–10%)
Locked compression plate (bridge mode)	✓ **Secondary (TO)** – "internal fixator"

Primary types (TO) — **contact healing** (ends apposed → osteoclast **cutting cones** cross, osteoblasts follow) · **gap healing** (small gap → transverse woven bone → lamellar) · also unicortical/greenstick fractures

Primary healing preferred — periarticular · both-bone forearm · simple ankle fractures with syndesmotic disruption (callus would impair rotation/joint)

Biomechanics of callus

Peripheral callus — ✓ **↑ moment of inertia → ↓ strain** · bending stiffness ∝ **diameter⁴** (2× diameter → **16×** stiffness) · ↑ torsional strength, peak torque

2 Mechanism

Stages of secondary healing

Stage	Events
Inflammation (haematoma)	Haematopoietic cells, macrophages, neutrophils, platelets → **PDGF, TNF-α, TGF-β, IL-1, 6, 10, 12** (from 24 h) · **BMPs ↑**, MSC migration (**SDF-1, G-CSF**) · granulation tissue · **COX-2 → Runx2/osterix** (NSAIDs act here)
Repair	**Primary callus within 2 wk** · stable → osteoblasts, unstable → chondrocytes · **type II collagen early → type I** · soft → hard callus (woven bone) · callus amount ∝ 1/immobilisation
Remodelling	Starts mid-repair, continues years · Ihh, PTHrP, FGF, BMP · **type X** collagen (hypertrophic chondrocytes, calcification) · chondrocyte apoptosis → **VEGF** → vessels · woven → **lamellar** (Wolff's law, piezoelectric)

Biologic stages & mediators

Mediator	Role
TNF-α	Pro-healing inflammatory cytokine · **deficiency (HIV) delays** endochondral + intramembranous ossification
BMP-2	↑ in **inflammatory phase** · recruits MSCs · differentiation
SDF-1 + G-CSF	✓ **MSC migration to fracture** (absence → nonunion)
VEGF	Neovascularisation
RANKL	Osteoclasts
IL-10	Anti-inflammatory
COX-2	**MSC → osteoblasts** · knockout mice heal **more slowly** (✗ Wrong options: faster) · converts arachidonic acid → PGH2 · ↑ in cancers
MMP-13	✓ From **terminally differentiated chondrocytes** → degrades **cartilaginous** ECM (soft → woven bone)

Collagen by phase (OB)	Type
Inflammation/clot	I, II
Fibroblasts / fibrous tissue	III / V
Soft (chondroid) callus	**II + IX**
Callus calcification	✓ **X** (hypertrophic chondrocytes)
Bone deposition	**I**

4 Diagnosis & Classification

Variables that influence healing

Blood supply — most important ↓ immediately → ↑ peaks **2 wk** → normal **3–5 months** · unreamed nails preserve endosteal supply · reaming devascularises inner **50–80%** · loose nails reperfuse faster

– – Active space – –

Head injury may ↑ osteogenic response

Mechanical soft-tissue attachments · stability/strain · location · bone loss · **segmental/ butterfly** → ↑ nonunion (intercalary blood supply)

Patient factor	Effect
Vitamin D / Ca deficiency	**84%** of nonunions metabolic · **>66% vitamin D deficient**
Protein malnutrition	↓ callus strength · amino acids ↑ albumin, body mass, muscle protein, callus mineralisation (✗ Wrong options: **not IGF-1 mRNA**)
Roux-en-Y gastric bypass	**Duodenum bypassed** → ↓ Ca/vit D → **2° hyperparathyroidism** → nonunion (✗ Wrong options: sleeve · band · colectomy · nephrectomy)
Diabetes	↓ callus cellularity, delayed endochondral ossification, weak callus · **1.6× longer**
Nicotine	↓ angiogenesis, weak callus · **spine pseudarthrosis +500%** · open tibia **~70% longer**
HIV	Fragility fractures, delay – ARVs, poor circulation, **TNF-α deficiency**, nutrition
Bisphosphonates ... 49	Longer healing (wrist ORIF) · atypical femur fractures
Corticosteroids	**6.5%** higher IT nonunion · ↓ IGF-1, TGF-β
NSAIDs	COX inhibition
Quinolones	Chondrocyte toxicity
High-dose radiation	↓ cellularity, long-term remodelling change

Detrimental (TO) age · malnutrition (**albumin <3.5, lymphocytes <1,500**) · smoking · **alcohol** · DM · cancer, osteomalacia, Cushing's, obesity · **anaemia** (iron for proline hydroxylation, mineralisation) · **PVD** · NSAIDs, chemo, steroids, BPs, quinolones · radiation · infection · open, intra-articular, segmental, comminuted, denervated fractures · **intact fellow bone** · soft-tissue interposition · inadequate reduction/fixation (too short, too stiff) · **nephrotic syndrome – not listed (TO)**

Vicarious blood supply (TO) femoral neck · scaphoid waist/proximal pole · talar body · paediatric lateral condyle · 5th MT base · distal ⅓ tibia · distal ⅓ ulna

5 Management

Nonunion (stability, metabolic causes)

Nonunion

- Hypertrophic (abundant callus) → ↑ **MECHANICAL STABILITY**
  (cause: inadequate stability, biology intact)
- Atrophic / oligotrophic (poor biology) → **TAKEDOWN + BONE GRAFT + STABILISE**
- No infection signs → METABOLIC WORK-UP (25-OH vit D) → **VITAMIN D + OBSERVE**

Oligotrophic distal tibia nonunion, no infection
✓ **observe + vitamin D** (keep weight-bearing)
✗ Wrong options: NWB · suppressive antibiotics

Lab of choice for vit D **25-hydroxyvitamin D**

Bone stimulation (LIPUS, capacitive coupling)

– – Active space – –

Modality	Mechanism
LIPUS (30 mW/cm² pulsed)	✓ **Nanomotion** (mechanical strain across cell wall) · ↑ vascularity, protein expression · ↑ callus torque/stiffness · delayed/nonunion healing ~**80%**
Capacitive coupling (AC)	✓ **Ca^{2+} translocation via voltage-gated channels** → calmodulin → ↑ BMP, cAMP, TGF-β1 · collagen, cartilage calcification
Direct current	↓ O_2, ↑ pH → ↓ osteoclasts, ↑ osteoblasts · inflammatory-like response
PEMF	Calcification of **fibrocartilage** (✗ Wrong options: fibrous tissue)

LIPUS Level I evidence
✓ radius shaft, distal radius, **casted tibia**, scaphoid
✗ Wrong options: **tibia treated with reamed IM nail**

Dynamisation

Term	Definition
Dynamisation	Gradually **destabilise** fixation to load bone
Reverse dynamisation	✓ **Early micromotion → callus**, then convert to **rigid** fixation to accelerate union
Accordion method	Alternate distraction/compression of regenerate

How it's asked

"Healing with locked compression plating (TO)?"	→ **secondary** (bridge mode)
"True regarding fracture healing (TO)?"	→ **rigid compression fixation → healing by cutting cones** ✗ Wrong options: cortical bone tolerates 10% · contact healing in LCP
"Femoral nail – healing type?"	→ **secondary** ✗ Wrong options: Haversian · cutting cones · extramembranous
"Definition of strain?"	→ **ΔL / original length** ✗ Wrong options: force/area (stress)
"Direct (primary) healing with?"	→ **lag screw + neutralisation plate** ✗ Wrong options: nails, ex-fix, bridge plate
"Peripheral callus biomechanics?"	→ **↑ moment of inertia, ↓ strain**
"Bridge plating of distal tibia relies on?"	→ **endochondral ossification via chondrocyte proliferation + hypertrophy**
"Construct healing by enchondral ossification?"	→ **bridge plate**
"Tibial nail healing – true?"	→ **soft callus replaced by woven bone** ✗ Wrong options: <2% strain · cutting cones
"Comminuted femur IMN – healing?"	→ **secondary, strain 2–10%**

– – Active space – –

"Radial shaft compression plate — true?"	→ **Haversian remodelling**
"Why comminuted vs transverse nailed femurs tolerate different gap deformation?"	→ **Perren's strain theory** ✗ Wrong options: moment of inertia · Young's modulus
"Medial malleolar screw fixation — healing?"	→ **Haversian remodelling**
"Relies on strain <2%?"	→ **transverse radial shaft + compression plating**
"NOT detrimental to healing (TO)?"	→ **nephrotic syndrome** ✗ Wrong options: PVD, DM, anaemia
"Hypertrophic humeral nonunion heals with?"	→ **↑ mechanical stability**
"Commonest cause of hypertrophic nonunion?"	→ **inadequate mechanical stability**
"Oligotrophic tibial nonunion, no infection — labs support?"	→ **observation + vitamin D**
"Surgery contributing to tibial nonunion?"	→ **Roux-en-Y gastric bypass**
"FALSE about COX-2?"	→ **"knockout mice heal faster"**
"Type X collagen characteristic of?"	→ **cartilage callus formation + calcification**
"Amino acid supplementation in malnourished rats ↑ all EXCEPT?"	→ **IGF-1 mRNA expression**
"Role of MMP-13 in early callus?"	→ **from terminally differentiated chondrocytes, degrades cartilaginous ECM**
"True about healing stages (clavicle)?"	→ **BMP ↑ in inflammatory phase** ✗ Wrong options: callus in inflammation · NSAIDs act in remodelling · TNF-α excess slows · woven bone final
"No SDF-1 or G-CSF → affects?"	→ **MSC migration to fracture**
"LIPUS mechanism?"	→ **nanomotion at fracture site**
"Capacitive coupling mechanism?"	→ **Ca^{2+} translocation via voltage-gated channels** ✗ Wrong options: ↓ O_2/↑ pH (direct current)
"Level I LIPUS benefit in all EXCEPT?"	→ **tibia shaft with reamed IM nail**
"Reverse dynamisation?"	→ **early micromotion then rigid fixation**
"Ex-fix of comminuted MC — healing like all EXCEPT?"	→ **Weber B ankle ORIF** (primary)

– – Active space – –

GENERAL AND REGIONAL ANESTHESIA IN ORTHOPAEDICS

– Basic Science · 39 / 108 –

1 Basics

Components of anaesthesia

Five A's amnesia · anxiolysis · analgesia · akinesia · attenuation of autonomic responses

General anaesthesia

Inhalational isoflurane, sevoflurane, desflurane · **nitrous oxide → gaseous abdominal (bowel) distension** (diffuses in, can't leave) → hinders fluoroscopy in pelvic/spine surgery

IV non-opioids propofol, etomidate, benzodiazepines, dexmedetomidine, ketamine · **opioids** – fentanyl family, morphine, hydromorphone

Neuromuscular blocker	Agents	Reversal
Depolarising (depolarises + blocks ACh-R)	**Succinylcholine** (short only)	–
Non-depolarising	Intermediate: **rocuronium, vecuronium**, atracurium, cisatracurium · long: pancuronium	**Neostigmine** · **sugammadex** (rocuronium)

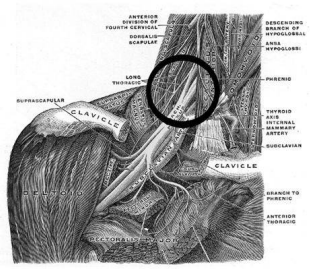
– – Active space – –

5 Management

Upper-limb blocks

Block	Site	Use	Key complication
Interscalene	Brachial plexus between **anterior + middle scalenes**, cricoid level	**Shoulder**, upper arm, elbow	**Phrenic block → hemidiaphragm paresis** (very common side effect) · **sensory neuropathy / temporary paraesthesia ≤6 mo (2.3%)** = commonest complication · Horner's · RLN · pneumothorax · vertebral artery · epidural/spinal spread
Supraclavicular	Above clavicle	Lower humerus → hand	**Hemidiaphragmatic paralysis > infraclavicular** · pneumothorax · subclavian artery
Infraclavicular	Below clavicle	Lower humerus → hand	Less phrenic involvement
Axillary	Axilla	Elbow, forearm, wrist, hand	–
Bier block	IV regional (lidocaine distal IV, tourniquet)	Short (<60 min) hand/ forearm (CTR)	**Deflate ≥30 min** → avoid LAST





Interscalene block

Interscalene contraindications

✓ **moderate–severe lung disease (FEV1 45%)** (diaphragm paresis ↓ lung volumes ~30%) · contralateral phrenic injury · LA allergy · infection · uncontrolled seizures · coagulopathy · pre-existing nerve injury
✗ Wrong options: age, aspirin, obesity alone, 1st-degree heart block

Hemidiaphragm paresis, SpO_2 97% on 2 L

✓ **observe + monitor**
✗ Wrong options: ABG, CTPA, chest tube, intubation

Postop dyspnoea after supraclavicular block, normal CXR

✓ **hemidiaphragmatic paralysis** (sniff test) · phrenic nerve (C3–5) on anterior scalene

Interscalene catheter → ptosis, miosis (TO)

✓ **Horner's syndrome → remove catheter** (reversible) · stellate ganglion paralysis from high LA concentration **(TO)**

Neuraxial

Technique	Notes
Spinal	Single shot, 24–27 G needle, morphine + bupivacaine · THA/TKA, hip fracture (= GA for delirium, 60-day mortality, ambulation) · spinal headache (↓ small needle), haematoma, opioid effects
Epidural	Indwelling catheter · hypotension, motor block
Combined spinal-epidural	Both

Lower-limb blocks

Block	Target / landmark	Use	Issue
Lumbar plexus (psoas compartment)	L1–4 (obturator, LFCN, femoral) · 3–5 cm lateral to L4 spinous process	Hip, thigh, knee	Retroperitoneal haematoma, epidural/ intrathecal spread, **falls**
Posterior quadratus lumborum	Behind QL → thoracolumbar fascia	Hip	Deep, technically hard
Fascia iliaca	Femoral, obturator, LFCN · 1 cm caudal to lateral ⅓ / medial ⅔ of pubic tubercle–ASIS line	Hip fracture, thigh, knee	✓ **↓ postoperative delirium in geriatric hip fracture**
Femoral	Lateral to femoral artery (L2–4)	Knee	**Prolonged quadriceps weakness, falls**
Adductor canal	Saphenous + nerve to VMO (distal to quad motor branches)	**TKA**	✓ **= pain control, better mobility** than femoral
iPACK	Between popliteal artery + posterior capsule	TKA	**Motor-sparing** · US-guided · ✓ **iPACK + ACB > ACB alone**
Sciatic	GT–PSIS / GT–sacral hiatus lines	Leg, ankle, foot	Heel ulcers, delayed nerve-injury diagnosis
Obturator	2 cm inferior + 2 cm lateral to pubic tubercle	Adductor spasm, hip OA pain, knee adjunct	Anterior branch → hip/ inner thigh · posterior → knee
Popliteal	Sciatic ~10 cm above popliteal crease	Foot/ankle (+ saphenous)	Foot drop, pressure sores
Saphenous	Behind sartorius	Medial knee/leg/ ankle	–
Periarticular injection	–	TKA/THA	Better TKA pain control than iPACK (Orthobullets)

Geniculate blocks chronic OA pain before TKA, not perioperative

– – Active space – –

Regional techniques – haematoma block & tourniquet (TO)

Haematoma block LA **injected directly into the fracture site** · distal radius, ankle fracture-dislocation · not "**around nerves at target site**" (false) **(TO)**

Tourniquet facts (TO)	
Cuff	**Widest possible** (arm 10 cm, thigh ≥15 cm) – effective at lower pressure
Pressure	Arm SBP + **100** · thigh SBP + **100–150** mmHg · best: **limb occlusion pressure (LOP)** + margin (LOP <130 → +40 · 131–190 → +60 · >190 → +80)
Time	**2 h** safe limit · if longer, deflate **10 min**
Commonest complication	**Nerve injury** · UL > LL · **radial nerve** most susceptible
Nerve injury cause	**Cuff pressure > duration** (✗ Wrong options: "duration more than pressure" (false)) · direct pressure > ischaemia · too low pressure (venous congestion) also
Post-tourniquet syndrome	**Ischaemia** · oedema, redness, pain, dysaesthesia, stiffness
Relative contraindications	PVD, DVT, severe infection, poor cardiac status · avoid exsanguination in infection/malignancy
Prep solutions	Prevent seepage under cuff (chemical burns)

Blood management

Hypotensive epidural anaesthesia (HEA) ✓ **best ↓ blood loss + transfusion** in bilateral THA · block to **T2** (cardio-accelerator fibres) → **MAP ~50 mmHg** · bradycardia → low-dose **epinephrine** · normal HR, CVP, SV, CO · usable in HTN, poor cardiac function, CKD · beats spinal, normotensive epidural, hypotensive TIVA, GA

Other preop EPO · autologous donation · acute normovolaemic haemodilution · cell saver

6 Complications

Malignant hyperthermia & local anaesthetic toxicity

	Malignant hyperthermia	**LAST**
Cause	**AD, RYR1** · triggered by **volatile agents + succinylcholine**	**Intravascular bupivacaine** (Na-channel block)
Features	Rigidity, **masseter spasm**, ↑ **$EtCO_2$**, ↑ temp, tachycardia, dark urine (myoglobin)	CNS: perioral numbness, metallic taste, seizures, coma · CVS: arrhythmia, **asystole/ VF**
Treatment	Stop triggers, hyperventilate O_2, cool + ✓ **dantrolene**	✓ **IV 20% lipid emulsion** + ACLS
Incidence	1 : 15,000–50,000	–

– – Active space – –

Antidote	For
Naloxone	Opioids
Flumazenil	Benzodiazepines
Protamine	Heparin
Neostigmine / sugammadex	Non-depolarising NMB (rocuronium)
Atropine, isoproterenol	β-blocker bradycardia

Bone cement implantation syndrome hypotension + hypoxaemia → IV fluids, vasopressors, 100% O_2

Epidural morphine + steroid paste after laminectomy ✓ **surgical site infection**

How it's asked

"Interscalene catheter, facial discomfort, ptosis/miosis (TO)?"	→ **Horner's – remove catheter** ✗ Wrong options: facial palsy
"Most likely complication of interscalene block?"	→ **sensory neuropathy** ✗ Wrong options: motor neuropathy, CRPS, pneumothorax
"Relative contraindication to interscalene block?"	→ **FEV1 45%** ✗ Wrong options: age, aspirin, obesity, 1st-degree block
"Most common complication after interscalene block?"	→ **temporary paraesthesia up to 6 months**
"Post-interscalene high-riding diaphragm, SpO_2 97%?"	→ **observe and monitor**
"Post-supraclavicular block dyspnoea, normal CXR?"	→ **hemidiaphragmatic paralysis** ✗ Wrong options: pneumothorax · LAST
"Best describes iPACK?"	→ **ACB + iPACK > ACB alone** ✗ Wrong options: femoral branches (ACB) · motor-sacrificing
"Block C (adductor canal) vs A (femoral)?"	→ **equivalent pain control, better mobility**
"Cardiac arrest during TKA, resuscitated with 20% lipid?"	→ **bupivacaine**
"Family history, rigidity, ↑ $EtCO_2$ after succinylcholine – antidote?"	→ **dantrolene**
"Nitrous oxide effect?"	→ **gaseous abdominal distension**
"Bilateral THA – best ↓ transfusion?"	→ **hypotensive epidural anaesthesia**
"Epidural morphine + steroid paste after laminectomy – complication?"	→ **surgical site infection**

– – Active space – –

"False about haematoma block (TO)?"	**→ "LA injected around nerves at target site"**
"False about tourniquet palsy (TO)?"	**→ "related more to duration than pressure"**

GENETIC PEARLS

– Basic Science · 40 / 108 –

1 Basics

Genes & lab markers – genetic conditions

Autosomal dominant	Gene	Defect
Achondroplasia	**FGFR3**	↓ chondrocyte proliferation
Apert	**FGFR2**	↓ chondrocyte proliferation
CMT (80–90%)	**PMP22**	Demyelination
Pseudoachondroplasia	**COMP**	Abnormal cartilage
SED congenita, **Kniest**	**COL2A1** (type II)	Cartilage matrix
MED type I	**COMP**	–
MED type II	**Type IX collagen**	–
Jansen metaphyseal chondrodysplasia	**PTHrP** receptor	PTH function
Schmid metaphyseal chondrodysplasia	**Type X collagen**	Cartilage matrix
Cleidocranial dysplasia	**RUNX2 / CBFA1**	↓ intramembranous ossification
OI types I–V	**COL1A1 / COL1A2**	Type I collagen
Milroy disease	FLT4	Lymphatics

Autosomal recessive	Gene
Diastrophic dysplasia	**DTD** (sulfate transporter – PG sulfation)
Friedreich ataxia	**Frataxin**
OI type VI	SERPINF1
McKusick metaphyseal chondrodysplasia	RMRP

X-linked recessive	Gene
SED tarda	**TRAPPC2**
CMT (10–20%)	**Connexin**

Tumour suppressors & translocations (RB, p53, Li-Fraumeni)

Tumour suppressor normally inhibits proliferation · loss → unregulated growth

Gene	Type	Facts
RB-1	✓ **Tumour suppressor** · **recessive** (both alleles)	Abnormal = **phosphorylated** (can't bind DNA) · 100% retinoblastoma · **35% osteosarcoma**
p53	Tumour suppressor · **dominant** (one allele)	Blocks entry to **S-phase** if DNA damaged · ~50% of all tumours · **osteosarcoma 20–65%** · chondrosarcoma · **Li-Fraumeni (germline, AD)**

– – Active space – –

Li-Fraumeni breast (25%) > **osteosarcoma (20%)** · soft-tissue sarcoma, leukaemia, brain, colon · homozygous p53 → 100% malignancy

Oncogenes **FAK** (focal adhesion kinase) · **Erb-2** (EGF receptor variant)

Sarcoma	Translocation	Fusion
Ewing's	**t(11;22)**	**EWS-FLI1**
Alveolar rhabdomyosarcoma	**t(2;13)** (PAX3 70%) / t(1;13) (PAX7)	**PAX3-FKHR**
Myxoid liposarcoma	**t(12;16)**	**TLS-CHOP**
Synovial sarcoma	**t(X;18)** (>90%)	**SYT-SSX1/2/4**
Clear cell sarcoma	**t(12;22)**	**EWS-ATF1**
Myxoid chondrosarcoma	**t(9;22)**	EWS-CHN
Osteosarcoma	✓ **None** (Rb/p53 mutations instead)	–

Translocations in up to 95% of sarcomas · allow expression of normally silent oncogenes

Other gene	Disease
IDH-1/2	Maffucci (Ollier)
GNAS (Gsα activating)	Fibrous dysplasia, **McCune-Albright**

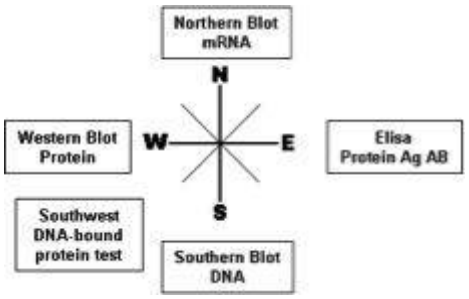
Epigenetics & siRNA

Epigenetics ✓ **inheritable alteration without DNA mutation** · **DNA methylation, histone modification, nucleosome location, non-coding RNA** · role in **OA** (methylation in chondrocytes/synoviocytes → proteases) · genomic imprinting (Prader-Willi, Angelman)

Term	Meaning
Transgene / transgenesis	Gene artificially introduced into embryo / another organism
Enhancer	Region ↑ transcription
Promoter	Controls transcription start
Transformation	Uptake of exogenous DNA (plasmid into bacterium)
Hybridisation	Annealing of complementary DNA/RNA

Laboratory techniques & markers

Assay	Detects
Southern blot	**DNA**
Northern blot	**RNA**
Western blot	**Protein**
Southwestern blot	DNA-binding proteins
RT-PCR	Low-copy RNA (→ cDNA) · very sensitive
siRNA	✓ **Blocks translation of mRNA** (mRNA degraded) · loss-of-function studies
Cytogenetics	✓ **Chromosome size/number → translocations, prenatal screening** (trisomy 21)
ELISA	Proteins in solution
Flow cytometry	Surface proteins → cell sorting
Gel electrophoresis	Separates DNA by size



Molecular biology assays

Tumour marker	Cancer
CEA	**Colorectal**
CA 19-9	Pancreatic
CA-125	Ovarian
CA 15-3	Breast
AFP	Hepatocellular

How it's asked

"RB-1 is best characterised as?"	→ **tumour suppressor** ✗ Wrong options: oncogene, proto-oncogene
"Translocation associated with osteosarcoma?"	→ **none** ✗ Wrong options: SYT-SSX1, CHOP-TLS, EWS-FLI1, FKHR-PAX3
"Correctly paired sarcoma + translocation?"	→ **clear cell sarcoma t(12;22)**
"9-yr-old, destructive distal femur lesion, p53 mutation — histology?"	→ **osteosarcoma**

– – Active space – –

"Inherited p53 mutation causes?"	→ **Li-Fraumeni** ✗ Wrong options: Maffucci (IDH) · FD/ McCune-Albright (GNAS)
"Heritable change without DNA mutation (methylation, histones)?"	→ **epigenetics**
"Function of siRNA?"	→ **blocks translation of mRNA** ✗ Wrong options: Western (protein) · Southern (DNA)
"DNA methylation in chondrocytes in OA?"	→ **epigenetics**
"Tool to detect translocations prenatally?"	→ **cytogenetic analysis**
"Serum marker for colorectal carcinoma?"	→ **CEA**

GLUTEUS MEDIUS

– Basic Science · 41 / 108 –

1 Basics

Anatomy

	Gluteus medius
Origin	Dorsal ilium below iliac crest
Insertion	Lateral + superior **greater trochanter**
Action	✓ **Major hip abductor** · anterior fibres IR · posterior fibres ER
Innervation	**Superior gluteal nerve (L4, L5, S1)**
Arterial supply	Superior gluteal artery



Gluteus medius

Hip abduction (TO, image)
✓ **gluteus medius**
✗ Wrong options: gluteus maximus · obturator externus/internus

3 Presentation

Trendelenburg sign after proximal femoral replacement

Mechanism
abductors detached and reattached to megaprosthesis · **tendon won't heal to metal** → abductor weakness

– – Active space – –

Trendelenburg weak abductor on **stance** side → **pelvis drops on the contralateral side** in single-leg stance · trunk shifts over stance leg · ↑ **hip instability** risk

Test	Detects
Trendelenburg	✓ **Abductor deficiency**
FADIR	Impingement
Ober	Tight ITB
Patrick (FABER)	Hip vs SI pathology
Thomas	Hip flexion contracture

How it's asked

"3 months after proximal femoral replacement for sarcoma – positive exam?"	→ **Trendelenburg sign** ✗ Wrong options: FADIR, Ober, Patrick, Thomas
"Muscle producing the hip movement shown – abduction (TO)?"	→ **gluteus medius**

GOUT

– Basic Science · 42 / 108 –

1 Basics

Epidemiology

Demographics	recurrent attacks in **men 40–60 yr**
Site	lower limb · **podagra (1st MTP)** · knee, ankle · hand/wrist possible · **tophi**: ear helix, eyelid, olecranon, Achilles
Risk factors	chemotherapy · alcohol

Pathophysiology

Primary gout	idiopathic disorder of **nucleic acid (purine) metabolism** → **hyperuricaemia** → MSU crystal deposition
Secondary gout	high cell turnover: **psoriasis, haemolytic anaemia, leukaemia, chemotherapy**
Inflammation	crystals → proteases, prostaglandins, **leukotriene B4**, free radicals · tophi/pannus release IL-6, IL-8, TNF-α
Associations	**renal stones**, urate nephropathy · **septic arthritis can coexist** (~1.5%) – crystals don't exclude infection

3 Presentation

Symptoms	acute severe joint pain, swelling, erythema · mimics septic arthritis · remits in 1–2 wk, recurs
Tophus aspirate	**white, toothpaste-like** · tophi often confused with infection

4 Diagnosis & Classification

Diagnostic criteria

Definitive ✓ **joint aspiration + polarised microscopy** (or tophaceous deposits)

Gout (MSU)	Pseudogout (CPPD)
Needle-shaped, thin, tapered, **intracellular**	**Rhomboid**
Strongly negatively birefringent	**Positively** birefringent
Yellow parallel to red compensator · blue perpendicular	Blue parallel
1st MTP	Knee (large joints)

Serum uric acid not diagnostic · **80%** with hyperuricaemia never get gout · may be **normal during an attack**

– – Active space – –

Key radiology findings

X-ray — **punched-out periarticular erosions with sclerotic overhanging edges** · **soft-tissue tophi** · joint space preserved until late



Gout — punched-out erosions and tophi

Foot X-ray	Diagnosis
Periarticular erosions + tophi, 1st MTP	**Gout**
Pencil-in-cup, bilateral	Psoriatic arthritis
Flattened 2nd MT head	Freiberg's
Hindfoot fragmentation	Charcot
MTP dislocations, aspherical 1st MT head	RA

Biopsy — acute + chronic inflammation, **prominent clefts** (crystals dissolved), no atypia

5 Management

Treatment & distinguishing from septic arthritis

Phase	Treatment
Acute	**NSAID – indomethacin** 50 mg tid (inhibits phagocytosis) · **colchicine** (if **peptic ulcer history**; can be IV) · steroids (oral, IA, IV) if NSAID/colchicine contraindicated (IA only if no sepsis)
Chronic	✓ **Allopurinol** (xanthine oxidase inhibitor) – not in acute attack
Prophylaxis after recurrent attacks	**Colchicine** (up to 85% effective)

Hot swollen joint, aspirate

- MSU crystals, synovial WBC <50k, well → **NSAIDs (± ALLOPURINOL LATER)**
- Crystals + WBC >50k / fever / infection risk → EMPIRIC IV ANTIBIOTICS → **I&D**
  (crystals do not exclude sepsis)
- CPPD crystals → **NSAIDs ± IA STEROID (not allopurinol)**

Fever 38.3, tachycardia, recent abdominal abscess, crystals on aspirate	✓ **empiric IV antibiotics** (cultures taken) → likely I&D
WBC 20k, normal serum markers, MSU crystals	✓ **oral NSAIDs** ✗ Wrong options: antibiotics · allopurinol · I&D · uric acid level
Isolated gout flare suitable for discharge	✓ **SFA 35k WBC + MSU** ✗ Wrong options: 75k + MSU (superimposed sepsis) · CPPD · 200k RBC (haemarthrosis)
Hand lytic lesions, uric acid 10.4, clefts on biopsy	✓ **colchicine + indomethacin** ✗ Wrong options: radiation, chemo, amputation, antibiotics

X vs Y

Aspirate image	Diagnosis
Negatively birefringent needles	**Gout**
Positively birefringent rhomboids	Pseudogout
Haemosiderin-laden giant cells	PVNS
Gram-positive cocci	Septic arthritis
Histiocytes, inflammatory cells	RA

How it's asked

"60-yr-old, 2 yr intermittent great-toe pain, erosions + soft-tissue densities?"	→ **MSU crystal deposition** ✗ Wrong options: CPPD, renal osteodystrophy, TB, sarcoid
"Foot radiograph most consistent with gout?"	→ **periarticular erosions + tophi** (B)
"Clinical + radiographic tophaceous lesion caused by?"	→ **uric acid deposition**
"Swollen hot 1st MTP — definitive diagnosis?"	→ **aspiration of 1st MTP** ✗ Wrong options: serum uric acid/CRP/ESR · MRI · RF
"Gout diagnosed by tophi or?"	→ **MSU crystals in synovial fluid** ✗ Wrong options: ↑ serum uric acid
"Hand lesion, negatively birefringent intracellular rods?"	→ **gout** ✗ Wrong options: CPPD, infection, GCT
"Alcoholic, recurrent knee swelling, colchicine — aspirate image?"	→ **negatively birefringent needles**
"Crystal analysis consistent with gout?"	→ **negatively birefringent needle-shaped**
"Line cook, tophi on fingers, acute wrist pain — SFA?"	→ **negatively birefringent needles**
"Recurrent medial forefoot pain responsive to NSAIDs, normal WBC?"	→ **needle-shaped negatively birefringent crystals**

– – Active space – –

"Febrile, recent diverticular abscess, crystals in knee aspirate — next?"	→ **empiric IV antibiotics**
"Elderly woman, finger lytic lesions, uric acid 10.4 — treatment?"	→ **colchicine + indomethacin**
"Knee aspirate WBC 20k, MSU, normal labs — next?"	→ **oral NSAIDs**
"Isolated gout flare suitable for discharge?"	→ **35k WBC + MSU**

HEMOPHILIA

– Basic Science · 43 / 108 –

1 Basics

Epidemiology & types

	Haemophilia A	Haemophilia B (Christmas)
Factor	**VIII**	**IX**
Incidence	**1 : 5,000** males	**1 : 30,000** males
Inheritance	**X-linked recessive**	**X-linked recessive**

Acquired haemophilia rare, autoimmune, not inherited

Family history affected **maternal uncles**

Pathophysiology VIII + IX needed for **thrombin generation in the intrinsic pathway** → unstable clot

Genes & lab markers

Coagulation ✓ **aPTT prolonged, PT normal** · LFTs normal · CBC (Hb)

Factor assays plasma factor VIII/IX activity + **inhibitor assay** (diagnosis)

Bethesda assay quantifies inhibitor antibody if inhibitor positive · none / **<5 BU low** / **>5 BU high**

Severity	Factor level
Mild	5–25%
Moderate	1–5%
Severe	**0–1%**

Bleeding disorder	Inheritance	Feature
Factor VIII / IX deficiency	✓ **X-linked recessive**	Spontaneous haemarthroses
Factor XI (haemophilia C)	Autosomal recessive	Surgical bleeding, no spontaneous haemarthrosis
von Willebrand (commonest bleeding disorder)	Autosomal dominant (chr 12)	**Mucosal** bleeding · ↓ platelet adhesion
Protein C / S deficiency, factor V Leiden	AD	**Thrombosis** (not bleeding)
Vitamin K deficiency	Acquired	Diet, malabsorption, liver disease

DIC (fulminant)

Labs ✓ **aPTT ↑** and PT ↑ · ↓ fibrinogen, ↓ platelets · ↑ FDPs, **↑ D-dimer** · heparin not proven · causes: massive transfusion, sepsis, burns, crush, liver disease, malignancy, obstetric

– – Active space – –

3 Presentation

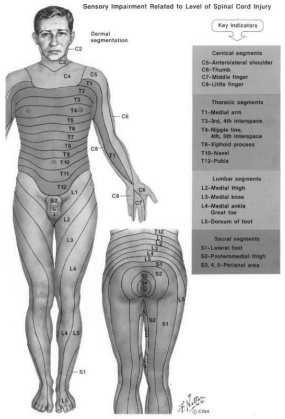
Bleeding joints (**knee most common** haemarthrosis; elbow, ankle, shoulder, spine) · muscles (quadriceps, hamstrings, **iliopsoas**, biceps, triceps) · mucocutaneous

Orthopaedic manifestations

Haemophilic arthropathy synovitis → cartilage destruction → deformity

Intramuscular haematoma (pseudotumour) ✓ **iliacus haematoma → femoral nerve** compression → paraesthesia in **L4 / saphenous** distribution (medial distal tibia) · groin pain after sport (forceful hip flexion)

Other LLD (**epiphyseal overgrowth**) · fractures (osteopenia, **heal in normal time**) · compartment syndrome



Dermatomes — L4/saphenous territory

4 Diagnosis & Classification

Key radiology findings

Modality	Finding / use
X-ray (AP, lateral, oblique, skyline)	Epiphyseal overgrowth · osteopenia · **squared patella + condyles (Jordan's sign)** · ballooned distal femur · **wide intercondylar notch** · long thin patella
CT	Pseudotumour location (bone vs soft tissue), fracture pattern
MRI	**Best soft tissue** · haemarthrosis, synovitis, **haemosiderin** · cartilage
US	Effusion vs pseudotumour · follow intramuscular haematomas

– – Active space – –



Knee — squared condyles, widened notch

5 Management

Management (factor replacement, synovectomy, haematoma)

Situation	Target factor level
Vigorous physiotherapy	20%
Acute haematoma	**30%**
Acute haemarthrosis, soft-tissue surgery	**40–50%** (OB: "within 60%")
Skeletal surgery	**100% for week 1 → >50% for week 2**

High-titre factor VIII inhibitor
✓ **factor VIIa** (bypasses VIII) for acute bleed, **compartment syndrome**, surgery

Femoral fracture + compartment syndrome + high inhibitor titre
✓ **factor VIIa immediately → fasciotomy + IMN**
✗ Wrong options: VIII replacement (won't work) · operate first · plate

Non-life-threatening bleed
analgesia, physio · no routine immobilisation or paracentesis

Synovectomy
✓ **recurrent haemarthrosis despite optimal medical management** · ↓ bleeding but doesn't restore ROM or reverse cartilage damage · not for radiographic narrowing, stiffness, prophylaxis

Radioactive synoviorthesis
↓ recurrence **more than surgical synovectomy**

End-stage
arthroplasty

VTE prophylaxis

Known bleeding disorder (factor VIII deficiency)
✓ **pneumatic compression only, no pharmacoprophylaxis (TO)** · also active liver disease
✗ Wrong options: protein C/S deficiency, factor V Leiden (thrombophilias)

6 Complications

Inhibitors (IgG)
4–20% · any time · **relative contraindication to surgery** · screen preop

– – Active space – –

Blood-borne infection HIV ~**10–15%** (this note) · up to 90% (arthropathy note, older cohorts) · recombinant factor VIII (1992) eliminated risk

Allergy to blood products

How it's asked

"Avoid pharmacoprophylaxis in favour of pneumatic compression (TO)?"	→ **factor VIII deficiency** ✗ Wrong options: protein C, S, factor V Leiden
"Haemophilic boy, iliacus haematoma, medial distal tibia paraesthesia — nerve?"	→ **femoral** ✗ Wrong options: obturator, genitofemoral
"Femur fracture + compartment syndrome + high VIII inhibitor?"	→ **factor VIIa now, then fasciotomy + IMN**
"Primary indication for synovectomy?"	→ **recurrent bleeding despite optimal medical management**
"Fulminant DIC finding?"	→ **↑ aPTT** ✗ Wrong options: ↑ fibrinogen · ↓ D-dimer · normal PT · heparin curative
"Factor VIII deficiency inheritance?"	→ **X-linked recessive**

HEMOPHILIC ARTHROPATHY

– Basic Science · 44 / 108 –

1 Basics

Epidemiology

Incidence ✓ **↓ significantly due to home factor treatment** (recombinant factor VIII since 1992 – also removed HIV/hepatitis risk)

Demographics young males, **3–15 yr**

Site **knee** (most common) · elbow, ankle, shoulder, spine

Prevention, factor replacement & inheritance

Cause	Inheritance	Factor
Haemophilia A	X-linked recessive	↓ **VIII**
Haemophilia B (Christmas)	X-linked recessive	↓ **IX**
von Willebrand	**Autosomal dominant**	Abnormal VIII + platelet dysfunction · mucosal bleeding, rare joint bleeds

X-linked recessive bleeding disorders ✓ **both factor VIII and IX deficiency**
✗ Wrong options: factor XI (AR) · vWD (AD)

Pathoanatomy repeated minor trauma → haemarthrosis → **synovitis (hypertrophic, hyperplastic)** → enzymatic cartilage destruction → deformity

Associations iliacus haematoma → **femoral palsy (L4 paraesthesia)** · LLD (epiphyseal overgrowth) · osteopenic fractures (normal healing) · **HIV up to 90%** (older cohorts)

3 Presentation

Haemarthrosis	Features
Acute	Painful, **tense** effusion
Subacute	After **2 prior bleeds**
Chronic	**Contractures, arthritis**

– – Active space – –

4 Diagnosis & Classification

Arnold–Hilgartner staging

Stage	Radiographic finding
1	**Soft-tissue swelling**
2	**Osteoporosis** (± epiphyseal overgrowth)
3	**Subchondral cysts**, joint grossly intact
4	**Cartilage loss – joint narrowing**
5	**Severe arthritis** (end-stage, fibrous ankylosis)



Arnold–Hilgartner stage 5

Key radiology findings

Knee	**squared patella + condyles (Jordan's sign)** · ballooned distal femur · **widened intercondylar notch** · narrowing · long thin patella
MRI	early degenerative change, haemosiderin · **US** – follow muscle haematomas

Labs

Factor VIII inhibitor (IgG) screen	✓ **before surgery** (negates factor therapy) · found in **5–25%** · relative contraindication to surgery
Exclude septic arthritis	exam + aspiration

5 Management

Indication	Factor VIII target
Vigorous PT	**20%**
Acute haematoma	**30%**
Acute haemarthrosis, soft-tissue surgery, synovectomy	**40–50%**
Skeletal surgery, TJA, arthrodesis	**100% wk 1 → >50% wk 2**

Before TKA (X-linked bleeding disorder)	✓ **factor VIII replacement** ✗ Wrong options: protein C/S · vit K · vWF
Desmopressin	mild–moderate haemophilia A
Non-operative	compressive dressing, analgesia, short immobilisation → rehab, steroids, splints, PT
Synovectomy	recurrent haemarthroses despite medical management · ↓ recurrence, pain, swelling
Synoviorthesis	intra-articular radioactive **colloidal P-32 chromic phosphate**
Arthroplasty	end-stage · **arthrodesis** – ankle

How it's asked

"Main reason for ↓ haemophilic arthropathy over 10–20 yr?"	→ **home factor treatment** ✗ Wrong options: PT, steroid injections, debridement, splints
"X-linked clotting disorder before TKR – replace?"	→ **factor VIII**
"Bleeding disorders that are X-linked recessive?"	→ **factor VIII and IX deficiency**
"Arnold–Hilgartner stage with subchondral cysts, joint intact?"	→ **stage 3**

– – Active space – –

HYPERPARATHYROIDISM

– Basic Science · 45 / 108 –

1 Basics

Epidemiology

Incidence	**0.1%** · **women** > men · with malignancy = most cases of **hypercalcaemia**
Cause (primary)	**single adenoma 80–90%** · hyperplasia 10%

Pathophysiology

PTH acts indirectly	binds **osteoblast** receptor → RANKL + M-CSF → osteoclasts → excess resorption · **cortical > cancellous**
PTH effects (OB)	↑ renal tubular Ca reabsorption · ↑ osteoclast activity · ↑ active vitamin D → ↑ gut Ca

Classification

Type	Cause	Ca	Phos	PTH
Primary	Adenoma / hyperplasia	**↑**	**↓**	**↑**
Secondary	Compensation for **↓ Ca / ↑ phosphate** – **CKD** (renal osteodystrophy), **malabsorption (coeliac)**, vit D deficiency	**↓ / N**	↑ (renal) / ↓ (malabsorption)	**↑**
Tertiary	Autonomous glands after long secondary HPT / renal transplant	**↑**	**↑**	**↑**
Malignancy (PTHrP)	Ectopic	**↑**	**↓**	**↓**

Genes & lab markers

Serum	↑ ALP · normal anion-gap metabolic acidosis (↓ HCO_3 reclamation)
Urine (primary)	**hypercalciuria (stones)** · ↑ cAMP
ECG	**short QT**

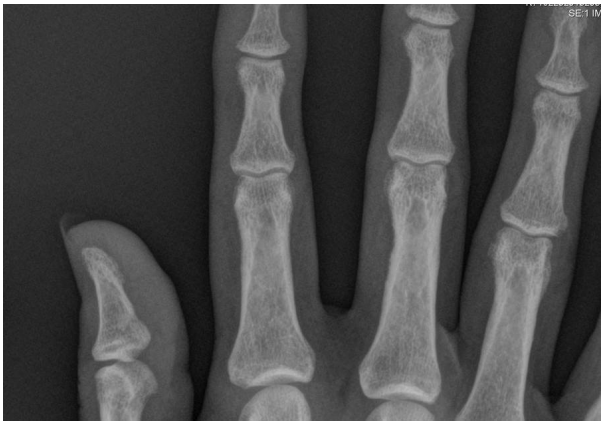
3 Presentation

"Stones, bones, groans"	renal stones · bone pain · constipation · weakness · often asymptomatic · rare 2° HTN · **fragility fracture** with abnormal labs

4 Diagnosis & Classification

Key radiology findings

Finding	Notes
Subperiosteal resorption	**Radial side of middle phalanges**, phalangeal tufts, distal clavicle, proximal tibia/humerus/femur
"Salt and pepper" skull	Calvarial trabecular resorption
Rugger-jersey spine	Sclerotic endplate bands (excess unmineralised osteoid) — **distinctive for HPT (TO)**
Brown tumours	Lytic, expansile → pathological fracture · **osteitis fibrosa cystica** (jaw)
Chondrocalcinosis	Menisci, articular cartilage
Looser zones	Renal osteodystrophy (osteomalacia)



Subperiosteal resorption of the phalanges

Brown tumour histology

Features	**bland spindled fibrous stroma** + **small multinucleated giant cells** · **haemorrhage + haemosiderin** · variable lytic/dense multiple lesions
vs GCT	GCT: larger giant cells, round stromal nuclei identical to giant-cell nuclei
vs myeloma/ metastases	serum markers (Ca, PTH) distinguish

5 Management

Acute hypercalcaemia	**IV fluids + loop diuretics**
Symptomatic	**parathyroidectomy** → watch **postop hypocalcaemia** (numbness, tingling, cramps) → **IV calcium gluconate**

6 Complications

Peptic ulcer (↑ gastrin) · **pancreatitis** (↑ lipase) · CNS (anxiety, confusion, coma) · **osteoporosis/ fragility fractures**

– – Active space – –

X vs Y

Lab panel	Diagnosis
↑ PTH, ↑ ALP, ↑ Ca, ↓ PO_4, normal renal function	**Primary HPT**
↑ PTH, ↑ ALP, ↓ Ca, ↑ PO_4	Secondary (renal)
↑ PTH, ↑ Ca, ↑ PO_4	Tertiary
Normal PTH, **↓ ALP**, ↑ Ca, ↑ PO_4, ↑ urinary phosphoethanolamine	Hypophosphatasia 171
↓ PTH, normal ALP, ↓ Ca, ↑ PO_4	Hypoparathyroidism ..169

How it's asked

"Rugger-jersey spine X-ray characteristic of (TO)?"	→ **hyperparathyroidism**
"Dialysis patient, Looser zone, rugger jersey, salt-and-pepper skull – labs EXCEPT?"	→ **hypophosphataemia** (phosphate ↑ in renal osteodystrophy)
"25-yr-old, brown tumours, normal GFR – labs?"	→ **↑ PTH, ↑ ALP, ↑ Ca, ↓ PO_4 (primary)**
"Coeliac sprue, brown tumour fracture – expected?"	→ **↑ PTH** ✗ Wrong options: ↑ Ca, ↑ 1,25-D, ↑ PO_4, ↓ ALP
"66-yr-old, multiple variable tibial lesions, spindle stroma + small giant cells + haemosiderin?"	→ **hyperparathyroidism (brown tumours)** ✗ Wrong options: myeloma, mets, GCT
"Fragility fracture, ↑ Ca, ↑ ALP, ↓ PO_4, ↑ PTH – aetiology?"	→ **PTH adenoma** ✗ Wrong options: age-related osteoporosis (normal labs)

HYPOPARATHYROIDISM

– Basic Science · 46 / 108 –

1 Basics

Pathophysiology

Cause ↓ PTH from chief cells · **thyroidectomy** (incl. parathyroids) = **most common**

Effects of ↓ PTH ↓ renal phosphate excretion → **↑ serum phosphate** · ↓ 1α-hydroxylation → **↓ 1,25-$(OH)_2D$ → ↓ calcium**

Genes & lab markers

↓	↑	Normal
PTH, calcium, 1,25-D, urinary Ca	**Serum phosphate**	**ALP**

Alkalosis ↑ albumin binding of ionised Ca → hypocalcaemia

ECG **prolonged QT**

PTH & vitamin D lab patterns

Condition	Ca	PO_4	PTH	Cause
Hyperparathyroidism 166	↑	↓	↑	Adenoma
Hypoparathyroidism	**↓**	**↑**	**↓**	**Thyroidectomy**
Ectopic PTH(rP)	↑	↓	↓	Malignancy
Vitamin D malabsorption	↓	↓	↑	Coeliac, GI disease
↓ Vit D + no renal PO_4 excretion	↓	↑	↑	**Renal failure**, pseudohypoparathyroidism

3 Presentation

Hypocalcaemia neuromuscular irritability: **Chvostek's sign**, **tetany**, **seizures** · **cataracts** · nail fungal infections · hair loss · vitiligo, blotchy skin

4 Diagnosis & Classification

Key radiology findings

Skull X-ray **basal ganglia calcification**

– – Active space – –



Basal ganglia calcification

5 Management

Calcium + vitamin D supplementation · monitor labs · **no hormone replacement** available (Orthobullets)

Renal disease & PTH

Chronic renal disease	✓ **↑ PTH production** (Ca wasted, PO_4 retained, ↓ 1,25-D) ✗ Wrong options: TSH unaffected
Renal osteodystrophy	high-turnover (↑ PTH, gland hyperplasia → osteitis fibrosa cystica, persists after correction) vs **low-turnover (more common on dialysis)** · metaphyseal enlargement, frontal bossing, bowing, genu varum

How it's asked

"Chronic renal disease – endocrine abnormality?"	→ **↑ PTH** ✗ Wrong options: ↓ PTH · TSH change
"Post-thyroidectomy tetany, ↑ PO_4, ↓ Ca, ↓ PTH?"	→ **hypoparathyroidism**

HYPOPHOSPHATASIA

– Basic Science · 47 / 108 –

1 Basics

Epidemiology

Severe (perinatal/ infantile) 1 : 100,000–300,000 · **mild (adult)** – 1 : 500–3,000 · diagnostic delay common

Genes & lab markers

Gene ✓ **loss-of-function TNSALP** (tissue-nonspecific alkaline phosphatase) in osteoblasts

Inheritance **severe forms autosomal recessive** · mild/adult often autosomal dominant

Pathophysiology ↓ ALP → extracellular ↑ **inorganic pyrophosphate (PPi)**, pyridoxal-5'-phosphate, **phosphoethanolamine** · **PPi inhibits hydroxyapatite** formation → hypertrophic-zone osteoid fails to mineralise → no zone of provisional calcification

Lab	Finding
Serum ALP	✓ **↓ (low)**
Urine phosphoethanolamine	✓ **↑ – diagnostic**
Urine PPi	↑
Ca, PO_4	Normal or ↑ (hypercalcaemia)
PTH	Normal

Low ALP in metabolic disease ✓ **hypophosphatasia**
✗ Wrong options: FHH, X-linked hypophosphataemia, secondary/tertiary HPT (all ↑ ALP)

3 Presentation

Rickets-like genu varum, short stature, ↑ fractures

Form	Features
Perinatal	Hypomineralisation, long-bone (pretibial) bowing, osteochondral spurs, caput membranaceum, **respiratory failure / lung hypoplasia**
Infantile	**Craniosynostosis**, rachitic ribs, flared metaphyses
Childhood	**Premature loss of deciduous teeth**, short stature, bone pain, fractures
Adult	**Femoral pseudofractures**, **metatarsal stress fractures**, chondrocalcinosis, poor healing, osteopenia

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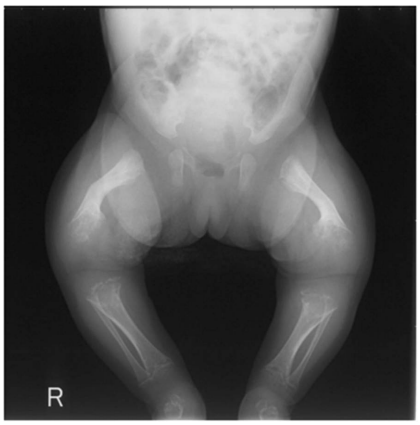


Abnormal dentition in hypophosphatasia

4 Diagnosis & Classification

Key radiology findings

X-ray — "deossification" adjacent to physis (tongues of lucency) · **physeal widening** · flared metaphyses



Abnormal bone formation near the physis

5 Management

Therapy — summary lists **enzyme replacement**, anti-inflammatories, multidisciplinary care · treatment section: "no approved therapies, phosphate under investigation" (Orthobullets) – read the options

Hypophosphataemic rickets – contrast

Child with marked genu varum + wide physes	✓ **first get serum phosphate, calcium, ALP** ✗ Wrong options: osteotomy, orthosis, scanogram, reassurance
X-linked hypophosphataemic (vit D-resistant) rickets	**X-linked dominant** · presents 18–24 mo · renal **phosphate wasting** · ↓ **PO_4, ↑ ALP**, normal Ca, normal vit D · short (<10th centile)
Treatment	**phosphate + calcitriol** together · calcitriol alone → ✓ **no effect on alignment or height** · phosphate alone → ↓ Ca → 2° HPT → more phosphaturia

X vs Y

	Hypophosphatasia	Hypophosphataemic rickets
ALP	↓	↑
PO_4	N / ↑	↓
Urine	**Phosphoethanolamine**	Phosphate wasting
Inheritance	AR (severe)	X-linked dominant

How it's asked

"Hypophosphatasia lab finding?"	→ **↓ serum ALP** ✗ Wrong options: ↓ urinary phosphoethanolamine · ↓ urinary PPi
"Metabolic disorder with ↓ ALP?"	→ **hypophosphatasia**
"2-yr-old, marked genu varum, wide physes – first step?"	→ **serum phosphorus, calcium, ALP**
"Hypophosphataemic rickets treated with calcitriol only?"	→ **no effect on alignment or height**

– – Active space – –

ILIACUS

– Basic Science · 48 / 108 –

1 Basics

Anatomy

	Iliacus
Origin	Upper ⅔ iliac fossa, inner lip of iliac crest, lateral sacrum, ventral SI + lower iliolumbar ligaments
Insertion	**Lesser trochanter** (with psoas → iliopsoas)
Action	Flexes thigh on trunk (and trunk on thigh)
Innervation	**Muscular branch of femoral nerve (L1–L3)**
Arterial supply	Iliolumbar (iliac branch) from internal iliac



Iliacus

Cross-sectional anatomy at the femoral heads

Structure	Position relative to iliacus
Psoas tendon	Immediately **medial**
Rectus femoris	Just anterior + slightly lateral
Sartorius 342	Superficial + anterior
Tensor fascia lata	Lateral + superficial
Obturator externus	Medial, deep to pectineus

Iliacus haematoma (haemophilia, anticoagulation) compresses **femoral nerve** (see Femoral nerve, Hemophilia chapters)

How it's asked

"Pelvic cross-section at femoral heads — arrowed muscle with psoas tendon just medial?"	→ **iliacus** ✗ Wrong options: rectus femoris, sartorius, obturator externus, TFL

– – Active space – –

IMMUNOLOGY

– Basic Science · 49 / 108 –

1 Basics

Innate & adaptive immunity

	Innate	Acquired (adaptive)
Specificity	Non-specific (structures common to microbes)	Pathogen/antigen-specific
Memory	No	Yes
Components	Skin barrier, inflammation, **complement**	Antigen processing + presentation
Cells	**NK cells, mast cells, eosinophils, basophils, macrophages, neutrophils, dendritic cells**	**CD8+ T cells**, T helpers, γδ T cells, B cells, plasma cells

First step of acquired response ✓ **antigen processing + presentation to T cells** by **APCs (B cells, dendritic cells)** → peptides on **MHC** → TCR recognises MHC/antigen
✗ Wrong options: T cells phagocytose antigen · disulfide cleavage occurs in APC

MoM bearing reactivity innate (cell-mediated) per OB · metal implants → **type IV** (Orthobullets)

Hypersensitivity

Type	Mediator	Examples
I – immediate/ anaphylactic	**IgE** → mast cells, basophils	Allergy, **urticaria** (rapid hives)
II – cytotoxic	IgG, IgM	Transfusion reactions
III – immune complex	IgG, IgM complexes deposit	–
IV – delayed (cell-mediated)	**T cells**, monocytes, macrophages · 2–3 days (7–10 for dermatitis)	**TB skin test** · **metal implants (plate + screws)** · **topical antibiotic (bacitracin) contact dermatitis**

Bacitracin contact dermatitis **5–10%** · erythema, oedema, **blistering**, no pus/fluctuance → stop agent ± steroid cream

Immunoglobulins

Ig	Key fact
IgG	**Most abundant**
IgM	✓ **First to appear** after antigen exposure
IgA	**Secretions** (gut, bronchial mucus, saliva, tears)
IgE	Parasites, allergy
IgD	Role unknown

Benign ethnic neutropenia

Most common neutropenia worldwide	**25–50% of people of African descent**, some Middle Eastern groups · lifelong low ANC **without ↑ infection risk**
ANC <1.5 × 10⁹/L	4.5% African-Americans · 0.79% white · 0.38% Mexican-Americans
More common in	**males**, **athletes**, children <5 · hypothesis: malaria protection

HLA associations & HIV arthropathy (TO)

Coeliac disease

✓ **HLA-DQ2 (TO)**
✗ Wrong options: DR4, B5, B47

HIV-associated arthritis (TO)	
Most common form	**Asymmetric oligoarthritis** (knees, ankles, male) (✗ Wrong options: "symmetrical polyarthritis most common")
Symmetrical polyarthritis	Mimics RA (ulnar deviation), acute onset, **non-erosive**
Course	Peak 1–6 wk, short-lived · some chronic destructive
Mucocutaneous / enthesopathy	**Rare**
X-ray	Usually non-erosive · may mimic RA · **new bone formation** (unusual in RA)
Other	Jaccoud arthropathy occasionally

How it's asked

"First step of TCR–MHC acquired response?"	→ **antigen processing + presentation to T cells**
"First antibody class after antigen exposure?"	→ **IgM** (IgG – most abundant)
"Type IV hypersensitivity seen with?"	→ **metal plate + screw placement** ✗ Wrong options: transfusions · red man syndrome
"Healthy patient, baseline WBC 1.5 for years?"	→ **more likely African than European descent** ✗ Wrong options: ↑ infection risk · female · non-athlete
"NOT innate immune cells?"	→ **CD8+ T lymphocytes**
"Finger laceration + bacitracin → erythema, blistering, no pus at 1 wk?"	→ **allergic contact dermatitis** ✗ Wrong options: cellulitis · IgE urticaria
"HIV arthropathy – all true EXCEPT (TO)?"	→ **"symmetrical polyarthritis is the most common form"**
"HLA in coeliac disease (TO)?"	→ **HLA-DQ2**

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INFORMED CONSENT & TIME OUT

– Basic Science · 50 / 108 –

1 Basics

Elective consent

Optimal	✓ **by the surgeon, in the office, a few days (within 7 days) before** surgery ✗ Wrong options: nurse/surgeon in pre-op holding · risk manager · scheduler
Disclose	who will operate · risks, benefits, indications, alternatives · surgeon's **HIV/HBV/HCV status** for exposure-prone procedures (at scheduling) · **origin of implants** (bovine – Hindu · porcine – Jewish, Muslim)
Adolescents	AAP: **>14 yr** who **dissent** treated as having capacity, even though parental legal consent needed

Emergency consent

Life-threatening, no legal consent available	proceed; **confirm + document necessity with a colleague**
Non-life-threatening	consent required first (language barrier, child)
Next of kin order	**spouse → children → parents → siblings → grandchildren**

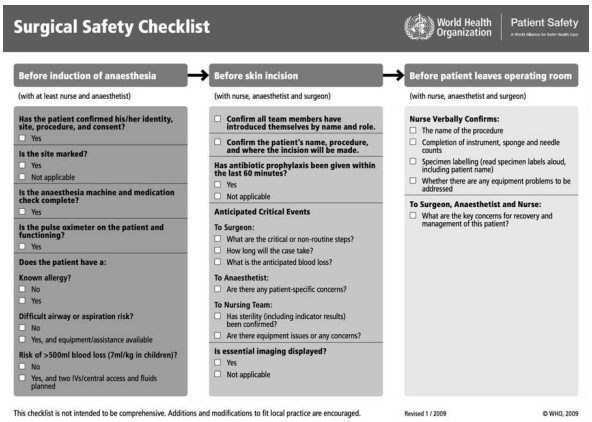
Research & privacy

HIPAA	no patient consent needed to share PHI with **other treating providers**
IRB	mandates clinical-trial consent process · levels: exempt (still needs chair certification), expedited, full board
No IRB review needed	✓ **internal quality-improvement studies** (OR efficiency, preop delays) even with identifiers · becomes research (IRB) if published/presented nationally or applied beyond the institution
IRB required	any study using PHI (**medical record number is PHI**) · **vulnerable populations**: prisoners, pregnant women, fetuses, neonates, cognitively impaired, **children**, students, employees

Surgical safety checklist

WHO checklist (2009)	✓ ↓ **surgical mortality**, ↓ in-hospital complications, ↑ adherence in OR crises (haemorrhage, arrest)

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WHO surgical safety checklist

How it's asked

"Optimal setting for elective written consent?"	→ **surgeon in office within 7 days prior**
"Which studies need no IRB review?"	→ **internal QI studies (2 and 4)** ✗ Wrong options: MRN retrospective study · children with CP
"Next of kin first in line?"	→ **spouse**

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INHERITANCE PATTERNS OF ORTHOPAEDIC SYNDROMES

– Basic Science · 51 / 108 –

1 Basics

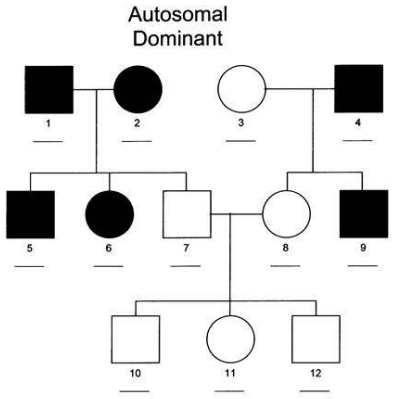
Mendelian patterns & pedigrees

Mendelian (monogenic) AD · AR · X-linked dominant · X-linked recessive · ~1% of humans

Rule of thumb ✓ **AD → structural protein defects** (e.g. FGFR3 in achondroplasia) · **AR → enzymatic/biochemical defects**

Pedigree key circle = female · square = male · filled = affected · horizontal line = mating · vertical = offspring

Pattern	Pedigree clue
AD	Every generation, both sexes, ~50% of offspring (no skipping if complete penetrance)
AR	Both sexes equally, **skips generations**, affected children of unaffected (carrier) parents, ¼ of offspring
X-linked recessive	Males; affected mother → **all sons affected**
X-linked dominant	Affected father → all daughters



Autosomal dominant pedigree

Autosomal dominant	Autosomal recessive	X-linked recessive	X-linked dominant
Syndactyly, polydactyly	**Diastrophic dysplasia**	**Duchenne, Becker** MD	**Hypophosphataemic rickets** (PHEX → ↑ FGF23)
Marfan (fibrillin)	Friedreich ataxia	**Hunter** syndrome	**Léri-Weill** (bilateral Madelung)
Cleidocranial dysplasia (RUNX2)	**Gaucher** (β-glucocerebrosidase)	**Haemophilia A, B**	–
MHE (EXT1 ~10% sarcoma, EXT2/3 ~1%)	SMA	**SED tarda**	–
Achondroplasia	**Sickle cell**	**Lesch-Nyhan**	–
MED · Schmid + Jansen chondrodysplasia · **Kniest**	**Hypophosphatasia**	–	–
Malignant hyperthermia	**Osteopetrosis – infantile malignant**	–	–
Ehlers-Danlos · **OI I–IV**	–	–	–
Osteopetrosis tarda (Albers-Schönberg) · **osteopoikilosis** · NF1	–	–	–

Multiple patterns CMT (AD, AR, X) · osteopetrosis (AD, AR) · OI (AD, AR) · NF (AD, AR) · SED (AD, X) · hypophosphataemic rickets (XD commonest, AD, AR)

No inheritance fibular hemimelia · **McCune-Albright / fibrous dysplasia** (GNAS – sporadic **mosaicism**, non-Mendelian)

Non-Mendelian inheritance (imprinting, anticipation, mosaicism)

Phenomenon	Definition	Example
Genomic imprinting	Expression depends on **parent of origin**	**Prader-Willi** (paternal defect – hypotonia, obesity, hypogonadism; chr 15) · **Angelman** (maternal – epilepsy, tremor, smiling)
Anticipation	**Earlier onset + more severe** in each generation · **trinucleotide repeats**	**Huntington's** (CAG, chr 4, AD), myotonic dystrophy, Friedreich ataxia, fragile X
Mosaicism	Post-zygotic mutation → mixed cell lines	Fibrous dysplasia, McCune-Albright
Variable penetrance/ expressivity	Same gene, varying degree	–
Genetic drift	Chance change in genotype frequency	–
Trisomy	Extra chromosome	Down (21)

Lesch-Nyhan syndrome

Defect **X-linked recessive** · deficient **HGPRT** (purine metabolism) → ✓ **↑ serum uric acid**

– – Active space – –

Features cognitive impairment · **choreoathetosis** · **self-mutilation** (biting fingertips), aggression · fractures, hip dysplasia, progressive scoliosis · surgery outcomes ≈ CP · ↑ HO, femur fixation failure

Lab	Condition
↑ Serum uric acid	**Lesch-Nyhan**
↑ Urine keratan sulfate	Morquio
↑ Ca, ↓ PO_4	Hyperparathyroidism.....166
↑ Glucocerebrosides	Gaucher
Chromosome 15 deletion	Prader-Willi

How it's asked

"Prader-Willi / Angelman — parent-of-origin expression?"	→ **genomic imprinting**
"Symptoms earlier each generation?"	→ **anticipation**
"Genetic defect NOT following Mendelian inheritance?"	→ **Gs-α defect (McCune-Albright)** ✗ Wrong options: FGFR3, sulfate transporter, fibrillin, glucocerebrosidase
"Chorea at 46, father at 55?"	→ **autosomal dominant with anticipation (Huntington's)**
"Example of anticipation?"	→ **more severe and earlier in subsequent generations**
"AR pedigree — which condition?"	→ **diastrophic dysplasia** ✗ Wrong options: DMD, Hunter, haemophilia B, SED tarda (XR)
"X-linked recessive disorder?"	→ **haemophilia A** ✗ Wrong options: Gaucher, diastrophic (AR) · Prader-Willi (imprinting) · hypophosphataemic rickets (XD)
"AD genes typically cause which defect?"	→ **structural**
"Pedigree for MHE?"	→ **autosomal dominant**
"Spotted juxta-articular sclerotic lesions in pregnant woman — inheritance?"	→ **AD (osteopoikilosis)**
"Radiograph suggesting X-linked dominant disease?"	→ **hypophosphataemic rickets**
"Lesch-Nyhan — lab finding?"	→ **↑ serum uric acid**
"Self-mutilating, cognitively impaired, X-linked recessive boy?"	→ **Lesch-Nyhan** ✗ Wrong options: Léri-Weill (XD) · Apert (AD)
"Bit off his fingertip, purine metabolism disorder?"	→ **Lesch-Nyhan**

INTRAMEMBRANOUS BONE FORMATION

– Basic Science · 52 / 108 –

1 Basics

Definition

Bone formed without a cartilage model	undifferentiated mesenchymal cells aggregate → osteoblast differentiation → organic matrix deposition
Also called	contact healing, Haversian remodelling (in fracture healing)

Sites of intramembranous ossification

✓ **Intramembranous (TO)**	**Not endochondral (TO)**
Embryonic flat bones – skull, maxilla, mandible, pelvis, **clavicle**, subperiosteal surface of long bones	Longitudinal physeal growth
Distraction osteogenesis	Embryonic long bones
Rigid fixation (compression plate)	Non-rigid (secondary) fracture healing
Perichondrial bone formation	Epiphyseal secondary centres
Blastema bone (children's amputations)	**Fracture callus**
One component of IM-nail healing	Degenerating OA cartilage, calcifying cartilage tumours, **DBM-induced bone**

Regulation

Signalling	canonical **Wnt** + Hedgehog · **β-catenin** → osteoblasts
Transcription factors	**CBFA1 / RUNX2** · **osterix (OSX)**

Sclerostin

Sclerostin	**Fact**
Gene	**SOST**
Source	**Osteocytes**
Action	✓ **Inhibits Wnt** (✗ Wrong options: activates) → ↓ osteoblasts
Overexpression	**↓ bone mass**
Loss	**Sclerosteosis, van Buchem disease** (high bone mass)

- - Active space - -

6 Complications

Genes & lab markers

Cleidocranial dysplasia	**defective intramembranous ossification** · **CBFA1 (RUNX2)** on **chromosome 6** · AD · absent clavicles, skull defects

How it's asked

"Sclerostin — all true EXCEPT?"	→ **"activates Wnt pathway"** ✗ Wrong options: SOST product · osteocyte-derived · sclerosteosis/van Buchem
"Intramembranous formation does NOT occur in (TO)?"	→ **callus formation during fracture healing** ✗ Wrong options: compression plating, distraction osteogenesis, flat bones

JUVENILE IDIOPATHIC ARTHRITIS

– Basic Science · 53 / 108 –

1 Basics

Definition & epidemiology

JIA	chronic autoimmune arthritis **>6 weeks** in a child **<16 yr** · **diagnosis of exclusion** (rule out infection, malignancy, rheumatic fever, IBD, SLE, Lyme)
Sex	**female > male**
Joints	**knee** > hand/wrist > ankle > hip > C-spine

Genes & lab markers

HLA	Subtype
DR4	Polyarticular
DR8, DR5, DR2.1	Pauciarticular

RF	✓ **seronegative common** (RF+ <15% · 97% normal RF) · RF+ → chronic, destructive, progresses to adult RA
ANA	"positive is diagnostic" (Orthobullets) · "so often false-positive that alone it has no diagnostic value" (OB) · **ANA+ → ↑ uveitis risk**
HLA-B27	positive in only **~8%**
ESR, CRP	frequently elevated (OB) · Orthobullets: basic serology often normal, not diagnostic · anaemia, ↓ albumin, leukocytosis
Serology does NOT confirm JIA	"definitive diagnosis by serology" (false)

3 Presentation

Symptoms	**morning stiffness**, limp that **improves during the day**, joint swelling, fever, rash, visual change
Diagnostic features (one required)	rash · RF · **iridocyclitis** · C-spine involvement · pericarditis · tenosynovitis · intermittent fever · morning stiffness

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Classification

Type	%	Joints	Features
Pauciarticular (oligoarticular)	**50% — most common**	**<5**, large (knee, ankle), asymmetric	**Girls 1–4 yr (peak 2–4)** · M:F 1:4 · **affected limb LONGER** · **highest uveitis/blindness risk (TO)** · best prognosis
Polyarticular	30%	**>5**, hand/wrist	Wrist ulnar deviation + flexion, MCP stiff in extension, radially deviated fingers · 60% remission
Systemic (Still's)	20%	Multiple	**Fever, rash, splenomegaly**, anaemia, ↑ WBC, ESR, CRP, platelets · M = F · 5–10 yr · **lowest uveitis risk** · worst prognosis

Onset early (pre-teen) vs late (teens)

4 Diagnosis & Classification

Uveitis & ophthalmology screening

Iridocyclitis (anterior uveitis) **often asymptomatic/indolent** · rapid irreversible vision loss if missed · up to 30% lose vision

Risk by type (TO) ✓ **pauciarticular > polyarticular > systemic** · pauciarticular + **ANA+** highest (~75%, OB2240 · ~20% in OB1331 — read the options)

Screening ✓ **urgent slit-lamp exam** · **every 4 months if ANA+**, twice yearly if ANA− · pupil asymmetry → immediate

Eye finding	Condition
Anterior uveitis	**JIA (pauci, ANA+)** — urgent, preventable
Superior lens dislocation	Marfan
Inferior lens dislocation	Homocystinuria
Corneal clouding	Hurler (not preventable), Hunter (mild)
Ptosis	Pompe

Key radiology findings

X-ray often **normal** early · late juxta-articular osteopenia, destruction

C-spine kyphosis, facet ankylosis, **atlantoaxial subluxation** → ✓ **flexion–extension C-spine films before intubation/surgery**

5 Management

First line **DMARDs** (etanercept — TNF inhibitor · rituximab — anti-CD20 · azathioprine — purine synthesis inhibitor · IA methotrexate/etanercept) · **intra-articular steroids** · NSAIDs/high-dose aspirin (less now) · gold occasionally · **frequent ophthalmology exams**

Painless elbow loss of motion + multiple joint pains (13-yr-old pitcher) ✓ **DMARDs** (JIA, not OCD — ✗ Wrong options: microfracture, drilling, immobilisation)

Operative synovectomy (failed medical) · **epiphysiodesis** for LLD (affected leg longer) · osteotomy deferred to maturity · arthrodesis/arthroplasty for severe disease

– – Active space – –

6 Complications

Prognosis 50% resolve · 25% mild disability · 25% crippling arthritis or blindness

Lead toxicity (related)

Lead inhibits ✓ **PTHrP** → growth inhibition · ↓ BMD (animal studies)

X vs Y

	Pauciarticular JIA	Lyme arthritis
Joints	Knee, oligo	Knee, oligo
Rash	No	Erythema migrans
Antibiotic response	No	Yes
Association	**Uveitis**	Bilateral facial palsy, carditis

How it's asked

"JRA type with highest iritis/blindness risk (TO)?"	→ **pauciarticular** ✗ Wrong options: systemic (lowest)
"4-yr-old girl, knee effusion, limp better later in day, ANA+ low titre — next workup?"	→ **ophthalmology evaluation** ✗ Wrong options: skeletal survey, MRI, bone scan, factor levels
"Highest risk of anterior uveitis needing ophthalmology?"	→ **pauciarticular JRA with +ANA** ✗ Wrong options: Marfan (lens) · Hunter (cornea)
"Urgent ophthalmology to prevent irreversible complication?"	→ **6-yr-old girl, 2 mo knee swelling, fever, ↑ ESR (JIA)**
"5-yr-old, oligoarthritis better at end of day, antibiotics failed, RF− — association?"	→ **uveitis** ✗ Wrong options: facial palsy (Lyme)
"FALSE about JIA?"	→ **"definitive diagnosis by serology"**
"4-yr-old with presumed JIA — likely serum result?"	→ **normal RF** ✗ Wrong options: normal ESR/CRP · ASLO · HLA-B27
"Peak onset of pauciarticular JRA?"	→ **2–4 years**
"14-yr-old with JIA before rectal surgery — required work-up?"	→ **cervical (flexion–extension) radiographs**
"Painless elbow stiffness + polyarthralgia in young pitcher — first line?"	→ **DMARDs**
"Lead toxicity inhibits?"	→ **PTHrP**

– – Active space – –

LEAD TOXICITY

– Basic Science · 54 / 108 –

1 Basics

PTHrP inhibition

Mechanism	lead inhibits ✓ **parathyroid hormone-related peptide (PTHrP)** signalling → **growth inhibition** (PTHrP normally delays chondrocyte hypertrophy in the physis)
Bone effect	↓ **bone mineral density** (animal studies)
Traps	✗ Wrong options: TSH · growth hormone · IGF · GnRH

How it's asked

"Lead toxicity affects growth by inhibiting?"	→ **PTHrP**

– – Active space – –

LEGAL CONSIDERATIONS IN ORTHOPAEDIC PRACTICE

– Basic Science · 55 / 108 –

1 Basics

Medical negligence – four elements

Element	Definition
Duty	Obligation to meet the professional standard of care (same specialty)
Breach of duty	✓ **Failure to exercise the diligence and care a reasonable, prudent person would under similar conditions** – deviation from standard of care
Causation	Breach **directly caused** the injury
Damages	Money awarded for injury

Standard of care (legal) ✓ **reasonable treatment showing knowledge, skill, diligence and care** (what a reasonable physician would do)
✗ Wrong options: majority norm · minimum · average · "what you'd want for family"

Corrected error, good outcome, lawsuit ✓ **defend: error occurred but was corrected, little/no damages**
✗ Wrong options: settle · claim no error · contact patient directly

Litigation statistics

Trend ortho suits ↑ **13%** 2003–2008 (ageing population) · ortho **7th** most sued specialty · **~33%** of claims pay out · defence ~$47,000

Causes "**improper performance**" 45% · procedures: joints (non-fusion), open/closed reductions · diagnosis: **OA (21%)**, joint disorder, **femur fracture**

Subspecialty ✓ **arthroplasty most sued (29.9%)** > spine (22%) > trauma (17.7%) · most cases won by surgeon · highest plaintiff success: **fasciotomy (48%)**, infection (43%)

Highest total indemnity by fracture ✓ **tibia** (> femur despite fewer cases)

Wrong-site claims **84%** pay indemnity vs 30% of other ortho claims

– – Active space – –

Legislation

Law	Content
Anti-Kickback Statute (1987)	**Criminal** – no remuneration to induce referrals for federal programmes
Stark law (Social Security Act §1877; 1989/1993)	✓ **No self-referral** to entities with physician financial relationship (PT, lab, radiology, DME) · fines to **$100,000** per scheme · "in-office" exceptions
Fee-splitting provisions	Sharing revenue for referrals
Certificate of Need	State control of new facilities
ACA (2010)	Insurance fairness/affordability (Sunshine Act passed within it)
Physician Payments Sunshine Act (2010)	✓ **Industry payments >$10 reported to CMS** (Open Payments)
HIPAA (1996)	PHI protection · **no consent needed to share with other treating providers** (referral records, treating chiropractor)

IRB

Origin — **Nuremberg Code** (Nazi physicians) · dignity, autonomy, informed consent, fair selection, equal burden/benefit · mandates trial consent process

Vulnerable populations — **fetuses, pregnant women, children, prisoners**, cognitively impaired, students, employees
✗ Wrong options: "prisoners are not"

No IRB needed — ✓ **internal QI study** (even with name, DOB, MRN) · IRB needed: MRN-based research, phase III device trial, CP children, RCT

5 Management

Informed consent (emergencies, minors, interpreters, competency)

Five elements (research + procedures) — **voluntariness** · **provision of information** (risks, benefits, alternatives, no-treatment consequences, lay terms) · **competency** · **understanding** · **decision**
✗ Wrong options: jurisdiction

Late-stage dementia excluded from trial ✓ **competency**

Situation	Action
Unconscious, life/limb-threatening, no family	✓ **Confirm + document necessity with a colleague of similar expertise**, then operate
GCS 10, fiancé present, IIIC tibia	**Two surgeons of similar expertise** (fiancé has no legal standing)
Intoxicated, compartment syndrome, refusing, no translator	✓ **Fasciotomy after documenting necessity** (lacks capacity; friend can't consent)
Non-emergent (closed femur), patient lacks capacity	✓ **Delay; obtain next-of-kin consent** (daughter before mother)
13-yr-old, urgent tibia fracture, parents abroad (German)	✓ **Call parents with a German interpreter** (✗ Wrong options: cousin · child translating)
Limited English	✓ **Certified live interpreter → certified telephone** (✗ Wrong options: family · minors · own limited Spanish)
Jehovah's Witness	✓ **Crystalloid, starch (Hetastarch), colloid** · (✗ Wrong options: whole blood, RBC, platelets, plasma, autologous/donor-directed) · cell saver = individual choice
Consent says resident only retracts, attending leaves before cementing	✓ **Ethically unsound — patient misled** ("ghost surgery" → battery)

Next of kin legal guardian → **spouse → children → parents → siblings → grandchildren**

Implant origin **bovine → Hindu** · porcine → Jewish, Muslim

Surgeon seropositive disclose **HIV, HBV, HCV** at scheduling of exposure-prone procedure · needlestick transmission: HIV ~0.3% · HCV ~1.8% · HBV 6–30% (30% if HBeAg+)

Wrong-site surgery & Universal Protocol

Joint Commission Universal Protocol (2004) **pre-procedure verification** (documents, labelled imaging displayed, blood/implants/equipment available) · **site marking** · **time-out**

Site marking surgeon's **initials (or "YES")**, permanent marker, **with the patient**, in pre-op holding, **visible in the surgical field after draping** · (✗ Wrong options: "X" · "NO" on other limb · plantar foot for TKA) · may be delegated to resident/PA/APRN (LIP accountable)

Correct sequence ✓ **read consent with patient → mark site → regional block → time-out → incise**

JCAHO time-out correct **patient, side, site, procedure** · antibiotic not mandated · **all team members speak** · any member can lead · attending present in room
✗ Wrong options: only surgeon

Two teams (ortho + plastics) both present **single time-out** sufficient · otherwise repeat

Late-arriving resident greet the team · no mandate to repeat time-out

Wrong side about to be incised (resident) ✓ **ask surgeon to revisit safety checklist**

Trend reported wrong-site surgeries ✓ **increased** since 1997/2003 protocols (↑ reporting) · 12.7/million · **ortho + podiatry 41%**

– – Active space – –

WHO checklist (2009) sign-in (before induction: ID, site, consent, airway, **blood-loss risk**) · time-out (team introductions, **antibiotics within 60 min**, **imaging displayed**) · sign-out · "implants + rep present" not a component · ↓ mortality, complications, ↑ crisis adherence · **surgeon most effective member**

Errors & disclosure

Error	Response
Wrong-side incision noticed	✓ **Immediately discuss with patient + family** (then risk management) · never alter record
Wrong knee incised in TKA	✓ **Close wound, complete correct TKA, tell family after surgery**
Wrong spinal level (T7 not T6)	✓ **Complete correct level**, document, apologise, inform after · (✗ Wrong options: abort)
Intraop MCL laceration	✓ **Tell the patient the MCL was injured and plan adjusted**
Documentation error	✓ **Addendum only** – altering record is illegal, no one can authorise it

Full disclosure + apology + responsibility + prevention plan ✓ **↑ patient satisfaction** (not ↑ litigation, not guaranteed prevention) · accept responsibility ≠ blame

Second opinion **ethical (not legal)** duty to disclose error/standard-of-care breach to **patient** · no legal duty to testify (but subject to subpoena) · no duty to tell original surgeon · only patient can transfer care

Impaired colleague (drinking attending) ✓ **express concern to the attending** → escalate to chief if needed
✗ Wrong options: operate unsupervised

Communication & patient safety

Communication breakdown ✓ **most common cause of errors harming patients** · leading cause of **wrong-side surgery, medication errors, diagnostic delays, loss to follow-up** · 70% of sentinel events · ↑ costs + complications
✗ Wrong options: SSI

Communication affects patient satisfaction, adherence, physician satisfaction, malpractice
✗ Wrong options: **Stark II litigation**

Best ↓ sentinel events ✓ **culture where all team members feel empowered to speak** (flattened hierarchy)

Retained sponge after shift change ✓ **OR-wide crew resource management training**
✗ Wrong options: blame nurses · single policy

CPOE ✓ **↓ medication errors** (incl. allergy errors)

Physicians interrupt **23 seconds**

Child lost to follow-up after casting (Spanish-speaking) ✓ **instructions not communicated** · discharge transitions need clear follow-up

Malpractice process

Summons received ✓ **notify liability carrier first** (assigns attorney)
✗ Wrong options: call patient · discard notes

Formal complaint **written response + all records to medical board**
✗ Wrong options: automatic suspension · mandatory attorney

– – Active space – –

Insurance ✓ **occurrence policy** (covers incidents during employment whenever filed) · claims-made needs **tail** · **nose** = prior acts if previous claims-made without tail

Terminating care written notice + **30–45 day** grace period · without it = **abandonment**

Workers' compensation & disability

Term	Definition
Maximum medical improvement	Further restoration not anticipated → ✓ **claim can be settled** (declared by treating physician)
Impairment	✓ **Loss of function from anatomic/physiologic derangement** (medical)
Disability	Limitation in meeting personal/social/occupational demands (legal)
Apportionment	Share of disability from each injury
FCE	Tests ability to function (work)

Choice of physician ✓ **state statute**

Suspected malingering / fear elicit with ✓ **empathy** (lean forward, open questions, few notes)

Team physician & athlete confidentiality

Employed team physician inform **team and player** of athletically relevant issues
✗ Wrong options: media

Life-threatening risk (transient quadriplegia + stenosis) ✓ **ban from participation** · no constitutional right to play · **scholarship kept** · waivers don't protect

Share without consent ✓ **treating chiropractor** (treatment)
✗ Wrong options: mother, coach, reporter, fiancée

Employment & billing

Independent contractor ✓ **contractor determines means + methods**; employer determines result · employer pays no taxes, insurance, retirement, training

Injection + separate osteoporosis management same day ✓ **procedure code + E&M with modifier 25**

Access to care

Paediatric access ✓ **type of insurance** (Medicaid vs private)

How it's asked

"Wrong-side incision recognised, correct ORIF done — next?"	**→ immediately discuss with patient and family**
"Wrong-knee ACL scope — prevented by?"	**→ surgeon initials operative limb with patient in pre-op holding**

– – Active space – –

"Before THA incision, entire staff must?"	→ time-out
"JCAHO time-out — all EXCEPT?"	→ correct preoperative antibiotic
"Wrong thoracic level resected — all EXCEPT?"	→ abort case and get more imaging
"Ortho + plastics both present — time-out?"	→ one is sufficient
"Resident sees wrong carpal tunnel about to be incised?"	→ ask surgeon to revisit safety checklist
"Wrong knee incised in TKA?"	→ close, do correct TKA, tell family afterwards
"Wrong-site surgeries since protocols?"	→ increased
"↓ mortality + ↑ crisis adherence?"	→ surgical checklist
"NOT in WHO checklist?"	→ correct implants/rep present
"NOT a way to prevent wrong-site surgery?"	→ marking an "X"
"NOT part of Universal Protocol?"	→ only staff surgeon speaks during time-out
"Correct sequence (shoulder scope)?"	→ consent read → mark → block → time-out → incise
"Late resident joins case?"	→ greet team, no repeat time-out
"Universal Protocol violation?"	→ TKA mark on plantar foot
"Unconscious IIIA tibia, no family?"	→ confirm + document necessity with a colleague
"Patient told resident only retracts, attending leaves before cementing?"	→ unsound — patient misled
"13-yr-old, parents in Germany?"	→ call parents with German interpreter
"Intoxicated, compartment syndrome, refusing?"	→ fasciotomy after documenting necessity
"Elderly Spanish speaker with granddaughter?"	→ certified live interpreter, then telephone
"GCS 10, fiancé present?"	→ surgeon + another physician of similar expertise
"Jehovah's Witness — acceptable?"	→ starch (Hetastarch)
"Dementia excludes trial enrolment — consent element?"	→ competency
"Head-injured, closed femur, daughter + mother present?"	→ delay until next-of-kin (daughter) consents
"Correct op-note antibiotic timing?"	→ illegal to alter; addendum
"Disclosure after intraop fracture?"	→ ↑ patient satisfaction
"Second opinion finds error?"	→ ethically must disclose to patient

"Corrected malrotation, good outcome, sued?"	→ defend — error corrected, little damage
"Intraop MCL laceration — tell patient?"	→ MCL injured, plan adjusted
"CPOE effect?"	→ ↓ medication errors
"Communication does NOT affect?"	→ Stark II litigation
"Communication breakdown leads to?"	→ delayed diagnoses + medication errors
"Most likely to ↓ sentinel events?"	→ empowered team culture
"Spanish-speaking child in cast 3 months?"	→ instructions not communicated
"Most common cause of harmful errors?"	→ communication breakdown
"Drunk attending naps after time-out?"	→ express concern to attending
"Physicians redirect patients after?"	→ 23 seconds
"Communication failure in surgery →?"	→ ↑ cost, ↑ complications
"Lost sponge after shift change — fix?"	→ OR-wide crew resource management
"7-yr-old casted DRF, discharged — true?"	→ poor communication harms care transitions
"Failure to exercise reasonable diligence?"	→ breach of duty
"Second-opinion obligations EXCEPT?"	→ legal duty to be plaintiff's expert witness
"Highest total indemnity fracture?"	→ tibia
"Legal standard of care?"	→ reasonable treatment with knowledge, skill, diligence, care
"New employed trauma surgeon — insurance?"	→ occurrence coverage
"Received summons — first step?"	→ notify liability carrier
"Most likely procedure in a claim?"	→ THA
"Infected TKA — why written response to board?"	→ formal complaint filed
"Workers' comp claim settled when?"	→ MMI declared by treating physician
"Loss of function from derangement?"	→ impairment
"No further restoration expected?"	→ MMI
"Choice of physician in workers' comp set by?"	→ state statute
"Elicit fear of confrontation?"	→ empathy
"Share spondylosis without consent with?"	→ treating chiropractor
"Scholarship athlete at life-threatening risk?"	→ ban from participation
"Transient quadriplegia + stenosis?"	→ do not return to football
"Pro ACL rupture — team physician informs?"	→ team and player
"IRB inaccurate statement?"	→ "prisoners not vulnerable"

– – Active space – –

"Reporting gifts >$10?"	→ **Sunshine Act**
"Send records to neurologist?"	→ **no extra consent**
"No IRB needed?"	→ **internal QI study**
"Seropositivity to disclose?"	→ **HBV, HIV, HCV**
"Co-worker refers to his own PT company?"	→ **Stark law**
"Injection + osteoporosis visit billing?"	→ **procedure + E&M with modifier 25**
"Independent contractor true?"	→ **determines means and methods**
"Limits paediatric ortho access?"	→ **type of insurance**
"Bovine grafts matter to?"	→ **Hinduism**

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LEVEL OF EVIDENCE

– Basic Science · 56 / 108 –

1 Basics

Steps of evidence-based medicine

Step	Action
1	**Formulate an answerable question**
2	**Gather** the evidence
3	✓ **Appraise** the evidence (levels of evidence used here)
4	**Implement**
5	**Evaluate** the process

Assigning level by study design & follow-up

Level	Therapeutic study (JBJS/Oxford)
I	**High-quality RCT** (randomised, blinded, **>80% follow-up**, narrow CI) · systematic review/meta-analysis of **homogeneous Level I RCTs** · all-or-none
II	**Lesser-quality RCT** (<80% follow-up, no blinding, improper randomisation, crossover) · **prospective cohort / prospective comparative** · SR of Level II or heterogeneous Level I
III	**Retrospective cohort** · **case-control** · retrospective comparative · SR/meta-analysis of Level III
IV	**Case series** (no or historical control)
V	**Expert opinion**, case report, personal observation, bench/animal work

Meta-analysis level	= **lowest level of included studies** (OB, Orthobullets) · one RCT with 70% follow-up in an SR → **Level II** · meta-analysis of retrospective cohorts → **Level III** · TO table: **meta-analysis of cohort studies = Level 2** (TO) – read the options
RCT with >20% loss to follow-up (TO)	✓ **downgraded to Level II** (attrition bias) even with flawless randomisation, concealment, ITT, blinded assessors **(TO)**
Reporting guidelines	**QUORUM / PRISMA** (meta-analyses of RCTs) · **MOOSE** (observational meta-analyses) · **CONSORT** (RCTs)
Study types	**therapeutic** (results of treatment) · **prognostic** (patient characteristic → outcome) · **diagnostic** (test efficacy) · economic/decision
Most common study type in ortho literature	✓ **Level IV case series** (58–64%)

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AAOS guideline recommendation strength

Strength	Evidence
Strong	≥2 high-quality studies
Moderate	1 high or 2 moderate
Weak	1 moderate or ≥1 low
Consensus	Expert opinion — only for interventions preventing **loss of life or limb**

GRADE framework (TO)

Baseline (TO) — **RCTs start High · observational studies start Low** · separates **quality of evidence** (high → very low) from **strength of recommendation** (strong / weak)

Downgrade (5) (TO)	Upgrade (3) (TO)
Risk of bias (no blinding, no concealment, attrition, early stopping)	**Large magnitude of effect** — RR >2 or <0.5 (+1) · RR >5 or <0.2 (+2)
Inconsistency (heterogeneity, high I^2)	**Dose–response gradient**
Indirectness (PICO mismatch, surrogate outcome e.g. BMD vs fracture)	✓ **All plausible confounders would reduce the observed effect**
Imprecision (small n, few events, wide CI)	–
Publication bias (funnel plot)	–

Not upgrade criteria (TO) — explained heterogeneity · PROSPERO registration / PRISMA adherence · precise CI around small RR 1.2

How it's asked

"Level of meta-analysis of cohort studies (TO)?"	→ **2**
"RCT, blinded, ITT, flawless randomisation, 22% loss to follow-up (TO)?"	→ **Level II — attrition >20%**
"Level II study?"	→ **prospective cohort** ✗ Wrong options: case-control (III) · high-quality RCT (I)
"Level III study?"	→ **retrospective case-control**
"Moseley sham-surgery RCT, >90% follow-up?"	→ **therapeutic Level I**
"SR of 15 homogeneous high-quality RCTs?"	→ **Level I**
"Meta-analysis of retrospective cohorts (QUORUM)?"	→ **Level III**
"Prospective non-randomised matched PRP cohort, >80% follow-up?"	→ **Level II**
"Prospective surgeon-preference IT fracture study, 85% follow-up, wide CI?"	→ **Level II**
"RCT, 20% crossover, unblinded surgeon assessor, 65% follow-up?"	→ **Level II**

"10 patients, new procedure, no control?"	→ **case series**
"Surgeon's 25-yr technique article?"	→ **Level V**
"SR of 6 RCTs, one with 70% follow-up?"	→ **Level II**
"Most common study type in ortho literature?"	→ **Level IV case series**
"Third step of EBM?"	→ **appraise the evidence**
"Upgrade Low → Moderate in GRADE (TO)?"	→ **all plausible confounders would reduce the effect**

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LIGAMENTS

– Basic Science · 57 / 108 –

1 Basics

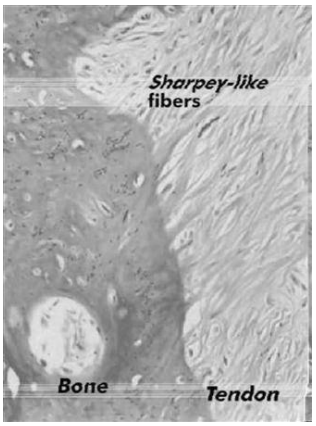
Function & composition

Function	restrict motion · stabilise joint · **proprioception** (mechanoreceptors, free nerve endings)
Composition	water · **type I collagen (70% dry weight)** · **elastin (more than tendon)** · lipids · proteoglycans · epiligament (some) · **fibroblasts** · low vascularity + cellularity
Ligament vs tendon	ligament: **less collagen**, **more PG + water**, less organised fibres, rounder fibroblasts
Blood supply	enters at **insertion** (unlike tendon) · uniform microvascularity · ACL/PCL: **middle geniculate artery**

Insertion anatomy (direct, indirect, Sharpey fibres)

	Indirect (fibrous)	**Direct (fibrocartilaginous)**
Frequency	**Most common**	–
Mechanism	Superficial fibres → **periosteum** · deep fibres → bone via **Sharpey fibres** (type I collagen)	4 transition zones of ↑ stiffness (force dissipation)
Example	**MCL into proximal tibia**	**Supraspinatus**, native ACL

Direct zone	**Content**
1 Ligament/tendon	Aligned type I collagen, decorin
2 Fibrocartilage (unmineralised)	Types **II + III**, small I, IX, X · aggrecan, decorin
3 Mineralised fibrocartilage	Type II + significant **type X**, aggrecan
4 Bone	Type I, high mineral



Sharpey fibres – indirect insertion

2 Mechanism

Viscoelasticity & mechanical behaviour

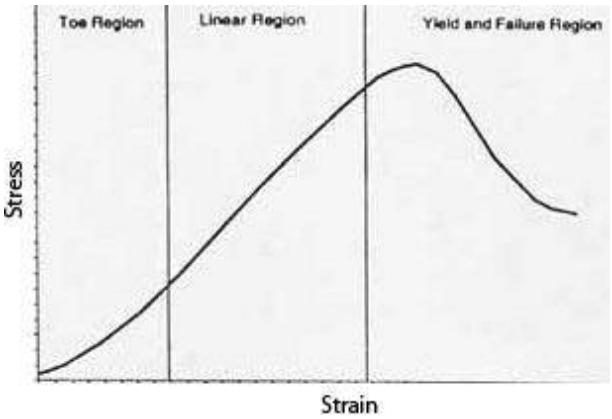
Viscoelastic ✓ behaviour depends on **loading rate** (faster → **stiffer + stronger**) → control load rate when testing prosthetic ligaments

Property	Definition
Creep	↑ deformation over time under **constant load**
Stress relaxation	↓ stress over time under **constant deformation**
Hysteresis	Unloading curve ≠ loading curve · area = energy dissipated

Load–elongation curve **toe region** (crimped fibres straighten, **non-linear**) → **linear region** (slope = stiffness / Young's modulus) → yield (elastic → plastic) → failure

Ligament vs tendon curve ligament recruits fibres gradually → **longer toe region** · tendon → shorter toe, higher loads

Anisotropic strong in **tension**, weak in compression · Young's modulus **< cancellous bone, > cartilage**



Stress–strain curve — toe, linear, failure

Failure

Mechanism sequential collagen fibre rupture · no plastic deformation

Site **adults: midsubstance** · **children: bony avulsion** at junction of **mineralised + unmineralised fibrocartilage**

Grade	Injury
I	Mild sprain
II	Moderate / partial tear
III	Complete tear

– – Active space – –

5 Management

Healing & graft incorporation

Phase	Time	Events
Inflammatory	**1–7 d**	Neutrophils, macrophages · **type III collagen** · TGF-β1, IGF, PDGF, BMP-12/13, bFGF
Proliferative	**7–21 d**	→ **type I collagen** · **weakest day 5–21**
Remodelling	>14 d	–
Maturation	≤18 months	–

Healed MCL at 1 year vs normal ✓ ↑ **number of fibres** · type III proportion back to normal · **cross-links only ~45%** · ↓ fibre mass + diameter · scar → ↓ ultimate strength, adhesions

Impair healing **intra-articular** (ACL < MCL) · age · **immobilisation** · smoking · **NSAIDs** · DM · alcohol · ↓ bFGF, NGF, IGF-1 · ↓ procollagen I, COMP, tenascin-C, tenomodulin, scleraxis

Improve (experimental) extra-articular · CD44 knockout · **IL-10** · IL-1 receptor antagonist · macrophage depletion · MSCs (tunnel healing) · PDGF-BB (↓ adhesions, no strength gain) · G-CSF (tunnel) · BMP-2/-12 (cuff) · collagen–PRP hydrogel for primary ACL repair · **CGRP** (MCL)

Allograft vs autograft ACL allograft: ✓ **prolonged inflammation**, slower incorporation, more large-diameter fibrils · tendon grafts heal to tunnel by **fibrovascular scar → Sharpey-like fibres** · intra-articular part: necrosis → host synovial repopulation → revascularisation → **ligamentisation**
✗ Wrong options: native 4-zone insertion

How it's asked

"Order of direct insertion zones?"	→ **ligament → fibrocartilage → mineralised fibrocartilage → bone**
"MCL tear – structure and insertion?"	→ **ligament; Sharpey fibres (indirect)**
"Why control loading rate testing a prosthetic ligament?"	→ **ligaments are viscoelastic**
"MCL sprain – mechanical property?"	→ **non-linear stress–strain on initial loading** ✗ Wrong options: isotropic · strongest in compression
"Grade III MCL at 1 yr vs normal – ↑?"	→ **gross number of fibres** ✗ Wrong options: type III · cross-links · diameter
"Allograft vs autograft ACL inflammatory phase?"	→ **prolonged**

LUMBOSACRAL PLEXUS

– Basic Science · 58 / 108 –

1 Basics

Nerve at risk – iliosacral screws

Danger screw anterior/superior on the **sacral ala** (above the alar slope / iliac cortical density on lateral view) → ✓ **L5 nerve root**

L5 deficit ✓ **loss of great-toe extension (EHL)** · also tibialis anterior, tibialis posterior, EDL, hamstrings, gluteus medius/maximus

Fluoroscopic view	Checks
Lateral	Below **iliac cortical density / alar slope** → protects L5
Outlet	**Above S1 foramen**
Inlet	Into **anterior** sacral body

Lateral femoral cutaneous nerve

Course lumbar plexus → **surface of iliacus** → ✓ **under the inguinal ligament** (near ASIS) → pierces fascia lata · variable path

Traps ✗ Wrong options: deep to psoas · medial to femoral vein · adjacent to femoral nerve · deep to iliopectineal fascia

How it's asked

"Iliosacral screw too anterior-superior on S1 ala – deficit?"	→ **loss of great-toe extension (L5)**
"Most common location of LFCN?"	→ **under the inguinal ligament**

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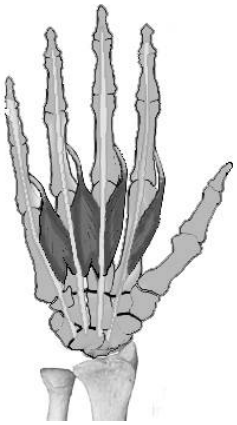
LUMBRICAL MUSCLES

– Basic Science · 59 / 108 –

1 Basics

Anatomy & function

	Lumbricals (4)
Origin	**FDP tendons** (radial side)
Insertion	**Radial lateral band** of extensor expansion
Action	**Flex MCP** (force palmar to MCP axis) + **extend PIP/DIP**
Innervation	**1st + 2nd: median** · **3rd + 4th: ulnar**
Arterial supply	Palmar metacarpal arteries



Lumbricals arising from FDP

Muscle	Pennation / origin
1st, 2nd lumbricals	✓ **Unipennate** · radial side of own FDP
3rd, 4th lumbricals	**Bipennate** · adjacent FDP tendons
Dorsal interossei (4)	Bipennate · adjacent metacarpals · **ABduct**
Palmar interossei (3)	Ulnar side index, radial side ring + small MC · **ADduct**

Intrinsic-plus hand tight lumbricals + interossei → MCP flexion with **↓ supple PIP/DIP flexion**

How it's asked

"Unipennate, radial side of FDP, inserts radial lateral band, palmar to MCP axis?"

→ **1st lumbrical**
✗ Wrong options: 4th lumbrical (bipennate) · dorsal interosseous

MATERIAL PROPERTIES

– Basic Science · 60 / 108 –

1 Basics

Definitions

Term	Definition	Units
Load	Force on a body	N
Stress	Internal force intensity = **force / area**	Pa (N/m²)
Strain	Relative deformation = **ΔL / L**	None
Elastic deformation	Reversible – returns to shape when unloaded	–
Plastic deformation	✓ **Irreversible** – doesn't return to original length	–
Toughness	✓ **Energy absorbed per volume before fracture = area under stress-strain curve**	J/m³
Creep	✓ **↑ deformation (strain) over time under constant load**	–
Load (stress) relaxation	✓ **↓ stress (peak load) over time at constant strain/elongation**	–
Hysteresis	Loading ≠ unloading curve (viscoelastic) · area = energy dissipated	–
Ductility	Ability to deform plastically under tension before rupture	–
Strain rate	Strain ÷ time	–
Finite element analysis	Complex shape split into triangles/quadrilaterals, forces balanced	–

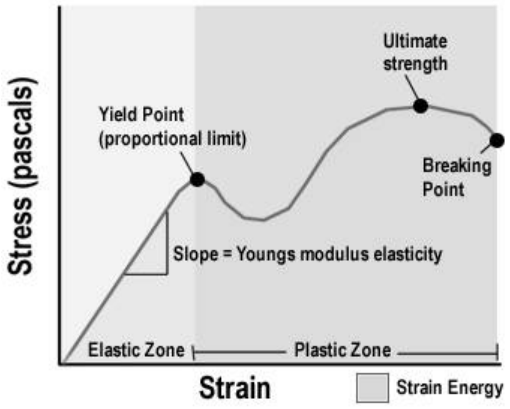
Creep in polyethylene liners plastic deformation under load **without wear debris** · TO note: creep "at high temperature, below elastic limit"

Creep (TO, FNB) answer key not provided by source · options: ↑ length under constant load, ↓ length under constant load, stiffness change from viscoelasticity, ↑ stiffness with time · Orthobullets definition = **increased deformation under constant load**

Stress–strain curve (yield, elastic, plastic, toughness, fatigue)

Axes (TO) ✓ **strain on X-axis**, stress on Y-axis

– – Active space – –



Stress–strain curve

Feature	Meaning
Elastic zone	Linear — returns to shape (A → B "elastic behaviour") · **Hooke's law**: stress ∝ strain
Young's modulus	**Slope in elastic zone = stiffness** (resistance to deformation)
Yield point / yield strength	✓ **Elastic → plastic transition** · stress producing a set permanent deformation
Plastic zone	Permanent deformation
Ultimate (tensile) strength	Highest point — **maximum stress / load to failure** · then necking (ductile)
Breaking (fracture) point	Failure
Toe region	Ligament/tendon — crimped fibrils straighten

Implant absorbing most energy before failure	✓ **high toughness** ✗ Wrong options: modulus · yield stress alone · brittle
Two plates load–displacement	larger area = **tougher** · higher peak = stronger · earlier break = more brittle
Hip screw returns to shape after bending	**elastic** deformation

Metal characteristics

Fatigue	✓ **failure below ultimate tensile strength after repetitive cycles** · depends on stress magnitude + cycles · ✓ mode of failure of a nailed femur that fails to unite (**cyclic fatigue**) ✗ Wrong options: static/catastrophic (single supraphysiological load)
Endurance (fatigue) limit	max cyclic stress never causing failure · arthroplasty implants designed above physiological loads · fracture fixation only until union (TO)
Bone vs implants (TO)	bone ultimate strength ~4× lower than steel → fails first in single load (periprosthetic fracture) · bone **resists fatigue better** (remodelling repairs microcracks)

2 Mechanism

Material descriptions

Type	Definition	Examples
Brittle	Linear to failure, little/no plastic deformation	**PMMA, ceramics**, hydroxyapatite
Ductile	Large plastic deformation before failure	**Metals**
Viscoelastic	Stress–strain depends on **rate + duration** of load (internal friction)	**Ligaments, bone, cartilage**
Isotropic	Same properties in all directions	**Metals (steel, titanium)**, golf ball
Anisotropic	Properties depend on load direction	**Ligaments, bone**

Viscoelasticity (creep, stress relaxation, anisotropy, strain rate)

Ligaments viscoelastic → **strength depends on rate of load** · creep, stress relaxation, non-linear elasticity · not **isotropic** · load–elongation region B = linear elastic after fibrils straightened (A = toe, C = ultimate load, D = energy, E = ultimate elongation)

Cortical bone – ↑ strain rate ✓ **↑ elastic modulus AND ↑ ultimate strength** · physiological rates (<0.1/s) elastic + ductile · high-impact rates brittle · modulus up to 2× in falls (TO)

Bone true statement (TO) ✓ **time-dependent viscoelastic properties**
✗ Wrong options: isotropic · less fatigue-resistant than steel · repetitive tension → oblique fracture (→ transverse)

Anisotropy of cortical bone (TO) compressive strength **50% > tensile** · longitudinal modulus **17 GPa** vs transverse **12 GPa** · longitudinal strength **193 MPa** vs 133 MPa

Articular cartilage biphasic viscoelastic (flow-dependent fluid drag + flow-independent matrix) · modulus changes with **duration + rate** of load

Allograft ACL may lengthen by **creep**

Bone failure (fracture patterns)

Bone strength ✓ **strongest in compression, weakest in shear (TO)**, intermediate tension

Load	Fracture
Tension	**Transverse** (muscle pull)
Compression	Crush · short oblique (fails at 45° plane of max shear) (TO)
Bending	**Butterfly fragment** (convex side fails in tension)
Torsion	**Spiral** · longer bone → ↑ outer-cortex stress
Bending + compression / bending + torsion	**Short oblique** (TO)

Bone composition collagen (low modulus, tensile strength) + hydroxyapatite (stiff, brittle, compressive) · remodels to ↑ **moment of inertia**

Ligaments/tendons tolerate **5–10% strain** (bone 1–4%)

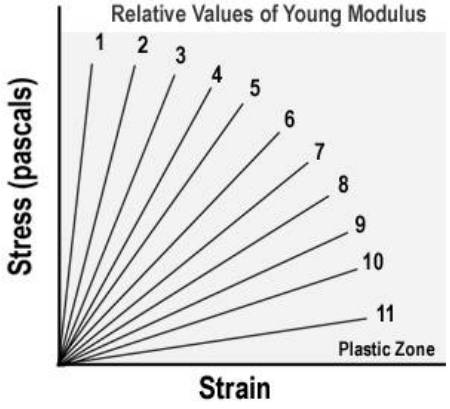
Modulus of elasticity of implant materials

Rank (stiffest → least)	Material	Approx GPa (OB)
1	**Ceramic (Al_2O_3)**	Alumina 340 · zirconia 248
2	**Co-Cr-Mo alloy**	240
3	**Stainless steel**	240
–	Tantalum	186
4	**Titanium**	115 – ✓ **closest to cortical bone**
5	**Cortical bone**	18
6	Matrix polymers	–
7	**PMMA**	3.1
8	**Polyethylene**	0.8–1.5
9	Cancellous bone	2
10	Tendon/ligament	–
11	Cartilage 90	–

Increasing modulus (TO) ✓ **polymers < bone cement < cortical bone < titanium < stainless steel < cobalt-chromium < ceramics**

PMMA modulus between cortical and cancellous (OB) · 10× lower than cortical bone (Webb & Spencer quote) – read the options

Ligament modulus < cancellous bone, > cartilage



Relative Young's modulus

Strength rankings	Order
Ultimate (tensile) strength	**Co-Cr > titanium > stainless steel > cortical bone**
Fatigue limit + tensile strength (TO)	Stainless steel < titanium < Co-Cr
Yield strength	Cast Co-Cr > titanium > wrought Co-Cr, stainless steel

Titanium vs cold-forged Co-Cr ✓ **↓ tensile strength** · ↓ fatigue + yield strength · **↑ ductility**

Specific materials

– – Active space – –

Material	Composition	Pros	Cons
Titanium (Ti-6Al-4V)	–	Most biocompatible · **self-passivation** (adherent oxide) → corrosion resistant · **low modulus** (≈ bone)	**Notch sensitivity, poor wear** → not for femoral head · more metal debris
316L stainless steel	**Iron + carbon (L = low, 0.03%) + chromium (~17–18%) + nickel (~12%) + molybdenum (~2.5–3%)** · **no cobalt (TO)**	Stiff, fracture resistant, cheap, ductile	**Pitting + crevice corrosion** · stress shielding
Cobalt alloy	**Co + Cr + Mo**	Very strong · better corrosion resistance than steel · TKA femoral components	Very stiff → stress shielding (✓ for cemented stems – protects mantle)
UHMWPE	–	Tough, ductile, resilient, wear-resistant, **high fracture toughness**	Abrasion (third body), thermoplastic, weak in tension · **gamma cross-linking ↑ wear resistance, ↓ fatigue + fracture resistance**
PMMA	Powder: polymer, **benzoyl peroxide (initiator)**, **barium sulfate (radio-opacifier)**, colour · Liquid: monomer, **DMPT (accelerator)**, hydroquinone (stabiliser)	**Interlocks with bone** (grout, not glue) · strongest in **compression** · ultimate strength at **24 h**	**Brittle**, notch-sensitive · poor **tension + shear** · **exothermic** polymerisation · hypotension on insertion · fails by microfracture
Silicones	–	Non-WB joint replacement	Poor strength/wear → **synovitis**
Ceramics	Alumina, zirconia	**Best wear vs PE**, high compressive strength, inert	**Brittle, low fracture toughness**, high modulus, low tensile strength, poor crack resistance

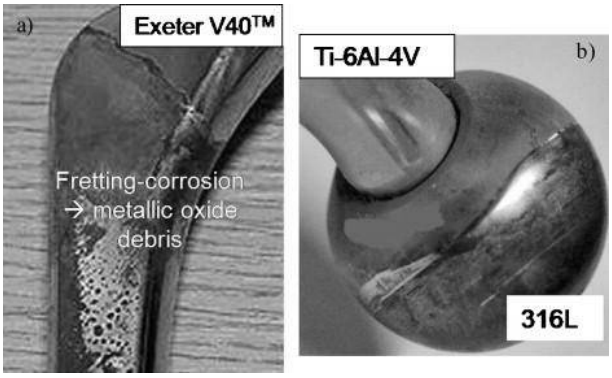
316L components (TO) chromium → oxide layer (anti-rust, passivation) · nickel → manufacturing (austenite) · molybdenum → protects oxide layer from pitting

5 Management

Corrosion (galvanic, fretting, crevice, passivation)

Type	Mechanism	Key facts
Galvanic	✓ **Electrochemical potential between two dissimilar metals in conductive medium** (anode corrodes)	**316L + Co-Cr = highest risk** · steel very susceptible · Co-Cr "most susceptible" of listed (✗ Wrong options: Ti, ceramics, PE) · cerclage wire SS on Ti stem · mixed-metal neck/stem
Crevice	**O_2 tension difference** in cracks/ crevices	**316L most prone** · **titanium least** · plate countersinks, cementless cups
Fretting	✓ **Micromotion at contact sites under load** → passive layer scraped off → pits, grooves, oxide debris	**Head–neck taper** (trunnionosis), ↑ serum Co/Cr, synovitis, ALTR · **commonest cause of modular mid-stem junction failure** · ↑ with more interfaces · worsens over time · ceramic heads ↓ it
Fretting fatigue	Cyclic load + small oscillation → cracks	✓ **Initiates mid-stem modular junction fracture** (completed by bending) · risk: high BMI, poor proximal bone support
Pitting	Local depassivation → holes	Stainless steel
Intergranular	Grain-boundary attack	–
Stress corrosion	High tensile stress + corrosive environment (manufacturing)	–
Degradation / oxidative	Polymers / free radical	–

Self-passivation ✓ **reason titanium is corrosion resistant** (oxide layer forms spontaneously)



Fretting corrosion at a modular junction

Ceramic properties & fracture

Ceramic disadvantage ✓ **low fracture toughness** (poor crack resistance)

Most serious CoC complication ✓ **fracture** (fragments embed → third-body wear) · squeaking (annoying, may herald failure)

CoC THA: new squeak + stiffness + LLD, fractured ceramic, normal CRP/ESR

✓ **thorough debridement + one-stage head + liner revision**
✗ Wrong options: two-stage (no infection)

Structural mechanics (moment of inertia, bending rigidity, tension band)

Area moment of inertia

✓ ∝ **distance² of material from the centre of mass** (tube > solid rod of same area)
✗ Wrong options: density · modulus

Implant	Bending rigidity ∝
Solid pin / rod	**radius⁴** (diameter⁴)
Hollow (cannulated) nail	~radius³
Plate	**thickness³** (2× thickness → 8× stiffness)
Ex-fix pin	diameter⁴ · pin stiffness ∝ 1/(bone–rod distance)³ · fragment contact most important

Plate on tension side

✓ **minimises bending stresses in the plate** (fracture closes under load, bone shares load)

Strain sharing

5 mm displacement across one 5 mm gap = 100% strain · shared across multiple gaps (each 5 → 6 mm) = 20% → **80% ↓ strain** (Perren)

Stress risers around plates/prostheses (TO)

✓ **overlap plate with stem**, gap <8 cm risky · interprosthetic distance <110 mm risky in weak bone · end-screw fracture 1–3% (non-locking), 2.6% (locking) · ✓ **oblique end screw** · ✓ **non-locking end screw in all-locked construct** · **unicortical end screw – NOT recommended** (weaker) · insert non-locking before locking screws

Screw mechanics (TO)

Term	Fact (TO)
Core (root/minor) diameter	= **drill size** · determines **strength, fatigue resistance, bending strength, tensile strength**
Major (thread) diameter	Thread depth (major − minor) → **pull-out strength**, stripping torque
Runout	Shaft–thread junction · **weakest part of partially threaded screw**
Pitch	Distance between threads · lead = advance per turn
Mechanical advantage	= **circumference / pitch** (✗ Wrong options: pitch/circumference)
Pull-out strength	↑ outer diameter, ↑ cortical thickness · (✗ Wrong options: thread density (if 5–6 threads in cortex)) · ✓ **↓ cortex thickness ↓ pull-out**
AO 3.5 mm cortical screw drill	✓ **2.7 mm** (TO key)

Screw types (TO)

non-tapping (tap needed) · self-tapping (cutting flutes) · self-drilling · cancellous (deep threads; partially threaded = lag) · locking (threaded head) · **Herbert headless** (variable pitch → compression) · interference (no compression, presses graft)

– – Active space – –

Plating principles (TO)

MIPO ideal (TO) ✓ **comminuted metaphyseal fractures of long bones**
✗ Wrong options: abraded skin · compound · diabetic

DHS barrel (TO) standard (long) **38 mm** · short **25 mm** · lag-screw thread **22 mm**

Bridge plating (TO) plate spans comminution, screws away from fracture

Working length (TO) ✓ **distance between the screws/pins closest to the fracture on either side** · ↑ working length → more flexible, ↓ stress risers · unicortical screw has shorter working length than bicortical · ex-fix: pin/cluster spacing · >3 diaphyseal screws add little
✗ Wrong options: "max distance between proximal-most and distal-most screws" = bridge span confusion

Bioabsorbable polymers (TO)

Polymer (TO)	Modulus	Strength retention	Full resorption
PGA	~7 GPa · **highest initial strength** · hydrophilic	**2–4 wk**	6–12 mo (TO) · 1–2 mo (Orthobullets)
PLLA	~3 GPa · hydrophobic methyl group	3–6 mo	**2–5 yr**
PLGA 50:50	~2 GPa · **most amorphous**	1–3 wk	✓ **1–2 mo – fastest of all**
PCL	~0.4 GPa · rubbery	–	2–4 yr
Magnesium	~45 GPa (≈ bone) · H_2 gas	3–6 mo	6–12 mo
β-TCP	Brittle	–	6–18 mo (matches bone formation)
Calcium sulfate	Very low	–	1–2 mo

PGA vs PLLA (TO) ✓ **PGA higher initial strength but degrades much faster**

X vs Y

	Creep	Stress relaxation	Hysteresis
Held constant	**Load**	**Strain/deformation**	–
Changes	↑ Strain	↓ Stress	Energy lost

How it's asked

"On a stress–strain curve (TO)?"	**→ strain on X-axis**
"Stress at which plastic deformation begins?"	**→ yield strength**
"Plastic deformation?"	**→ change in length that doesn't recover**
"Fatigue?"	**→ failure below tensile strength after many cycles**

"Yield point on figure?"	→ **C** (A fracture, B ultimate, D modulus)
"Hip screw returns to shape after bending?"	→ **elastic**
"Plate B larger area under curve?"	→ **tougher**
"Implant absorbing most energy?"	→ **high toughness**
"A → B straight portion?"	→ **elastic behaviour**
"Point separating elastic and plastic?"	→ **yield point**
"Energy per volume before fracture on plot?"	→ **area under curve (E)**
"Nailed femur fails 8 months later?"	→ **cyclic fatigue**
"True of bone (TO)?"	→ **time-dependent viscoelastic**
"Ligament region after fibrils straightened?"	→ **B (linear)**
"Viscoelastic → tensile strength affected by?"	→ **rate of load**
"ACL properties EXCEPT?"	→ **isotropism**
"Load relaxation?"	→ **↓ peak loads over time at same elongation**
"Creep?"	→ **progressive deformation under constant force**
"Cortical bone ↑ strain rate?"	→ **↑ modulus and ↑ ultimate strength**
"Greater longitudinal than transverse compressive strength?"	→ **anisotropy**
"Melted stainless-steel cube?"	→ **isotropic**
"Allograft creep?"	→ **↑ strain at constant load**
"Viscoelasticity of cartilage?"	→ **modulus changes with duration + rate of load**
"Increasing modulus order (TO)?"	→ **cortical bone < Ti < SS < Co-Cr < ceramic**
"Stainless steel description?"	→ **iron-carbon, stiffer than titanium**
"Modulus closest to cortical bone?"	→ **titanium**
"Cr, Mo, Co components of?"	→ **cobalt alloy**
"Ti vs cold-forged Co-Cr?"	→ **↓ tensile strength**
"PMMA true?"	→ **modulus between cortical and cancellous** ✗ Wrong options: $BaSO_4$ initiator · ductile · shear-resistant · endothermic
"Highest modulus?"	→ **ceramic**
"Galvanic corrosion?"	→ **electrochemical potential between dissimilar metals in conductive medium**
"Most susceptible to galvanic corrosion?"	→ **Co-Cr** (of listed)

– – Active space – –

"Cause of fretting corrosion?"	→ micromotion at head–neck junction
"Modular revision stem junction failure mechanism?"	→ micromotion at contact (fretting)
"Titanium corrosion resistance?"	→ self-passivation
"Initiates mid-stem modular junction failure?"	→ fretting fatigue
"Pits, grooves, oxide debris from micromotion?"	→ fretting
"True of 316L?"	→ highly susceptible to pitting/crevice corrosion
"Fretting corrosion true?"	→ small motion at modular junctions
"Black debris at Co-Cr head / Ti neck taper?"	→ fretting corrosion
"Low fracture toughness bearing?"	→ ceramic
"Most serious CoC complication?"	→ fracture
"Fractured CoC, normal labs?"	→ debridement + one-stage head/liner revision
"Bones weakest in (TO)?"	→ shear
"Not recommended to ↓ stress-riser fracture (TO)?"	→ unicortical end screw
"Bending rigidity of solid pin ∝ radius to?"	→ 4th power
"Area moment of inertia influenced most by?"	→ distance from centre of mass
"Plate on tension side because?"	→ minimises bending stress in plate
"Strain change at Y (comminuted) vs X?"	→ 80% decrease
"316L composition (TO)?"	→ iron, carbon, chromium, nickel, molybdenum
"Not a component of 316L (TO)?"	→ cobalt
"NOT true of cortical screw (TO)?"	→ "MA = pitch/circumference"
"↓ pull-out strength (TO)?"	→ thinner cortex
"Bending strength of cortical screw (TO)?"	→ inner core
"AO drill for 3.5 mm cortical screw (TO)?"	→ 2.7 mm
"Ideal fracture for MIPO (TO)?"	→ comminuted metaphyseal
"Long DHS barrel (TO)?"	→ 38 mm
"Plating principle in picture (TO)?"	→ bridge
"Working length – all true EXCEPT (TO)?"	→ "max distance between proximal-most and distal-most screws"
"PGA vs PLLA (TO)?"	→ PGA stronger initially but degrades faster
"PLGA ratio with fastest degradation (TO)?"	→ 50:50

MEDIAN NERVE

– Basic Science · 61 / 108 –

1 Basics

Branches in the hand (TO)

Carpal tunnel — median nerve enters hand deep to flexor retinaculum → divides into **lateral** and **medial** divisions **(TO)**

Division (TO)	Supplies
Lateral (stouter)	**Thenar 2½: APB, opponens pollicis, superficial head FPB** · **1st lumbrical** · sensation thumb + radial index
Medial (mainly sensory)	Rest of lateral 3½ digits · **2nd lumbrical**

Recurrent motor branch — innervates **thenar muscles** → ✓ **at risk excising a thenar mass** (e.g. lipoma)
✗ Wrong options: deep ulnar · AIN · palmar cutaneous ulnar · no "recurrent ulnar branch"

AIN — FPL, pronator quadratus, FDP index (± middle) · terminal branch → wrist capsule only

Median nerve palsy – thenar function & opposition (TO)

Loss of opposition (TO) — ✓ **median nerve**
✗ Wrong options: ulnar · radial · medial division

Thumb abductors (TO) — ✓ **median + radial nerve both** (APB – median · APL – radial/PIN)

3 Presentation

High median palsy vs AIN

Lesion	Attempted fist
High median (arm)	✓ **Thumb + index can't flex** (FPL, FDP + FDS index lost) – "benediction" attitude · middle partly flexes
AIN syndrome	Loss of **DIP flexion thumb, index, middle** (FPL, FDP) – no sensory loss
PIN syndrome	Finger drop at MCPs, wrist extends in **radial deviation**
Radial nerve (arm)	**Wrist + finger drop**
Median at wrist (CTS)	Thenar weakness + sensory loss, long flexors intact

How it's asked

"Excising thenar lipoma – nerve at risk?" → **recurrent motor branch of median**

– – Active space – –

"Can't flex thumb + index on making a fist?"	→ **median nerve lesion in the arm** ✗ Wrong options: AIN · PIN · CTS
"Nerve supply to thumb abductors (TO)?"	→ **median and radial both**
"Nerve injury causing loss of opposition (TO)?"	→ **median**

MOLECULAR BIOLOGY BASICS

– Basic Science · 62 / 108 –

1 Basics

Nucleic acids & cell cycle

Chromosomes **46** (22 autosome pairs + 1 sex pair) · **haploid** = DNA content of egg/sperm (half)

	DNA	RNA
Bases	**A, T, G, C** (A–T, G–C)	**A, U, G, C** – no thymine
Strands	Double	Usually single (double in some viruses/interactions)
2' position	**Hydrogen** (deoxyribose)	**Hydroxyl** (ribose) → less stable
Helix	**B-form**	**A-form**
Location	Nucleus	Nucleus + cytoplasm

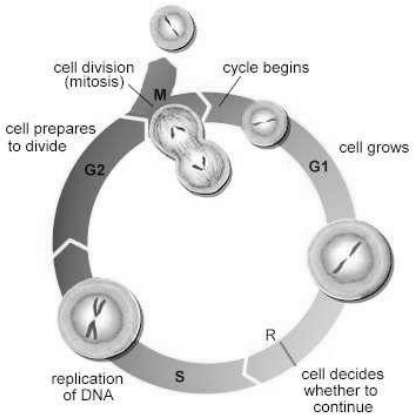
RNA	Role
mRNA	Carries DNA code for protein
rRNA	Ribosome
tRNA	Brings amino acids to mRNA

Term	Definition
Gene	DNA segment encoding a protein
Exon	Coding – **ex**pressed
Intron	Non-coding
Codon	3 nucleotides → 1 of 20 amino acids
Promoter	Controls transcription initiation
Enhancer	Transcription-factor binding site, regulates transcription
Transcription	✓ **RNA polymerase reads DNA → complementary mRNA**
Translation	mRNA codons → polypeptide
Replication	DNA synthesis in S phase
Recombination	Meiotic haploid product with new allele combination
Translocation	Exchange between non-homologous chromosomes (Ewing, rhabdo, synovial sarcoma)

– – Active space – –

Cell-cycle phase	Event	Ploidy
G0	Stable, not cycling	2N
G1	Growth	2N
S	✓ **DNA replication**	→ 4N
G2	Gap	4N
M	Mitosis	→ 2N daughters

No "R phase"



Cell cycle

Apoptosis programmed cell death via intracellular signalling · (✗ Wrong options: lysis (contents spilled)) · loss = hallmark of cancer · **TNF receptor 1** → ✓ **apoptosis**

Laboratory techniques (RT-PCR, GWAS)

Technique	Use
Agarose gel electrophoresis	Separates **DNA by size** (negative charge, small runs faster)
Southern blot	**DNA** sequence
Northern blot	**RNA**
Western blot (SDS-PAGE)	**Protein**
DNA ligation	**Recombinant DNA**
Plasmid	✓ **Extrachromosomal circular DNA replicating independently** · antibiotic resistance · vector (transformation)
PCR	DNA → many DNA copies · primers + heat-stable DNA polymerase
RT-PCR	RNA → cDNA → amplified · product = ✓ **DNA** · ✓ **measures mRNA level** of a gene
ELISA	Antigen / antibody detection
Immunocytochemistry	Localise protein in cells
DNA microarray	Expression profiling
DNA profiling	Forensic identity

Sequencing	Scope
Whole-genome (WGS)	Entire genome, coding + non-coding
Whole-exome (WES)	**Protein-coding exons only** (1% of genome, 85% of disease variants, 1/6 cost)
GWAS	✓ **SNPs across many patients with a phenotype** → mutational variability

6 Complications

Herniated nucleus pulposus pain

✓ **TNF-α** main inflammatory mediator (also IL-1, IL-8)

How it's asked

"DNA replication phase?"	→ **S**
"RNA statements EXCEPT?"	→ **"RNA uses thymine, DNA uracil"**
"Plasmid?"	→ **extrachromosomal circular DNA replicating independently**
"Transcription?"	→ **RNA polymerase reads DNA → mRNA**
"RT-PCR product?"	→ **DNA**
"Assay for mRNA level?"	→ **RT-PCR** ✗ Wrong options: Western, Southern, ELISA
"GWAS true?"	→ **uses SNPs to evaluate variability among patients**
"TNF receptor 1 involved in?"	→ **apoptosis**
"Inflammatory modulator of disc herniation pain?"	→ **TNF-α**

– – Active space – –

MULTIPLE SCLEROSIS

– Basic Science · 63 / 108 –

1 Basics

Epidemiology & pathophysiology

Incidence	5 / 100,000 (US) · **20–40 yr** · **women > men** · **northern latitudes**
Risk factors	stress · smoking · **↓ sunlight / low vitamin D** · not considered hereditary
Mechanism	CNS **demyelination + axonal injury** · **T-cell–mediated autoimmune** · genetic + environmental + infectious factors
Orthopaedic associations	**↑ fracture risk** (falls + ↓ BMD) · **osteoporosis** (inactivity, ↓ vit D, immunomodulators) · gait abnormality (foot drop) · spasticity, contractures

3 Presentation

Clinical definition	**≥2 episodes of neurological dysfunction separated in time and space** (brain, cord, optic nerve)
Symptoms	**optic neuritis**, diplopia, nystagmus · fatigue, depression · dysarthria, dysphagia · weakness, ataxia, falls · paraesthesia, **peculiar sensations (wetness)** · bowel/bladder dysfunction · relapsing–remitting **transient limb numbness** (non-dermatomal)
Exam	wide-based gait, limb ataxia, slapping foot · **spasticity, hyperreflexia, Babinski, Hoffmann** · **internuclear ophthalmoplegia (MLF)** · **Lhermitte's sign**

4 Diagnosis & Classification

Key radiology findings

MRI brain + spinal cord with gadolinium	✓ **multiple focal demyelinating (T2) lesions** · **asymmetric periventricular plaques** · cord lesions commonest in cervical spine



MS – periventricular plaques

Genes & lab markers

CSF	**↑ IgG (80%)** · **oligoclonal bands** · mononuclear pleocytosis (25%) · routine bloods normal (exclude mimics)
Differential	cervical myelopathy · CNS mass · B12 deficiency · sarcoidosis · CNS infection

5 Management

Young woman, transient arm numbness, UMN signs, multifocal T2 cord lesions	✓ **neurology referral** ✗ Wrong options: laminoplasty · plasmapheresis/IVIG (GBS) · B12 (subacute combined degeneration) · lumbar MRI
Acute relapse	**corticosteroids** · **prophylaxis** – **interferon-β** (↓ relapses, slows RRMS)
Spasticity	baclofen, gabapentin, clonazepam, **botox** · PT: gentle stretching, progressive resistance · osteoporosis treatment

6 Complications

Course	✓ **relapsing–remitting (commonest)** · primary progressive · secondary progressive · progressive relapsing
Life expectancy	5–10 yr shorter

How it's asked

"29-yr-old woman, relapsing arm numbness, brisk reflexes, bilateral Hoffmann, multifocal cord lesions – next?"	→ **neurology referral**

– – Active space – –

MUSCLE BIOLOGY & PHYSIOLOGY

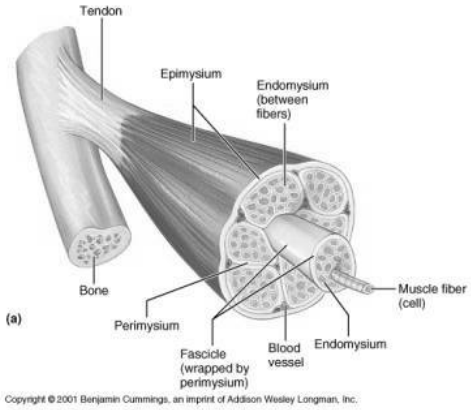
– Basic Science · 64 / 108 –

1 Basics

Gross anatomy

Sheath	Surrounds
Epimysium	Whole muscle (bundles)
Perimysium	**Fascicles**
Endomysium	**Individual fibres**

Myotendinous junction infolded membrane ↑ attachment area · ✓ **weak link – commonest site of strains** (eccentric)



Muscle connective-tissue layers

Sarcomere

Units fibre (cell, from myoblasts) → myofibril (chain of sarcomeres) · **thick = myosin**, **thin = actin**

Band	Content	In contraction
A band	Actin + myosin	**Unchanged**
I band	Actin only	Shortens
H zone	Myosin only	Shortens
Z line	Actin attachment, sarcomere border	Move closer

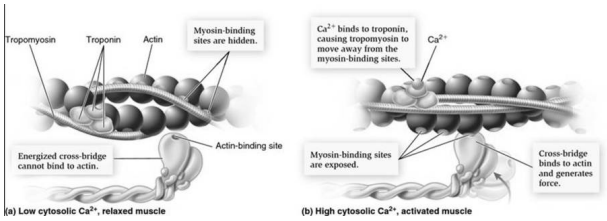
Action potential & excitation–contraction coupling

Nerve action potential ✓ **Na^+ through voltage-gated channels** (positive-feedback depolarisation)

Motor end plate ACh released → binds ACh receptor · **myasthenia gravis** = ↓ ACh receptors · **botox** = blocks ACh release

Coupling depolarisation → sarcoplasmic reticulum **releases Ca^{2+} → Ca^{2+} binds troponin** → tropomyosin shifts → myosin sites on actin exposed → cross-bridges + ATP hydrolysis → sliding · **low Ca^{2+}: tropomyosin blocks** binding sites

– – Active space – –



Excitation–contraction coupling

2 Mechanism

Contraction types (concentric, eccentric, isotonic)

Type	Definition	Example
Isometric	Constant **length**	**Push against wall**
Isotonic	Constant **tension** – ✓ **concentric and eccentric**	**Biceps curl up AND down**
Concentric	Shortens while contracting	✓ **Upward biceps curl**
Eccentric	Lengthens while contracting · **highest forces**	Lowering phase · ✓ **distal biceps rupture (eccentric phase of isotonic exercise)**
Isokinetic	Constant **speed** (Cybex machine)	–
Plyometric	Rapid lengthening → contraction	Box jumps

Force generation & fibre types (cross-sectional area, recruitment)

Force / maximal tension ✓ **most dependent on cross-sectional area** (number of contracting myofibrils)
✗ Wrong options: fibre type · length (Blix curve) · morphology (pennation angle)

Speed + duration of contraction ✓ **fibre type**

Progressive overload (strength training) ✓ **muscle fibre hypertrophy** (↑ fibre size), ↑ coordination

	Type I (slow twitch) – "slow red ox"	Type II (fast twitch)
Metabolism	**Aerobic, oxidative**	**Anaerobic, glycolytic** (IIa oxidative-glycolytic, IIb glycolytic)
Energy	Krebs / oxidative phosphorylation	ATP-CP
Use	**Endurance** · low strength, low speed	**Sprinting** · high strength + speed
Features	↑ mitochondria, vascular	↑ ATPase, low triglyceride · **fatigue rapidly** (IIb)
Atrophy	**First to atrophy with deconditioning**	–

– – Active space – –

Recruitment order (size principle)	✓ **Type I (slow, fatigue-resistant) → Type IIa (fast, fatigue-resistant) → Type IIb (fast, fatigable)** · no "slow fatigable" units

Metabolic systems

System	Duration	Example	Notes
ATP-CP (phosphagen)	**<20 s**	100 m sprint	Anaerobic, **no lactate** · creatine supplement (cramps)
Lactic anaerobic	**20–120 s**	400 m	Glucose → lactic acid
Aerobic	Long, low intensity	Distance running	Krebs cycle, glucose + fatty acids

6 Complications

Muscle injury & healing

Soreness (DOMS)	**oedema + inflammation** of connective tissue · worse with **unaccustomed eccentric** exercise · **peaks 24–48 h** · ↑ CK
First cells after injury (contusion)	✓ **neutrophils** (free radicals may ↑ damage)
Strain	✓ **myotendinous junction** (eccentric) · inflammation → fibrosis · contraction bands seal membrane defects · basal lamina scaffolds preserved
Healing phases	necrosis → inflammation → repair → fibrosis · regeneration needs macrophage clearance (NSAIDs may impair)
Fibrosis	✓ **TGF-β1** · regeneration trophic factors **IGF-1, bFGF** · **angiotensin-II receptor blockade ↓ fibrosis**
Atrophy	disuse/denervation → fatty infiltration, ↑ fatigability · **single-joint muscles atrophy faster** · ↓ CSA → ↓ force

How it's asked

"Upward biceps curl – contraction?"	→ **concentric**
"Distal biceps rupture during dumbbell curl – phase?"	→ **eccentric phase of isotonic exercise**
"Isotonic contraction?"	→ **both upward and downward biceps curl**
"True of skeletal muscle?"	→ **duration + speed depend on fibre type** ✗ Wrong options: force depends on length/fibre type
"Progressive overload in adults →?"	→ **fibre hypertrophy**
"Cross-sectional area determines?"	→ **maximal tension**
"Force most dependent on?"	→ **cross-sectional area**
"Motor-unit recruitment order?"	→ **slow fatigue-resistant → fast fatigue-resistant → fast fatigable**

"First cells after thigh contusion?"	→ **neutrophils**
"Promotes fibrosis after muscle injury?"	→ **TGF-β1** ✗ Wrong options: IGF-1, bFGF (regeneration)
"Calf strain — structure affected?"	→ **musculotendinous junction (C)**
"Na^+ through voltage-gated channel →?"	→ **nerve action potential**

– – Active space – –

– – Active space – –

MUSCULOCUTANEOUS NERVE

– Basic Science · 65 / 108 –

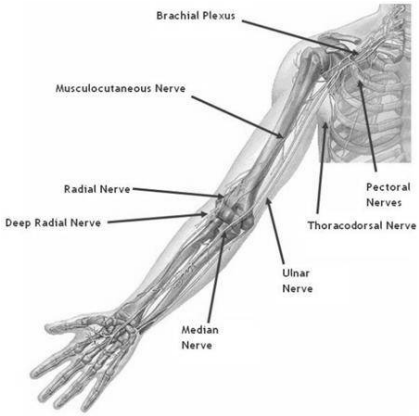
1 Basics

Origin & course

Origin — **C5–7** · superior trunk → **lateral cord** (terminal branch) · leaves axilla at inferior border of pectoralis minor **(TO)**

Course — ✓ **pierces coracobrachialis 3–8 cm** distal to coracoid (Orthobullets) · **5–8 cm** (TO) · 1.5–9 cm (brachial plexus note) → runs **between biceps (superficial) and brachialis (deep)** → pierces deep fascia **lateral to biceps tendon** above elbow → **lateral antebrachial cutaneous nerve**

Motor	Sensory
Coracobrachialis, biceps, brachialis (C5–6) · articular branches to humerus, elbow **(TO)**	**Lateral antebrachial cutaneous** – lateral (+ volar) forearm



Musculocutaneous nerve piercing coracobrachialis

6 Complications

Injury after coracoid transfer

Mechanism — iatrogenic in **deltopectoral approach** · dissection/retraction **medial to the conjoint tendon** · ✓ **vigorous retraction of conjoint tendon (short head biceps + coracobrachialis) → neurapraxia (TO)** · Latarjet / coracoid transfer, open shoulder reduction

Lateral forearm numbness after coracoid transfer — ✓ **musculocutaneous nerve**
✗ Wrong options: axillary (lateral arm) · radial (dorsal wrist) · median

Sensory & motor deficit (TO)

Expected finding (TO)	Not expected (TO)
↓ elbow flexion + supination	Not **weak pronation** (pronators – median)
Lateral (+ volar) forearm sensory loss	Not elbow extension weakness (triceps – radial)
Weak/absent biceps reflex	–
Small-amplitude LABCN SNAP, ↓ biceps CMAP	–
Biceps atrophy	–

Pitfall (TO) complete division can be missed: **brachioradialis** still flexes elbow → palpate biceps during testing

How it's asked

"Vigorous conjoint-tendon retraction endangers (TO)?"	→ **musculocutaneous nerve** ✗ Wrong options: axillary · anterior circumflex humeral artery
"Lateral forearm numbness 1 wk after coracoid transfer?"	→ **musculocutaneous**
"After open shoulder reduction, lateral forearm numbness – unlikely finding (TO)?"	→ **weak supination AND pronation** (pronation normal)

– – Active space – –

MYASTHENIA GRAVIS

– Basic Science · 66 / 108 –

1 Basics

Epidemiology & pathophysiology

Bimodal	**young women <40**, **older men >50**
Risk factors	**HLA-B8** · drugs: **penicillamine, aminoglycosides**
Mechanism	✓ **autoantibodies (Ig) at the neuromuscular junction** against **nicotinic ACh receptor** (commoner) or **MuSK** · **type II hypersensitivity** · T cells stimulate B-cell antibodies
Associations	**thymoma**, thymic hyperplasia

Genes & lab markers

First test	✓ **anti-AChR antibodies** · anti-MuSK
EMG	**decremental CMAP** on repetitive stimulation

3 Presentation

Hallmark	**fluctuating weakness worse with use / end of day**, better with rest · true fatigue (↓ contractile force)
Commonest presenting symptoms	**ocular: ptosis, diplopia** (ptosis on sustained upgaze)
Bulbar	dysphagia, dysarthria, fatigable chewing · **proximal** weakness
Bedside tests	**ice-pack test** (cold ↓ acetylcholinesterase → ptosis improves) · **edrophonium (Tensilon)** (short-acting AChE inhibitor; only with ptosis/ophthalmoparesis)

4 Diagnosis & Classification

Key radiology findings

CT chest	exclude **thymoma**

Differential

Condition	Distinguishing feature
Myasthenia gravis	**Worse with use** · antibodies to AChR (postsynaptic)
Lambert–Eaton (LEMS)	**Improves with use** · autonomic features
Botulism	**Poor pupillary light response** · ↓ ACh release
Non-depolarising NMB	↓ ACh binding
Muscular dystrophy	**Absent dystrophin**, excess sarcolemmal Ca^{2+}, young onset
CMT	**Demyelination**, distal long-axon degeneration
Thyroid ophthalmopathy	–

5 Management

First-line symptomatic	**AChE inhibitor: pyridostigmine** (side effects: cramps, diarrhoea, bradycardia, sweating, secretions)
Chronic immunotherapy	**oral prednisone**
Myasthenic crisis (respiratory weakness)	**IVIG or plasmapheresis**
Thymectomy	any **thymoma** (with or without MG)

6 Complications

Myasthenic crisis	respiratory failure · most ocular MG progresses to generalised

How it's asked

"45-yr-old woman, best in morning, fatigable arms, ptosis on upgaze – pathology?"	→ **Ig antibodies at NM junction** ✗ Wrong options: ↓ ACh release (botulinum) · demyelination (CMT) · absent dystrophin
"Weakness that improves with use?"	→ **Lambert–Eaton**

– – Active space – –

NONUNION AND BONE DEFECTS

– Basic Science · 67 / 108 –

1 Basics

Definitions

Nonunion	arrest of fracture repair, no further capacity to heal without intervention
FDA definition (TO)	✓ **≥9 months since injury + no visible progressive healing for 3 months**
Delayed union	not healed in expected time · **failure to unite by 6 months** (Orthobullets) · **3–6 months** (TO) · varies with site and soft-tissue injury
Large segmental defect	functional nonunion

Causes

Commonest	**inadequate stabilisation + poor blood supply** · infection · **smoking (biggest modifiable risk)** · location (scaphoid, distal tibia, 5th MT base) · segmental/butterfly (intercalary blood supply)
Smoking	nicotine vasoconstriction + **CO binding Hb** · prior smokers also at risk · > NSAIDs, BMI >40, delay >24 h · **reamed nailing ↓ nonunion** ✗ Wrong options: risk
Spine fusion in smoker	✓ **maintain smoking cessation** best ↓ pseudarthrosis

4 Diagnosis & Classification

Classification & biology

Type	Biology	Mechanics	Appearance	Treatment
Hypertrophic	**Adequate** vascularity	**Inadequate stability** (too much motion)	**Abundant callus**, no bridging ("elephant foot")	✓ **↑ mechanical stability only** (e.g. exchange nail, compression plate) – rarely needs graft
Oligotrophic	Usually adequate	**Malreduction/ distraction** (displacement) · or **overly stiff construct**	Little/no callus	Improved fixation ± biological stimulation
Atrophic	**Inadequate blood supply**	Inadequate immobilisation (Orthobullets) · excessively stiff fixation (OB)	**No callus, sclerotic/ rounded ends**	✓ **Revision fixation + bone graft** (autologous ICBG gold standard · BMP · DBM, cancellous chips) + healthy soft tissue
Septic	Infected	–	Variable callus	Staged: debride, beads/VAC, flaps, transport · eradicate infection
Synovial pseudarthrosis	Sealed canals + fluid-filled pseudomembrane	–	False joint · **bone scan "cold cleft"**	Resect non-viable ends, stable fixation

Classification axes (TO) epiphyseal / metaphyseal / diaphyseal · septic / aseptic · hypertrophic / oligotrophic / atrophic · pseudarthrosis



Hypertrophic nonunion – abundant callus, no bridging

Biopsy of aseptic nonunion after ex-fix ✓ **bland cartilage callus** (failed endochondral healing)
✗ Wrong options: PMNs (infection) · Haversian/lamellar bone (healed)

Locked plate distal femur – short working length, high screw density ✓ **high stiffness → low strain → oligotrophic nonunion → fatigue failure** · revised with long working length, healed **without graft**

– – Active space – –

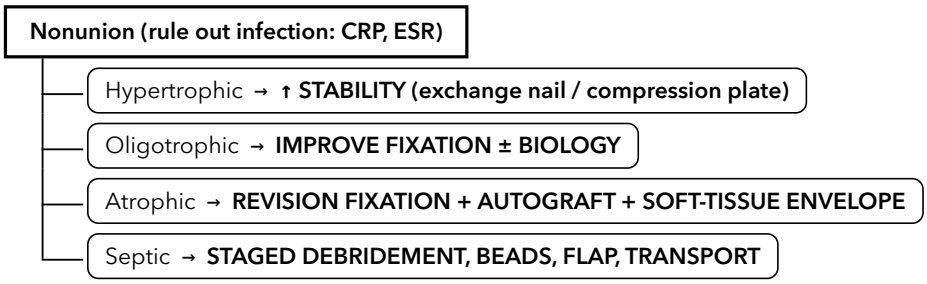
Work-up

History	mechanism, treatments, metabolic/nutritional/immune status, **NSAIDs, nicotine** · pain on axial loading
Exam	neurovascular, soft tissue, nonunion mobility, deformity
Labs	✓ **CBC, ESR, CRP** (infection) · total protein, albumin · **vitamin D (deficient in 60–70%)**, TSH, PTH
Most accurate test for septic nonunion	✓ **CRP** (rises in 6 h, peaks 2–3 days, normalises 5–21 days with treatment) ✗ Wrong options: IL-1, IL-6, WBC, ESR
First step when nonunion + risk of infection (open fracture, erythema)	✓ **labs: vitamin D, TSH, inflammatory markers** before revision

Key radiology findings

X-ray	cornerstone · **4 views** · full-length standing films if LLD
CT	if union uncertain (hardware artefact)

5 Management



Atrophic ulna nonunion 6 mo after BBFA ORIF, normal CRP/ESR	✓ **revision ORIF with autograft** ✗ Wrong options: observe, cast, IM nail, revision without graft
Atrophic radius nonunion with gap	✓ **supportive soft-tissue envelope** essential · plating + cancellous graft treats defects **up to 6 cm** · **3.5 mm plates** adequate · poor envelope → vascularised graft + soft tissue
Atrophic tibial nonunion 9 mo, normal labs	cause ✓ **lack of biology** → exchange nail + graft
Non-operative	brace · **bone stimulators** ✗ Wrong options: synovial pseudarthrosis · mobile nonunion · gap >1 cm

Electrical stimulation

Induction coupling **↑ BMP-7, BMP-2, TGF-β1, ↑ osteoblast proliferation · not ↓ osteoclast differentiation**

Mode	Effect
Direct current	↓ O_2, ↑ pH → ↓ osteoclasts, ↑ osteoblasts
Capacitive coupling (AC)	cAMP, collagen, cartilage calcification
PEMF	Fibrocartilage calcification
Combined magnetic	–

How it's asked

"Ulna fracture 12 mo, pain, abundant callus on X-ray (TO)?"	**→ hypertrophic nonunion**
"Nonunion after ex-fix, cultures negative – histology?"	**→ bland cartilage callus**
"Least likely to need added biology?"	**→ abundant callus without bridging (hypertrophic)**
"Tibial nail 9 mo, no bone formation, normal labs – cause?"	**→ lack of biology (atrophic)**
"Humerus ORIF, little callus, broken plate, normal labs?"	**→ atrophic**
"Distal femur locked plate nonunion healed with revision, no graft – initial problem?"	**→ high stiffness**
"Smoker having spine fusion – ↓ pseudarthrosis by?"	**→ smoking cessation**
"Obese smoker, NSAIDs, delayed nailing – highest nonunion risk?"	**→ smoking**
"Open subtroch femur, 9-mo pain, erythema, nonunion – next?"	**→ vitamin D, TSH, inflammatory labs**
"Septic nonunion – most accurate lab?"	**→ CRP**
"Atrophic ulna nonunion, normal CRP/ESR?"	**→ revision ORIF with autograft**
"Atrophic radius nonunion with gap – required besides revision + graft?"	**→ supportive soft-tissue envelope**
"Induction coupling effects EXCEPT?"	**→ ↓ osteoclast differentiation**

– – Active space – –

NORMAL BONE METABOLISM

– Basic Science · 68 / 108 –

1 Basics

Regulators

Hormones	PTH · calcitonin · sex hormones (oestrogen, androgens) · growth hormone · thyroid hormone
Steroids	vitamin D · glucocorticoids

Peak bone mass & oestrogen

Peak bone mass	hip **16–25 yr** · greater in **men, African-Americans** · ✓ **timing varies by anatomic site** (appendicular before axial · women earlier at hip, later at spine; spine PBM may be 3rd–4th decade) · strong genetic + environmental influence · **correlates poorly** with postmenopausal BMD
Bone loss	**0.3–0.5%/yr** after maturity · **2–3%/yr** in untreated women years 6–10 after menopause
Most important sex steroid for peak bone mass (men and women)	✓ **oestrogen**
Osteoporosis risk factors	age · female · early menopause · fair skin · family history of hip fracture · low weight · smoking · glucocorticoids · alcohol · low protein · anticonvulsants, antidepressants

Calcium & phosphate

	Calcium	Phosphate
In bone	**99%** (hydroxyapatite)	**86%**
Intracellular	1%	14%
ECF	0.1%	0.08%
Serum forms	**45% protein-bound · 45% ionised** · 10% anion-bound	Mostly inorganic $H_2PO_4^-$, unbound
Regulated by	Gut absorption, bone resorption, renal reabsorption	Renal reabsorption/excretion · ↑ PO_4 → ↑ PTH
Daily need	See below	**1000–1500 mg**

Group	Calcium (mg/day)
Lactating	**2000**
Pregnant, postmenopausal, healing fracture	**1500**
Adolescents / young adults	1300
Adults	750
Children	600

Hormones

Hormone	Source	Net effect	Key mechanisms
PTH (84 aa)	**Chief cells** of 4 parathyroids	**↑ Ca, ↓ PO_4**	Bone: osteoblasts → IL-1, IL-6, **RANKL, M-CSF** → osteoclasts (**osteoclasts lack PTH receptor**) · kidney: ↑ **1α-hydroxylase** (25-OH → 1,25-$(OH)_2D$), ↑ Ca reabsorption, ↑ PO_4 excretion · gut: indirect via 1,25-D · **intermittent = anabolic**
PTHrP	Tumours	↑ Ca	Same receptor (osteoblast, kidney) · metaphyseal dysplasia link
Calcitonin (32 aa)	Thyroid **parafollicular C cells**	Limited role · ↓ osteoclast number + activity	Osteoclasts **have** calcitonin receptor · ↑ in **medullary thyroid Ca / MEN II** · used in Paget's, osteoporosis, malignant hypercalcaemia
Vitamin D (secosteroid)	Skin UVB + diet	↑ Ca (gut absorption, osteoclast activation) · ✓ **mineralisation of osteoid**	Liver → **25-OH-D** · kidney → **1,25-$(OH)_2D$** (↑ by ↑ PTH, ↓ Ca, ↓ PO_4) · or inactive **24,25-$(OH)_2D$** (↓ PTH, ↑ Ca, ↑ PO_4) · terminal osteoclast differentiation
Oestrogen	Ovaries	**Prevents bone loss** (↓ resorption + remodelling frequency)	↑ femoral-neck BMD, ↓ hip fracture
Growth hormone	Pituitary	↑ Ca (gut absorption, ↓ urinary loss)	Excess → **gigantism (proliferative zone)**
Thyroid hormone	Thyroid	Physeal growth (chondrocytes, **type X collagen**, ALP) · ↑ resorption	Excess thyroxine → **osteoporosis**
Glucocorticoids	–	↑ bone loss	↓ gut Ca absorption · ↓ formation (**cancellous > cortical**) – ↓ collagen, ↓ osteoblasts

Phenytoin	impairs vitamin D metabolism
Vitamin D deficiency	osteomalacia, rickets

Therapeutic oestrogen

Benefit	↓ bone loss if started within **5–10 yr of menopause** · BMD gain **2–4%/yr** for first 2 yr
Risks/effects	↑ **heart disease, breast cancer** · ↓ **hip fracture**, ↓ **endometrial cancer** (with cyclic progestin) · ↓ **urinary pyridinoline**, ↓ **serum ALP**

Anabolic vs antiresorptive

Anabolic	✓ **intermittent PTH / teriparatide** (receptor on osteoblasts, preosteoblasts) · sustained PTH → resorption
Antiresorptive	bisphosphonates · oestrogen · SERMs · **denosumab** (RANKL inhibitor) · **calcitonin**

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4 Diagnosis & Classification

Bone turnover markers (TO)

Formation (TO)	Resorption (TO)
Bone-specific ALP	**Urine + serum N-telopeptides**
Osteocalcin	**Urine + serum C-telopeptides**
P1CP (carboxy-terminal procollagen-1 propeptide)	Urine **hydroxyproline**
Type I collagen extension peptide (P1NP)	✓ **Urine deoxypyridinoline** · pyridinoline
–	Urine hydroxylysine glycosides
–	**Serum TRAP**, bone sialoprotein

How it's asked

"Peak bone mass most dependent on which sex steroid?"	→ **oestrogen**
"True of peak bone mass?"	→ **timing varies by anatomic site** ✗ Wrong options: always end of 2nd decade · independent of environment · correlates with postmenopausal BMD
"Hormone over-secretion affecting proliferative zone (picture)?"	→ **gigantism**
"True of anti-osteoporosis drugs?"	→ **intermittent PTH is anabolic** ✗ Wrong options: calcitonin anabolic · denosumab ↑ osteoclasts · SERMs anabolic
"NOT a bone formation marker (TO)?"	→ **urine deoxypyridinoline**

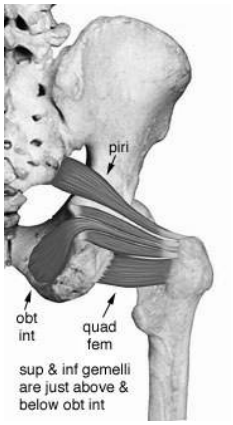
OBTURATOR INTERNUS

– Basic Science · 69 / 108 –

1 Basics

Anatomy

	Obturator internus
Origin	Internal surface of obturator membrane + posterior bony margin of obturator foramen
Insertion	**Medial greater trochanter** (with superior + inferior gemelli)
Action	External rotation · abducts flexed thigh
Innervation	**Nerve to obturator internus (L5, S1)**
Arterial supply	Internal pudendal, superior + inferior gluteal arteries



Obturator internus

Cross-section at the greater trochanters

Obturator internus projects **posteriorly around the ischium** (lesser sciatic notch) → posterior proximal femur **just below piriformis**

Structure	Position
Obturator externus	**Anterior** to ischium
Pectineus	Posterior, deep to femoral vessels
Adductor longus / brevis	Just appearing anteriorly / behind longus
Adductor magnus	More distal (not at this level)

How it's asked

"Pelvic cross-section at greater trochanters – muscle wrapping posteriorly around ischium?"

→ **obturator internus**
✗ Wrong options: obturator externus (anterior)

– – Active space – –

OCCUPATIONAL HEALTH

– Basic Science · 70 / 108 –

1 Basics

Radiation safety (fluoroscopy technique, lead, pregnancy, cataract dose)

↑ Radiation exposure	↓ Radiation exposure
Standing **close to emitter** · on the **emitter (tube) side**	**Distance** (inverse square law) · stay >2 m (6 ft)
Larger body part / large patient (↑ kVp, mA, scatter)	**Smaller part** · **mini C-arm**
Limb **close to X-ray source**	Limb/patient **close to image intensifier** (detector)
Large C-arm, **continuous fluoro**, cine	Short **beam-on time**, pulsed shots, **surgeon controls pedal**
No collimation	✓ **Collimation** (↓ scatter, better image)
Geometric magnification (moving intensifier away / tube toward patient)	Electronic magnification only when needed
Intraop CT (O-arm) – patient cancer risk ~2× (2.3/10,000)	**Inverted beam** (tube beneath patient, intensifier above)
–	Lead apron (≥0.5 mm), thyroid shield (2.5×), **lead glasses ≥0.15 mm**, **sleeves + axillary supplements**

Highest scatter with a lateral shot ✓ **3 ft from the emitter side** (patient attenuates beam toward collector)

False statement (TO) false: "**image magnified by moving X-ray tube away from patient**" · (geometric magnification = intensifier away → ↑ dose) · large patients need more exposure, mini C-arm less, collimation ↓ scatter **(TO)**

Lead aprons (TO) ↓ exposure 4× (lateral), 16× (PA)

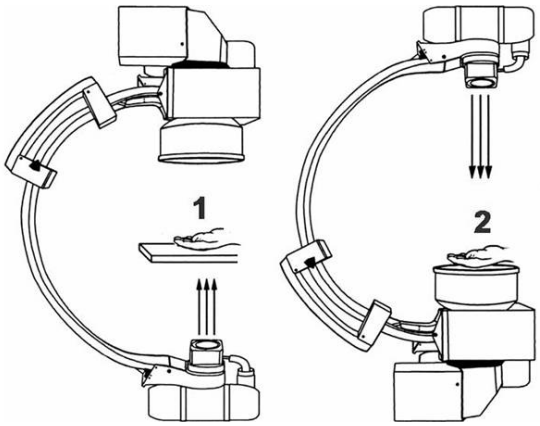
Distance (TO) at 18 inches from beam only **0.1%** of direct dose · scatter <1 mSv outside 2 m · in-beam **1,200–4,000 mrem/min**

Dose limits

Site	Annual limit
Whole body (TO)	**20 mSv**
Thyroid / eyes (TO, ICRP)	**150 mSv**
Hands (TO)	**500 mSv**
Lens – cataract threshold (NCRP)	✓ **50 mGy** cumulative/yr (1 mSv = 1 mGy in diagnostic imaging)

Surgeon cancer risk ↑ **breast cancer** (**upper outer quadrant** – commonest site, poorly covered by vests → ✓ **sleeves + axillary supplements**) · lateral images expose breast more · breast as radiosensitive as marrow, colon, lung · **cataracts** (lead glasses)

– – Active space – –



Inverted C-arm reduces surgeon exposure

Bloodborne pathogen transmission

Exposure	Risk
HIV needlestick	✓ **~0.3%** (↑ with blood volume, viral titre) · PEP with NRTIs protective
HIV mucous membrane	~0.09%
HIV frozen bone allograft	**<1 per million** · donor screening key · none since **2001** (this note)
HIV blood transfusion	**1 / 500,000 units** (window period)
Hepatitis B needlestick	**37–62% seroconvert** · 22–31% clinical hepatitis
Hepatitis C needlestick	**0.5–1.8%**

Resident duty hours (ACGME)

Limits	**≤80 h/week** (4-week average) · +10% if educationally justified · **1 day off in 7** · in-house call **≤ every 3rd night** · **10 h off** between shifts · includes clinical, academic, admin, call
Concern	**continuity of care** (handoffs) · patient safety
Handoff failures	omitted current condition, no anticipatory guidance, no rationale, no tasks assigned · **signing out to physician extenders** is **not** cited

Occupational lung disease (TO)

Miller's asthma (TO)	**mill-workers' / grain workers' asthma** · hypersensitivity to **granary weevil** (*Sitophilus granarius*)
Millar's asthma (TO)	**spasmodic croup** of **infants** (<2 yr, nocturnal glottic spasm) – TO keys "seen in infants" for this spelling – read the options

How it's asked

"NOT true about OT radiation (TO)?"	→ **"magnify by moving the X-ray tube away from patient"**
"Annual limits body / thyroid-eyes / hands (TO)?"	→ **20 / 150 / 500 mSv**
"↑ patient exposure on C-arm?"	→ **imaging a larger body part** ✗ Wrong options: limb near intensifier · collimation · mini C-arm

– – Active space – –

"Least surgeon exposure in hand surgery?"	**→ thumb against image intensifier, mini C-arm**
"Best ↓ exposure getting a lateral for distal locking?"	**→ image intensifier as close to patient as possible**
"↑ exposure?"	**→ continuous fluoroscopy**
"Lead garments — true?"	**→ sleeves + axillary supplements ↓ breast dose**
"Commonest breast cancer site?"	**→ upper outer quadrant**
"Pregnant resident — highest scatter on lateral?"	**→ 3 ft from emitter side**
"Cataract threshold (NCRP)?"	**→ 50 mGy/yr**
"HIV needlestick risk?"	**→ 0.3%**
"Sign-out concern EXCEPT?"	**→ signing out to physician extenders**
"Miller's asthma due to / seen in (TO)?"	**→ granary weevil / grain workers** · "Millar's asthma seen in" **→ infants**

OCHRONOSIS

– Basic Science · 71 / 108 –

1 Basics

Alkaptonuria – homogentisic acid oxidase

Defect	✓ **deficiency of homogentisic acid oxidase** (rare inborn error, tyrosine/phenylalanine pathway) → **homogentisic acid** accumulates in fibrillar collagen
Ochronosis	**bluish-black discoloration of cartilage + connective tissue** · acid polymerises in joints → **early degenerative arthritis**
Tendons	deposition in **Achilles, patellar tendons** → spontaneous rupture

3 Presentation

Black urine (homogentisic acid polymerises on standing) · early large-joint OA · back pain

Ochronotic spondylitis	4th decade · progressive degeneration

4 Diagnosis & Classification

Key radiology findings

Spine	**disc calcification + disc-space narrowing** (multilevel)
Knee	chondrocalcinosis, early OA



Ochronotic spondylitis – calcified, narrowed discs

Genes & lab markers

Diagnosis	**homogentisic acid in urine**

– – Active space – –

5 Management

No specific medical treatment (Orthobullets) · treat as OA: observation, NSAIDs, tramadol, steroids · **early arthroplasty** for severe disease

X vs Y

Finding	Disease
Black cartilage, homogentisic acid	**Alkaptonuria (ochronosis)**
Urate crystals	Gout 155
Lead	Plumbism
Hepcidin failure → iron overload	Haemochromatosis
LDLR mutation	Familial hypercholesterolaemia
Flowing anterior ossification	DISH
Syndesmophytes, SI fusion	Ankylosing spondylitis

How it's asked

"Back pain, calcified discs, chondrocalcinosis, dark urine?"	→ **ochronosis (alkaptonuria)** ✗ Wrong options: DISH, AS, homocystinuria
"Black cartilage at TKA – cause?"	→ **homogentisic oxidase deficiency**

OPERATIVE BLOOD LOSS & TRANSFUSION

– Basic Science · 72 / 108 –

1 Basics

Risk factors for transfusion

Strongest predictor ✓ **preoperative haemoglobin**

Others female · older age · larger patient · longer surgery · **lateral release in TKA**

5 Management

Preoperative optimisation

Measure	Facts
Labs	CBC + **MCV** (microcytic → iron; macrocytic → folate/B12), folate, B12
Iron / vitamin regimen	**Iron 256 mg + vit C 1 g + folate 5 mg daily × 30–45 d** → ↓ transfusion (only if deficient) · iron s/e: constipation, reflux, pain
Erythropoietin	Renal pericapillary cells · > autologous donation efficacy · cost ≈ 4 units blood · high-risk patients · regimens: **600 IU/kg × 4 (days −21, −14, −7, 0)** · 300 IU/kg × 15 · 150 IU/kg × 9
Preop autologous donation	1–2 units **≥3 weeks** before · Hb >11, weight >50 kg (best if Hb >14) · not cost-effective, many units wasted
Preop embolisation	Vascular tumours (**renal cell**) within **24 h** of surgery

– – Active space – –

Intraoperative

Method	Facts
Acute normovolaemic haemodilution	Blood drawn at surgery, replaced with colloid, reinfused · less planning · rarely used (OR time)
Tourniquet	SBP + 100–150 mmHg · **≤2 h** · Esmarch (✗ Wrong options: infection, tumour) · caution PVD: arterial insufficiency, absent pedal pulses, popliteal aneurysm, arterial calcification → vascular consult
TXA	**Lysine analogue** · blocks plasmin(ogen) binding sites on fibrin → ↓ **fibrinolysis**, stabilises clot · ↓ blood loss + transfusion (TKA <3%, THA <10%) · **IV = oral = topical** · no ↑ DVT/PE · PT/PTT unchanged · s/e: **seizures** (glycine/GABA)
Aminocaproic acid	Lysine analogue, plasmin inhibitor · cheaper · no seizure-threshold effect
Collagen (microfibrillar)	Activates **intrinsic** pathway · dry powder, dry field + dry instruments
Fibrin sealant	Fibrin + factor XIII + thrombin + Ca^{2+}
PRP / platelet-poor plasma	Haemostasis, healing / autologous fibrin sealant with thrombin + Ca
Gelfoam	Animal-skin gelatin · mechanical for low-pressure bleeders · ± thrombin · **remove excess** (impairs bone healing)
Gelatin–thrombin matrix (Floseal, Surgiflo)	Porcine/bovine gelatin + human thrombin · **spine epidural bleeding** · end of cascade
Cell saver	Filter, wash, reinfuse **within 6–8 h** · complications: **coagulopathy** (FDPs, TNF-α, ILs – esp. unwashed), wound drainage, contamination, cost

Topical haemostats not cost-effective in TJA · useful in spine (near neural tissue)

TXA dosing & contraindications (TO)

TXA in TKR (TO) ✓ **oral and IV equal efficacy**
✗ Wrong options: "topical for 2 min better than IV" · ↑ DVT · "IV best"

Doses (TO) IV **10–15 mg/kg or 1 g** (20 min before tourniquet inflation ± 15 min before deflation) · topical **1.5–3 g in 100 mL** for **5 min** before tourniquet release · oral **2 g 2 h preop** + 2 doses postop (cheapest)

Absolute contraindications (FDA) ✓ **current subarachnoid haemorrhage, active intravascular clotting, severe hypersensitivity** only · embolic stroke, DVT, AF, factor V Leiden are **not** absolute

Relative (OB/TO) pregnancy, OCP, seizures, **coronary stent <6 mo**, prior VTE/CVA/MI (TO list) · renal dosing

Topical TXA alternative route if MI, CAD, CVA history

Postoperative transfusion

1 unit PRBC ↑ Hb **~1 g/dL**

Threshold consider at **Hb <8 g/dL** · no routine transfusion >8 (no ↑ mortality, even cardiac patients) · transfuse if **symptomatic** (orthostasis, fluid-resistant tachycardia)

Allogeneic blood **immunomodulation** → ↓ cell-mediated immunity, ↑ infection

Platelet count	Action
<5,000	Transfuse (spontaneous bleeding)
<50,000	Transfuse **before surgery**
<100,000	Consider before surgery
>100,000	Not needed

Platelets ✓ **blood product most likely contaminated** (stored at **room temperature**) · 10/million units · **gram-positive** commonest

Mild postop anaemia oral iron

Jehovah's Witness

Allogeneic blood forbidden (some individuals consent)

Acceptable ✓ **continuous closed-loop cell salvage** · colloids, coagulation factors (personal choice) · **bovine-derived haemostats** generally OK · human-plasma products individual decision
✗ Wrong options: **batch-processed** (breaks continuity)

Profound intraop haemorrhage ✓ **continuous closed-loop cell saver**
✗ Wrong options: TXA infusion · EPO (too slow)

How it's asked

"True about TXA in TKR (TO)?"	→ **oral and IV equal efficacy**
"TXA true EXCEPT?"	→ **"increases fibrinolysis"**
"Absolute contraindication to TXA in THA?"	→ **none of these** (embolic stroke, DVT, AF, FVL)
"Jehovah's Witness, profound intraop bleeding?"	→ **continuous closed-loop cell salvage**
"Blood product most likely contaminated?"	→ **platelets**
"Strongest predictor of postop transfusion?"	→ **preoperative haemoglobin**

– – Active space – –

OPIATES & ANALGESIC MEDICATIONS

– Basic Science · 73 / 108 –

1 Basics

Pain physiology

Definitions acute (tissue damage) · chronic (3–6 months, no damage needed) · pathological (neuropathic)

Pathway nociceptors (**substance P** – depletion ↑ pain threshold; **capsaicin** depletes it) → **A and C fibres** → dorsal column + spinothalamic tract → thalamus · brainstem = endogenous opioid modulation

Analgesic mechanisms (bupivacaine, gabapentin, acetaminophen)

Drug	Mechanism
Acetaminophen	Central · ✓ **inhibits PGE2 production via IL-1β** · minimal anti-inflammatory · = aspirin for pain · **hepatotoxic** (avoid in liver disease)
NSAIDs	↓ pain transduction (COX)
Local anaesthetics (lidocaine, bupivacaine)	✓ **Block intracellular portion of voltage-gated Na^+ channel** → no depolarisation
Gabapentin / pregabalin	✓ **Bind α2δ subunit of presynaptic voltage-gated Ca^{2+} channels** → ↓ excitatory neurotransmitter release, ↑ GABA · effective: postherpetic neuralgia, diabetic neuropathy, fibromyalgia · gabapentin = pregabalin, cheaper · **taper** to stop
Opioids (fentanyl)	**μ** (μ1 supraspinal analgesia, μ2 respiratory depression/dependence) + **κ** (sedation, spinal analgesia)
Tramadol	μ-opioid agonist
Midazolam	↑ GABA-A Cl^- influx → hyperpolarisation
Rocuronium	Competitive ACh-receptor block
Phentolamine	α-adrenergic blocker
Cyclobenzaprine	Central · ↑ noradrenaline, serotonin receptors · helps first 2 weeks · not for neuromuscular spasticity · interacts with MAOIs, alcohol

Local anaesthetic	Duration (± adrenaline)	Max dose (± adrenaline)
Lidocaine (amide)	2 h (4 h)	**4.5 mg/kg (7 mg/kg)**
Bupivacaine (amide)	4 h (8 h)	**2.5 mg/kg (3 mg/kg)**

Amides lidocaine, bupivacaine (liver metabolism, renal excretion) · **esters** – procaine, cocaine, benzocaine

Opioid metabolism

CYP2D6 activates oxycodone, hydrocodone, tramadol, **codeine** · **SSRIs/SNRIs block CYP2D6 → analgesic failure**

Morphine milligram equivalents	**hydrocodone 1 · oxycodone 1.5 · hydromorphone 4** (3–7× morphine) → ✓ **hydromorphone > 2× oxycodone**
Methadone	synthetic, 3× morphine bioavailability, cheap · CYP450 · distribution 2–3 h, **elimination 15–60 h** → accumulates · **respiratory depression, torsades** → pain specialist

5 Management

Special populations

Morbidly obese	✓ **IV PCA dosed on ideal body weight** ✗ Wrong options: actual weight · IM/SC (poor fat vascularity)
Red hair (MC1R mutation)	✓ **need more anaesthetic (MAC) + higher postoperative opioid doses** (esp. women)
Elderly	↑ opioid sensitivity (↑ fat %, ↓ hepatic/renal function)
Children	AAP dosing guidelines

Opioid prescribing & consumption

Bony > soft-tissue procedures	ORIF distal radius uses **more** than cubital tunnel release
Upper limb consumption	75% need ≤15 pills · 45% <5 · reasonable: **15 pills + 1 refill** (3–4% use it) · ✓ **>50% of prescribed pills unused**
Carpal tunnel release	✓ **ibuprofen 600 mg ≈ oxycodone 5 mg** pain scores · most use <10 tablets for ≤2 days · opioids → more adverse events · open = endoscopic · standardised protocol ↓ prescribing

Hand prescribing guide	Tablets
Trigger finger, lumps/bumps	0
Mucous cyst, CTR, de Quervain, Dupuytren	10
Wrist ganglion, hand ORIF, LRTI, tendon transfer	20
Large trauma, wrist fusion, DRUJ reconstruction	40

Chronic opioids	pain specialist, written contract, in-person refills · for OA → ↑ surgical complications
Weaning opioids 50% before THA	✓ **better outcome than not weaning, similar to opioid-naïve**

6 Complications

Pain-pump chondrolysis

Continuous intra-articular local anaesthetic infusion	**FDA warning** · **glenohumeral chondrolysis** (onset ≥2 months, mean diagnosis 8.5 months) · pain, stiffness, ↓ ROM after shoulder scope → ✓ **shoulder radiographs to assess chondrolysis**

– – Active space – –

How it's asked

"Morbidly obese postop analgesia?"	→ **IV PCA by ideal body weight**
"Distal radius ORIF vs cubital tunnel release — opioid use?"	→ **ORIF patient uses more**
"Opioid user weans 50% before THA?"	→ **better than not weaning, similar to opioid-naïve**
"True about CTR pain management?"	→ **over half of prescribed pills unused**
"CTR analgesia — most accurate?"	→ **ibuprofen 600 mg ≈ oxycodone 5 mg**
"MME comparison?"	→ **hydromorphone > 2× oxycodone**
"Binds marked site on voltage-gated Na channel?"	→ **bupivacaine**
"Binds presynaptic Ca channels?"	→ **gabapentin**
"Inhibits PGE2 via IL-1β?"	→ **acetaminophen**
"Red-haired patient — postoperative implication?"	→ **higher analgesic doses**
"Pain + stiffness 3 mo after Bankart with intra-articular pump?"	→ **radiographs for chondrolysis**

ORTHOPAEDIC IMPLANTS

– Basic Science · 74 / 108 –

1 Basics

Determinants

Implant behaviour **material properties** (modulus, strength) + **structural properties** (shape, size, moment of inertia)

Screws

Term	Definition
Pitch	Distance between threads
Lead	Distance advanced per revolution
Working length (distance) of a screw	✓ **Length of bone traversed by the screw**
Root / core diameter	Narrowest diameter
Bending strength	∝ **inner (minor) diameter³**
Pull-out strength	∝ **outer (major) diameter²** · ↑ with large major–minor difference, **fine pitch**

Pull-out in osteoporotic bone ↑ by cortical purchase · placement **parallel to trabeculae** · **locking (fixed-angle)** construct · **PMMA augmentation**
✗ Wrong options: **tapping** (↓ pull-out)

Pedicle screw pull-out most affected by **bone quality** · pedicle screws need the **largest force** to disrupt bone–implant interface (vs sublaminar cables, laminar/pedicle hooks)

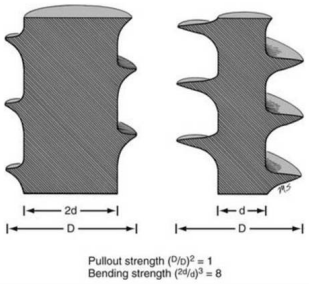


FIGURE 291-4 The theoretical bending strength of a screw is directly proportional to the third power of the minor screw diameter (d). The pullout strength of a screw is proportional to the square of the major diameter (D) and is affected by the density of the bone beneath the screw threads. Theoretically, a twofold increase in the minor screw diameter with no change in the major diameter increases the bending strength eightfold without affecting the pullout strength.

Pull-out vs bending strength vs minor diameter

Plate biomechanics (compression, bridging, working length, stress shielding)

Load-bearing device best on **tension side** · bending rigidity ∝ **thickness³** · titanium modulus closest to cortical bone

– – Active space – –

Plate working distance ✓ **distance between the two screws closest to the fracture on either side** · ↓ working distance → ↑ stiffness

Plate function	Mechanism
Compression (DCP)	✓ **Cortical screw placed eccentrically (away from fracture) in oval hole** → head slides to centre → compression · transverse fractures
Concave pre-bend	Compresses **near and far cortices** (transverse fracture)
Neutralisation / protection	With lag screws — resists shear, rotation (absolute stability)
Buttress / antiglide	Resists shear
Tension band	Converts tension to compression
Bridging	Spans comminution, screws far from fracture · **relative stability**, length + alignment · preserves blood supply · **flexible but axially stable** (long working length) · indirect reduction (distraction) · locking not required

Stability	Healing	Requirement
Absolute	Primary (Haversian/ intramembranous)	**Lag screw ± neutralisation / compression plate** · **low strain (<2%), high stiffness** · no micromotion
Relative	Secondary (endochondral)	Strain **<10%** (else fibrous union) · nail, bridge plate, ex-fix, cast

Strain residual 2 mm gap after lag screws (simple oblique) → **greater interfragmentary strain** than bridged comminuted fracture (strain shared)

Open vs submuscular plating open plating ✓ **compromises periosteal and medullary blood flow more**

LC-DCP vs DCP ✓ **less contact-induced osteopenia** (less plate–bone contact)

Stress shielding ✓ **↓ physiological stress in bone because a stiffer structure shares load** → resorption (Wolff)

Locked bridge plate biomechanically ≈ **external fixator without compression** ("internal fixator")

Locking plates (indications, hybrid, unicortical, stiffness)

Indications indirect reduction · **osteoporotic** diaphyseal/metaphyseal fractures · bridging severe comminution · **short periarticular segments** · can't plate tension side · periprosthetic · proximal humerus, distal radius, distal femur, proximal tibia

Not indicated **compression plating of transverse fractures** · simple transverse **both-bone forearm** · oligotrophic diaphyseal nonunion · lateral malleolus (least evidence)

Advantages less angulation in comminuted metaphyseal fractures · **axial load borne by the plate** (no plate–bone friction needed) · preserves periosteum · ↓ screw toggle

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↑ Locked construct stiffness	↓ Stiffness
Bicortical locking screws (vs unicortical – biggest difference in **torsion**)	Unicortical screws
More screws	**Far cortical locking screws** (flexible shaft, unthreaded near cortex)
Longer plate	Overdrilling near cortex
Screw divergence from hole **<5°**	Angular cross-threading
Screws **closer to fracture** (short working length)	Screws far from fracture
Stainless steel > titanium · supplemental medial plate	–

Unicortical locked vs bicortical non-locked ✓ **less torsional, greater axial** strength

Locked vs non-locked rigid construct ✓ ↓ **interfragmentary strain** (stiffer)

Hybrid plating **non-locking screws first** (lag plate to bone = reduction aid) → then **locking screws** (fixed angle, resist **varus collapse**; osteoporotic diaphysis)

Closed MIPO locked articular fracture risk ✓ **non-anatomic articular reduction** (articular surface must be reduced under vision) · percutaneous → less stripping, ↑ malunion

2 Mechanism

Intramedullary nails (rigidity, curvature, dynamisation)

Load-sharing device

Property	Proportional to
Torsional rigidity	**radius⁴** (shear modulus, polar moment) · ↑ by reaming (larger nail) · ↓ by **slotting**
Bending rigidity – solid nail	**radius⁴**
Bending rigidity – hollow nail	~**radius³**

6 mm vs 12 mm solid nail ✓ **1/16 torsional rigidity** ($½^4$)

Radius of curvature ✓ **nail ROC > femur ROC** (straighter) → **anterior distal cortical perforation** risk · closed-section stiffer than slotted · start point affects varus/valgus · healing mostly endochondral

Locking dynamic (axially + rotationally stable) · static (unstable) · **secondary dynamisation**: remove proximal screw or move it into the **dynamic (long) slot** – hole A

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External fixator stiffness

↑ Stability (unilateral)	Notes
Fracture-end contact	Most important overall
✓ **Larger pin diameter**	**Greatest single factor** – stiffness ∝ r^4 (2× diameter → **16×**)
More pins	–
✓ **↓ Bone–rod distance**	Quick, safe fix (✗ Wrong options: oversizing small-bone pins (fracture risk))
Pins in different planes · multiplanar	–
Larger / stacked rods · rods in different planes	–
↑ Spacing between pins in each fragment · near pin close to fracture	–
Bicortical pins · HA-coated	–

Circular (Ilizarov) – ↑ stability	
Larger wires · olive wires · extra wires	–
Smaller ring diameter · more rings · 2 central rings near fracture	–
Wires crossing at **90°** · **↑ tension**	✓ **Tensioned wires → more axial compression, less transverse shear** than half pins

5 Management

Total hip implants

Stem rigidity length + radius

Femoral component neutral or slight **valgus** (↓ moment arm, ↓ cement stress)

↑ Femoral offset ✓ ↑ abductor moment arm, ↓ joint reaction force, ↓ abductor force needed
✗ Wrong options: ↑ implant strain + medial cement-mantle strain

Additive manufacturing (TO)

Technology (TO)	Material	Use
FDM	PLA, ABS, PEEK filament	Anatomical models, splints · **PLA Tg ~60 °C → warps in autoclave**
SLA	Photopolymer resin (UV)	Highest resolution · models, autoclavable PSI (HDT >135 °C resins)
SLS	Polyamide (nylon), metal powders	✓ **Autoclavable PSI cutting guides** (HDT >135 °C)
SLM / DMLS	**Ti-6Al-4V**, Co-Cr (full melt)	Custom implants, cages, tumour prostheses · ✓ **gradient porosity → ↓ effective modulus → ↓ stress shielding**
EBM	Titanium · **vacuum, >700 °C**	✓ **Porous acetabular cups / cones (pores 400–600 µm)** · in-situ annealing ↓ residual stress
Bioprinting	Cells in bioink	Experimental

AM limits (TO) doesn't eliminate fatigue (micro-voids → need HIP/polishing) · introduces build-axis anisotropy · no intrinsic anti-biofilm property

Solid titanium modulus (TO) ~110 GPa vs cortical 10–20, cancellous <3 GPa

How it's asked

"Open vs submuscular plating – blood flow?"	**→ open compromises periosteal + medullary more**
"LC-DCP advantage?"	**→ less contact-induced osteopenia**
"Distance of bone traversed by a screw?"	**→ working length**
"How DCP compresses?"	**→ eccentric cortical screw in oval hole**
"Lag screws with 2 mm gap vs bridge plate?"	**→ greater interfragmentary strain**
"Ulna plating technique in figure?"	**→ bridging**
"Stress shielding?"	**→ ↓ bone stress due to stiffer structure sharing load**
"Absolute stability construct?"	**→ lag screws + neutralisation plate (clavicle)**
"Low strain + high stiffness humerus?"	**→ lag screw + locked neutralisation plate**
"Compression at near + far cortices?"	**→ concave-bent compression plate**
"Locked bridge plate ≈?"	**→ external fixator without compression**
"Plate working distance?"	**→ between screws closest to the fracture**
"Bridge plating true?"	**→ long working length → flexible, axially stable**

– – Active space – –

"MC plate in figure acting as?"	→ **bridge plate**
"MIPO healing due to?"	→ **flexible and stable construct**
"Bridging plate image?"	→ **C**
"Fibula: compression + shear/rotation protection?"	→ **lag screw + neutralisation plate (E)**
"↑ locking construct stiffness?"	→ **more screws** ✗ Wrong options: far cortical, unicortical, overdrilling
"Locking plate indications EXCEPT?"	→ **compression plating of transverse fracture**
"Hybrid plate – screw A vs B?"	→ **A locked (fixed angle), B non-locked (reduces plate to bone)**
"No relative indication for locking?"	→ **oligotrophic diaphyseal nonunion**
"Closed locked-only distal femur articular fracture – risk?"	→ **non-anatomic articular reduction**
"Least evidence for locking?"	→ **lateral malleolus (E)**
"Biggest unicortical vs bicortical locking difference?"	→ **torsion**
"Unicortical locked vs bicortical unlocked?"	→ **less torsional, greater axial strength**
"Locked vs non-locked?"	→ **↓ interfragmentary strain**
"Least indicated for locking?"	→ **transverse both-bone forearm**
"Hybrid plating advantage?"	→ **non-locked first as reduction aid, locked resist varus**
"Axial forces in locked construct borne by?"	→ **plate**
"Locking plates true?"	→ **stiffness ↑ with plate length**
"↓ stiffness of distal femur locked plate?"	→ **far cortical locking screws**
"Most stiffness added (periprosthetic)?"	→ **bicortical locking screw at B (near fracture)**
"↑ ex-fix strength?"	→ **↓ bone–rod distance**
"Greatest impact on unilateral frame stiffness?"	→ **pin diameter**
"Pin twice the diameter – bending stiffness?"	→ **16×**
"Strongest influence on ex-fix stiffness?"	→ **pin diameter**
"Tensioned fine wires vs half pins?"	→ **↑ axial compression, ↓ transverse shear**
"Distal radius ex-fix losing reduction – next?"	→ **move rods closer to bone** ✗ Wrong options: 6 mm MC pins
"6 mm vs 12 mm solid nail torsional rigidity?"	→ **1/16**
"Nail planning – true?"	→ **nail ROC > femur ROC**
"Dynamise tibial nail – hole?"	→ **A (dynamic slot)**
"Pull-out in osteoporotic bone ↑ by all EXCEPT?"	→ **tapping**

"Spinal fixation needing largest force to disrupt?"	→ **pedicle screws**
"Modulus closest to cortical bone?"	→ **titanium**
"Porous Ti cup manufacture minimising thermal stress (TO)?"	→ **EBM Ti-6Al-4V, 400–600 µm lattice**
"PSI guide surviving autoclave (TO)?"	→ **SLS/SLA nylon or resin with HDT >135 °C**
"DMLS/SLM advantage over CNC (TO)?"	→ **gradient porosity ↓ modulus → ↓ stress shielding**

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OSTEOPENIA & OSTEOPOROSIS

– Basic Science · 75 / 108 –

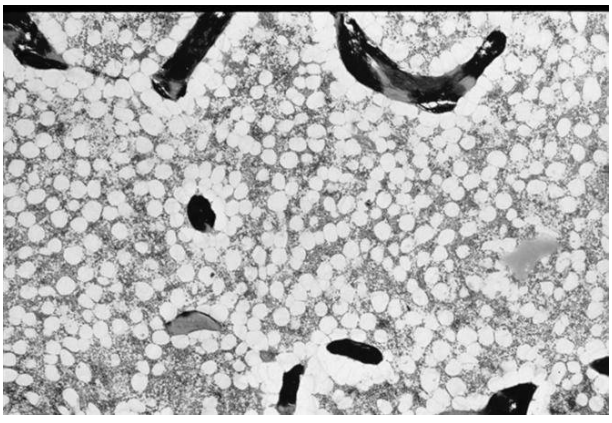
1 Basics

Epidemiology

Prevalence	**10 million** Americans · **200 million** worldwide · 34 million US osteopenia
Fragility fractures / yr	**1.5 million** → 700,000 vertebral · 300,000 hip · 200,000 wrist
Commonest fragility fracture	✓ **vertebral body > peritrochanteric (hip) > distal radius** ✗ Wrong options: "distal radius more common than VCF"
Peak age	wrist 50–60 · vertebral 60–70 · hip 70–80
M:F	**1:4** · men → **secondary osteoporosis in 60%** (hypogonadism, glucocorticoid excess, alcoholism)
Bone mass	peak bone mass attained **before 30 yr** · failure to attain it = major cause of osteoporosis
Ageing	✓ **trabecular (cancellous) bone loses strength more than cortical** · cancellous ↓ **8.5%/decade** vs cortical **5%/decade** · volume fraction (apparent density) explains most of the loss

Pathophysiology

Osteoporosis	✓ **quantitative** defect · **↓ bone quantity, normal bone quality (mineralisation)** ✗ Wrong options: qualitative (= osteomalacia)
Causes	failure to build peak mass · bone loss later in life · ↓ loading (non-weight-bearing)
Histology	**thinned trabeculae** · **↓ osteon size** · enlarged Haversian + marrow spaces



Osteoporotic bone – thinned trabeculae

Osteoclast ruffled border	Agents
↑ (activated)	**PTH · $1,25(OH)_2$ vit D_3 · prostaglandin E**
Flattened	**Bisphosphonates · calcitonin**

Disuse osteopenia — Wolff's law

Wolff's law ✓ bone densifies under repetitive load, ↓ density with ↓ load → **disuse osteopenia** after non-weight-bearing (e.g. bilateral calcaneus fractures) · dense calcar = load path

✗ Wrong option	What it actually describes
Hydraulic stiffening	Tissue fluid stiffens skeleton → ↑ strength
Hooke's law	Stress–strain of a solid
Hueter-Volkmann law	Compression inhibits physeal growth
Delpech's law	Growth across physis

Genes & lab markers

Gene polymorphisms **calcitonin receptor** · **estrogen receptor-1** · **vitamin D receptor** · **COL1A1** (+ >45 others) · not **COMP** (= multiple epiphyseal dysplasia) · no single cause

Labs generally **normal** in osteoporosis · **↓ 25-OH vitamin D** = most likely abnormal lab after low-energy fracture · other labs exclude secondary causes
✗ Wrong options: hydroxyproline, PTHrP, Sox-9, LRP5

Vitamin D D_3 → 25-OH D (liver) → **1,25-$(OH)_2$ D** (active) or 24,25-$(OH)_2$ D (inactive) in kidney · deficient **<20 ng/mL** · not diagnostic of osteoporosis

Vitamin D replacement (TO) ✓ **↑ 25-OH D and ↑ 1,25-$(OH)_2$ D, ↓ sclerostin (TO)** · osteocytes have 1,25-D receptors → ↓ sclerostin synthesis → ↑ bone formation **(TO)**

Bone remodelling (TO) peak mass at **30 yr** · resorption ~**10 days**, formation ~**3 months** · up to **20%** of skeleton remodelled/yr **(TO)**

Bone turnover markers (TO)	Examples
Formation (osteoblast)	**P1NP / P1CP** (type 1 collagen propeptides) · **ALP** (total, bone-specific) · **osteocalcin**
Resorption (osteoclast)	**CTX-1, NTX-1** (telopeptides) · hydroxyproline · pyridinoline / deoxypyridinoline · bone sialoprotein · **TRAP** · **cathepsin K**
Osteocyte activity	RANKL · OPG · Dickkopf-1 · sclerostin
Reference markers (IOF)	✓ **serum CTX-1** (resorption) + **serum P1NP** (formation) **(TO)**

BTM uses (TO) fracture risk (↑ BTM ∝ risk, independent of BMD) · **monitor compliance/response** — resorption markers **fall >40% within 3 mo** of bisphosphonate, formation markers fall over 6–12 mo · denosumab → very rapid fall in resorption markers · guide restart after drug holiday · **circadian variation** — time assays **(TO)**

Response thresholds (TO) **>30–40% ↓ CTX** (antiresorptive) or **>30% ↑ P1NP** (anabolic) at 3–6 mo = response **(TO)**

– – Active space – –

3 Presentation

Risk factors for osteoporosis increasing age · postmenopausal · long-term ↓ calcium · steroids · alcohol · smoking · low BMI

Drugs ↑ osteoporosis risk	Not associated
Oral glucocorticoids · androgen deprivation · aromatase inhibitors · **protease inhibitors** · **SSRIs** · prolactin-raising **antiepileptics (phenytoin)** · cytotoxics	✓ **NSAIDs** · cephalexin · simvastatin · glipizide · allopurinol

Phenytoin mechanism **hepatic CYP450 induction → ↑ vit D catabolism** · direct osteoblast inhibition · ↓ Ca absorption · ↑ homocysteine · PTH resistance · ↓ sex hormones · ↓ vit K

Disease associations hyperparathyroidism · thyrotoxicosis · type I DM · hypogonadism · malnutrition / malabsorption · immobilisation · **RA** · alcoholism · vit D deficiency · multiple myeloma

Pelvic ring insufficiency fracture risk: **osteoporosis, corticosteroids, RA, THA with constrained liner** (redistributes force to pelvis) · not **ceramic bearings** · groin pain in elderly · treat bed rest + analgesia

4 Diagnosis & Classification

Key radiology findings

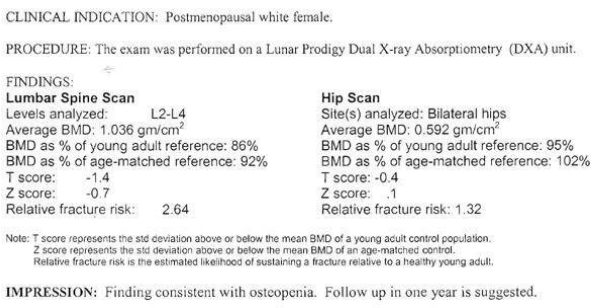
X-ray thinned cortices · loss of trabecular bone · kyphosis · **codfish vertebra** · views: lateral spine, AP pelvis/hip

X-ray sensitivity ✓ loss not apparent until **30–40%** (Orthobullets/OB5227) · **40%** (OB8542) – read the options · normal X-ray does not exclude osteopenia

DEXA ✓ **most accurate, least radiation** · **lumbar L2–L4** + hip (neck, trochanter, intertrochanter) · measures **axial AND appendicular** skeleton
✗ Wrong options: "axial only"

Other densitometry **SPA** reliable in appendicular only (spine unreliable) · **QCT ~10× DEXA radiation** · conventional **MRI** ≈ QCT accuracy without radiation · **QCT** also used to monitor therapy (cost/radiation limit use) **(TO)**

After low-energy fracture ✓ **DEXA** (✗ Wrong options: MRI pelvis, urine electrophoresis, CT, bone scan) · old multilevel VCFs on MRI, no acute pain → ✓ **DEXA** then medical Rx (✗ Wrong options: brace, cement, fusion)



CLINICAL INDICATION: Postmenopausal white female.

PROCEDURE: The exam was performed on a Lunar Prodigy Dual X-ray Absorptiometry (DXA) unit.

FINDINGS:

Lumbar Spine Scan		Hip Scan	
Levels analyzed:	L2-L4	Site(s) analyzed:	Bilateral hips
Average BMD:	1.036 gm/cm^2	Average BMD:	0.592 gm/cm^2
BMD as % of young adult reference:	86%	BMD as % of young adult reference:	95%
BMD as % of age-matched reference:	92%	BMD as % of age-matched reference:	102%
T score:	-1.4	T score:	-0.4
Z score:	-0.7	Z score:	.1
Relative fracture risk:	2.64	Relative fracture risk:	1.32

Note: T score represents the std deviation above or below the mean BMD of a young adult control population.
Z score represents the std deviation above or below the mean BMD of an age-matched control.
Relative fracture risk is the estimated likelihood of sustaining a fracture relative to a healthy young adult.

IMPRESSION: Finding consistent with osteopenia. Follow up in one year is suggested.

DEXA report – postmenopausal woman

Diagnostic criteria (WHO 1994)

Term	Definition
T-score	BMD vs **young normal controls** (peak bone mass, 25–30-yr-old same sex/ethnicity)
Z-score	BMD vs **age-, sex-, race-matched** controls
Normal	**T ≥ −1.0** (e.g. T −0.7 = normal)
Osteopenia	**T −1.0 to −2.5**
Osteoporosis	**T ≤ −2.5** (e.g. −2.6, −3.5 = osteoporosis regardless of other factors)

Which score — **T-score for postmenopausal women** (lowest of spine, femoral neck, trochanter, total hip; not peripheral sites) · **Z-score for premenopausal women, young men, children** → metabolic bone disease in young
✗ Wrong options: osteopenia/osteoporosis "by Z-score"

WHO limits — no data to define criteria for men or other ethnic groups · T-score tells nothing about metastases

Vit D level — low but not diagnostic · DEXA remains the diagnostic test

Biopsy — after **tetracycline labelling** · rules out osteomalacia

Type I vs Type II

	Type I (postmenopausal)	Type II (senile)
Age	50–70	>70
Bone	Almost exclusively **trabecular**	Trabecular > cortical
Fractures	**Distal radius, vertebral**	**Hip, pelvis**
Calcium	Net negative (↓ gut absorption, ↑ urinary loss)	Poor absorption
Vit D	↓ total (not free) 1,25-D	–

5 Management

Risk factors & fracture prediction

Strongest predictor of future fragility fracture — ✓ **prior fragility fracture**
✗ Wrong options: T < −2.5, low vit D, family history, prednisone

Future vertebral fracture — ✓ **prior vertebral fracture** (~**4×**; risk ↑ with number) > prior hip/wrist · **2 previous VCFs = strongest predictor** in postmenopausal women
✗ Wrong options: T −2.8, hip fracture in 6 mo, age >75, bisphosphonate

After 1 VCF — **5× 2nd vertebral** · **5× hip** · **15% ↑ 5-yr mortality** · morbidity: back pain, height loss, balance, restrictive lung disease, pneumonia

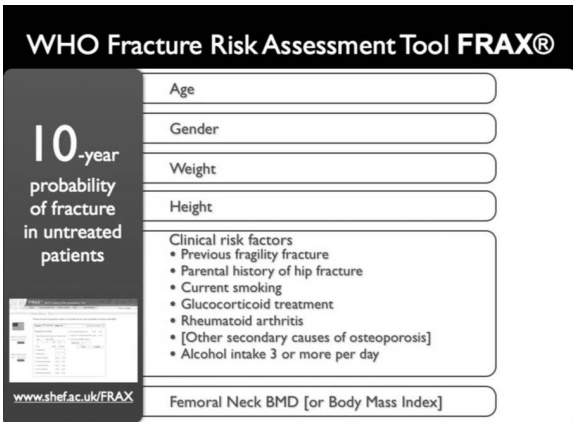
After hip fracture — **20% ↑ mortality** · **men > women** · only **1/3** regain prior function · up to **10× 2nd hip fracture**

Baseline risk — 65-yr-old white woman, no risk factors → **10% 10-yr fragility fracture risk**

FRAX — ✓ **Fracture Risk Assessment tool** · WHO **2008** (Kanis, Sheffield) · **10-yr probability** of **hip** and **major osteoporotic fracture** (proximal humerus, wrist, hip, clinical vertebral) · with/without **femoral neck BMD (TO)**

– – Active space – –

FRAX inputs	age · sex · race · height · weight · BMI · prior fragility fracture · **parental hip fracture** · oral glucocorticoids · **RA** · secondary osteoporosis · current smoking · alcohol **≥3 units/day (TO)**
FRAX does not include	falls risk · **bone turnover markers** · dietary calcium · vitamin D deficiency **(TO)** · developed because **~50%** of fragility fractures occur with BMD above osteoporosis threshold **(TO)**
Monitoring response to treatment (TO)	✓ **DEXA** (guideline tool) · **QCT** (reliable, costly) · **BTMs** ✗ Wrong options: **FRAX — least useful for assessing adequacy of therapy (TO)**
DXA monitoring (TO)	every **2 yr** (1 yr on anabolics) · BMD ↓ **>5% spine or >4% hip** (beyond 1–2% precision error) = failure → look for secondary causes · **new fragility fracture on treatment = failure** regardless of BMD **(TO)**

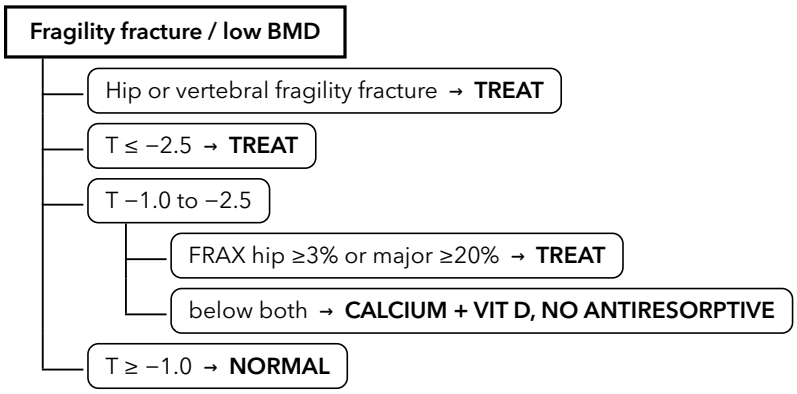


FRAX tool

Screening (USPSTF / OB)	**women ≥65** + younger women with risk ≥ 65-yr-old white woman · BMD testing also postmenopausal women with ≥1 risk factor · men ≥70 (OB922) · evidence insufficient for men (USPSTF)

Operative criteria — when to start pharmacological treatment (NOF 2008)

Postmenopausal women / men ≥50 with ANY of	**Treat?**
Hip or vertebral fracture (even without DEXA)	Yes
T ≤ −2.5 femoral neck / spine	Yes
T −1.0 to −2.5 AND **10-yr hip ≥3%**	Yes
T −1.0 to −2.5 AND **10-yr major ≥20%**	Yes
Major ≥10%	not a criterion **(TO)**
FRAX hip ≥3% **without** documented osteopenia	No
T −2.0, hip 1.5%, major 8.9%	**no antiresorptive**
T −1.5, hip **4.8%**, major 8% (smoker on phenytoin)	✓ **Ca 1200 + vit D 800 + bisphosphonate + neuro review of phenytoin + repeat DEXA 1 yr**



Calcium & vitamin D

Age	Daily calcium
1–3 yr	500 mg
4–8 yr	800 mg
9–18 yr (e.g. 13-yr-old)	**1000–1500 mg**
Adults >50 / osteoporosis	**1200–1500 mg**

Vitamin D
800–1000 IU/day (NOF, >50 yr) · prevention dose **400–800 IU** (OB418/OB6265) · on 1200 mg Ca alone after DRF → ✓ **add 800–1000 IU vit D**
✗ Wrong options: 1600–2000 IU · ↑ Ca to 2000 · calcitonin

Healthy 52–55-yr-old woman
✓ **1000–1500 mg Ca + 400–800 IU vit D** · ✓ 1500 mg + 1000 IU
✗ Wrong options: semiannual BMD · estrogen (HRT not approved for treatment – OB6265)
✗ Wrong options: 2200 mg · 5000–10,000 IU

Premenopausal, T −2.2, DRF
✓ **1200 mg calcium** (prevention dose, not enough for T < −2.5)

Effects
Ca ↓ fracture risk **34%** · vit D ↓ hip 10%, non-vertebral 7% · high-dose (~800 IU) ↓ hip **24%**, non-vertebral **30%** · best for **type II (senile)**

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Osteoporosis drugs

Drug	Mechanism	Key facts
Non-N bisphosphonates (etidronate, clodronate, tiludronate)	**Toxic ATP analogue** → osteoclast apoptosis	Esophagitis, ulcers, ONJ, atypical fractures
N-bisphosphonates (pamidronate, alendronate, risedronate, zoledronate, ibandronate)	**Inhibit farnesyl pyrophosphate synthase** (mevalonate)	**1st line** · renal excretion, unmetabolised · bind hydroxyapatite → detectable **years** later
HRT (estrogen-progestin)	–	Type I within 6 yr of menopause · ↓ hip fracture but ↑ breast ca, CAD/MI, stroke, PE/DVT, Alzheimer's
Estrogen only	ER on osteoblasts + osteoclasts	**Prior hysterectomy only** · unopposed → endometrial hyperplasia / uterine ca
Testosterone	–	Hypogonadal men · not FDA-approved
Salmon calcitonin	**Binds osteoclast receptors** → ↓ resorption (from thyroid **parafollicular cells**)	Women **>5 yr postmenopause** · ✓ ↓ **pain of acute VCF** (neurotransmitter) · nasal/injection (gastric acid destroys) · rhinitis; nausea, flushing
Raloxifene (SERM)	**ER agonist bone, antagonist breast** (competitive partial agonist)	**Vertebral fractures only** · not anabolic · hot flashes, leg cramps · **CI: VTE**
Teriparatide	**PTH 1–34**, intermittent → osteoblasts	**Anabolic** · daily SC · transient ↑ Ca
Denosumab	**Anti-RANKL** IgG2 (like OPG)	68/40/20% · **CI severe hypocalcaemia**
Romosozumab	**Anti-sclerostin** → Wnt ↑	↓ vertebral **73%** at 12 mo, **75%** at 24 mo · **CV events**
Strontium ranelate	Dual action (↑ formation, ↓ resorption)	not FDA-approved · ↑ MI

FDA-approved for active treatment (OB418) alendronate, risedronate, raloxifene, estrogen, calcitonin → ↓ fragility fractures up to **50%**

Bisphosphonates (indications, atypical fractures, ONJ, drug holiday)

Mechanism incorporate at remodelling sites → released in acid resorption pit → taken up by osteoclasts → **flattened ruffled border**, ↓ function + **survival** of osteoclasts
✗ Wrong options: formation

Agent	Fracture reduction
Alendronate	Hip, spine, wrist **~50%** · vertebral **48%**, non-vertebral **47%** · 70 mg weekly or 10 mg daily
Risedronate	Vertebral **65%** / non-vertebral **39%** (table) · **40%** each over 3 yr (text)
Ibandronate	Vertebral **77%**, hip 41%, non-vertebral 15% (Orthobullets) · ✓ **failed to significantly ↓ hip / non-vertebral fracture in BONE trial (TO)** – read the options
Zoledronic acid (IV annual)	Spine **70%**, hip **40%** over 3 yr · ✓ **only bisphosphonate ↓ all-cause mortality** – **HORIZON-RFT**, infusion within **90 days** of hip fracture repair **(TO)**

Fragility VCF / hip fracture ✓ start antiosteoporotic medication regardless of DEXA → ✓ **inhibit farnesyl pyrophosphate synthase**
✗ Wrong options: calcitonin receptor inhibition · RANK activation · ER inhibition · continuous PTH

Atypical femoral fracture (AFF) long-term BP → **extremely low turnover** (↓ urinary **NTx**) → microdamage unrepaired in lateral tension cortex

AFF radiographic features
Focal lateral cortical thickening (beaking)
Transverse orientation
Medial spike
Simple, non-comminuted pattern
Subtrochanteric / diaphyseal (✗ Wrong options: distal femur)



Atypical subtrochanteric fracture – transverse, lateral beak, medial spike

ONJ **IV bisphosphonates** (✗ Wrong options: oral) · also **denosumab** · rare · stop inciting drug · long-term IV zoledronate for Paget → ✓ **osteonecrosis of mandible** (✗ Wrong options: sarcoma (teriparatide risk))

Long-term alendronate ✓ **diaphyseal femoral insufficiency fractures** · ONJ only jaw
✗ Wrong options: femoral head AVN · retinal detachment · uterine ca · scurvy

– – Active space – –

Drug holiday	not in **high risk / declining BMD / severe osteoporosis / chronic steroids** in first **10 yr** → continue (e.g. 8 yr BP, steroids, T −2.6) · after 10 yr → holiday **1–2 yr** or until BMD ↓ / fracture (OB)
Drug holiday (TO)	alendronate skeletal half-life **~10 yr** · reassess after **3–5 yr oral or 3 yr IV ZA** · moderate risk → holiday **2–3 yr** (oral) / **3 yr** (ZA) with DXA + BTM · high risk → continue or switch · common practice holiday after 3–5 yr **(TO)**
No drug holiday (TO)	**denosumab** (rebound) · **anabolics** (BMD lost) · **SERMs** (no residual effect) **(TO)**

Anabolic agents (teriparatide, abaloparatide, romosozumab)

Teriparatide	recombinant **N-terminal PTH 1–34** (of 84 aa; chief-cell hormone) · activates **adenylate cyclase** · **20 µg/day SC** (abdomen/thigh) · max **2 yr** · ✓ **stimulates osteoblast function** · **only US anabolic** (OB, older)
Dosing pattern	**intermittent (daily) → anabolic** · **continuous infusion → net resorption** + hypercalcaemia (e.g. 2-wk infusion)
Intermittent PTH effects	↑ osteoblast number · activates existing osteoblasts · lining cells → osteoblasts · ↓ osteoblast apoptosis · osteoblasts secrete RANKL + IL-6 (coupling)
Anabolic mediators (TO)	↑ **IGF-1** · ↑ **FGF-2** · ↓ **sclerostin** (Wnt/β-catenin ↑) · ↑ **Runx2** · not **RANKL** (= catabolic pathway, with M-CSF) **(TO)**
Signalling (TO)	PTH1R (GPCR) on osteoblasts, osteocytes, renal tubule → Gs-cAMP-**PKA** (main) + Gq-PKC · **both anabolic and catabolic** effects **(TO)**
FDA indications (TO)	postmenopausal high-risk · **men with primary or hypogonadal osteoporosis** · **glucocorticoid-induced (TO)**

Teriparatide contraindications (TO)
Paget disease / unexplained ↑ ALP
Skeletal malignancy / **bone metastases**
Prior skeletal radiation
Open epiphyses (children)
Hypercalcaemia
Hypercalciuria / urolithiasis

Osteosarcoma	**rats**, dose/duration dependent (3–60× human exposure) · not shown in humans · black-box (since removed per TO; 24-mo course still standard – plateau) **(TO)**
Adverse effects	**transient hypercalcaemia** · hypercalciuria · nausea, headache, dizziness, orthostatic hypotension · no hyperphosphataemia · **↑ ALP** (formation marker, not ↓)
Monitoring (TO)	serum + urine Ca before · serum Ca at **1 mo** · hypercalcaemia → ↓ calcium supplement first, then alternate-day dosing **(TO)**
Fracture effect	↓ vertebral + non-vertebral (if anything, preferential **vertebral**) · Neer: non-vertebral ↓ 35% (20 µg) / 40% (40 µg)

With bisphosphonates **concomitant use beneficial** (NOT true) — BPs **blunt** anabolic effect **(TO)** · IV ZA concomitant does not blunt; **denosumab + teriparatide → enhanced BMD (TO)** · prior antiresorptive blunts response **(TO)** · OB213138: may be given alone or as adjunct; long-term BP users → stop BP
✗ Wrong options: abolishes

Refractory osteoporosis (5 yr BP, prior cement augmentation, new VCFs) — ✓ **recombinant N-terminal PTH**

Abaloparatide (TO) synthetic **PTHrP 1–34** (41% homology) · ✓ **binds more selectively to RG state** (transient cAMP → formation) · **8.5×** lower R0 affinity · → less resorption, less cortical porosity, better radius/femoral neck BMD, **less hypercalcaemia**
✗ Wrong options: PTHR2 agonist **(TO)**

PTH1R conformation (TO)	Signal	Effect
R0	Prolonged	Osteoclast → resorption
RG	Transient	Osteoblast → formation

Sclerostin **SOST** gene, **osteocytes** · binds **LRP5/6**, blocks Wnt → β-catenin degraded → ↓ osteoblast differentiation · loss of SOST → **↑ bone mass** (**van Buchem**, **sclerosteosis**) · (✗ Wrong options: hereditary osteoporosis · antagonises BMP) · ✓ **anti-sclerostin antibodies ↑ BMD**

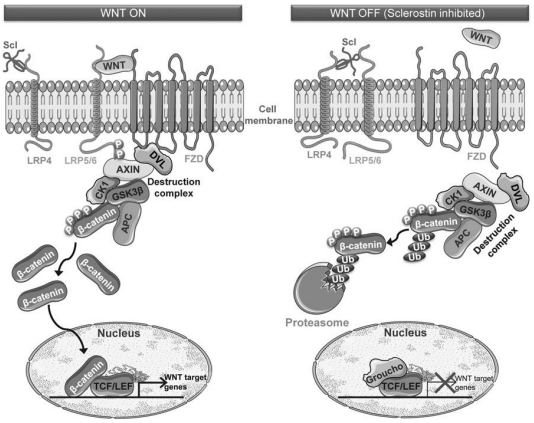
Romosozumab ✓ **inhibits an antagonist of Wnt/β-catenin** · humanised IgG2 · **dual action** (↑ formation, ↓ resorption via ↓ RANKL, ↑ OPG) · ✓ **most rapid + greatest ↑ lumbar BMD in 12 mo** (**13.3%** spine, 6.9% hip) **(TO)**

Romosozumab dose **210 mg SC monthly** (2 × 105 mg) × **12 mo (TO)** · **120 mg monthly** × 12 mo (TO17130219) — read the options

Romosozumab indication FDA **Apr 2019** · **treatment** of postmenopausal women at high risk · not **prevention (TO)** · very high risk = T < −2.5 + vertebral/hip fracture in 24 mo or >2 vertebral fractures, or FRAX major >30% / hip >4.5% **(TO)**

Romosozumab trials (TO) **FRAME** ↓ vertebral **73%** at 12 mo · **ARCH** vs alendronate: 48% fewer vertebral, 38% fewer hip · **CV events 2.5% vs 1.9%** → black box: avoid if MI/stroke within 1 yr **(TO)** · must follow with antiresorptive (BMD → baseline 1 yr after stopping) **(TO)**

Blosozumab investigational anti-sclerostin



Sclerostin–Wnt pathway

– – Active space – –

Denosumab

Mechanism	fully human **IgG2 anti-RANKL** · blocks RANKL–RANK (like **OPG**) · ✓ **inhibits bone resorption** · ↓ osteoclast **formation, function, survival** (BPs do not affect formation) **(TO)** ✗ Wrong options: RANK · OPG · TNF
Pharmacology	does not bind hydroxyapatite · cleared within **6 mo**, undetectable at **1 yr** · **fully reversible** · usable with **bone metastases** · also **GCT of bone**
Dose	**60 mg SC every 6 mo** (100% compliance) **(TO)**
FREEDOM (TO)	↓ vertebral **68%**, hip **40%**, non-vertebral **20%** over 3 yr · 10-yr extension: spine BMD **+21.7%**, **no plateau** (BPs plateau 3–5 yr) **(TO)**
Indications (TO)	BP alternative (**eGFR <30–35**, oesophageal disease, intolerance) · second line **(TO)**
Adverse	arthralgia · nasopharyngitis · back pain · **ONJ** (✓ antibody-based therapy linked to ONJ; ~0.04%) · cellulitis · hypocalcaemia (renal impairment, vit D deficiency) · CI **severe hypocalcaemia**
Abrupt discontinuation	✓ **rebound turnover → multiple vertebral fractures (TO)** ✗ Wrong options: hypercalcaemia · ONJ · AFF

Sequential therapy & treatment transitions (TO)

Anabolic first	✓ **anabolic → antiresorptive** (DATA / DATA-Switch) · antiresorptive first blunts later anabolic response **(TO)**

Sequence (decreasing BMD gain) (TO)
Romosozumab 12 mo → denosumab (greatest)
Teriparatide/abaloparatide 24 mo → denosumab
Romosozumab 12 mo → bisphosphonate
Denosumab alone
Bisphosphonate 3–5 yr then holiday (moderate risk)

Situation (TO)	✓ Correct	✗ Wrong option
After 24 mo teriparatide	✓ **Immediately (1–3 mo) → denosumab** (continued ↑ spine + hip BMD) or BP (ZA 1–2 yr / alendronate 3–5 yr)	drug holiday (lose almost all gain within 12 mo) · "effects long-lasting, no antiresorptive"
Stopping denosumab	✓ **Bisphosphonate — ZA exactly 6 mo** after last dose (≤8 mo if CTX-guided) · >2.5 yr use → repeated ZA / 1–2 yr alendronate, monitor CTX at 3 + 6 mo	ZA at 12 mo · **teriparatide after denosumab** (transient hip/radius bone loss, no rebound protection)
Stress reaction on ZA (AFF precursor)	✓ **Stop ZA, consider teriparatide** · protected WB · **contralateral femur imaging** (bilateral **20–50%**) · correct vit D >30 ng/mL	switch to **denosumab** (absolutely avoid) · oral alendronate · continue ZA

Incomplete AFF (TO)	Management
Prodromal thigh/groin pain OR lucent lateral cortical line ("dreaded black line")	✓ **Prophylactic full-length cephalomedullary nail**
Asymptomatic, beaking only, no lucent line	Protected WB **2–3 mo** + serial X-rays

Thigh pain + lateral cortical thickening on alendronate

✓ **cephalomedullary nailing** (protects femoral neck) · (✗ Wrong options: standard antegrade nail · CT staging / oncology referral · observation) · nail pearl: exaggerated anterior bow → avoid anterior cortical perforation **(TO)**



Impending atypical femoral fracture

Fracture liaison & post-fracture management

Fragility fracture work-up

hip + spine densitometry · labs for secondary causes · Ca **1500 mg** + vit D **400–800 IU** · **protein-enriched diet**
✗ Wrong options: **low protein diet**

Inpatient hip ORIF with new osteoporosis

✓ **metabolic work-up as inpatient + osteoporosis clinic appointment** · treatment rates after hip fracture surprisingly low
✗ Wrong options: PCP follow-up · PT · nutritionist · orthopaedist starting BP and sending to PCP

Fracture liaison service

✓ **more likely to receive DEXA** + referral to metabolic bone clinic
✗ Wrong options: fewer labs / PCP / endocrine visits · changes fracture care

Vertebral compression fracture & vertebroplasty

Osteoporotic VCF

commonest fragility fracture · T likely < −2.5 · no CT if neuro intact with adequate films · neoplasm unlikely in typical demographic

Vertebroplasty

✓ **no benefit in RCTs** · kyphoplasty for pain persisting **>6 wk** nonop (limited AAOS strength)
✗ Wrong options: kyphoplasty in first week for narcotic-level pain

Bone density for implant fixation (TO)

Location	Minimum BMD for implant stability (mg/cm³) (TO)
Proximal femur	✓ **250**
Proximal humerus	95
Vertebral body (pedicle)	95–150
Vertebral body	220

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6 Complications

Bisphosphonates	esophagitis · dysphagia · gastric ulcer · **ONJ (IV)** · **atypical subtrochanteric fracture**
Teriparatide	transient hypercalcaemia · rat osteosarcoma · Paget sarcoma risk (contraindicated)
Denosumab	ONJ · rebound vertebral fractures on stopping · hypocalcaemia
Romosozumab	**CV events** · hyperostosis · OA · cancer · ONJ · AFF
Raloxifene	VTE · hot flashes · leg cramps
HRT	breast ca · CAD/MI · stroke · PE/DVT · Alzheimer's

Osteoporosis vs Osteomalacia

	Osteoporosis	**Osteomalacia**
Defect	↓ **mass, normal mineralisation** (quantitative)	Variable mass, ↓ **mineralisation** (qualitative)
Age	Postmenopausal / elderly	Any
Cause	Endocrine, age, inactivity, alcohol, ↓ Ca	Vit D deficiency/pathway, ↓ PO_4, hypophosphatasia, RTA
Symptoms	Pain at fracture site	Generalised bone pain
X-ray	**Axial** fractures	**Appendicular**, symmetric, **Looser zones**
Serum Ca / PO_4	Normal	Low or normal
ALP	**Normal**	↑ (except hypophosphatasia)
Urinary Ca	High or normal	Normal or low
Tetracycline biopsy	Normal	**Abnormal**

Antiresorptive vs Anabolic

	Bisphosphonate	**Denosumab**	**Teriparatide**	**Romosozumab**
Target	FPP synthase / ATP analogue	RANKL	PTH1R	Sclerostin
Action	Antiresorptive	Antiresorptive	**Anabolic**	**Anabolic + antiresorptive**
Dosing	Oral weekly / IV yearly	60 mg SC 6-monthly	20 µg SC daily, 2 yr	SC monthly, 12 mo
Persistence	Years in bone	Gone by 6–12 mo	BMD lost without relay	BMD lost without relay
Drug holiday	Yes (moderate risk)	No (rebound)	No	No
Signature risk	ONJ, AFF	Rebound VCFs, ONJ	Osteosarcoma (rats), ↑ Ca	CV events

How it's asked

"T-score −3.5 / −2.6?"	→ **osteoporosis** ✗ Wrong options: needs more information · normal for elderly
"Osteoporosis T-score?"	→ **−2.5 or below** ✗ Wrong options: −3.0 · +2.5
"Postmenopausal, T −0.7, Z −0.4?"	→ **normal** ✗ Wrong options: osteopenia by Z-score
"Osteopenia definition?"	→ **T −1 to −2.5** ✗ Wrong options: Z-score definitions · T vs age-matched
"X-ray shows loss after what %?"	→ **40%** ✗ Wrong options: 5%, 20%, 90%
"BMD assessment EXCEPT?"	→ **DEXA measures axial skeleton only** (false)
"Next investigation after fixation of low-energy hip / old VCFs on MRI?"	→ **DEXA scan** ✗ Wrong options: bone scan · cement augmentation
"Strongest predictor of future low-energy fracture?"	→ **prior fragility fracture** ✗ Wrong options: T < −2.5
"Greatest predictor of future vertebral fracture?"	→ **2 previous VCFs** ✗ Wrong options: T −2.8 · recent hip fracture
"Which elderly woman at greatest VCF risk?"	→ **prior T6 compression fracture** ✗ Wrong options: prior hip · T −3.0
"Bone that loses more strength with age?"	→ **trabecular more than cortical**
"Drug NOT associated with osteoporosis?"	→ **NSAIDs** ✗ Wrong options: SSRI, steroids, phenytoin, protease inhibitors
"Medication ↑ non-traumatic hip fracture?"	→ **phenytoin** ✗ Wrong options: simvastatin, glipizide, allopurinol
"Pelvic insufficiency fracture risk EXCEPT?"	→ **THA with ceramic bearings** ✗ Wrong options: constrained liner
"Gene polymorphism NOT linked to osteoporosis?"	→ **COMP** ✗ Wrong options: calcitonin receptor, ER-1, VDR, COL1A1
"Lab most likely abnormal after low-energy DRF?"	→ **25-OH vitamin D**
"Vit D replacement – expected change?"	→ **↑ 25-OH D and 1,25-D, ↓ sclerostin (TO)**
"Calcium for a 13-yr-old?"	→ **1000–1500 mg**
"Calcium for 52-yr-old with T −3.0?"	→ **1000–1500 mg**

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"On 1200 mg Ca alone after DRF – adjust?"	→ **add 800–1000 IU vitamin D**
"Premenopausal DRF, T −2.2 – treatment?"	→ **1200 mg calcium**
"Healthy 52-yr-old, no risk factors?"	→ **1000–1500 mg Ca + 400–800 IU vit D** ✗ Wrong options: estrogen · semiannual DEXA
"FRAX stands for?"	→ **Fracture risk assessment tool (TO)**
"Least useful tool to assess response to bisphosphonate?"	→ **FRAX (TO)** ✗ Wrong options: BTMs, DEXA, QCT
"NOT an indication to treat?"	→ **10-yr major fracture risk ≥10% (TO)**
"T −2.0, hip 1.5%, major 8.9%?"	→ **do not start antiresorptive**
"T −1.5, hip 4.8%, on phenytoin?"	→ **Ca + vit D + bisphosphonate + replace phenytoin + DEXA in 1 yr**
"NOF 2008 bisphosphonate indication?"	→ **T −1.0 to −2.5 AND hip risk >3%** ✗ Wrong options: major >10%
"Teriparatide – NOT true?"	→ **concomitant bisphosphonate beneficial (TO)**
"Teriparatide anabolic effect NOT mediated by?"	→ **RANKL (TO)**
"Intermittent PTH effect?"	→ **anabolic**
"Recombinant PTH that builds bone?"	→ **teriparatide (Forteo) – hormone of chief cells** ✗ Wrong options: SOST protein · P-C-P backbone · parafollicular hormone
"Net bone resorption?"	→ **teriparatide continuous infusion** ✗ Wrong options: daily injection
"Osteoporosis drug with osteosarcoma risk?"	→ **recombinant PTH 1–34**
"Teriparatide – true?"	→ **stimulates osteoblast function** ✗ Wrong options: ↓ ALP · hyperphosphataemia · inhibits osteoclastogenesis
"Refractory osteoporosis after 5 yr BP?"	→ **recombinant N-terminal PTH**
"Abaloparatide vs teriparatide?"	→ **binds more selectively to RG state (TO)**
"Fastest, greatest ↑ lumbar BMD in 12 mo?"	→ **romosozumab (TO)**
"Romosozumab – NOT true?"	→ **approved for prevention (TO)**
"Anti-sclerostin mechanism?"	→ **inhibits an antagonist of Wnt/β-catenin** ✗ Wrong options: RANKL · FPP synthase

"Sclerostin – accurate?"	→ **anti-sclerostin antibodies ↑ bone density** ✗ Wrong options: SOST → hereditary osteoporosis
"Direct RANKL inhibitor?"	→ **denosumab** ✗ Wrong options: romosozumab, blosozumab
"Antiresorptive, no hydroxyapatite binding, undetectable at 1 yr, OK with metastases?"	→ **denosumab**
"Denosumab interferes with RANKL → ?"	→ **inhibits bone resorption (TO)**
"Antibody therapy with ONJ?"	→ **denosumab** ✗ Wrong options: pamidronate (not antibody)
"Abrupt denosumab cessation risk?"	→ **multiple vertebral fractures (TO)**
"True about anti-osteoporotic treatment?"	→ **denosumab effect short-lived – follow with bisphosphonate (TO)**
"Very high risk – best sequence?"	→ **anabolic then antiresorptive (TO)**
"After 24 mo teriparatide?"	→ **switch immediately to denosumab (TO)**
"Femoral stress reaction on ZA?"	→ **stop ZA, consider teriparatide (TO)**
"Bisphosphonate failing to ↓ hip fracture?"	→ **ibandronate (TO)**
"Therapy ↓ all-cause mortality after hip fracture?"	→ **zoledronic acid (TO)**
"Fragility VCF – mechanism of indicated drug?"	→ **inhibits farnesyl pyrophosphate synthase**
"Fracture of long-term bisphosphonate?"	→ **transverse subtrochanteric, lateral beak, medial spike**
"Long-term alendronate associated with?"	→ **diaphyseal femoral insufficiency fracture**
"Long-term IV zoledronate for Paget – complication?"	→ **osteonecrosis of mandible** ✗ Wrong options: sarcoma
"Thigh pain, lateral cortical thickening, on alendronate?"	→ **cephalomedullary nailing**
"8 yr BP + chronic steroids, T −2.6?"	→ **continue, no drug holiday**
"Fragility fracture management NOT included?"	→ **low protein diet**
"Best osteoporosis care after hip ORIF?"	→ **inpatient metabolic work-up + osteoporosis clinic**
"Benefit of fracture liaison service?"	→ **more likely to get DEXA**
"L1 osteoporotic VCF – true?"	→ **↓ bone quantity, normal quality** ✗ Wrong options: kyphoplasty in week 1 · T −2.1 diagnostic
"Vertebroplasty for VCF?"	→ **no benefit in RCTs**

– – Active space – –

"Disuse osteopenia principle?"	→ **Wolff's law** ✗ Wrong options: Hueter-Volkmann · Hooke
"Minimum BMD for proximal femoral implant fixation?"	→ **250 mg/cm³ (TO)**

OUTCOME MEASURE TOOLS

– Basic Science · 76 / 108 –

1 Basics

SF-36 & patient-reported outcomes

SF-36	**generic** short-form health survey, **36 questions** · most widely applied general health instrument · **self-administered** (office, mail, phone) · validated with **normative population data** + multilingual translations
SF-36 structure	**8 scaled scores** (each 0–100, equal item weight) · physical functioning · role physical · bodily pain · general health · vitality · social functioning · role emotional · mental health · measures functional ability, well-being, overall health
SF-36 limitation	✓ **ceiling and floor effects** · bias **lower > upper extremity** ✗ Wrong options: "no normative data" · ignores patient perspective · must be on-site
SF-36 use	compare disease burden · **polytrauma with foot injury → lower SF-36** than polytrauma without foot injury (≥12 mo)
SF-12	12-item short version · physical + mental health · pre-injury pain-related disability predicts pain at 6 mo
VR-12	Veterans Rand 12-item, derived from SF-36 · physical + mental summary scores · HRQOL, disease burden
EQ-5D	generic QOL · **5 dimensions** (mobility, self-care, usual activities, pain/discomfort, anxiety/depression) + **VAS** · **243 health states** · index **0 = death, 1 = perfect health** · 170+ languages
PROMs vs physician-reported	✓ **better assessment of patient satisfaction** ✗ Wrong options: cost-effective care · ambulation · disposition · analgesia needs

PROMIS

PROMIS	NIH (2004 Roadmap) · **general + disease-specific** · adult + paediatric · domains **physical, mental, social**
Method	**computer adaptive testing (CAT)** + **item response theory (IRT)** → next item chosen from previous answers → **fewer questions**, high precision
Scoring	**T-scores**, population mean **50, SD 10**
Advantage	✓ **reduced floor and ceiling effects** · quickest, broadest population, highly reliable · high correlation with ODI + SF-36 ✗ Wrong options: more items · low reliability/validity
Validated in	✓ **spine · foot & ankle · upper extremity**
Foot & ankle	PROMIS ✓ **superior** to FAOS / OMAS at detecting clinically significant change at **6 and 12 mo** after ankle fracture ✗ Wrong options: needs 24 mo follow-up
Legacy measures	SF-36, EQ-5D, HHS, KOOS, DASH, QuickDASH, FAAM, ODI · not IRT-based · validated only in specific populations

- - Active space - -

4 Diagnosis & Classification

Joint-specific scores

Region	Scores
Knee	**KSS** · IKDC · **KOOS** · **WOMAC** · Lysholm · Tegner · Oxford knee **(TO)**
Hip	Merle d'Aubigné–Postel · **Harris Hip Score** · modified HHS · HOOS · Oxford hip · Lequesne · WOMAC **(TO)**
Shoulder	**DASH** · Constant · Oxford shoulder · WOSI / WORC / WOOS · Simple shoulder test · UCLA · SPADI · ASES **(TO)**

"Not a knee score" ✓ **DASH** (upper limb) **(TO)**

WOMAC hip + knee **OA** · **24 items**: **pain 5 · stiffness 2 · function 17** · sum 0–20 / 0–8 / 0–68 · **higher = worse** · TO key: **knee** (OA commoner in knee than hip) **(TO)** · also back pain, RA, JRA, SLE, fibromyalgia

KOOS young/middle-aged, post-traumatic OA (also TKA) · **5 domains** (pain, symptoms, ADL, sport/recreation, QOL) · 42 items 0–4 → 0–100 (**100 = no problems**)

Harris Hip Score (THA)	Points
Pain (none)	**44**
Function (gait: no limp, no aid, >6 blocks)	**33**
Functional activities	14
Physical exam (ROM)	9
Rating	**90–100 excellent · 80–90 good · 70–79 fair · 60–69 poor · <60 failed**

I. PAIN (44 POSSIBLE)	
A. None or ignores it	44
B. Slight, occasional, no compromise in activities	40
C. Mild pain, no effect on average activities, rarely moderate pain with unusual activity	30
D. Moderate pain, tolerable but with limitations in ordinary work or life	20
E. Marked pain, serious limitation of activities	10
F. Totally disabled, crippled, pain in bed	0
II. FUNCTION (47 POSSIBLE)	
A. Gait (33 possible)	
1. Limp	
a. None	11
b. Slight	8
c. Moderate	5
d. Severe	0
2. Support	
a. None	11
b. Cane for long walks	7
c. Cane most of the time	5
d. One crutch	3
e. Two canes	2
f. Two crutches	0
g. Not able to walk	0
3. Walking distance	
a. > 1 km	11
b. 0,5-1 km	8
c. 100-500 m	5
d. Only inside the house	2
e. Confined to chair or bed	0
B. Activities (14 possible)	
1. Stairs	
a. Normally without using a railing	4
b. Normally using a railing	2
c. In any manner	1
d. Not able to do stairs	0
2. Shoes and socks	
a. With ease	4
b. With difficulty	2
c. Unable	0
3. Sitting	
a. Comfortably in ordinary chair for 1 hour	5
b. On a high chair for one-half hour	3
c. Unable to sit comfortably in any chair	0
4. Uses public transport	1
TOTAL	

* Harris WH. Traumatic arthritis of the hip after dislocation and acetabular fractures: treatment by mold arthroplasty. An end-result study using a new method of result evaluation. J Bone Joint Surg Am. 1969 Jun;51(4):737-55.

Harris Hip Score

FAOS lateral ankle instability, Achilles tendinosis, plantar fasciitis · **5 subscales** (pain, symptoms, ADL, sport, QOL) · last week · 0–4 → normalised **100 = no symptoms**

– – Active space – –

Constant shoulder score (0–100)	Max
Pain (subjective)	**15**
Function / ADL (subjective)	**20**
Range of motion (objective)	**40**
Strength (objective)	**25**



Constant Score

Subjective Shoulder Assessment (35 total points)

Criteria	Points	Patient Score
Pain (15 points)		
None	15	= 15
Mild	10	
Moderate	5	
Severe	0	
Activities of daily living (10 points)		
Ability to work	0-4	= 4
Ability to engage in recreational activities	0-4	= 4
Ability to sleep	0-2	= 2
Ability to work at a specific level (10 points)		
Waist	2	
Chest	4	
Neck	6	
Head	8	
Above head	10	= 10

Objective Shoulder Assessment (65 points)

Criteria	Points	
Flexion and abduction (scored separately)		
>150	10	Flexion= 10
121°-150°	8	Adduction= 10
91°-120°	6	
61°-90°	4	
31°-60°	2	
0°-30°	0	
Combined active external rotation (10 points)		
Hand behind head, elbow forward	2	= 2
Hand behind head, elbow back	2	= 2
Hand on top of head, elbow forward	2	= 2
Hand on top of head, elbow back	2	= 2
Full elevation from top of head	2	2
Combined active internal rotation (10 points)		
Interscapular region	10	= 10
Inferior tip of scapula	8	
Twelfth rib	6	
Lumbosacral junction	4	
Buttock	2	
Lateral thigh	0	
Strength (25 points)	1/lb	= 25

Score
100

FROM: Constant CR, CORR 1987;214:160
*example is a patient with normal shoulder function. Change values as indicated under "Patient Score" to determine patients Constant Score.

Constant score

UCLA shoulder **5 sections**: pain · function · active forward flexion · strength of forward flexion · **patient satisfaction** · max **35** · **>27 = good/excellent**

DASH **30-item** self-report upper limb · disability/symptom (30 items, 1–5) + optional sport/music/work (4 items) · score = **[(sum/n) − 1] × 25** · not calculable if **>3 missing** · optional module not scored if any missing

QuickDASH **11 items**, 5-point Likert · not scored if **>1 missing** · 0–100, higher = more disability

Oxford Shoulder Instability Score 12 questions · **12 (best) to 60 (worst)** · uni- + multidirectional instability

Spine scores

ODI **gold standard** low back functional outcome · **10 sections × 5** · % = scored / possible × 100 (16/50 = **32%**; one section missed → 16/45 = **35.5%**)

NDI modified ODI · 10 items · commonest neck pain measure (chronic neck pain, whiplash, radiculopathy) · sum / 50 × 100 · 0% none → 100% complete disability

ASIA spinal cord injury

Cervical myelopathy	Based on
mJOA (17 points)	UE motor · LE motor · sensory · **bladder**
Nurick	**Gait** · employment · cord symptoms

– – Active space – –

TABLE 1 Japanese Orthopaedic Association Criteria* for the Evaluation of Operative Results in Patients with Cervical Myelopathy*

I. Upper extremity function
- Impossible to eat with either chopsticks or spoon (0 points)
- Possible to eat with spoon, but not with chopsticks (1 point)
- Possible to eat with chopsticks but inadequate (2 points)
- Possible to eat with chopsticks but awkward (3 points)
- Normal (4 points)

II. Lower extremity function
- Impossible to walk (0 points)
- Need cane or aid on flat ground (1 point)
- Need cane or aid only on stairs (2 points)
- Possible to walk without cane or aid, but slow (3 points)
- Normal (4 points)

III. Sensory
- Upper extremity
  - Apparent sensory loss (0 points)
  - Minimal sensory loss (1 point)
  - Normal (2 points)
- Lower extremity
  - Apparent sensory loss (0 points)
  - Minimal sensory loss (1 point)
  - Normal (2 points)
- Trunk
  - Apparent sensory loss (0 points)
  - Minimal sensory loss (1 point)
  - Normal (2 points)

IV. Bladder function
- Complete retention (0 points)
- Severe disturbance (1 point)
  - Inadequate evacuation of bladder
  - Straining
  - Dribbling of urine
- Mild disturbance (2 points)
  - Urinary frequency
  - Urinary hesitancy
- Normal (3 points)

*Total normal score = 17 points.

mJOA classification

5 Management

MCID – statistical vs clinical significance

p-value — chance of the observed difference if no true difference · **p < 0.05** = statistically significant · says nothing about clinical relevance

MCID — ✓ must be identified to judge **clinical significance** (e.g. KOOS difference, p = 0.04)
✗ Wrong options: "significant → better outcomes" · larger sample needed

ODI 38.4 vs 34.1, p = 0.002 — ✓ **statistically significant but likely below MCID** (4.3 points) · (✗ Wrong options: NNT (applies to risk)) · 80% power / β 0.2 = standard · non-randomised → **confounding**
✗ Wrong options: Hawthorne = behaviour change when observed

Predictors of pain & patient satisfaction

Predicts moderate/severe pain at 6 mo after MSK injury	Not a predictor
Failure to complete high school · pre-injury pain-related disability · disability compensation · moderate/severe pain at discharge	✓ **Age, gender (male)**

↑ Satisfaction AND quality — ✓ **interpreter for limited English proficiency**
✗ Wrong options: requested hydrocodone · early MRI · waiting-room amenities · early pain referral

HCAHPS — inpatient satisfaction survey · concern it may not reflect quality

Generic vs Specific

	Generic	Disease/joint-specific
Examples	SF-36, SF-12, VR-12, EQ-5D, PROMIS	WOMAC, KOOS, HHS, DASH, Constant, ODI, NDI
Strength	Compare burden across diseases	Sensitive to joint change
Weakness	Ceiling/floor effects (SF-36)	Population-limited

How it's asked

"Instrument with general + disease-specific measures, validated in spine, foot & ankle, upper extremity?"	→ **PROMIS** ✗ Wrong options: SRS-22, ODI, SF-12, PHQ-9
"PROMIS vs other tools in spine?"	→ **reduced floor and ceiling effect** ✗ Wrong options: more items · poor correlation
"IRT-based, quickest, broadest, reliable PROM?"	→ **PROMIS** ✗ Wrong options: DASH, QuickDASH, FAAM, SF-36
"PROMIS vs legacy in foot & ankle?"	→ **superior at detecting change at 6 and 12 mo**
"ODI difference 4.3, p = 0.002?"	→ **statistically significant, likely below MCID**
"KOOS improvement, p = 0.04 — next?"	→ **identify MCID to judge clinical significance**
"Polytrauma with Lisfranc injury vs without at 12 mo?"	→ **lower SF-36**
"Limitation of SF-36?"	→ **ceiling and floor effects**
"Benefit of PROMs (WOMAC) over physician-reported?"	→ **better assessment of patient satisfaction**
"NOT a predictor of pain 6 mo after MSK injury?"	→ **male gender**
"↑ Both satisfaction and quality?"	→ **interpreter**
"NOT a knee functional score?"	→ **DASH (TO)** ✗ Wrong options: WOMAC, KSS, KOOS
"WOMAC is used for which joint?"	→ **knee** (TO key; hip + knee OA) **(TO)**

– – Active space – –

PECTORALIS MAJOR

– Basic Science · 77 / 108 –

1 Basics

Anatomy

	Pectoralis major
Origin	**Clavicular head** – anterior medial half of clavicle · **sternocostal head** – sternum, superior 6 costal cartilages, external oblique aponeurosis
Insertion	**Lateral lip of intertubercular (bicipital) groove**
Action	**Adducts + medially rotates** humerus · draws scapula anteriorly/inferiorly · clavicular head flexes, sternocostal head extends
Innervation	**Lateral + medial pectoral nerves (C5–T1)**
Arterial supply	Pectoral branch of **thoracoacromial trunk**



Pectoralis major

Humeral insertion

Insertion relative to long head of biceps	Muscle
Just lateral	✓ **Pectoralis major**
Medial (floor / medial lip)	**Teres major, latissimus dorsi**
Lesser tuberosity, medial	**Subscapularis**
✗ Wrong option: humerus	Short head of biceps – **originates on coracoid**

How it's asked

"Tendon inserting just lateral to long head of biceps?" → **pectoralis major**
✗ Wrong options: teres major, latissimus dorsi (medial) · subscapularis (lesser tuberosity)

PERONEUS TERTIUS

– Basic Science · 78 / 108 –

1 Basics

Anatomy

	Peroneus tertius
Origin	With EDL from **distal third of extensor (medial) fibular surface** + anterior intermuscular septum
Insertion	**Dorsal base of 5th metatarsal** (± fascia over 4th interosseous space)
Action	✓ **Dorsiflexion + eversion** (with EDL; also abducts foot)
Innervation	**Deep peroneal nerve (L5, S1)** · anterior compartment
Arterial supply	Anterior tibial artery



Peroneus tertius

Function

Absent in **10%** of population · unique to humans · proximally migrated deep extensor of 5th toe

Clinical use **arthroscopic portal landmark** · insertion point for tendon transfers to assist dorsiflexion

How it's asked

"Function of peroneus tertius (portal landmark)?" → **dorsiflexion and eversion**
✗ Wrong options: dorsiflexion alone · eversion alone · 5th / lesser toe extension

– – Active space – –

PIRIFORMIS

– Basic Science · 79 / 108 –

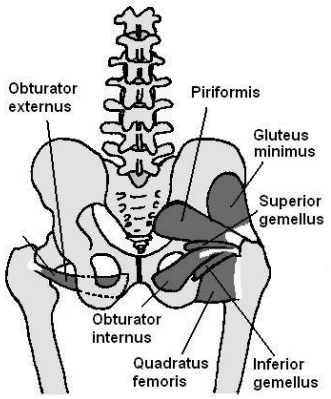
1 Basics

Anatomy & gluteal nerves

	Piriformis
Origin	**Anterior (ventral) surface of sacrum** (3 digitations) + margin of greater sciatic notch · lesser: SI capsule, sacrotuberous ligament
Insertion	**Superior border / apex of greater trochanter** (piriformis fossa)
Action	**External rotator** · abducts flexed hip
Innervation	**Nerve to piriformis (L5, S1, S2)**
Arterial supply	Superior + inferior gluteal, internal pudendal arteries

Only short rotator from ventral sacrum
✓ piriformis
✗ Wrong options: obturator internus/externus (pubic ramus, obturator membrane) · quadratus femoris (near ischial tuberosity) · superior gemellus (near acetabulum)

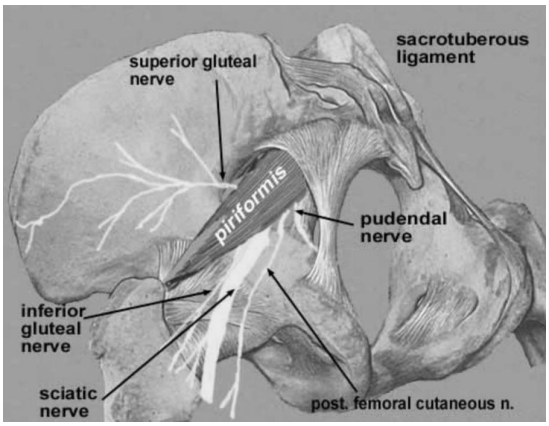
Sacral plexus forms on its **anterior surface**



Posterior hip muscles – piriformis

Gluteal nerves named by relation to
✓ **piriformis**
✗ Wrong options: gluteus maximus · sciatic nerve · sacrospinous ligament

Nerve	Roots	Exit	Supplies
Superior gluteal	L4, L5, S1	**Above piriformis**	Gluteus medius, minimus, **TFL** (runs between medius + minimus)
Inferior gluteal	L5, S1, S2	**Below piriformis**	**Gluteus maximus** (sole motor)



Gluteal nerves and piriformis

Greater sciatic foramen contents (TO)

Greater sciatic foramen notch closed by **sacrotuberous + sacrospinous** ligaments · piriformis fills upper part, divides into two **(TO)**

Compartment (TO)	Vessels	Nerves
Above piriformis	**Superior gluteal vessels**	✓ **Superior gluteal nerve**
Below piriformis	Inferior gluteal · **internal pudendal**	Inferior gluteal · **pudendal** · **sciatic** · **posterior femoral cutaneous** · nerve to obturator internus · nerve to quadratus femoris

Lesser sciatic foramen (TO) **obturator internus tendon** · nerve to obturator internus · internal pudendal vessels · pudendal nerve **(TO)**

✗ Wrong options: **Through sciatic notch** – ✓ **obturator vessels** (obturator canal) **(TO)**

How it's asked

"Superior and inferior gluteal nerves named for relation to?"	→ **piriformis**
"Muscle originating from ventral sacrum?"	→ **piriformis** ✗ Wrong options: obturator internus/ externus, quadratus femoris, superior gemellus
"Through greater sciatic foramen above piriformis?"	→ **superior gluteal nerve (TO)** ✗ Wrong options: internal pudendal, posterior femoral cutaneous, nerve to obturator internus
"Structure NOT passing through sciatic notch?"	→ **obturator vessels (TO)**

– – Active space – –

PLATELET-RICH PLASMA

– Basic Science · 80 / 108 –

1 Basics

Definition & biology

PRP	plasma from own blood enriched with **autologous platelets** · controversial use to stimulate bone + soft-tissue healing
Optimal concentration	**3–5× whole blood** · **>5× inhibits healing** · not 10–15× (OB) · therapeutic working definition **≥1 million platelets/mL (TO)**
Yield (TO)	**30 mL** venous blood → **3–5 mL** PRP **(TO)**
Proposed actions	↑ ECM deposition · ↓ pro-apoptotic signals · ↓ joint inflammation · acts via **growth factors** ✗ Wrong options: aggrecan / structural polypeptide production
Knee OA mechanism	✓ **↓ COX-2 and NF-κB** + **↑ hyaluronic acid by synoviocytes** · ↓ cartilage degradation · ↑ chondrocyte proliferation ✗ Wrong options: NSAIDs (no HA ↑) · steroid (nuclear receptor) · HA injection (rooster comb/bacterial lubricant)

Genes & lab markers

Growth factors released	**PDGF** · **TGF-β** · **VEGF** · **IGF-1** · **EGF** · **CTGF** · **FGF-2** · >30 bioactive proteins; 7 fundamental GFs **(TO)**
Cell adhesion molecules (TO)	**fibrin, fibronectin, vitronectin (TO)**

Platelet granule (TO)	Contents
α-granules (unique to platelets, most abundant, **50–80/platelet**)	✓ **Growth factors** (VEGF, FGF) · αIIbβ3, P-selectin · PF4, fibrinogen · vWF, factor V · chemokines (CXCL1, IL-8) · >300 proteins
Dense (δ) granules (2nd)	**Serotonin, histamine** · adenine nucleotides · polyphosphates · **calcium**

Degranulation (TO)	secretory proteins activated (histones, carbohydrate side chains) → bind receptors on MSCs, osteoblasts, fibroblasts, endothelial + epidermal cells **(TO)**
pH dependence (TO)	✓ GF effect **IS pH dependent** · PDGF highest at **pH 5.1** · TGF-β lower at pH 5.0 than 7.1/7.6 (Liu) **(TO)**
Release kinetics (TO)	**~70% of GFs in first 10 min** after activation · **~100% within 1 h** · small amounts over platelet lifespan (**8–10 days**) **(TO)**

4 Diagnosis & Classification

Types of PRP (TO)

Type (TO)	Platelets / leukocytes	Notes
Pure PRP / leukocyte-poor (LP-PRP)	High / minimal	Tendon, cartilage — less inflammation · neutrophils ~**100 cells/mL**
Leukocyte-rich (LR-PRP / L-PRP)	High / high	Acute inflammatory response · **↑ IL-1β, TNF-α, MMPs** antagonise anabolic effect · neutrophils ~**5000 cells/mL** · most commercial products
Platelet-rich fibrin (PRF)	–	**No anticoagulant** → fibrin scaffold → slow sustained GF release
Autologous conditioned plasma (ACP)	✓ **High platelet, low leukocyte**	Leukocyte-poor form · tendon, cartilage, OA
PRGF	Balanced GFs, minimal leukocytes	Controlled healing

Preparation (TO)

Anticoagulant
✓ **citrate-based**: **acid citrate dextrose (ACD) A/B** · CTAD · sodium citrate → prevent platelet activation, preserve function
✗ Wrong options: **EDTA damages platelet membrane** (EDTA still listed as used by one TO source) · heparin · hirudin **(TO)**

Differential centrifugation
sediments constituents by specific gravity · **soft spin 1500 rpm × 5 min → hard spin 6000 rpm × 15 min (TO)**

Method (TO)	Steps	PRP location
PRP method	ACD tubes · **don't chill** · soft spin → transfer supernatant → hard spin	**Lower 1/3 = PRP**, upper 2/3 = PPP · platelet pellet at bottom → resuspend in 2–4 mL plasma
Buffy coat method	Store **20–24 °C** · **high-speed** spin first · remove supernatant · low spin / leucocyte filter to remove WBC	Bottom **RBC** · ✓ **middle = platelets + WBC (buffy coat)** · top **PPP**

Activation
✓ **calcium chloride** (Orthobullets/OB3629 key) · **thrombin + calcium chloride** both initiate activation (OB, TO)
✗ Wrong options: citrate dextrose (= anticoagulant) · TGF-β, IL-8, PDGF (= released products)

Unanswered TO item (TO24050608)
"agent commonly used to activate platelets?" · options TGF-α / IL-2 / **calcium chloride** / **thrombin** · **answer key not provided by source** · sources support both $CaCl_2$ and thrombin — read the options

Point of care
in-office / in-OR devices, **minutes**
✗ Wrong options: 24-h lab processing

Dosing (TO)
up to **3 injections, 3–6 wk apart**, **5–10 mL (TO)**

– – Active space – –

Regulation

FDA — PRP + preparation devices **"cleared" (510(k)), not "approved"** · not a drug · **not FDA-approved for OA (TO)** · blood products exempt from HCT/P pathway (21 CFR 1271) → not regulated like other cellular/tissue products (OB) · one TO table labels PRP "FDA approved" vs ACS not – read the options

5 Management

Clinical evidence

Indication	Evidence
Lateral epicondylosis (tennis elbow)	✓ **Potential benefit** – PROs + pain
BPTB donor sites (patellar + tibial plug)	✓ **Improved PROs of knee function**, ↓ patellar tendon gap
ACL graft ligamentisation / maturation	Not supported
Rotator cuff repair augmentation	**No added benefit** (possibly detrimental) · safe
Meniscal repair	No clear evidence
Achilles tendinopathy (midsubstance/insertional)	Not supported
Acute ligament/tendon/muscle injury, chronic tendinopathy	No consensus
Osteoarthritis	No consensus, lack long-term follow-up
Fracture healing / fusion	Limited evidence, some detrimental

TO indication list — knee OA · partial tears (RC, hamstring, quadriceps, common extensor/flexor origin, triceps) · Achilles + chronic tendinosis · meniscus grade 1–2 ± parameniscal cyst · non-union / delayed union · partial ACL / ATFL tears **(TO)**

Orthokine – autologous conditioned serum (ACS) (TO)

Orthokine — blood incubated with **pyrogen-free glass spheres** at **37 °C × 24 h** → leukocytes stimulated → ✓ **↑ IL-1 receptor antagonist (IL-1Ra)** + IGF-1, PDGF, TGF-β1 → cell-free serum injected
✗ Wrong options: platelet concentration · stem cells · steroid **(TO)**

Process (TO) — 10–60 mL blood · spin 3000–3500 rpm · serum stored **−20 °C** · injected once–twice weekly · hypoxia-like (HIF-1α) response, not EPO **(TO)**

WADA (TO) — not banned; may need **TUE** (M1/M2 blood manipulation categories) **(TO)**

ACS (Orthokine) vs PRP (TO)

	ACS	PRP
Source	Whole blood + glass spheres, incubated	Centrifuged whole blood
Mechanism	↑ **IL-1Ra** (anti-inflammatory)	Platelet GFs (PDGF, TGF-β, VEGF)
Best for	OA, lumbar stenosis, disc prolapse, muscle injury	Tendon, ligament, cartilage, post-surgical
Processing	**24 h** incubation	**15–30 min**
Injections	Fewer	Multiple over weeks
FDA (per TO table)	Not approved	"Approved" (✗ Wrong options: per other TO key)

How it's asked

"Ideal anticoagulant for PRP?"	→ **acid citrate dextrose (TO)** ✗ Wrong options: EDTA, heparin, hirudin
"PRP – all true EXCEPT?"	→ **use in OA is FDA approved** (✓ α-granules, citrate, 1 million/mL) **(TO)**
"Buffy coat method – layer with PRP?"	→ **middle layer (TO)**
"PRP – all true EXCEPT?"	→ **GF effect NOT pH dependent** (false – it is pH dependent) **(TO)**
"Autologous conditioned plasma has?"	→ **high platelet, low leukocyte (TO)**
"Orthokine therapy – true?"	→ **↑ IL-1Ra to block inflammation (TO)**
"Agent used to activate platelets in prepared PRP?"	→ **calcium chloride** ✗ Wrong options: citrate dextrose, TGF-β, IL-8, PDGF
"PRP in acute rotator cuff repair – true?"	→ **may give no added benefit** ✗ Wrong options: 10–15× concentration · 24-h processing · regulated like HCT/Ps
"Attenuates COX-2 + NF-κB, ↑ HA production in knee OA?"	→ **PRP** ✗ Wrong options: NSAIDs, steroids, HA
"PRP potential benefit?"	→ **tennis elbow, BPTB donor sites**

– – Active space – –

POPLITEUS

– Basic Science · 81 / 108 –

1 Basics

Popliteus anatomy

	Popliteus
Origin	**Lateral femoral condyle** – anterior popliteal groove (main)
Insertion	**Posterior (posteromedial) tibia**, fan-like, just above popliteal (soleal) line
Action	**Internally rotates tibia on femur → unlocks knee** at start of flexion
Innervation	**Tibial nerve (L4, L5, S1)**
Arterial supply	Medial inferior genicular (popliteal) + muscular branch of posterior tibial artery



Popliteus

Tripartite proximal tendon (TO)

✓ **tripartite** · main attachment **lateral femoral condyle** · + **head of fibula** · + **posterior lateral meniscus (TO)**
✗ Wrong options: bipartite · medial femoral condyle

Popliteus action (TO)	Foot position
External rotation of femur	Foot plantigrade (weight-bearing)
Internal rotation of tibia	Foot free
"Unlocking of the knee"	Both

5 Management

Posteromedial exposure

Extended posteromedial knee approach

interval **medial gastrocnemius ↔ posterior semimembranosus** → posteromedial capsule (inside-out meniscal repair)

Deeper exposure	posteromedial tibia / **PCL fossa** → ✓ **subperiosteal elevation of popliteus** off posterior tibia ✗ Wrong options: plantaris, semitendinosus, gracilis (superficial) · soleus (distal)
Uses	meniscal repair · tibial plateau ORIF · **PCL tibial avulsion** fixation · PCL tibial inlay · Baker's cyst · posterior capsular release
At risk	superficial **saphenous nerve + vein** · popliteal artery branches (inferior medial genicular often spared)

– – Active space – –

How it's asked

"True about popliteus?"	→ **tripartite, main proximal attachment lateral femoral condyle (TO)**
"Extended posteromedial approach — dissect what for PCL fossa / posterior tibia?"	→ **popliteus** ✗ Wrong options: plantaris, soleus

– – Active space – –

POSTERIOR INTEROSSEOUS NERVE

– Basic Science · 82 / 108 –

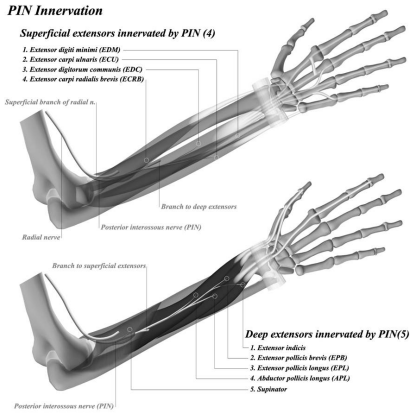
1 Basics

Anatomy & injury

Course middle + lower trunks → **posterior cord** → radial nerve → divides at elbow into **superficial radial nerve (SBRN, sensory)** + **PIN (motor)**

PIN motor supply	Muscles
Superficial extensors (4)	**EDM · ECU · EDC · ECRB**
Deep extensors (5)	**Extensor indicis · EPB · EPL · APL · supinator**
✗ Wrong option: PIN (radial nerve proper)	**ECRL**

Sensory **dorsal wrist capsule** only (no skin)



PIN innervation – superficial and deep extensors

Wrist – 4th dorsal compartment

Dorsal compartment	Contents
1	APL, EPB
2	ECRL, ECRB
3	EPL
4	**EDC, EIP, terminal PIN**
5	EDM
6	ECU

3 Presentation

PIN palsy pattern (✗ Wrong options: MCP extension of thumb + fingers) · ✓ **wrist extension with radial deviation** (ECRL intact, ECRB/ECU out) · ✓ sensation radial forearm + dorsal thumb (SBRN intact) → lesion **distal to SBRN/PIN split**

Tenodesis intact	passive wrist flexion → digits extend → ✓ **tendons in continuity** → nerve lesion, not tendon

– – Active space – –

5 Management

Proximal forearm stab, can't extend fingers/thumb, tenodesis intact	✓ **explore + repair PIN** ✗ Wrong options: NCS/EMG first · splint · tendon repair · observation
Dorsal Thompson approach	identify / protect PIN · **pronation** moves PIN away · radial nerve splits proximal to the approach · median / AIN / musculocutaneous not encountered

Radial nerve vs PIN palsy

	Radial nerve (proximal)	PIN
Wrist extension	No (wrist drop)	✓ **with radial deviation**
Finger/thumb MCP extension	No	No
Dorsal radial sensation	No	✓ intact

How it's asked

"Tendon in same dorsal compartment as PIN?"	→ **extensor indicis proprius** ✗ Wrong options: EDM 5th, ECRB 2nd, EPL 3rd, APL 1st
"Stab to proximal forearm, can't extend digits, tenodesis intact — management?"	→ **explore and repair PIN**
"After dorsal Thompson approach — no thumb/index MCP extension, wrist extends radially, sensation intact?"	→ **posterior interosseous nerve** ✗ Wrong options: radial · AIN

– – Active space – –

PROPHYLACTIC ANTIBIOTICS

– Basic Science · 83 / 108 –

1 Basics

Overview

Share of antibiotic use	**25–50%** of all antibiotics used are prophylactic
Indications	**implanted foreign body** (e.g. IM nail – not "no antibiotics needed") · bone grafting · large dissection / dead space / haematoma · significant blood loss
Not needed	**carpal tunnel release** · diagnostic arthroscopy controversial
Commonest pathogens	**S. aureus** · **S. epidermidis** · aerobic streptococci · anaerobic cocci · gram-positive skin flora
Most effective SSI measure	✓ **IV antibiotics within 1 h before incision** · lumbar spine: no iodine vs chlorhexidine difference, no benefit from >10 min scrub, laminar flow helps TJA but not shown in spine ✗ Wrong options: MRSA screening/decolonisation · horizontal laminar flow · body exhaust suits

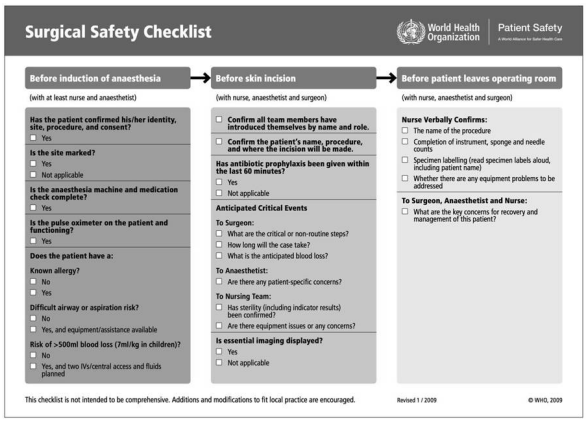
5 Management

Timing before incision

Agent	Timing
Cefazolin	✓ **Within 1 h before incision** · (✗ Wrong options: after incision) · **before tourniquet inflation**
Vancomycin	**Within 2 h** before incision
Redosing	**3–4 h** after first dose for long cases (OB8335) · 2017 CDC: insufficient evidence on intra-op redosing

WHO Surgical Safety Checklist	3 checkpoints: **before induction · before skin incision · before leaving OR** · ✓ **antibiotics given within 60 min before incision** ✗ Wrong options: blood products on standby · VTE prophylaxis ordered · implant rep · credentialing

– – Active space – –


Surgical Safety Checklist

World Health Organization | Patient Safety

Before induction of anaesthesia
(with at least nurse and anaesthetist)

Has the patient confirmed his/her identity, site, procedure, and consent?
☐ Yes

Is the site marked?
☐ Yes
☐ Not applicable

Is the anaesthesia machine and medication check complete?
☐ Yes

Is the pulse oximeter on the patient and functioning?
☐ Yes

Does the patient have a:

Known allergy?
☐ No
☐ Yes

Difficult airway or aspiration risk?
☐ No
☐ Yes, and equipment/assistance available

Risk of >500ml blood loss (7ml/kg in children)?
☐ No
☐ Yes, and two IVs/central access and fluids planned

Before skin incision
(with nurse, anaesthetist and surgeon)

☐ Confirm all team members have introduced themselves by name and role.
☐ Confirm the patient's name, procedure, and where the incision will be made.

Has antibiotic prophylaxis been given within the last 60 minutes?
☐ Yes
☐ Not applicable

Anticipated Critical Events

To Surgeon:
☐ What are the critical or non-routine steps?
☐ How long will the case take?
☐ What is the anticipated blood loss?

To Anaesthetist:
☐ Are there any patient-specific concerns?

To Nursing Team:
☐ Has sterility (including indicator results) been confirmed?
☐ Are there equipment issues or any concerns?

Is essential imaging displayed?
☐ Yes
☐ Not applicable

Before patient leaves operating room
(with nurse, anaesthetist and surgeon)

Nurse Verbally Confirms:
☐ The name of the procedure
☐ Completion of instrument, sponge and needle counts
☐ Specimen labelling (read specimen labels aloud, including patient name)
☐ Whether there are any equipment problems to be addressed

To Surgeon, Anaesthetist and Nurse:
☐ What are the key concerns for recovery and management of this patient?

This checklist is not intended to be comprehensive. Additions and modifications to fit local practice are encouraged.

Revised 1 / 2009

© WHO, 2009

WHO Surgical Safety Checklist

Agent selection & allergy (TO)

Situation	Agent
No β-lactam allergy	✓ **Cefazolin** (or cefuroxime) — **1st-generation cephalosporin** (S. aureus cover, low toxicity, penetration, half-life, cost) **(TO)**
Dose	**Double cefazolin if >80 kg**
Penicillin / β-lactam allergy	✓ **Vancomycin** (TO key) or clindamycin **(TO)**
MRSA history / high-prevalence area	**Vancomycin + cefazolin** · judicious — ↑ **VRE** risk
β-lactam allergy + MRSA risk	Vancomycin ± **gentamicin or aztreonam**
Trauma IM nail, no allergy, no MRSA	✓ **Cefazolin** (MRSA colonisation in trauma only **1–4%**) · (✗ Wrong options: vancomycin · cipro · metronidazole)

Antibiotic class	Mechanism
Vancomycin (glycopeptide)	✓ **Bactericidal, blocks peptidoglycan cross-linkage · red man syndrome**
Cephalosporin	Bactericidal, peptidoglycan cross-linkage, no red man
Aminoglycoside	Bactericidal, irreversible **30S**
Macrolide / clindamycin	Bacteriostatic, reversible **50S**
Fluoroquinolone	Bactericidal, **DNA gyrase**

Other prophylaxis ciprofloxacin: GU surgery, brackish/salt/fresh water wounds · metronidazole: GI surgery, soil/faecal contamination

CDC guidelines — duration

AAOS (TJA) prophylaxis **should NOT exceed 24 h** after surgery

CDC 2017 (Category IA) ✓ **clean + clean-contaminated: no additional doses after closure in OR, even with a drain** → single perioperative dose sufficient
✗ Wrong options: continue until drains out · 24 h for drains · duration by length of surgery

CDC 2002 start within 1 h · appropriate agent · stop within **24 h**

– – Active space – –

Dental prophylaxis in TJA patients (AAOS & ADA)

At-risk TJA patients
All patients for first 2 yr after TJA
Immunocompromised – drug- or radiation-induced
Inflammatory arthropathy (RA, SLE)
Previous PJI · **type I DM** · malnourishment · haemophilia · HIV · malignancy

Evidence **limited / inconclusive** → clinical judgment

Dental regimen (1 h before)	Drug
No penicillin allergy	**Amoxicillin or cephalexin 2 g PO** · IV/IM: cefazolin 1 g or ampicillin 2 g
Penicillin allergy	**Clindamycin 600 mg** PO (or IV)

Splenectomy prophylaxis

Vaccines **pneumococcal** (all, incl. functional hyposplenism) · **Hib** + **meningococcal C conjugate** (if not immunised) · **influenza annually**

Antibiotics **lifelong** oral phenoxymethylpenicillin or erythromycin

Sterilisation & disinfection (TO)

Sterilisation vs disinfection ✓ **endospores**: disinfection leaves spores · sterilisation destroys spores + pathogens **(TO)**

Arthroscopic equipment ✓ **plasma sterilisation (TO)**
✗ Wrong options: autoclave (damages delicate scopes) · ETO (**12-h cycle**, can't turn around same day) · formaldehyde dip

Plasma systems (TO) **Sterrad-100**: vaporised **H_2O_2** (45 min) + **radiofrequency** (13.56 MHz, 300 W, 0.5 Torr) · **Plazlyte**: **peracetic acid** vapour + microwave plasma (O_2/H_2/Ar) **(TO)**

Spaulding classification (1957) ✓ **principles of disinfection and sterilisation** of reusable devices by risk **(TO)**
✗ Wrong options: skull, pelvic fractures, DDH

Spaulding (TO)	Definition	Examples	Reprocessing
Non-critical	Intact skin only	BP cuffs, stethoscopes, toys, walking aids	Clean ± **low-level disinfection**
Semi-critical	Non-intact skin / intact mucosa, no penetration	Trans-rectal probes, specula	**High-level disinfection** minimum, sterilisation preferred
Critical	Penetrate tissue / bloodstream / sterile area	**Surgical + dental instruments**	**Sterilisation** after every use

Semi-critical touching open lesions treat as **critical (TO)**

Methods (TO)	Agents
Disinfection — oxidising	Sodium hypochlorite · povidone iodine · H_2O_2 · peracetic acid
Disinfection — non-oxidising	Chlorhexidine · **70% alcohol** · quaternary ammonium · **glutaraldehyde**, OPA
Sterilisation	**Autoclave** · **ETO** · H_2O_2 / peracetic acid vapour or **plasma**

How it's asked

"On WHO Safe Surgery Checklist?"	→ **antibiotics within 60 min before incision**
"Most reduces infection after lumbar spine surgery?"	→ **IV antibiotics before incision** ✗ Wrong options: chlorhexidine vs iodine · laminar flow
"Optimal antibiotic timing for TJA?"	→ **within 1 h before incision** ✗ Wrong options: within 2 h · after incision
"Most notable impact on SSI?"	→ **IV antibiotics within 1 h of incision** ✗ Wrong options: MRSA decolonisation · laminar flow · exhaust suits
"Cephalosporin generation used before THR?"	→ **1st (TO)**
"Penicillin allergy — preferred prophylaxis?"	→ **vancomycin (TO)** ✗ Wrong options: cefazolin
"Cefazolin anaphylaxis + prior MRSA, THA — antibiotic?"	→ **bactericidal, peptidoglycan cross-link, red man syndrome** (vancomycin)
"Femur IMN, no allergy, no MRSA?"	→ **cefazolin** ✗ Wrong options: none · vancomycin
"2017 CDC strong recommendation?"	→ **clean-contaminated: single perioperative dose sufficient** ✗ Wrong options: continue with drains
"Most effective sterilisation of arthroscopic equipment?"	→ **plasma sterilisation (TO)** ✗ Wrong options: autoclave · ETO
"Spaulding classification deals with?"	→ **disinfection and sterilisation (TO)**

– – Active space – –

PSEUDOGOUT (CPPD)

– Basic Science · 84 / 108 –

1 Basics

Definition inflammatory arthritis from intra-articular **calcium pyrophosphate dihydrate (CPPD)** crystal deposition

Demographics **elderly (>60)** · rare in young unless associated disease

vs gout **older** · **more proximal joints** · **positively** birefringent crystals

Associated conditions

Associations
Hyperparathyroidism
Haemochromatosis
Wilson's disease
Gout · RA · SLE
Haemophilia
Long-term **haemodialysis** (pyrophosphate-like deposition)

Chondrocalcinosis present in **7%** · high association with knee **OA** (relationship unclear)

3 Presentation

Acute attack warm, erythematous, tender, **monoarticular** · mimics gout + septic arthritis · ± superficial mineral deposits

Joint frequency	
1st	✓ **Knee**
2nd	**Wrist**
3rd	2nd + 3rd MCP joints

Course can cause rapidly progressive joint destruction · inflammatory markers normal / minimally ↑

4 Diagnosis & Classification

Key radiology findings

Chondrocalcinosis ✓ **linear calcification of fibrocartilage + articular cartilage** · **menisci (knee)** · **TFCC (wrist)** · distinguishes CPPD from other secondary OA (gout, haemochromatosis, RA, ochronosis)



Chondrocalcinosis of the knee menisci

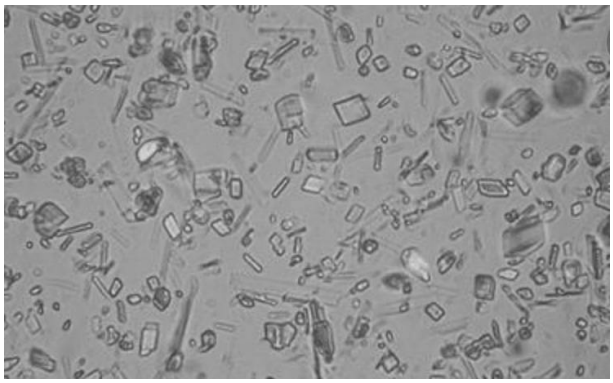


TFCC chondrocalcinosis

Diagnostic criteria — aspiration

Crystals

✓ **weakly positively birefringent, rhomboid-shaped** · WBC moderate (**<50,000**) · culture negative



Weakly positive birefringent rhomboid crystals

5 Management

Phase	Treatment
Acute	**NSAIDs** · splint (comfort) · **intra-articular steroid** · rest
Chronic / recurrent	**Colchicine 0.6 mg PO bid** (prophylaxis ↓ recurrence) · **intra-articular yttrium-90**

– – Active space – –

6 Complications

Complications permanent joint damage · renal disease

CPPD vs differentials

Finding	Disease
CPPD crystals, chondrocalcinosis	✓ **Pseudogout**
Monosodium urate, **strongly negative needle** crystals, punched-out erosions	Gout.....155
Destructive **pannus**, symmetric narrowing, osteopenia	RA
Empty osteocyte lacunae	Osteonecrosis
Recurrent haemarthroses, wide notch, big condyles	Haemophilia
Iron deposition (cirrhosis, pigmentation, DM, cardiac)	Haemochromatosis
Homogentisic acid (homogentisic acid oxidase deficiency)	Ochronosis.....241

How it's asked

"Knee pain, hyperparathyroidism, meniscal calcification – finding?"	→ **CPPD crystals** ✗ Wrong options: urate · pannus · empty lacunae · haemarthroses
"Linear cartilage calcification on knee AP?"	→ **chondrocalcinosis (pseudogout)** ✗ Wrong options: gout, haemochromatosis, ochronosis
"TFCC calcification – other joint most involved?"	→ **knee** ✗ Wrong options: 1st MTP

PSOAS

– Basic Science · 85 / 108 –

1 Basics

Anatomy

	Psoas
Origin	Transverse processes **L1–L5** + bodies and discs **T12–L5**
Insertion	**Lesser trochanter**
Action	Flexes torso and thigh on each other
Innervation	**Direct fibres of lumbar plexus (L1–L3)**
Arterial supply	Lumbar branch of iliopsoas branch of internal iliac artery



Iliopsoas

Nerves at risk in lumbar approaches

Nerve	Relation to psoas
Genitofemoral (L1–L2)	✓ **Emerges through psoas at L3–L4 → runs on anteromedial surface** – met in anterior retroperitoneal approach
Ilioinguinal, iliohypogastric, LFCN	**Posterior** to psoas → on quadratus lumborum / iliacus
Femoral	**Posterior** to psoas → wraps laterally onto iliopsoas distally

Lumbar plexus in transpsoas (lateral) approach

✓ **more dorsal at L1–2, more ventral at L4–5**
✗ Wrong options: constant position

How it's asked

"Anterior retroperitoneal approach – nerve on anteromedial psoas?"

→ **genitofemoral**
✗ Wrong options: ilioinguinal, femoral, LFCN, iliohypogastric

– – Active space – –

"Lumbar plexus position in psoas?"	**→ dorsal in upper lumbar, ventral in lower lumbar**

PSORIATIC ARTHRITIS

– Basic Science · 86 / 108 –

1 Basics

Definition	**seronegative spondyloarthropathy** · DIP-predominant hand arthritis · silvery plaques · uveitis · nail dystrophy
Incidence	**5–20%** of psoriasis patients (Orthobullets, OB210216) · 5–10% (OB3340) · ~20% (OB3581) – read the options ✗ Wrong options: "most psoriasis patients develop arthritis"
Sex	M = F
Skin first	plaques precede arthritis by years in **80–85%** · arthritis first in **~20% (TO)**

Genes & lab markers

HLA-B27	positive in **~50%** ✗ Wrong options: all patients
RF and ANA	usually **negative**

Patterns (Wright & Moll) (TO)

Pattern (TO)	Features
Symmetrical polyarthritis	✓ **Most common** · resembles RA but milder · RF-negative **(TO)**
Asymmetrical oligoarthritis	**<5 joints**, small/medium · **2nd commonest** (DIP, PIP, MCP) **(TO)**
Distal (DIP-predominant)	DIP + PIP · **acro-osteolysis** of tufts + nail changes
Arthritis mutilans	Rare · extreme phalangeal + MCP destruction → telescoping · **opera-glass hand (doigt / main en lorgnette)** · pencil-in-cup
Spondyloarthropathy	~1/3 · SI (usually **unilateral**) + spine · **cervical > thoracolumbar** (distinguishes from AS) **(TO)**

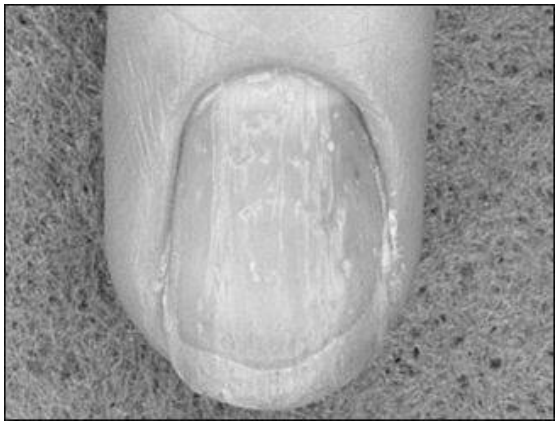
3 Presentation

Skin	**silvery plaques** on extensor surfaces (elbows, knees) · scalp
Hands	**dactylitis (sausage digit)** – up to **50%**, toe > finger **(TO)** · **nail pitting (onychodystrophy)** · **onycholysis** (distal nail-plate lifting) · opera-glass hands
Opera-glass hand	shortened phalanges → excess skin folds transversely, digits retract like telescoping binoculars
Enthesitis	up to **40% (TO)** · **Achilles**, **posterior tibial tendonitis**, **plantar fasciitis** · heel, foot, hip pain
Other	chronic **uveitis** · ankylosis of small hand/foot joints more than RA **(TO)**

– – Active space – –



Dactylitis



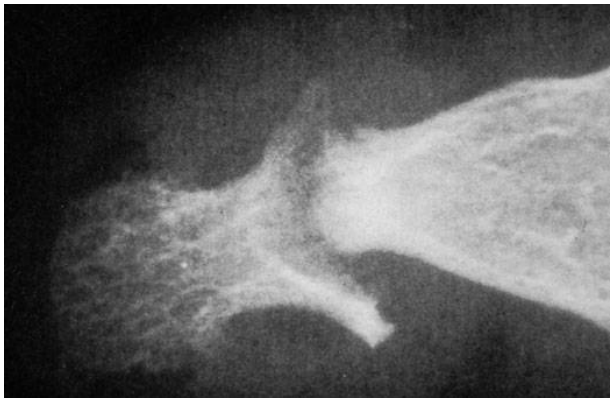
Nail pitting

4 Diagnosis & Classification

Key radiology findings

Pencil-in-cup

✓ **DIP**: destruction of middle phalanx head + expansion of distal phalanx base · **centripetal erosions → joint-space widening**
✗ Wrong options: DJD



Pencil-in-cup deformity

Other hand signs

acro-osteolysis (distal tuft resorption) · **fluffy periostitis** (periosteal ossification) · small-joint erosions or **fusions** (PIP, MCP, wrist) · flail digits

Spine

sacroiliitis · syndesmophytes · paravertebral ossification · destructive discovertebral lesions

Diagnostic criteria – CASPAR

Requirement inflammatory articular disease (joint, spine, entheseal) + **≥3 points**

CASPAR criterion	Points
Current psoriasis	**2**
Personal or family history of psoriasis	1
Psoriatic nail dystrophy (onycholysis, pitting, hyperkeratosis)	1
Negative RF	1
Dactylitis (current or recorded)	1
Juxta-articular new bone formation on hand/foot X-ray (✗ Wrong options: osteophytes)	1

Unanswered TO item (TO24057301) clinical photo: nail dystrophy, DIP swelling, skin plaques · options psoriatic arthropathy / RA / OA / reactive arthritis · **answer key not provided by source**

5 Management

Medical (mainstay, like RA) NSAIDs · methotrexate · sulfasalazine · cyclosporine · **TNF-α inhibitors** · rheumatology referral

Treatment – TNF inhibitors

Failed NSAIDs, active disease ✓ **biologic DMARD – TNF inhibitor** (or IL-17, IL-12/23) = ACR first line → ↓ joint destruction + surgery

✗ Wrong option	Why not
NSAIDs	Symptomatic only, not disease-modifying
Weight loss / bariatric	does not treat inflammation
Hydroxychloroquine	not approved · **precipitates / worsens psoriasis**
Oral prednisone	Weak disease modification · **pustular flare on withdrawal**
csDMARDs (MTX, leflunomide, SSZ)	Limited effect on joint damage + skin

Operative advanced disease → **digit fusion vs resection arthroplasty** · osteotomy, arthroplasty · **high infection rates** reported

– – Active space – –

Psoriatic vs other hand arthritis

	Psoriatic	RA	Erosive OA	Gout
Joints	**DIP**	PIP, **MCP (2nd–3rd)**, ulnar styloid, triquetrum · no DIP	DIP, PIP, 1st CMC	Any
Erosions	**Centripetal**, pencil-in-cup	Marginal, subchondral cysts	**Central "gull wing"**, not marginal	**Punched-out "rat-bite"**, overhanging edges, sclerotic margins
Other	Periostitis, acro-osteolysis, telescoping	Ulnar drift, swan-neck, boutonnière	Osteopenia, fusiform swelling	Tophi
Serology	RF− · HLA-B27 ~50%	RF+	–	–

How it's asked

"Most common form of psoriatic arthritis?"	→ **symmetrical polyarthritis (TO)** ✗ Wrong options: asymmetric oligoarthritis (2nd)
"Sausage digits + pencil-in-cup?"	→ **psoriatic arthritis** ✗ Wrong options: gout · RA
"Toe dactylitis, posterior tibial tendonitis, nail pitting?"	→ **psoriatic arthritis** ✗ Wrong options: scleroderma · pseudogout
"Nail pitting + onycholysis — matching radiograph?"	→ **telescoping / opera-glass destruction** ✗ Wrong options: RA MCP ulnar drift · erosive OA gull-wing
"Psoriatic arthritis — true?"	→ **DIP erosions often centripetal** ✗ Wrong options: most psoriasis patients get arthritis · all HLA-B27+ · RF/ANA positive · antifungal
"Severe PsA after NSAID failure — what would have prevented TKA?"	→ **TNF inhibitor** ✗ Wrong options: hydroxychloroquine · prednisone

PTH & VIT D PHYSIOLOGY

– Basic Science · 87 / 108 –

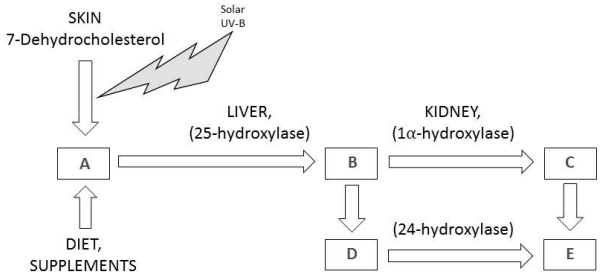
1 Basics

Overview

Organs in Ca homeostasis skin · liver · parathyroid · kidney · bone · small intestine · ↑ PTH + ↑ vit D → ↑ serum Ca

Vitamin D metabolism & serum markers

Step	Site	Product
7-dehydrocholesterol + **UV-B**	**Skin**	Previtamin D_3 → **cholecalciferol (D_3)**
Diet / supplements	Gut	D_2 (**ergocalciferol** – vegetables, supplements) · D_3 (oily fish, fortified foods) – both inert
Carried on **vitamin D-binding protein** → **25-hydroxylase**	**Liver**	**25-(OH) D_3 = calcidiol**
1α-hydroxylase (↑ by PTH, ↓ Ca, ↓ PO_4)	**Kidney**	✓ **1,25-$(OH)_2$ D_3 = calcitriol – active (TO)**
24-hydroxylase	Kidney	**24,25-$(OH)_2$ D_3 – inactive** · 1,24,25-$(OH)_3$ D_3



Vitamin D pathway – A D_3, B 25-OH D, C 1,25-$(OH)_2$ D, D 24,25-$(OH)_2$ D, E 1,24,25-$(OH)_3$ D

Self-regulation high 1,25-D → ↑ **24,25-D** (inactive) · negative feedback on itself

Skin conversion ↓ with age · nearly absent in darkly pigmented individuals

Vitamin D actions ↑ intestinal Ca + PO_4 absorption · ↑ bone resorption (Ca + PO_4 release; osteoclastic, TO) → **↑ serum Ca AND ↑ PO_4** · OB219175: vit D (with **FGF23**) **inhibits osteoclast activity** → ↑ bone mass – read the options

– – Active space – –

PTH functions (TO)

Source **chief cells** of parathyroid · trigger **hypocalcaemia / hypomagnesaemia**

PTH action	Effect
Bone	↑ **osteoclastic** resorption – receptor on **osteoblasts** → **IL-1** → osteoclasts
Kidney	↑ **Ca reabsorption (DCT)** · ↓ **phosphate reabsorption**
Kidney	↑ **1α-hydroxylation** of 25-OH D → ↑ 1,25-D
Net	**↑ serum Ca, ↓ serum PO_4**

PTH function – EXCEPT answer (TO) **"↑ osteoblastic resorption of bone"** – no such thing; resorption is osteoclastic **(TO)**

Genes & lab markers

Best test for deficiency ✓ **25-OH vitamin D** – long half-life, circulating level **~1000×** 1,25-D
✗ Wrong options: 1,25-D (estimates stores poorly; only a fraction converted)

Half-life	25-OH D	$1,25\text{-}(OH)_2$ D
OB4653	**2–3 wk**	**4–6 h**
OB210168	**15 days**	**15 h**

Unit conversion 1 nmol/L = 0.4 ng/mL · 1 ng/mL = 2.5 nmol/L

25-OH D level	Status
<30 nmol/L (<12 ng/mL)	At risk of **deficiency** (NIH, OB217497)
30–50 nmol/L (12–20 ng/mL)	Possible inadequacy
≥50 nmol/L (≥20 ng/mL)	Sufficient for most
75 nmol/L (30 ng/mL)	✓ **Normal** (OB217497) · Endocrine Society: **>75 nmol/L (>30 ng/mL)** needed for max bone/muscle effect
<20 ng/mL	**Deficient** (OB219590)
20–30 ng/mL	✓ **Insufficient** (OB219590)
30–80 ng/mL	Normal range (OB7541)

RDA vitamin D_3 **400 IU** infants <1 yr · **600 IU** age 1–70 (incl. pregnancy/lactation) · **800 IU** adults >70

5 Management

– – Active space – –

Supplementation & fracture healing

Vitamin D in bone healing	✓ **regulates inflammatory response** at injury site (macrophages, neutrophils, T cells express **VDR**, convert vit D locally) · ↑ **VEGF**-mediated vasculogenesis · ↑ **MSC commitment to osteoblasts** (matrix adhesion, integrins) · ↑ ECM: **collagen I, osteocalcin, osteopontin** · prevents infection, ↑ muscle/soft tissue
Deficiency	↑ fracture + arthrodesis **nonunion** · ↑ falls in elderly (supplement ↓ falls) · atrophic nonunion work-up → **25-OH D**
Insufficiency fracture + 25-OH D 14 ng/mL, normal Ca	✓ **high-dose vitamin D** ✗ Wrong options: observation (at risk elsewhere) · calcium alone · bisphosphonate before correcting mineral imbalance · surgery
Supplement form	**D_3 preferred** · D_2 most available by prescription (US) · hypervitaminosis D rare, high doses tolerated

Vitamin D vs PTH

	1,25-$(OH)_2$ vit D	**PTH**
Serum Ca	↑	↑
Serum PO_4	**↑**	**↓**
Gut	↑ Ca + PO_4 absorption	Indirect (via 1,25-D)
Kidney	–	↑ Ca reabsorption, ↓ PO_4 reabsorption, ↑ 1α-hydroxylase
Disorders	Rickets / osteomalacia	Hypoparathyroidism · pseudohypoparathyroidism · renal osteodystrophy

How it's asked

"Most active vit D metabolite?"	→ **1,25-dihydroxycholecalciferol (TO)** ✗ Wrong options: 25-OH · 24,25
"Correct regarding vit D?"	→ **24,25-dihydroxycholecalciferol is inactive** ✗ Wrong options: 1,25-D best lab test · 1,25-D longer half-life
"Serum marker after vit D repletion (pathway letter)?"	→ **B – 25-hydroxyvitamin D**
"Post-hip fracture 25-OH D 75 nmol/L (30 ng/mL)?"	→ **normal level**
"Atrophic humeral nonunion – value showing insufficiency?"	→ **25-OH D 25 ng/mL** ✗ Wrong options: 35 / 45 ng/mL · any 1,25-D value
"Sacrococcygeal insufficiency fracture, 25-OH D 14 ng/mL?"	→ **high-dose vitamin D**

– – Active space – –

"Primary function of vit D in osteoporotic fracture healing?"	**→ regulates inflammatory response at injury site** ✗ Wrong options: activates RANKL · ↓ VEGF · inhibits osteocalcin
"NOT a function of PTH?"	**→ ↑ osteoblastic resorption of bone (TO)**

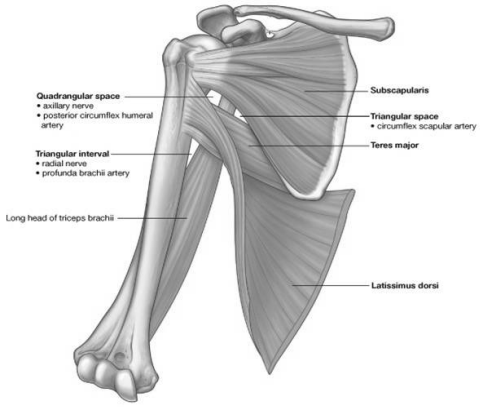
QUADRANGULAR SPACE, TRIANGULAR SPACE, TRIANGULAR INTERVAL

– Basic Science · 88 / 108 –

1 Basics

Contents of shoulder spaces

Space	Borders	Contents
Quadrangular (quadrilateral) space	Medial **long head triceps** · lateral **humeral shaft** · superior **teres minor** · inferior **teres major**	✓ **Axillary nerve + posterior humeral circumflex artery**
Triangular space	Lateral **long head triceps** · superior teres minor · inferior teres major	**Scapular circumflex artery** (no major nerve)
Triangular interval	Medial long head triceps · lateral **lateral head triceps / humerus** · superior **teres major**	**Radial nerve + profunda brachii artery**



Quadrangular space, triangular space and triangular interval

How it's asked

"Correct pairing of shoulder space and contents?"

→ **quadrangular space – axillary nerve + posterior humeral circumflex artery**
✗ Wrong options: quadrangular (radial nerve/profunda) · quadrangular (scapular circumflex) · triangular space (axillary nerve)

– – Active space – –

RADIAL NERVE

– Basic Science · 89 / 108 –

1 Basics

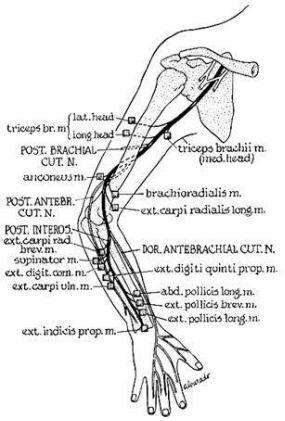
Anatomy

Course

posterior cord → exits **triangular interval** (long head triceps ↔ humerus) with **profunda brachii artery** → **spiral groove** → pierces lateral intermuscular septum **7.5 cm proximal to elbow** → at lateral humeral condyle gives cutaneous branches (superficial radial, posterior cutaneous) → **PIN** through **supinator**, winds around **radial neck**

Accompanying artery

✓ **profunda brachii**
✗ Wrong options: posterior humeral circumflex (axillary nerve/ quadrangular space) · scapular circumflex (triangular space) · brachial (medial with median/ulnar) · lateral thoracic (serratus anterior)



Radial nerve motor branch order

3 Presentation

Injury level

Deficit	Level
✗ Wrong option: wrist, thumb, finger extension (wrist drop)	**Radial nerve** – spiral groove / distal-third humeral shaft
✗ Wrong option: digit + thumb extension · ✓ **wrist extension with radial deviation**	**PIN at radial neck** (✗ Wrong options: spiral groove · midshaft radius · humeral necks)

Holstein-Lewis fracture

distal-third humeral shaft fracture + **radial nerve neuropraxia** · **80–90% recover** with observation → not an indication for surgery

– – Active space – –



Holstein-Lewis fracture

5 Management

Reinnervation order (TO)

Recovery direction **proximal → distal (TO)**

Recovery (TO)	Muscle / function
First muscle	**Brachioradialis**
First function (wrist/finger drop after humeral shaft fracture)	✓ **Wrist extension (radial deviation)** (✗ Wrong options: supination · elbow extension · MCP extension)
Last function	**MCP joint extension**
Last muscle	✓ **Extensor indicis proprius** (✗ Wrong options: brachioradialis, ECRB, APL, EPL)

Critical period of delay (TO)

Critical period time beyond which motor return isn't expected if repair delayed **(TO)**

Nerve (TO)	Critical period
Radial	✓ **15 months**
PIN	9 months
Ulnar – high	9 months
Ulnar – low	15 months
Median – high	9 months
Median – low	12 months
Sciatic	12–15 months
CPN	12 months

How it's asked

"Radial palsy after humeral shaft ORIF, nerve intact – first function to recover?" → **wrist extension (TO)**

– – Active space – –

"Critical period of delay in radial nerve palsy?"	→ **15 months (TO)** ✗ Wrong options: 9, 10, 12
"Last muscle reinnervated in radial palsy?"	→ **extensor indicis proprius**
"GSW — can't extend digits/thumb, wrist extends with radial deviation — level?"	→ **radial neck** (PIN)
"Holstein-Lewis fracture, wrist drop — artery with injured nerve?"	→ **profunda brachii**

– – Active space – –

REHAB & PROSTHETICS

– Basic Science · 90 / 108 –

1 Basics

Principles

Goal	restore function close to original · multidisciplinary team
Upper limb	**limb salvage ideal** (prosthesis lacks sensation) · residual length matters for socket suspension
Lower limb goals	comfortable · easy on/off · light · durable · cosmetic · functional
Most important factor choosing LL prosthesis	✓ **patient functional status** (current + potential) ✗ Wrong options: cost, comorbidities, age, gender

Metabolic cost of amputation

Amputation	↑ Metabolic cost of walking
Syme's	**15%**
Traumatic transtibial	✓ **25%** – least of the options
Vascular transtibial	40%
Traumatic transfemoral	68%
Vascular transfemoral	100%

Rule	traumatic < vascular (ongoing vascular disease above level) · more proximal → higher cost
Other costs	**wheelchair propulsion +9%** vs normal walking · **crutch walking > walking with prosthesis**

Gait aids & crutch gait (TO)

Antalgic gait any pain-related gait · **shortened stance** on affected limb

Gait	Pattern
3-point	Crutches + injured limb advance together (weight on crutches) → weight onto uninjured limb
4-point	One crutch → opposite leg → other crutch → other leg
Swing-to	Crutches advanced, legs swung **to** crutches · slower, ↓ fall risk
Swing-through	Crutches advanced, legs swung **past** them · faster, ↑ energy, ↑ fall risk · ✓ **paraplegia / SCI with paraparesis (TO)**

– – Active space – –

Indications for and Characterization of Five Types of Supported Gait

	Four-point Gait	Three-point Gait	Two-point Gait	Swing-to Gait	Swing-through Gait
Indications	Weakness in both legs or poor coordination	Inability to bear full weight on one leg	Weakness in both legs or poor condition	Weakness of bilateral lower extremities	Inability to fully bear weight on both legs
Sequence	Left crutch, right foot, right crutch, left foot. Repeat.	Move both crutches and the weaker limb forward, bear all weight down through the crutches, move the stronger lower limb forward, and repeat.	With two aids, advance the right aid and left foot, followed by the left aid and right foot.	Both crutches are advanced forward, followed by swinging the feet slightly off the floor to an imaginary line just behind crutches.	Both crutches are advanced together, followed by swinging the feet to beyond the line of the crutches.
Advantages	Excellent stability with three points of contact at all times	Eliminates all weight bearing from the affected leg	Reciprocal motion promoted, speed maintained, closely simulates normal gait	Easy to learn	Fastest gait pattern (faster than normal gait)
Disadvantages	Slow walking speed	Requires good balance	Difficult to learn the pattern, less stable	Requires good upper extremity strength	Requires the most floor space, the ability to balance the trunk/limbs confidently, and large energy consumption. There are marked oscillations in the center of mass.

Gait patterns with assistive devices

Crutch-walking muscles latissimus dorsi · triceps + biceps · quads · hip extensors · hip abductors

Cane held **contralateral** → shifts centre of gravity toward affected side

TTWB / NWB one leg ✓ **2 axillary crutches**
✗ Wrong options: single forearm crutch · double cane · single platform crutch

5 Management

Upper limb prostheses (myoelectric vs body-powered)

Timing fit **ASAP**, even before full wound healing · better if **within 30 days**

	Myoelectric	Body-powered
Best candidate	✓ **Midlength transradial** (only terminal device to drive)	**Heavy labour**
Mechanism	Surface electrodes on residual muscles → motor	Harness: **shoulder flexion + abduction** · ring just off **C7 midline toward non-amputated side**
Pros	**Cosmesis** · **more proximal function/coverage** · any position incl. **overhead** · **less gross limb movement** · sedentary use	**Most durable** · moderate cost · **lighter** · **more sensory feedback** · less maintenance · less training
Cons	**Heavier, expensive, more maintenance**, **less sensory feedback** · clean environment only	Poor cosmesis · **more harnessing** · more gross movement
Rejection (Silcox)	50% rejected myoelectric	32% rejected conventional (of those with both)

Elbow disarticulation / transhumeral two joints → heavy · best = **hybrid** myoelectric + body-powered + switch components

Proximal transhumeral / shoulder disarticulation — no lever arm → **universal shoulder joint** positioned with contralateral arm + lightweight hybrids · many choose **cosmetic only**

Component	Detail
Passive terminal device	More cosmetic, less functional
Active terminal device	More functional · hooks / hands with cables or myoelectric
Hand-like device	Thumb, index, long finger · gloved · **office worker**
Non-hand (hook, pincer)	Task tools, quick release · **physical labour**
Voluntary opening	Closed at rest, opens with proximal contraction · **more common**
Voluntary closing	Open at rest · **heavier, less durable**
Quick-disconnect wrist	Swap specialised devices
Locking wrist	Prevents rotation during lift
Wrist flexion unit	**Bilateral amputees**, on **longer** limb → midline tasks (shaving, buttoning)
Rigid elbow hinge	**Short transradial**, no pronation/supination, elbow flexion kept
Flexible elbow hinge	Wrist disarticulation / long transradial with pronation/supination

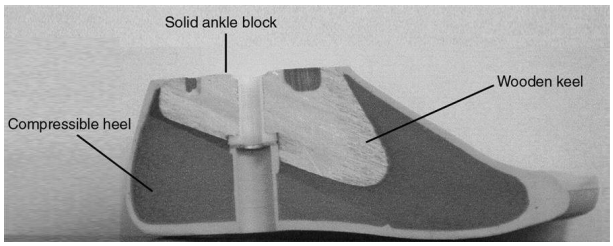
Grips — precision (pincer) · tripod (3-jaw chuck) · lateral (key) · hook power · spherical

Prosthetic foot selection

Foot	Features
Single axis	Ankle hinge → **DF + PF only** · poor durability, cosmesis
SACH (solid ankle cushioned heel)	Rigid wood keel, foam · **low activity / transfers** · being phased out · **overloads contralateral foot**
Dynamic response (energy storing)	Keel = spring → DF, ↓ contralateral load, push-off · standard for most
Articulating dynamic response	✓ **Inversion, eversion, rotation** · absorbs shear · **uneven terrain** (multi-axis — BK amputee walking uneven ground)
Non-articulating dynamic response	Spring but no inversion/eversion
Short keel	Less responsive · **moderate activity, hiking**
Long keel	**High-demand athletes**
Shank	Structural support · endo/exoskeletal · **lever arm after transmetatarsal amputation**

SACH vs dynamic response — SACH: **less dorsiflexion** · ↑ **contralateral GRF** · ↓ **max gait velocity** · **less frequent replacement** · not longer midstance (dynamic foot has **longer midstance**)

– – Active space – –



SACH foot

Hiker on SACH

✓ **short-keel articulated dynamic-response foot**
✗ Wrong options: single axis · non-articulated · long keel

Partial foot / toe amputation, short stance, poor toe-off

✓ **full-length steel shank + rocker sole** (shank restores lever arm, rocker → heel–toe transfer) · (✗ Wrong options: solid AFO, HKAFO, UCBL) · rocker may destabilise knee

Prosthetic knees

Knee	Indication	Features
Polycentric (4-bar)	Transfemoral, **knee disarticulation**, **bilateral**	**Variable centre of rotation** · controlled flexion · moderately fast pace · supports more weight · shortens limb at step initiation (↓ stumble) · **heavier, costly, more maintenance**
Stance-phase control (weight-activated)	**Older patients**, proximal amputation, uneven terrain	Constant friction in swing · locks with weight-bearing
Fluid (hydraulic / pneumatic)	**Active** patients accepting weight	**Variable cadence** (piston) · ↓ excess flexion · extends earlier
Constant friction (single axis)	General · uneven terrain · **commonest paediatric** · not old/weak	Hinge + friction · **light, durable, cheap** · **single speed** · stance stability from alignment
Variable friction (cadence control)	–	Multiple pads · variable speed · not durable
Manual locking	–	Constant friction + extension lock

Polycentric centre of rotation

posterior to weight line → stance control, harder flexion · anterior → easier flexion, less control
✗ Wrong options: **"fixed anterior to weight-bearing line"**



Polycentric knee

Transfemoral sockets & osseointegration (OPRA) (TO)

Socket — stump–prosthesis interface · preparatory socket re-adjusted as oedema resolves · **PTB socket commonest for BKA** · transfemoral/quadrilateral → **hard to keep femur adducted** (allows 10° adduction, 5° flexion)

Transfemoral socket (TO)	Weight bearing	Suspension
Quadrilateral	**Ischial tuberosity** on posterior shelf	Suction ± belts
Ischial containment	Whole stump surface · medial wall contains **ischial ramus** · more comfort/control	Suction

Quadrilateral socket used in — ✓ **transfemoral amputation (TO)**
✗ Wrong options: Syme's, BKA, transhumeral

Suspension — belts, wedges, straps, suction · **silicone liner = airtight seal**

Pylon — tube linking socket and terminal device · exoskeleton (hard shell) / endoskeleton (metal frame, soft cover)

Osseointegration (OPRA) — stem in residual bone + percutaneous post → direct skeletal attachment **(TO)**

OI advantages (TO) — no socket problems (skin breakdown, fit, pain) · may ↓ disuse **osteoporosis** + contralateral arthritis · **short residual limb**, HO, scar, multiple amputations · **osseoperception** (terrain sensing) · ↑ wear time · **easier** donning/doffing **(TO)**
✗ Wrong options: "no donning/doffing needed"

OPRA contraindications (FDA) (TO)
Incomplete skeletal growth · atypical anatomy · <2 mm cortex around implant
Osteoporosis
Age >65 or <22 · **weight >220 lb** (with prosthesis) · pregnancy
Non-compliance · severe PVD · complicated DM · residual-limb skin disease
Neuropathy, severe phantom pain · active infection / dormant bacteria
✓ **NOT a contraindication: malignancy as cause of amputation**

Orthoses & splints (TO)

GRAFO — ✓ **all**: **Newton's third law** · **accelerates knee extension in midstance** (weak quadriceps) · **compensates weak gastrocnemius–soleus (TO)**

Turnbuckle splint — gradual stretch of contracted joint · **Volkmann's ischaemic contracture, stiff elbow** · ↓ rehab time, good compliance **(TO)**

Physical modalities – TENS (TO)

TENS frequencies — **<10 Hz, high intensity → muscle contraction** · **>50 Hz, low intensity → paraesthesia** without contraction **(TO)**

Indications — ✓ **generalised muscle pain**, LBP, spasm, myofascial, arthritic, neurogenic, post-surgical, diabetic neuropathy, dysmenorrhoea · not acute **fracture pain** (no proven benefit) **(TO)**

– – Active space – –

Contraindications — **pregnancy, epilepsy, pacemaker** · pads not over eyes, trans-cerebral, anterior neck, ant + post chest together, broken skin, tumour, spine · ✓ **metal implants are NOT a contraindication** (insufficient heating) **(TO)**
✗ Wrong options: "relative"

6 Complications

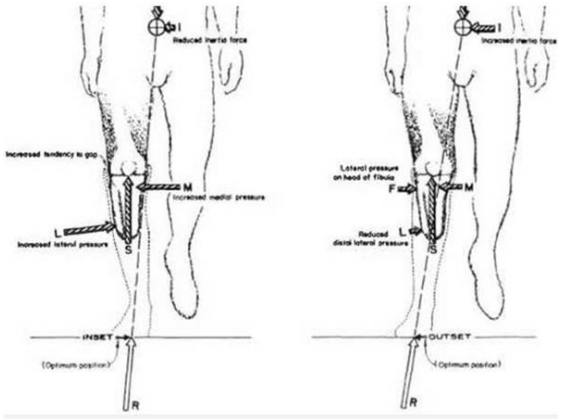
Choke syndrome — socket too tight → venous obstruction · acute: red, indurated, **orange-peel** skin · chronic: **haemosiderin**, venous stasis ulcers

Skin — **contact dermatitis** (liner, socks, suspension → remove item) · cysts + sweating = **excess shear / poor fit** · scar massage

Painful residual limb — HO · bony prominence · poor fit · **neuroma** · thin soft tissue

Transtibial prosthesis alignment & gait problems

Problem	Cause / effect
Swing-phase pistoning	**Ineffective suspension**
Stance-phase pistoning	**Poor socket fit** / volume change (change stump sock)
Inset (medialised) foot	**Varus strain** · **circumduction** · pain **medial-proximal + lateral-distal**
Outset (lateralised) foot	**Valgus strain** · **broad-based gait** · pain **lateral-proximal + medial-distal** (e.g. lateral knee + medial distal stump → ✓ **reduce lateral offset**)
Anterior foot	Stable ↑ **knee extension** + patellar pain
Posterior foot	Unstable ↑ knee flexion · **drop-off / buckling**
Dorsiflexed foot	↑ patellar pressure · drop-off
Plantar-flexed foot	Drop-off + ↑ patellar pressure (per Orthobullets)
Heel too soft	↑ **knee extension**
Heel too hard	↑ **knee flexion** + toe external rotation



Inset vs outset foot

Drop-off (knee buckling in terminal stance) — body weight over MTPs too early · caused by ↓ anterior support: DF foot, **posterior foot**, soft keel, **posterior toe break** → ✓ **move foot anterior**

How it's asked

– – Active space – –

"Inset BKA foot — EXCEPT?"	→ **increased knee extension in stance** ✗ Wrong options: varus strain · circumduction · medial-proximal pain
"Drop-off in terminal stance — fix?"	→ **move foot more anterior** ✗ Wrong options: dorsiflex · move posterior · softer keel
"Toe amputation, short stance, irregular toe-off?"	→ **full-length steel shank + rocker sole**
"Lateral knee + medial distal stump pain after BKA?"	→ **reduce lateral offset of foot**
"Hiking BKA amputee struggling on SACH?"	→ **short-keel articulated dynamic-response foot**
"SACH vs dynamic response — EXCEPT?"	→ **longer midstance** ✗ Wrong options: less replacement · less DF · ↑ contralateral GRF
"Multi-axis articulated foot recommended for?"	→ **BKA walking on uneven ground**
"Most important factor choosing LL prosthesis?"	→ **functional status**
"Polycentric fluid knee vs constant friction — EXCEPT?"	→ **lighter weight**
"Polycentric hydraulic knee — EXCEPT?"	→ **centre of rotation fixed anterior to weight line**
"Body-powered vs myoelectric — EXCEPT?"	→ **decreased harnessing** ✗ Wrong options: heavy labour, lighter, less therapy, less maintenance
"Myoelectric advantage — EXCEPT?"	→ **more sensory feedback**
"Least ↑ metabolic demand?"	→ **traumatic transtibial**
"Swing-through gait used by?"	→ **SCI with paraparesis (TO)**
"Crutch gait in video (legs swung to crutches)?"	→ **swing-to gait**
"TTWB after pilon ORIF — device?"	→ **double axillary crutches**
"GRAFO?"	→ **all of the above** (Newton's 3rd law, knee extension, weak calf) **(TO)**
"Identify splint (gradual stretch of contracture)?"	→ **turnbuckle splint (TO)**
"Quadrilateral socket used in?"	→ **transfemoral amputation (TO)**
"NOT a contraindication to OPRA?"	→ **malignancy as cause of amputation (TO)**
"NOT a feature of bone-anchored prosthesis?"	→ **no donning/doffing required (TO)**
"True about TENS?"	→ **used for generalised muscle pain (TO)** ✗ Wrong options: CI with metal · fracture pain · pacemaker relative

– – Active space – –

RENAL OSTEODYSTROPHY

– Basic Science · 91 / 108 –

1 Basics

Definition metabolic bone disease of **chronic renal insufficiency** · deficient mineralisation from electrolyte + endocrine abnormalities → osteomalacia, osteonecrosis, pathological fracture

Labs & pathophysiology (TO)

Key step ✓ **uraemia-related phosphate retention**
✗ Wrong options: PTH receptor sensitivity · metabolic alkalosis · dialysis electrolyte shifts

Chronic renal failure

- ↓ PO_4 excretion → hyperphosphataemia → Ca-PO_4 precipitates → **↓ Ca**
- ↓ 1α-hydroxylase (PO_4 inhibits it) → ↓ 1,25-D → ↓ Ca → **OSTEOMALACIA**
- ↓ Ca + ↑ PO_4 → chief cell hyperplasia → **SECONDARY HYPERPARATHYROIDISM**
  - autonomous gland → hypercalcaemia → **TERTIARY HYPERPARATHYROIDISM**

Tertiary hyperparathyroidism (TO) prolonged stimulation → gland autonomous, insensitive to Ca/PO_4/vit D → **hypercalcaemia (TO)**

Classification

	High turnover (high PTH)	Low turnover (normal PTH)
Cause	Chronic ↑ PO_4 → **secondary hyperparathyroidism** (PO_4 ↓ Ca, ↓ 1α-hydroxylase, directly ↑ PTH)	**Aluminium deposition** in bone
Mechanism	Chief cell hyperplasia	↓ osteoblast differentiation/ proliferation · **↓ PTH release** · disrupted mineralisation
Ca	**↓**	**↑**
PO_4	**↑**	Normal
ALP	**↑**	**↑**
PTH	**↑**	**Normal**

Genes & lab markers

Newly diagnosed ✓ **↑ PTH, ↑ PO_4 (hyperphosphataemia), ↓ Ca** · ↓ active vit D · low urinary Ca + PO_4 **(TO)**

ALP ↑ in (TO) ✓ **renal osteodystrophy** · rickets (all) · osteomalacia · Paget's · primary hyperparathyroidism · CRF with 2° HPT · lytic neoplasms **(TO)**

ALP normal (TO) osteoporosis · **osteopetrosis** · fibrous dysplasia · **multiple myeloma** · hypoparathyroidism **(TO)**

ALP ↓ (TO) achondroplasia · **hypophosphatasia** · cretinism **(TO)**

Lab pattern (Ca / PO_4 / ALP / PTH)	Diagnosis
↓ / ↑ / ↑ / ↑	✓ **High-turnover renal osteodystrophy**
↑ / N / ↑ / N	Low-turnover renal osteodystrophy
↓ / ↓ / ↑ / ↑	Nutritional (vit D deficiency) rickets
↓ / ↑ / N / ↓	Hypoparathyroidism 169
↑ / N / N–↑ / ↑	Primary hyperparathyroidism

Histology thinned trabeculae · extreme bone turnover · **amyloid** pink on **Congo red**

3 Presentation

Symptoms weakness · bone pain · **pathological fracture (commonest complication)** · deformity · growth retardation (children)

Hypocalcaemia abdominal pain · cramps · dyspnoea · seizures · mental status change · **Trousseau** (carpopedal spasm with BP cuff) · **Chvostek** (facial twitch on tapping facial nerve)

Orthopaedic manifestations
Osteomalacia (adults) · growth retardation (children)
AVN
Tendinitis, **tendon rupture**
Carpal tunnel – **β2-microglobulin amyloid**
Pathological fracture – **brown tumours** (HPT) or amyloid
Osteomyelitis, septic arthritis

Deformity in children (TO) ✓ **genu valgum** commonest (TO; ROD occurs later than nutritional rickets' physiological-varus age) · Orthobullets lists **femoral varus** – read the options · others: genu varum, **SCFE**, ankle valgus/varus, coxa vara, bowing, scoliosis, osteonecrosis, OCD, brown tumour **(TO)**

4 Diagnosis & Classification

Key radiology findings

X-ray **Looser zones** · **brown tumour** · **osteosclerosis** (mineralised osteomalacic bone) → **rugger jersey spine** (alternating sclerotic + lytic bands) · soft-tissue / **tumoral calcinosis** · chondrocalcinosis · osteopenia · fracture

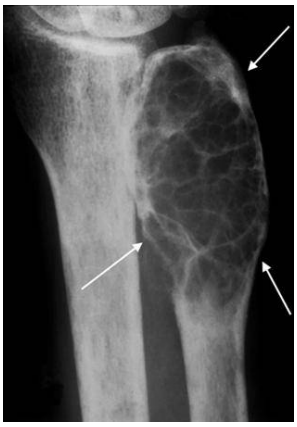
– – Active space – –



Rugger jersey spine

Children — **widened physis + zone of provisional calcification** · physeal **widening, bowing, cupping** · varus femurs (e.g. 9-yr-old, <5th centile height, femoral bowing → ✓ **renal osteodystrophy**; ✗ Wrong options: XLH less likely at that onset · fibrous dysplasia – no ground glass, OI – no fractures)

CT — osseous resorption · may **mimic bone tumour**



Brown tumour

Osteitis fibrosa cystica — multiple cysts from PTH resorption **(TO)**

Metaphyseal bands (TO)

Dense metaphyseal bands – causes (TO)
Chronic anaemia – sickle cell, thalassaemia
Chemotherapy – methotrexate · treated leukaemia
Growth arrest/recovery lines – asthma, DM, CF, JIA, malnutrition
Heavy metals – **lead (commonest)**, bismuth, arsenic, phosphorus, fluoride, mercury
Healing **renal osteodystrophy** · healing rickets · **bisphosphonates** · neonatal normal variant
✓ **Hypervitaminosis D** – rarest → "not seen" answer

5 Management

Treatment — treat underlying renal disease / relieve urological obstruction

How it's asked

"Metaphyseal bands NOT seen in?"	→ **hypervitaminosis D (TO)** ✗ Wrong options: lead, methotrexate, sickle cell
"Serum ALP increases in?"	→ **renal osteodystrophy (TO)** ✗ Wrong options: osteopetrosis, myeloma, hypophosphatasia
"Labs in new renal osteodystrophy?"	→ **↑ PTH, hyperphosphataemia, hypocalcaemia**
"Key step causing osteomalacia in renal osteodystrophy?"	→ **uraemia-related phosphate retention**
"High-turnover renal osteodystrophy labs?"	→ **↓ Ca, ↑ PO_4, ↑ ALP, ↑ PTH**
"Commonest deformity in renal osteodystrophy?"	→ **genu valgum (TO)**
"Dialysis patient, bilateral hip fractures, ↑ ALP + PTH, thin trabeculae?"	→ **renal osteodystrophy** ✗ Wrong options: metastasis, myeloma
"9-yr-old, short, bilateral femoral bowing, cupped wide physes?"	→ **renal osteodystrophy** ✗ Wrong options: diastrophic dysplasia, OI, fibrous dysplasia

– – Active space – –

RHEUMATOID ARTHRITIS

– Basic Science · 92 / 108 –

1 Basics

Epidemiology

Commonest inflammatory arthritis	**3% women, 1% men**
Genetics	**HLA-DR4**, HLA-DW4 · ~**15%** monozygotic twin concordance · trigger: infection or HLA locus

Pathophysiology & serology (RF, anti-CCP, synovium)

Immunology	**IgM cell-mediated (T cell–MHC class II)** response → soft tissue (early) → cartilage, bone (later) ✗ Wrong options: IgA/IgD/IgE/IgG-mediated
Primary destructive cell	**mononuclear cells** · cytokines **IL-1, TNF-α** (+ IL-6)
Cascade	Ag–Ab reactions → microvascular proliferation/obstruction → **pannus** (prominent **intimal hyperplasia**) → subluxation, chondrocyte death → **tenosynovitis + tendon rupture**

RA synovium	
✓ **Intimal hyperplasia**	2 → **10–20 cell layers**
✓ **↓ Apoptosis** of lining	
✓ **↑ Angiogenesis**	
✓ **Abundant lymphocytes**	Perivascular lymphoid follicles
✗ Wrong option: **basement membrane disruption**	Synovium **has no true basement membrane**

Pannus	invasive granulation tissue · fibroblast-like synoviocytes, few inflammatory cells · anchorage-independent growth → directly destroys cartilage
Bone erosion	✓ **TNF + IL-6 (+ IL-1) blockade slows erosion** · TNF mobilises osteoclast precursors · **M-CSF + RANKL** drive **osteoclast** differentiation · **denosumab** slows erosion, not inflammation · TNF ↑ **Dkk-1** → blocks **Wnt** → ↓ repair (Wnt interference ↑ erosion) · erosion ∝ synovitis · steroids, MTX, leflunomide bone-sparing via ↓ synovitis ✗ Wrong options: "synovial fibroblasts resorb"

Genes & lab markers

Rheumatoid factor	✓ **IgM antibody against Fc of native IgG** · immune complexes deposit (kidney) ✗ Wrong options: IgG vs IgM · IgA vs IgG
RF positivity	**75–80%** (Orthobullets) · **80–90%** (OB6543) – read the options · sensitive, not specific · also non-RA autoimmunity

Anti-CCP ✓ **most sensitive + specific** · sensitivity **50–75%**, specificity **90–95%** · as sensitive as, more specific than RF · **high levels → erosive disease** · may precede arthritis · anti-CCP ⊂ **ACPA** · ACPA bind citrullinated vimentin on osteoclasts → TNF → osteoclastogenesis · **independently predicts erosion**
✗ Wrong options: "synovitis but not erosion"

Other **anti-MCV** · ↑ ESR · ↑ CRP · **ANA in 30–40%** · joint fluid: **↓ complement**, ± RF

Autoantibody	Disease
✓ **Anti-IgG (RF)**	**RA**
Anti-dsDNA	SLE
Anti-ACh receptor	Myasthenia gravis
Anti-topoisomerase I	**Scleroderma**
Anti-GAD	Type 1 DM

3 Presentation

Symptoms insidious **morning stiffness** · symmetric **polyarthropathy** of hands + feet · **DIP spared** · also knees, C-spine, elbows, ankles, shoulders

Rheumatoid nodules **commonest extra-articular manifestation** · **20%** (Orthobullets presentation) / **25%** (finger section) · **strongly RF+**, **aggressive disease** · over IP joints, **olecranon**, ulnar forearm · may form sinus · treat steroid injection / excision (cosmesis, pain, biopsy)

Systemic rheumatoid vasculitis · pericarditis · pulmonary disease

Syndrome	Features
Felty	RA + **splenomegaly + leukopenia**
Still's	Acute RA + **fever, rash, splenomegaly**
Sjögren	Exocrine glands → ↓ saliva + tears · lymphoid proliferation

4 Diagnosis & Classification

Key radiology findings

X-ray **periarticular erosions** + **osteopenia** · symmetric joint-space narrowing · **central glenoid erosion** · **protrusio acetabuli**

Protrusio acetabuli femoral head medial to **teardrop** · RA (bilateral + multijoint on bone scan → ✓ RA) · Marfan · Paget · Otto pelvis (**arthrokatadysis** = primary protrusio, with osteomalacia, no other joints)
✗ Wrong options: DDH

– – Active space – –



RA hand — periarticular erosions, MCP subluxation

Diagnostic criteria — 2010 ACR/EULAR (TO)

Target ≥1 joint with definite synovitis not better explained · ✓ **score ≥6/10 = RA** · <6 → reassess, may accumulate **(TO)**
✗ Wrong options: ">8"

Domain (TO)	Points
Joints — 1 large	0
2–10 large	1
1–3 small	2
4–10 small	3
>10 joints (≥1 small)	**5** — greatest weight
Serology — RF− and anti-CCP−	0
Low +ve RF or anti-CCP (1–3× ULN)	2
High +ve (≥3× ULN)	3
Acute phase — abnormal CRP or ESR	1
Duration ≥6 wk	1

Excluded joints (TO) **hand DIP**, **1st CMC**, **1st MTP (TO)**

1987 ARA criteria ≥4 of 7 for **6 wk**: morning stiffness **≥1 h** · swelling **≥3 joints** · nodules · hand X-ray erosions/decalcification · **symmetric** arthritis · serum RF · hand (MCP, PIP) + wrist arthritis

Post-viral (chikungunya) arthritis (TO)

Phases (TO) acute <3 mo (viraemic 5–10 d, sub-acute 6–21 d) · **chronic >3 mo** — RA-like, SpA, fibromyalgia patterns · chronic in **14–25%** (age >45, viral load $>10^9$/mL) · neuropathic pain ~20% **(TO)**

Tests (TO) RT-PCR days 0–7 · IgM from 5–10 d (peak 3 wk, lasts 2–3 mo) · IgG after 1 wk, lasts years · chronic → check **RF, anti-CCP, HLA-B27 (TO)**

– – Active space – –

Phase (TO)	Treatment
Acute	Rest, fluids, **paracetamol** · avoid NSAIDs until **dengue excluded**, aspirin, steroids <3 wk
Sub-acute	Low-dose prednisolone (10 mg × 5 d, taper) for synovitis · HCQ for persistent symptoms
✓ **Chronic (>3 mo, e.g. 18 mo)**	✓ **MTX + sulfasalazine + HCQ** (combination > HCQ alone) · TNFi if resistant

5 Management

DMARDs & biologics (TNF, IL-1, CTLA4, CD20)

Approach aggressive early DMARDs > pyramid · advances → ↓ surgery

Drug	Mechanism
Methotrexate	**Folate analogue – inhibits DHFR** · ↓ neovascularisation · adenosine release · ✓ **+ tetracycline (doxycycline) more effective** (anti-collagenase)
Leflunomide	**Pyrimidine synthesis inhibitor**
Sulfasalazine	Unknown · ↓ ESR/CRP · ↓ prostaglandins, leukotrienes
Hydroxychloroquine	Blocks **toll-like receptors** → ↓ dendritic cells
Gold	Inhibits monocytes
D-penicillamine	–
Etanercept	**TNF receptor (p75/type II)–IgG1 Fc fusion protein** binding TNF-α · SC · RA, JRA, PsA, psoriasis, AS
Infliximab	**Human–mouse chimeric anti-TNF-α mAb** · opportunistic infection, malignancy
Adalimumab · golimumab	Human anti-TNF-α mAb
Certolizumab	Pegylated anti-TNF-α
Anakinra	**Recombinant IL-1 receptor antagonist** (IL-1RI) · refractory / 2nd–4th line
Rituximab	**Anti-CD20 → B cells** · not TNF · with MTX after TNF failure
Abatacept	**Binds CD80/CD86 (co-stimulation) → T cells**
Ustekinumab	Anti-**IL-12/IL-23**
Tocilizumab	**IL-6 receptor inhibitor** · 2nd line after TNF failure
Cyclosporine	Calcineurin inhibitor

TNF blockers revolutionised RA (also JRA, SpA, CRMO, histiocytosis X) · ↓ pannus, cartilage loss, osteoclastic resorption · risks: **latent TB reactivation**, infection, **lymphoma**

– – Active space – –



Ulnar drift at the MCP joints

Perioperative medication management & infection risk

Medication	Perioperative rule
NSAIDs	Stop **5 half-lives** before (aspirin 7–10 d; naproxen 3 d)
Steroids	Stress dose by surgical stress
Methotrexate · sulfasalazine · HCQ · penicillamine	✓ **Continue**
Leflunomide	Continue (minor) · stop 1–2 d before major, restart 1–2 wk
Etanercept	Hold **1 wk** (OB3092 key) · **2 wk** before THA (2017 ACR/AAHKS, OB212995; Orthobullets complications) · 4 wk (Keith) · plan at end of dosing interval +1 wk – read the options
Adalimumab	Stop **10 days** before
Infliximab	Lowest level **2 wk before next infusion**
Rituximab	Stop **7 months** before major surgery
Ustekinumab	Skip next dose + 1 wk (~**13 wk** from last) · restart **>14 d → ↓ wound complications**
Anakinra	Continue minor · stop 1–2 d major · restart 10 d

Infection ✓ **prior SSI = most significant risk factor** (✗ Wrong options: age, hardware, sulfasalazine, infliximab) · ✓ **etanercept** shown to ↑ post-op infection (✗ Wrong options: naproxen, leflunomide, sulfasalazine, aspirin) · immunosuppression effect overall controversial

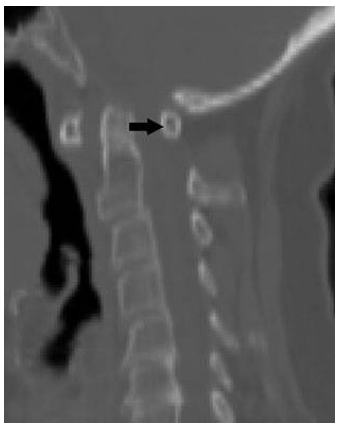
Cervical spine instability & preoperative screening

C-spine involved in **90%**, often missed · ✓ **pre-op C-spine X-rays with flexion–extension** before any arthroplasty (anaesthetic death risk > shoulder MRI) · JRA (polyarticular/systemic) → **C-spine assessment most important**

Pattern	
Atlantoaxial subluxation	**Commonest**
Basilar invagination	Cranial migration of dens
Subaxial subluxation	Multilevel listhesis (staircase), midcervical kyphosis

PADI (posterior atlanto-dens interval) ✓ **<14 mm → fusion** even if asymptomatic (predicts neuro progression)

– – Active space – –



Atlantoaxial instability – PADI

Hand & wrist (extensor tendon rupture, tendon transfers, wrist fusion) (TO)

Upper-limb reconstruction order (TO) ✓ **hand/wrist > shoulder > elbow** · lower limb before upper · LL order **forefoot > hip > knee > hindfoot > ankle (TO)**

Ulnar drift (wind-swept) (TO) ✓ **ulnar displacement of extensors + radial collateral ligament attenuation** · radial sagittal band fails → EDC into ulnar valleys · wrist radial deviation → ulnar bowstring (zigzag) · ECU subluxes volar-ulnar **(TO)**

Ulnar drift at MCP

- Early, flexible → **SYNOVECTOMY + EXTENSOR CENTRALISATION + INTRINSIC RELEASE (cross-intrinsic)**
- Late / fixed → **SILICONE (SWANSON) MCP ARTHROPLASTY**
  (correct radially deviated wrist at same time · satisfaction ~70%)
- Thumb MCP without IP disease → **MCP FUSION**

Finger deformity	Pathoanatomy	Treatment
Boutonnière	PIP synovitis → **central slip** attenuation · lateral bands **volar** · **oblique** retinacular contracture → DIP hyperextension	Splint (flexible) · central slip imbrication / **Fowler distal tenotomy** (moderate) · PIP fusion / arthroplasty (rigid)
Swan neck	**Terminal tendon** rupture (DIP) · **FDS, volar plate, collaterals** attenuate · lateral bands **dorsal** · **transverse** retinacular attenuation, triangular ligament contracture	Splint · **FDS tenodesis** / proximal Fowler (flexible) · releases (rigid)
Arthritis mutilans	RA or PsA · pencil-in-cup, wind-chime fingers	Interposition graft + fusion

– – Active space – –

Nalebuff thumb	Deformity	Treatment
I – Boutonnière	✓ **Most common** · **MCP flexion + IP (hyper)extension** · EPB attenuation, EPL subluxes volar **(TO)**	✓ **Restore MCP extension + stabilise IP (TO)** · synovectomy + EPL rerouting (early) · **MCP fusion 10–15°** (late, most reliable) · IP fusion if fixed
II – Boutonnière + CMC	Uncommon / very rare	As I and III
III – Swan neck	**CMC adduction/subluxation + MCP hyperextension + IP flexion**	**Address CMC first** – trapeziectomy/LRTI · MCP >30° → volar capsulodesis or fusion · first web release
IV – Gamekeeper	Adductor contracture + **UCL laxity** · P1 radial deviation	Flexible: synovectomy, UCL reconstruction, adductor release · fixed: MCP arthroplasty / fusion
V	MCP hyperextension – **volar plate laxity** (swan neck with MCP disease)	Volar capsulodesis (MCP in flexion)
VI	**Skeletal collapse (mutilans)**	Arthrodeses

Type I vs III (TO) CMC normal → I · CMC subluxed + adducted metacarpal → III **(TO)**

Flexor problem	Treatment
Triggering	Synovectomy + **FDS resection**
Mannerfelt – **FPL** rupture (commonest flexor) on **scaphoid spur** in carpal tunnel	**FDS4 → FPL** transfer + spur excision · graft · IP fusion
FDP rupture	Synovectomy + DIP fusion
FDS rupture	Observe

Extensor rupture frequency **EDM > EDC ring > EDC small > EPL** · ✓ **extensor digiti quinti most at risk** (silent – EDC still extends small finger)

Vaughan-Jackson ✓ **rupture of digital extensors ulnar → radial** · **DRUJ instability + volar carpal subluxation → dorsal ulnar head** → attrition · **EDM first** · **painless**, no warning · treat **EIP → EDC transfer + distal ulna resection**

Ruptured	Transfer
EPL	**EIP → EPL**
EDQM alone	Leave
EDQM + EDC5	**EIP → EDC5** or EDC piggyback
EDQM + EDC5 + EDC4	EIP → EDQM + EDC4 side-to-side to EDC3
Multiple	Palmaris graft, **FDS**

EIP transfer ✓ **no index extension deficit** (cut proximal to sagittal band)

FDS (middle/ring) → EDC (Nalebuff–Patel) ✓ **radially around radius, dorsally** (easy, avoids dorsal synovitis, corrects ulnar deviation)
✗ Wrong options: through interosseous membrane

Radial sagittal band failure extensor slips ulnar + volar to MCP axis
✗ Wrong options: active extension but ✓ **holds if passively extended** → ✓ **sagittal band (extensor hood) reconstruction** (Watson – distally based central-third slip)

Loss of MCP extension	Clue
Extensor rupture	not tenodesis · can't hold passive extension
Sagittal band rupture	Holds passive extension
PIN palsy (elbow/ radiocapitellar synovitis)	✓ **tenodesis intact**, can't hold, wrist extends radially, no ulnar drift → ✓ **EMG/NCS confirms** (✗ Wrong options: exploration · elbow MRI)
MCP volar subluxation	Ulnar deviation, X-ray
Trigger finger	Catching

Caput ulnae **DRUJ synovitis** → ECU subsheath stretch → ECU subluxes → carpus supinates away from ulna → **volar carpal subluxation** → extensor rupture · treat **Darrach** (+ ECU stabilisation / retinacular flap) · hemiresection · **Sauvé-Kapandji** (preserves TFCC, younger)

Radiocarpal destruction **supination**, radial deviation, **ulnar + volar translocation** of carpus · zigzag with MCP ulnar drift

Wrist	Indication
Synovectomy ± **ECRL → ECU (Clayton)**	Early
✓ **Radiolunate (Chamay) / RSL fusion**	**Radiocarpal (radioscaphoid) arthritis, preserved midcarpal, good arc (e.g. 80°)**
Four-corner fusion	Isolated midcarpal
Total wrist fusion	Pancarpal, limited arc (20°), poor bone · **gold standard** · often + Darrach
Total wrist arthroplasty	Sedentary, good bone, reasonable motion

Elbow (synovectomy, total elbow arthroplasty)

Elbow procedure	Indication
Medical + **steroid injection**	First line
Arthroscopic / open synovectomy	Pain **without instability**, ROM preserved (arc >90°) · **early** (Larsen 3 → ↑ reoperation; 5-yr reoperation-free 77%)
Synovectomy + radial head excision (lateral approach)	**Radiocapitellar** focus · **PIN compression from radial head synovitis** (wrist/thumb extension weakness)
Interposition arthroplasty (Achilles, fascia, dermis)	**Young active**, not TEA candidates
✓ **Semiconstrained TEA**	**End-stage**, pain, loss of motion, instability · elderly · **5 lb lifting limit** · (✗ Wrong options: unconstrained in inflammatory arthritis)
Arthrodesis / resection	Salvage only

Synovectomy indications preserved joint space · young on DMARD · early disease · heavy work (rules out TEA) · **female gender** is not an indication

– – Active space – –

Shoulder · hip · knee

Shoulder	**>90%** develop symptoms · **rotator cuff tears** (large, irreparable → poorer arthroplasty, glenoid loosening) · central glenoid wear, osteopenia, cysts
Hip	**protrusio acetabuli**
Knee synovectomy	↓ pain + swelling · does not halt radiographic progression, prevent TKA or ↑ ROM · synovium re-forms then becomes rheumatoid
TKA	✓ **resurface patella** in RA · unresurfaced → ↑ **anterior knee pain** · resurfaced → ↑ extensor rupture, patellar clunk

Forefoot & midfoot reconstruction

Rheumatoid foot	bilateral, symmetric · **forefoot first** · HLA-DR4 · ~90% develop forefoot deformity (hallux valgus, hammer toes, lesser MTP dislocation)
Earliest forefoot sign	**MTP synovitis** → MTP hyperextension · distal fat-pad migration · plantar callosities · ulcers
Gold standard	✓ **1st MTP arthrodesis + lesser metatarsal head resections** (+ hammertoe correction – **Hoffman-Clayton**) · Coughlin & Mann **90%** good/excellent

✗ **Wrong option**	**Why**
Keller arthroplasty	Unstable platform, recurrence, poorer function
1st MTP replacement	Unreliable long-term
Bunionectomy / chevron / realignment	Doesn't stabilise ray or address arthritis · osteotomies only if no erosions
Lapidus	TMT fusion – does not address MTP
Weil osteotomies	do not address erosive lesser MTPs
Lesser MTP fusion	Lateral foot needs motion
Lesser MTP arthroplasty	Resection preferred



1st MTP fusion + lesser metatarsal head resections

Talonavicular arthritis	isolated **symmetric** TN degeneration, no trauma → ✓ **RA** · treat **fusion** ✗ Wrong options: Köhler, OA, OCD

How it's asked

"RA synovium EXCEPT?"	→ **basement membrane disruption** ✗ Wrong options: intimal hyperplasia, ↓ apoptosis, angiogenesis, lymphocytes
"RF targets which immunoglobulin?"	→ **IgG**
"RF best described as?"	→ **IgM antibody against native IgG**
"Anti-CCP used for?"	→ **rheumatoid arthritis** ✗ Wrong options: AS, PsA, SLE, Reiter
"Bone erosion in RA — true?"	→ **TNF + IL-6 blockade slows erosion** ✗ Wrong options: Wnt interference ↓ erosion · independent of synovitis
"% adults RF positive?"	→ **80–90%**
"Ulnar drift + MCP subluxation, anti-CCP+ — immune mechanism?"	→ **IgM cell-mediated**
"Symmetric polyarthropathy + nodules — autoantibody?"	→ **against IgG**
"TNF-α therapy revolutionised?"	→ **RA**
"Biologic NOT targeting TNF?"	→ **rituximab**
"TNF blocked by?"	→ **etanercept** ✗ Wrong options: tocilizumab, anakinra, abatacept, rituximab
"Added to MTX, better than MTX alone?"	→ **doxycycline**
"IL-1 antagonist, 2nd line?"	→ **anakinra**
"Infliximab mechanism?"	→ **TNF inhibitor** ✗ Wrong options: DHFR · CD20 · calcineurin
"Etanercept mechanism?"	→ **inhibitory binding to TNF-α**
"DMARD on IL-1 receptor (figure)?"	→ **anakinra**
"Binds CD80/CD86?"	→ **abatacept**
"Blocks TNF–receptor binding (figure)?"	→ **infliximab**
"TKA in 2 wk on sulfasalazine, penicillamine, etanercept?"	→ **continue SSZ + penicillamine, stop etanercept 1 wk before**
"THA on etanercept + HCQ?"	→ **continue HCQ, hold etanercept 2 wk**
"Greatest SSI risk in RA foot reconstruction?"	→ **prior MSSA wound infection**
"Perioperative drug shown to ↑ infection?"	→ **etanercept**
"Why hold ustekinumab ~13 wk pre-op, restart >14 d?"	→ **wound complications**
"Most important pre-op evaluation in JRA?"	→ **cervical spine assessment**

– – Active space – –

"PADI threshold for stabilisation?"	→ **14 mm**
"RA shoulder — work-up before surgery?"	→ **C-spine flexion–extension X-rays** ✗ Wrong options: shoulder MRI
"Multilevel subaxial listhesis + kyphosis?"	→ **rheumatoid arthritis**
"Upper-limb reconstruction order?"	→ **hand > shoulder > elbow (TO)**
"Wind-swept ulnar drift mechanism?"	→ **ulnar extensor displacement + RCL attenuation (TO)**
"Nalebuff I reconstruction goal?"	→ **restore MCP extension, stabilise IP (TO)**
"Can't actively extend ring finger but holds passive extension?"	→ **sagittal band reconstruction**
"Vaughan-Jackson syndrome?"	→ **rupture of digital extensors** ✗ Wrong options: FPL = Mannerfelt · caput ulnae
"Extensor most at risk in RA wrist?"	→ **extensor digiti quinti**
"EIP transfer — effect on index?"	→ **no effect**
"FDS → EDC transfer route?"	→ **radially around the radius**
"Can't extend MCPs, tenodesis intact, can't hold passive extension — confirm with?"	→ **electrodiagnostic studies** (PIN palsy)
"Painless loss of ring + small extension — true?"	→ **ulnar head prominence from DRUJ instability** ✗ Wrong options: ring first · pain/ catching · Frohse
"Indication for RL/RSL fusion?"	→ **isolated radioscaphoid arthritis, 80° arc**
"RA forefoot, failed nonop?"	→ **1st MTP fusion + lesser metatarsal head resections** ✗ Wrong options: Keller · MTP replacement · Lapidus
"Symmetric isolated TN arthritis, no trauma?"	→ **RA**
"End-stage RA elbow after hand reconstruction?"	→ **total elbow arthroplasty**
"NOT an indication for synovectomy ± radial head resection?"	→ **female gender**
"Early RA elbow, arc 15–110°, stable?"	→ **arthroscopic synovectomy**
"Elbow synovitis + wrist/thumb extension weakness?"	→ **synovectomy + possible radial head excision**
"Unresurfaced patella TKA in RA — more likely?"	→ **anterior knee pain** ✗ Wrong options: clunk · tendon rupture
"Bilateral protrusio + multijoint bone scan uptake?"	→ **rheumatoid arthritis** ✗ Wrong options: arthrokatadysis · DDH

"False about 2010 RA criteria?"	→ **score >8 needed** (✓ ≥6) **(TO)**
"18 mo post-chikungunya small-joint arthritis — treatment?"	→ **MTX + sulfasalazine + HCQ (TO)**

– – Active space – –

– – Active space – –

RICKETS

– Basic Science · 93 / 108 –

1 Basics

Definition — defective mineralisation of osteoid/cartilage from inadequate **Ca + PO_4** **before physeal closure** · after closure = **osteomalacia**

Primary problem — ✓ **low calcium** available for mineralisation (Ca/PO_4 homeostasis)
✗ Wrong options: primarily a physeal (proliferative zone) disorder · type I collagen (OI) · EXT-1 (MHE) · dysplastic fibrous bone (FD)

Site — poor calcification at **zone of provisional calcification → ↑ physeal width** + cortical thinning/bowing · fastest-growing bones (tibia, distal femur, distal radius) · rarely hand tubular bones

Causes — vit D deficiency · vit D malabsorption · **renal osteodystrophy** · **X-linked hypophosphataemia**
✗ Wrong options: **vitamin K deficiency**

Activation — ✓ **PTH** activates renal **1α-hydroxylase** (low Ca) → calcitriol · most potent effect: ↑ intestinal Ca absorption

Classification

Type	Inheritance / gene	Mechanism	Age
Vit D-resistant (familial hypophosphataemic, XLH)	✓ **X-linked dominant – PHEX** (commonest **heritable** rickets)	**Renal tubules can't reabsorb PO_4** · ↑ **FGF23** → PO_4 wasting + suppressed 1α-hydroxylase · GFR normal	**1–2 yr**
ADHR	**AD – FGF23** (cleavage site)	↓ FGF23 degradation · **iron deficiency triggers** expression **(TO)**	–
ARHR	**AR – DMP1** (also ENPP1, FAM20C)	Impaired osteocyte maturation, ↑ FGF23	–
Nutritional (vit D-deficient)	–	↓ intake → ↓ gut Ca → ↑ PTH → resorption → ↑ ALP	**6 mo–3 yr**
VDDR I	**AR – renal 1α-hydroxylase** (12q14; CYP27B1)	↓ **calcitriol**	–
VDDR II	**AR – vit D receptor** (inactivating)	↑ **calcitriol** · end-organ resistance · alopecia (2A)	–
HHRH	**SLC34A3** (NaPi-IIc)	FGF23-**independent** PO_4 wasting **(TO)**	–

XLH inheritance — affected father → **100% daughters carriers/affected**, sons unaffected · more severe in males

Nutritional risk factors premature infants · ✓ **dark-skinned (African American) breastfed >6 mo without vit D** · malabsorption (coeliac), TPN · Asian immigrants · vegetarian diets · rare since milk fortified

FGF23-dependent hypophosphataemia (TO)	FGF23-independent (TO)
XLH, ADHR, ARHR · tumour-induced osteomalacia · epidermal naevus · osteoglophonic dysplasia · fibrous dysplasia · iron-infusion · **α-klotho translocation** · **Dent's** (per TO key) · NPT2a, NHERF1 mutations · post-transplant / hepatic resection	✓ **Fanconi syndrome** · ✓ **HHRH (hereditary hypophosphataemic rickets with hypercalciuria)**

FGF23 (TO) **high** → ↓ PO_4, ↓ 1,25-D, osteomalacia · ✓ **low** → ↑ PO_4, ↑ 1,25-D, soft-tissue calcification, **hyperostosis (TO)** · signals via FGFR1 + **Klotho**; phosphaturia partly PTH-dependent **(TO)**

Genes & lab markers

Disease (Orthobullets chart)	Ca	PO_4	ALP	PTH	25-D	1,25-D
XLH (vit D-resistant)	**N**	**↓↓**	↑	**N**	N	N / inappropriately low
Nutritional	N–↓	↓	↑	**↑**	**↓**	↓
VDDR I	↓	↓	↑	↑	N (TO table: ↑↑)	**↓↓**
VDDR II	↓	↓	↑	↑	N	**↑↑** (>1000 pg/mL)
Hypophosphatasia 171	↑	↑	**↓↓**	N	N	–
Renal osteodystrophy	↓	**↑**	↑	↑↑	N / ↓	↓↓
Hyperparathyroidism 166	↑	↓	↑	↑	–	–
HHRH (TO)	N/↑	↓	↑	**↓**	N	**↑** · **↓ FGF23** · **hypercalciuria**

Key nutritional vs XLH discriminator ✓ **PTH ↑ in nutritional**, **normal in XLH** · normal Ca + PTH → ✓ **XLH**

XLH PTH normal (OB, Orthobullets) · N/↑ (TO table) – read the options

Urine (TO) XLH/renal tubular: **↑↑ urine PO_4**, ↓ urine Ca · nutritional: ↓ urine Ca + PO_4 **(TO)**

Routine vit D test ✓ **25(OH)D** · 1,25-D uninformative (normal/↑ with secondary HPT) **(TO)**

Histology disordered elongated **proliferative zone** · poorly defined provisional calcification · **widened osteoid seams** · "Swiss cheese" trabeculae · collagen perpendicular to Haversian canals

3 Presentation

Symptoms listlessness · irritability · weakness

Signs **tibial bowing** (wide proximal tibial physes) · **genu varum** (commonest) · **rachitic rosary** (costochondral enlargement) · **widened wrists** + ankles · skull flattening · **kyphosis (cat back)** · ligamentous laxity · hypotonia · waddling gait · delayed dentition, enamel defects · pathological fractures · short stature

– – Active space – –



Tibial bowing in rickets

XLH short stature · **genu varum** · windswept deformity · medial tibial torsion · genu valgum in some (e.g. 4-yr-old knock knees) · no myopathy

VDDR like nutritional but **more severe** · type I: hypotonia, seizures, fractures in infancy · type II: + bone pain, dental caries/hypoplasia, **alopecia**

Calcipenic vs phosphopenic (TO)	**Calcipenic**	**Phosphopenic**
Muscle weakness	Present	**Absent** (present in TIO)
Bone pain	Common	Uncommon
Limbs	All equal	**Predominantly lower limbs**
Tetany · enamel hypoplasia	May occur	Absent
Dental abscess	Absent	May occur
Ca	Low/normal	Normal
ALP	Markedly ↑	Mild–moderate ↑
PTH	↑	Normal / minimally ↑
Osteopenia, osteitis fibrosa	Present	✓ **Absent**

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) ✓ **physeal widening** · **metaphyseal cupping**, irregular metaphyses · loss of zone of provisional calcification definition · osteopenia · **Looser zones** (compression side) · ✓ **anterior rib widening (rachitic rosary)** · **genu varum** · codfish vertebrae · dorsal kyphosis



Rachitic rosary

Radiographic mimic	Feature
Hyperparathyroidism	**Punched-out** lesions, **subperiosteal resorption**, erosion around physis edge
Jansen metaphyseal chondrodysplasia	**Rounded epiphyses**, wide epiphysis–metaphysis separation · ligand-independent **PTHR1** activation
Renal osteodystrophy	**Rugger jersey** sclerotic vertebral endplates

Extreme physeal widening + osteopenia + deformity
✓ pathognomonic for rickets → **labs (Ca, PO_4, ALP, vit D) + endocrinology first**
✗ Wrong options: osteotomy / guided growth before medical control (high recurrence)

Looser zones (TO)

Looser zone pseudofracture; indentation of soft bone by vascular crossings **(TO)**

Seen in (TO)	Not seen
Osteomalacia (characteristic) · renal osteodystrophy · fibrous dysplasia · hyperthyroidism · **Paget's** · **XLH** · OI	✓ **Lead poisoning**

Oncogenic (tumour-induced) osteomalacia (TO)

TIO paraneoplastic · adolescents/young adults · mesenchymal tumours secrete **FGF23 ("phosphatonin")** → phosphaturia **(TO)**

Tumours (TO) **haemangiopericytoma (commonest)** · GCT · chondromyxoid fibroma · chondroblastoma · osteosarcoma · osteoblastoma · NOF / metaphyseal fibrous defect · fibrous dysplasia · MFH · fibrosarcoma · haemangioma
✗ Wrong options: **multiple myeloma (TO)**

TIO labs (TO) ✓ **↓ PO_4 (severe), normal Ca, low/normal 1,25-D, ↑ FGF23** · ↑ ALP · low TmP · PTH less raised than dietary osteomalacia **(TO)**

TIO work-up + Rx (TO) localise: **octreotide scan**, FDG-PET/CT, MRI/CT · **excise tumour** → PO_4 normalises within hours · supplement PO_4 + vit D until normal **(TO)**

Renal tubular rickets & Fanconi syndrome (TO)

Renal tubular rickets (TO) ✓ **↑ urine PO_4, ↓ urine Ca, normal 1,25-D, ↓ serum PO_4, normal PTH** · normal Ca · low urinary pH · early infancy, severe deformity, no myopathy or hypocalcaemia · RTA, Fanconi, XLH **(TO)**

– – Active space – –

Fanconi syndrome (TO) proximal tubular defect → wasting of **PO_4, amino acids, glucose, bicarbonate (TO)**

Fanconi (TO)	Finding
Blood	**Metabolic acidosis** · ↓ PO_4 · ↓ 25(OH)D · ↑ ALP · ↑ PTH · (✗ Wrong options: **↑ serum Ca**)
Urine	↑ PO_4 · ↑ K^+ · calciuria
Bone	↓ BMD, osteoporosis

5 Management

Type	Treatment
XLH	**Calcitriol + phosphate** (phosphate must be with calcitriol – prevents 2° HPT) · **burosumab**
Nutritional	**Vitamin D 5000 IU/day × 6–10 wk**
VDDR I	**Calcitriol** (physiological dose) + calcium – excellent response **(TO)**
VDDR 1B (25-hydroxylase)	Calcidiol **(TO)**
VDDR II	Partial resistance → **vit D** (very high calcitriol) · total → **calcium** · severe hypocalcaemia → ✓ **prolonged high-dose IV calcium** (alopecia doesn't respond) **(TO)**
HHRH	**Oral phosphate ONLY** · **active vit D contraindicated** (worsens hypercalciuria, stones) **(TO)**
ADHR	PO_4 + calcitriol · **iron repletion** may normalise FGF23 **(TO)**
ARHR	PO_4 + calcitriol · risk enthesopathy, vascular calcification (ENPP1) **(TO)**

Doses calcitriol **20–30 µg/kg/day** (children, as printed) · 0.5–0.75 µg/day adults · phosphate **20–40 mg/kg/day** in 3–5 doses (children) · 750–1000 mg/day adults

Burosumab ✓ **fully human mAb neutralising circulating FGF23** → restores PO_4 reabsorption + 1α-hydroxylase · (✗ Wrong options: PHEX chaperone · Klotho agonist **(TO)**) · approved **≥1 yr** (Orthobullets) · first line **>6 months** + adults (TO) – read the options

Conventional XLH therapy risks (TO) **nephrocalcinosis**, tertiary HPT → renal US · nephrocalcinosis → ✓ **switch to burosumab (TO)**
✗ Wrong options: loop diuretic · more phosphate

XLH medical therapy early calcitriol + phosphate ✓ ↓ **need for osteotomies**, improves height, ↓ recurrence · (✗ Wrong options: normalises mineralisation completely) · continue into **adulthood** (pain, insufficiency fractures) · (✗ Wrong options: calcium alone) · OB2930 (Loeffler): adding phosphate to vit D gave no extra benefit – read the options

Operative criteria

Deformity	Surgery
Growth remaining (**<11 yr**)	**Hemiepiphysiodesis / guided growth** – effective once medically treated
>11 yr / severe tibial bowing	**Multilevel corrective osteotomy** (K-wires, plates, IM nails, ex-fix)
Before metabolic control	No – high recurrence

XLH outcome staged surgery better, but early degenerative joint disease

Rickets types compared

	Nutritional	XLH	VDDR I	VDDR II
Defect	↓ Vit D intake	PHEX → ↑ FGF23	1α-hydroxylase	VDR
Inheritance	–	XLD	AR	AR
PTH	↑	**N**	↑	↑
1,25-D	↓	N/↓	**↓↓**	**↑↑**
Treatment	Vit D	Calcitriol + PO_4 / burosumab	Calcitriol	High-dose calcitriol + Ca / IV Ca

How it's asked

"Low circulating FGF23 associated with?"	→ **hyperostosis (TO)**
"NOT FGF23-mediated hypophosphataemic rickets?"	→ **HHRH (TO)**
"XLD vs nutritional rickets lab pattern?"	→ **answer key not provided by source** · options: ↓P/↑ALP/normal PTH · ↓P/↑ALP/↑PTH · normal P with low ALP · gene mapping needed · per OB, XLH = ↓PO_4, ↑ALP, **normal PTH**
"Hypophosphataemia, hyperphosphaturia, low FGF23, ↑ 1,25-D, hypercalciuria – gene?"	→ **SLC34A3 (TO)** ✗ Wrong options: PHEX, FGF23, CYP27B1
"Burosumab mechanism?"	→ **human mAb neutralising FGF23 (TO)**
"XLH on phosphate + calcitriol develops nephrocalcinosis?"	→ **switch to burosumab (TO)**
"Labs confirming nutritional over X-linked rickets?"	→ **↓ PO_4, ↑ ALP, ↑ PTH**
"Normal Ca + PTH found in?"	→ **XLH** ✗ Wrong options: nutritional · VDDR
"Low PO_4 + normal Ca – common hereditary rickets?"	→ **X-linked hypophosphataemic**
"4-yr-old knock knees, ↓ PO_4, ↑ ALP, ↓ 1,25-D, normal Ca/25-D/PTH?"	→ **familial hypophosphataemic rickets** ✗ Wrong options: nutritional · renal OD

– – Active space – –

"Deformity unresponsive to vitamins, normal Ca/PTH/vit D — inheritance?"	→ **X-linked dominant**
"Genu varum, ↓↓ PO_4, ↑ ALP, normal PTH + vit D?"	→ **vitamin D-resistant rickets** ✗ Wrong options: hypophosphatasia · renal OD
"XLH gene?"	→ **PHEX** ✗ Wrong options: GNAS1, CLCN7, TGFBR2, COL1A1
"3-yr-old genu valgum, coxa vara, wide physes, mother had surgeries — next?"	→ **Ca, PO_4, ALP, vit D + endocrinology** ✗ Wrong options: osteotomy · guided growth first
"Familial hypophosphataemic rickets — true?"	→ **renal tubules can't absorb phosphate** ✗ Wrong options: 2nd commonest · ↓ physeal width · hyperphosphataemia · calcitriol normalises mineralisation
"AD pedigree, ↓ serum PO_4, ↑ urine PO_4 — mutation?"	→ **FGF23** ✗ Wrong options: PHEX (XLD) · DMP1 (AR)
"Man with hypophosphataemic rickets planning family — true?"	→ **100% daughters carriers/affected** ✗ Wrong options: sons carriers · FGFR3 · PTH therapy
"PHEX child on treatment with bowing — true?"	→ **Ca + PO_4 supplementation ↓ need for osteotomy** ✗ Wrong options: stop at maturity · always progresses · AR
"Genu varum, ↓ PO_4, ↑ ALP, normal Ca + 25-D — gene?"	→ **PHEX** ✗ Wrong options: COMP, COL5A1, MECP2, TNSALP
"Vit D form measured routinely?"	→ **25(OH)D (TO)**
"Commonest complication of stoss vit D injections?"	→ **answer key not provided by source** · options hypercalcaemia / intermittent hypercalcaemia / hyperphosphataemia / hypophosphataemia · no explanation given
"Breastfed, no vit D, spongy gums, frontal bossing, painful chest nodules, white line of Frankel?"	→ **answer key not provided by source** · options scurvy / nutritional rickets / hypophosphataemic rickets / Barton's disease · stem mixes rickets (bossing, rosary) and scurvy (gums, Frankel line) features
"Lab consistent with nutritional rickets?"	→ **↑ PTH** ✗ Wrong options: ↑ Ca, ↑ PO_4, ↓ ALP, ↑ vit D
"Breastfed Haitian 3-yr-old with bowing — associated finding?"	→ **widened wrist physes** ✗ Wrong options: ↓ PTH · ↓ ALP

"Best radiographic characterisation of rickets?"	→ **widening of anterior ribs** ✗ Wrong options: narrow physes · rounded epiphyses (Jansen) · punched-out (HPT) · rugger jersey (renal OD)
"NOT a cause of rickets?"	→ **vitamin K deficiency**
"US nutritional rickets commonest in?"	→ **African American, breastfed** >6 mo
"Primary problem in rickets?"	→ **low calcium**
"Nutritional rickets labs?"	→ **↓ Ca + ↑ PTH** ✗ Wrong options: normal 25-D with ↓ 1,25-D (VDDR I)
"Vegetarian toddler with bowing + cupped physes?"	→ **low vit D, low Ca, ↑ PTH, ↑ ALP**
"1α-hydroxylase deficiency — rickets type?"	→ **answer key not provided by source (TO)** · OB507: loss of 1α-hydroxylase = **VDDR type I**
"VDDR 2A with alopecia, 1,25-D >1000, severe hypocalcaemia?"	→ **prolonged high-dose IV calcium (TO)** ✗ Wrong options: rhPTH · burosumab
"Loss of 1α-hydroxylase gene causes?"	→ **hereditary VDDR type I**
"Hormone activating 1α-hydroxylase?"	→ **PTH**
"VDDR I vs II?"	→ **I = renal enzyme deficiency · II = inactivating VDR mutation**
"Looser zone NOT seen in?"	→ **lead poisoning (TO)**
"Oncogenic osteomalacia — EXCEPT tumour?"	→ **multiple myeloma (TO)**
"TIO biochemical profile?"	→ **↓ PO_4, normal Ca, low vit D, ↑ FGF23 (TO)**
"ADHR phenotypic trigger?"	→ **iron deficiency (TO)**
"NOT true of Fanconi syndrome?"	→ **↓ serum PO_4 and ↑ serum Ca (TO)**
"Renal tubular rickets profile?"	→ **↑ urine PO_4, ↓ urine Ca, normal 1,25-D, ↓ serum PO_4, normal PTH (TO)**
"NOT a feature of phosphopenic rickets?"	→ **osteopenia (TO)**
"Commonest deformity in renal osteodystrophy?"	→ **genu valgum (TO)**

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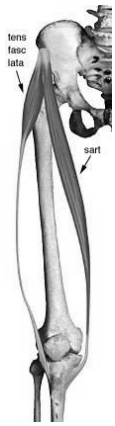
SARTORIUS

– Basic Science · 94 / 108 –

1 Basics

Anatomy

	Sartorius
Origin	**ASIS**
Insertion	Superior medial tibial shaft near tuberosity – **pes anserinus** (with gracilis + semitendinosus)
Action	**Flexes + laterally rotates hip** · flexes knee
Innervation	**Femoral nerve (L2, L3, L4)**
Arterial supply	Muscular branches of femoral artery



Sartorius

Femoral nerve innervation

Femoral nerve supplies	**sartorius** · **rectus femoris** (quadriceps) ✗ Wrong options: obturator · tibial · peroneal · "sartorial nerve" (doesn't exist)

How it's asked

"Nerve to rectus femoris?"	→ **femoral** ✗ Wrong options: obturator · tibial · peroneal

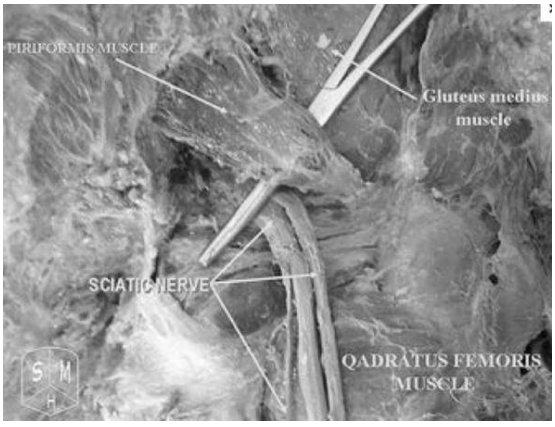
SCIATIC NERVE

– Basic Science · 95 / 108 –

1 Basics

Anatomy at the hip & posterior approach

Exit	✓ **greater sciatic notch** · emerges **anterior (deep) to piriformis** · passes between **piriformis and superior gemellus** (>80%) → posterior to superior gemellus, obturator internus, inferior gemellus, quadratus femoris → deep to biceps femoris
Variants	passes above or **pierces piriformis** · commonest rotator variant: runs **between capsule and short external rotators**, exiting below superior gemellus
Posterior THA	✓ **deep to piriformis, superficial to short external rotators** · primary THA: identify by **palpation** · revision: palpation + direct visualisation
Lesser sciatic notch	**obturator internus tendon** ✗ Wrong options: sciatic nerve



Sciatic nerve course relative to piriformis and short rotators

Peroneal division innervation

Thigh	**tibial division → all hamstrings** (long head biceps, semitendinosus, semimembranosus) · ✓ **peroneal division → short head of biceps femoris only** · both gastrocnemius heads = tibial
Peroneal division	roots **L4–S3** (✗ Wrong options: L3–S1) · separates from tibial in upper popliteal fossa · crosses **posterior** to lateral gastrocnemius head · **deep peroneal** starts **lateral** to anterior tibial artery → crosses medial · commonest injury: **lateral** (✗ Wrong options: anterior) knee dislocation
Localising	thigh-level lesion → **denervates short head biceps + tibialis anterior** · knee-level lesion **spares short head biceps** → test short head biceps (EMG)
Transection	definitive repair at **1–3 wk** (zone of injury declares)

– – Active space – –

How it's asked

"Posterior THA — sciatic nerve location?"	**→ deep to piriformis, superficial to short external rotators**
"Sciatic nerve exit from pelvis?"	**→ greater sciatic notch, between piriformis and superior gemellus** ✗ Wrong options: lesser notch · obturator foramen
"Posterolateral approach — nerve passes between?"	**→ piriformis and superior gemellus**
"Peroneal division supplies in thigh?"	**→ short head of biceps femoris**
"Peroneal division — true?"	**→ thigh injury denervates short head biceps + tibialis anterior** ✗ Wrong options: anterior knee dislocation · L3–S1

SCLERODERMA

– Basic Science · 96 / 108 –

1 Basics

Definition autoimmune disease of **collagen deposition + fibrosis** → hardening of skin, subcutis, muscle, internal organs

Demographics **F > M (3–14:1)** · African American > Caucasian · **30–50 yr** · ✓ **rapidly progressive in children**

Triggers genetic predisposition · **silica**, solvents (benzene), radiation

Pathophysiology triad **collagen deposition + fibrosis** · **non-inflammatory vasculopathy** (microvascular fibroproliferation) · **autoimmunity**

Genes & lab markers

Genetics usually sporadic · HLA · **STAT4 → limited** · **IRF5 → diffuse**

Antibody / lab	Association
ANA	**90–95%** · speckled / centromere · **nucleolar = specific** for systemic sclerosis
Anti-centromere	**Limited (CREST)** · ~50%
Anti-Scl-70 (anti-topoisomerase I)	**Systemic (diffuse)** · ~30%
Anti-histone	Also drug-induced lupus, SLE, RA
ESR, CRP	↑
Creatinine	Renal monitoring
CXCL4	↑ → pulmonary fibrosis
NT-proBNP	↑ → early pulmonary hypertension

Classification

	Limited	Diffuse (systemic)
Skin	**Face + hands**	Widespread
Organs	–	**Kidney, lung, heart**
Course	–	**Rapidly progressive**, high mortality
Antibody	**Anti-centromere**	**Anti-Scl-70**
Subtype	**CREST** – **C**alcinosis · **R**aynaud · **E**sophageal dysmotility · **S**clerodactyly · **T**elangiectasia	–

Linear scleroderma (en coup de sabre) child with linear skin lesions + **limb atrophy** + **Achilles contracture / toe-walking** → ✓ **scleroderma** · early surgery + immunosuppression
✗ Wrong options: arthrogryposis (non-progressive, no skin lesions) · JIA · eczema · abuse

– – Active space – –



Linear scleroderma — en coup de sabre

3 Presentation

Skin	**diffuse pruritus** · **Raynaud** (white → blue → red; cold/stress) · tightness, induration, shiny loss of wrinkles · **sclerodactyly** · digital ulcers · diuretic-resistant oedema · **hyper- + hypopigmentation** · telangiectasia · calcinosis
Hand contractures	**PIP flexion + compensatory MCP hyperextension** · **1st web adduction contracture**



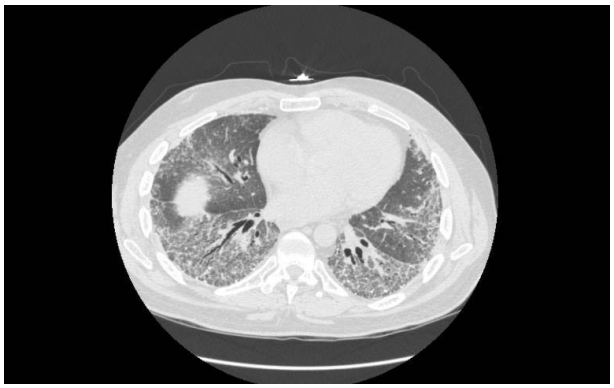
Sclerodactyly

MSK	myalgia · arthralgia · **osteolysis (calcaneus)** · contractures, atrophy, LLD, angular malalignment (children)
Systemic	**acid reflux** (smooth muscle atrophy → ↓ LES pressure) · dry cough, dyspnoea (**restrictive lung disease**), dry rales · palpitations · cor pulmonale · **renal hypertension**

4 Diagnosis & Classification

Key radiology findings

CT chest	**ground glass = early fibrosis** · honeycombing + bronchiolectasis = established interstitial fibrosis



Honeycomb + ground glass – lung fibrosis

Angiography loss of digital flow

Manometry **distal oesophageal aperistalsis + ↓ LES tone** · PFTs for early fibrosis · routine ECG

Differential nephrogenic systemic fibrosis · eosinophilic fasciitis · GVHD

5 Management

Problem	Treatment
Progression (esp. lung)	**Methotrexate · mycophenolate · cyclophosphamide** if refractory
Renal involvement	**ACE inhibitor**
Pruritus	Antihistamines
Raynaud	**Calcium-channel blockers**
Pulmonary hypertension	**Ambrisentan + tadalafil**

Operative criteria

Deformity	Surgery
Symptomatic calcinosis	I&D
Fixed PIP flexion	**PIP arthrodesis**
Fixed MCP hyperextension	**MCP resection arthroplasty**
1st web contracture	**Adductor pollicis release ± trapeziectomy**
Digital gangrene	Amputation

6 Complications

Complications digital infarct/gangrene · **pulmonary hypertension** · **pulmonary fibrosis** · renal failure · pulmonary/renal/cardiac involvement = worse prognosis

How it's asked

"7-yr-old rapidly progressive toe-walking, limb atrophy, linear skin lesions?" → **scleroderma**
✗ Wrong options: arthrogryposis · JIA · abuse

– – Active space – –

"Anti-topoisomerase I antibodies?"	**→ scleroderma (diffuse)**

SCURVY

– Basic Science · 97 / 108 –

1 Basics

Definition severe **vitamin C** deficiency

Epidemiology deficiency in 8% men, 6% women (US) · M:F 4:3 · bimodal: **infants 5–10 mo** (rare <7 mo if breastfed – breast milk has vit C) + **men >60**

Risk factors elderly men living alone ("tea and toast") · overcooking · alcoholism · smokers · malabsorption (Whipple's, IBD, chemotherapy) · infants on **evaporated/condensed milk**

Pathophysiology

Biochemistry humans lack functional **L-gluconolactone oxidase** → can't make ascorbic acid · ✓ ↓ **chondroitin sulfate + collagen synthesis** · impaired **hydroxylation** of collagen peptides · ↓ carnitine → myalgia

Site greatest effect on **metaphysis** (physeal region) · defective **primary + secondary spongiosa** · type I collagen defect → ↓ **osteoblast** function · fragile capillaries → bleeding
✗ Wrong options: osteoclast (osteopetrosis)

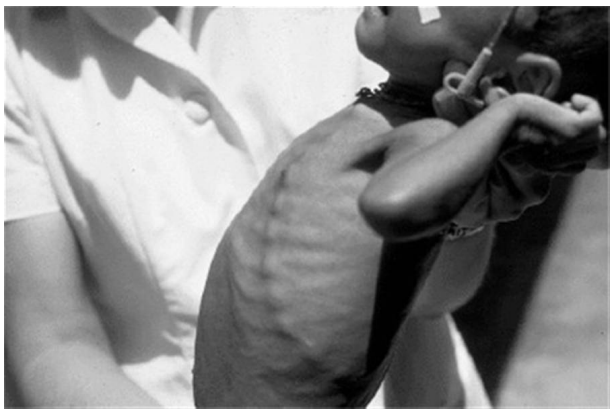
Histology primary trabeculae replaced by **granulation tissue** · haemorrhage · ✓ **widened zone of provisional calcification**

3 Presentation

Symptoms malaise · bone pain · myalgia · **bleeding gums, loose teeth** (bluish-purple gums) · haematuria · haematemesis · iron deficiency

Signs petechiae, ecchymoses · **joint effusions** · **swelling over long bones (subperiosteal haemorrhage)** · massive thigh swelling in infant

Scorbutic rosary costochondral separation with **angular step-off** · not rachitic rosary (**rounded, nodular**)



Scorbutic rosary

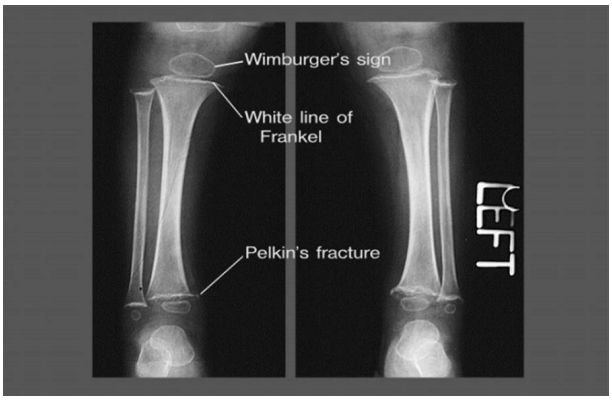
Sites wrists · knees · sternal rib ends (rapid growth)

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4 Diagnosis & Classification

Key radiology findings

Sign	Description
✓ **White line of Frankel**	Dense band = **widened zone of provisional calcification** between epiphysis + metaphysis
Trümmerfeld zone (scurvy line)	Transverse **lucent** metaphyseal band next to Frankel line
Wimberger ring	Dense ring around **epiphysis**
Pelkin spur	Metaphyseal spurs + fractures
Corner sign of Park	Metaphyseal clefts
Pencil-point cortex	Thin cortices
Ground-glass osteopenia	↓ trabeculae
Others	Subperiosteal elevation · **epiphyseal separation** · fractures, dislocations



White line of Frankel and Wimberger ring

Views wrists · knees · sternal rib ends

Diagnostic criteria

Diagnosis history + clinical + radiological picture + **response to vitamin C** · labs usually unhelpful · low fasting serum ascorbic acid

5 Management

Treatment prompt **vitamin C** · adults **250 mg qid × 1 wk** · excellent prognosis if early

Unanswered TO item (TO24050610) infant, swollen bleeding gums + massive thigh swelling, no trauma/fever · options Ewing / osteosarcoma / **scurvy** / haemophilia · **answer key not provided by source** · bleeding gums + subperiosteal long-bone swelling are scurvy features (Orthobullets)

Scurvy vs Rickets

	Scurvy	Rickets
Deficiency	Vitamin C	Vitamin D / Ca / PO_4
Rosary	**Angular step-off**	**Rounded, nodular**
ZPC	**Widened, dense (Frankel line)**	Poorly calcified, wide physis, cupping
Bleeding	Gums, subperiosteal, joints	No

How it's asked

"Somali 7-yr-old, purple bleeding gums, distal femur pain — true?"	→ **widening of zone of provisional calcification** ✗ Wrong options: ↓ osteoclast · vit D · ↑ chondroitin/collagen · epiphysis primarily
"Infant with bleeding gums + massive thigh swelling, no trauma/fever?"	→ **answer key not provided by source** (TO) · options Ewing / osteosarcoma / scurvy / haemophilia

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SEMIMEMBRANOSUS

– Basic Science · 98 / 108 –

1 Basics

Anatomy

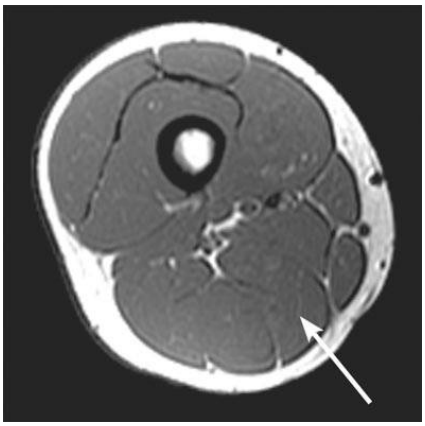
	Semimembranosus
Origin	**Superolateral quadrant of ischial tuberosity**
Insertion	**Posterior medial tibial condyle**
Action	Extends hip · flexes knee · medially rotates tibia (esp. knee flexed)
Innervation	**Tibial nerve (L5, S1, S2)**
Arterial supply	Perforators of profunda femoris · inferior gluteal · superior muscular branches of popliteal



Semimembranosus

Cross-sectional anatomy – mid-thigh (medial → lateral, posterior)

Structure	Position
Gracilis	**Most medial**, superficial to adductor magnus
Adductor magnus	Medial + **deep** to semimembranosus
✓ **Semimembranosus**	**Medial** to semitendinosus
Semitendinosus	Lateral to semimembranosus
Biceps femoris	Most lateral



Mid-thigh cross-section

How it's asked

"Mid-thigh cross-section — posteromedial muscle superficial to adductor magnus, medial to semitendinosus?"	→ **semimembranosus** ✗ Wrong options: semitendinosus · biceps · gracilis

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SICKLE CELL ANEMIA

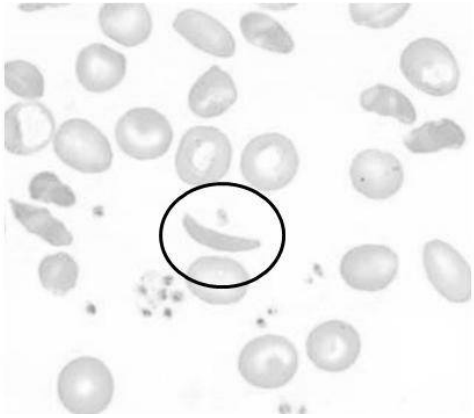
– Basic Science · 99 / 108 –

1 Basics

Epidemiology ~**2 million** Americans carry the gene · **1 in 12 African Americans**

	Sickle cell anaemia (disease)	Sickle cell trait
Genotype	**Two HbS alleles**	One HbS + one HbA
Severity	**Most severe**	Usually asymptomatic · **most common**
Key risk	Crises, infarcts, AVN, infection	**Exertional sickling collapse** → sudden death

Pathophysiology low O_2 → **sickle-shaped RBCs** → can't traverse vessels → vascular occlusion



Sickle cells

Infection susceptibility **hyposplenism / functional asplenia** · sluggish circulation · ↓ opsonisation / impaired complement · GI micro-infarcts → Salmonella entry

Orthopaedic manifestations

Manifestation
Sickle cell crisis
Osteomyelitis · septic arthritis
Osteonecrosis of femoral + humeral heads
Bone infarcts
Growth retardation / skeletal immaturity
Dactylitis (acute hand/foot swelling)

3 Presentation

Sickle cell crisis severe bone pain · begins ~**age 2–3** · mediated by **substance P** · may cause infarcts · **hydroxyurea** → pain relief

Sickle cell trait in athletes

✓ **oxygen + oral/IV hydration + rest** for exertional sickling collapse (leading cause of death in college football) · other SCT risks: compartment syndrome, lumbar myonecrosis, splenic infarct, haematuria, hyposthenuria, VTE
✗ Wrong options: warm clothes · extra analgesia / rest days (disease, not trait) · NSAID avoidance

4 Diagnosis & Classification

Diagnosis **haemoglobin electrophoresis**

Key radiology findings

X-ray **bone infarcts · biconcave "fish-tail" vertebrae**



Fish vertebrae in sickle cell

Bone infarction vs osteomyelitis & Salmonella

Test	Infarct	Osteomyelitis
Bone scan	Abnormal	Abnormal
Bone marrow scan	✓ **↓ marrow uptake**	**Normal** marrow uptake
MRI	–	**Gadolinium T1** identifies infection
CBC, ESR, CRP	May be ↑	May be ↑
Definitive	–	**Aspirate + culture**

Infarct confirmed (↓ marrow uptake)

✓ **symptomatic care: NWB, IV hydration, haematology**
✗ Wrong options: biopsy · debridement · oral antibiotics · steroid injection

– – Active space – –

Osteomyelitis organisms

S. aureus + Salmonella (which is commoner is unclear; S. aureus still most likely per OB3500/OB4399) · ✓ **Salmonella osteomyelitis occurs most commonly / almost exclusively in sickle cell** · often **diaphyseal** · may spread from gallbladder / gut micro-infarcts

Osteomyelitis population	Organism
Sickle cell	✓ **Salmonella** (characteristic) + S. aureus
Neonates <4 mo	S. aureus, Enterobacter, group A/B Strep
Sneaker puncture	**Pseudomonas**
IVDU, dialysis	Staph, Enterobacter, Pseudomonas, Strep

Salmonella osteomyelitis → associated finding

✓ **sickled cells** on smear
✗ Wrong options: OI, child abuse, renal failure, thalassaemia (one case)
✗ Wrong options: myeloma plasma cells · synovial sarcoma · GCT giant cells · Ewing pseudo-rosettes

5 Management

Septic arthritis

fever, bone pain, ↑ CRP/ESR · aspirate + culture · **I&D** – consider **pre-op oxygenation + exchange transfusion**

Osteonecrosis

Femoral head AVN

may be **bilateral** · mechanism ✓ **HbS → sickling → vascular occlusion**
✗ Wrong options: SCFE slip · Salmonella · GAG accumulation · COL5A1

Screening

✓ **haemoglobin : haematocrit ratio** (proxy for **MCHC**) = **strongest predictor** of femoral head ON · high-risk → hip MRI for early ON / hip preservation

AVN stage	Treatment
Initial	**Partial weight-bearing + ROM**
Failed nonop / intractable pain	**THA** – **poor results** (ongoing bone remodelling)

Humeral head ON prognosis

✓ **sickle cell = most benign course / best prognosis**
✗ Wrong options: steroids (commonest non-traumatic cause) · alcohol · radiation · malunion

How it's asked

"Sickle cell thigh pain, abnormal bone scan, ↓ marrow uptake?"	→ **symptomatic care, NWB, IV fluids, haematology** (infarct)
"Salmonella osteomyelitis – associated histology (figure)?"	→ **sickled red cells** ✗ Wrong options: plasma cells · pseudo-rosettes
"Salmonella haematogenous osteomyelitis commonest in?"	→ **sickle cell disease** ✗ Wrong options: neonates · IVDU · dialysis · sneaker puncture
"3-yr-old with Salmonella osteomyelitis also has?"	→ **sickle cell anaemia** ✗ Wrong options: thalassaemia · OI

"Sickle cell, 2-yr progressive hip pain, AVN — mechanism?"	→ **abnormal HbS alleles** ✗ Wrong options: SCFE · Salmonella
"Humeral head ON with best prognosis?"	→ **sickle cell disease** ✗ Wrong options: steroids, alcohol, radiation
"Purpose of Hb : Hct ratio in sickle cell?"	→ **screen for femoral head osteonecrosis risk**
"Football players with sickle trait may need?"	→ **oxygen + hydration**

– – Active space – –

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STATISTIC DEFINITIONS

– Basic Science · 100 / 108 –

1 Basics

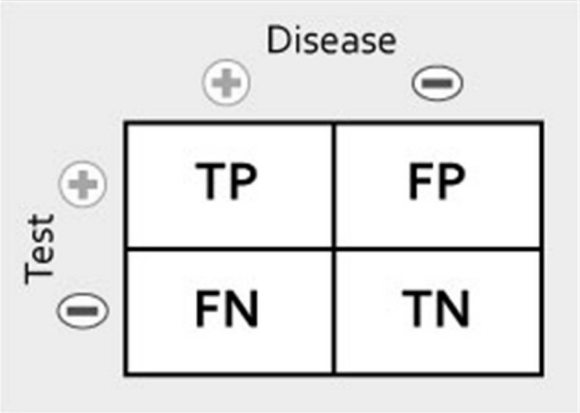
Central tendency & spread

Measure	Definition	Best for
Mode	Most frequent value	**Nominal** (categorical) data
Median	Middle value · unaffected by extremes	**Skewed** data · all levels except nominal
Mean	Arithmetic average · uses all values · most used	Sensitive to extremes
Variance	Variability of each data point from the mean · = SD^2 (average of squared deviations)	–
Standard deviation	Spread of the group	±1 SD = **68%** · ±2 SD = **95%** · ±3 SD = **99.7%**

Diagnostic test properties (sensitivity, specificity, PPV, ROC)

	Disease +	Disease –
Test +	**a** = true positive	**b** = false positive
Test –	**c** = false negative	**d** = true negative

Measure	Formula	Meaning
Sensitivity	**TP / (TP + FN)** = a/(a+c)	Positive in those **with** disease · **screening** (few false negatives)
Specificity	**TN / (TN + FP)** = d/(b+d)	Negative in those **without** disease · **confirmation** (avoid treating the unaffected)
False-positive rate	b/(b+d) = 1 – specificity	–
False-negative rate	c/(a+c)	–
PPV	**TP / (TP + FP)** = a/(a+b)	Positive test truly diseased · **depends on prevalence**
NPV	**TN / (TN + FN)** = d/(c+d)	Negative test truly disease-free · depends on prevalence
LR+	**Sensitivity / (1 – specificity)**	How a positive result changes likelihood
LR–	(1 – sensitivity) / specificity	How a negative result changes likelihood



2 × 2 table

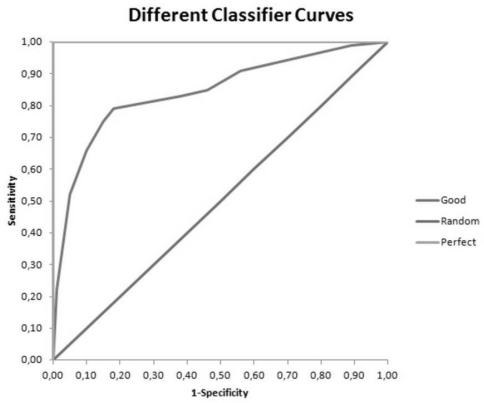
Worked examples 6 of 10 diseased test + → sensitivity **60%** · 5 false + of 90 healthy → specificity 85/90 = **94.4%** · sens 90%, spec 95%, prevalence 10% → PPV 9/(9 + 4.5) = **67%** · 80/100 flu +, 5/100 healthy + → NPV 95/(20 + 95) = **83%** · 86/(86 + 4) = 95.5% → **specificity**

Trade-off sensitivity and specificity usually traded off · false negative worse than false positive → choose **more sensitive** test · drug with nasty side effects, must not treat non-diseased → ✓ **highest specificity**

Precision closeness of repeated measurements (random error) · precise ≠ accurate

Post-test pre-test odds = p/(1 − p) · post-test odds = pre-test odds × LR · post-test probability = odds/(odds + 1)

ROC curve x-axis **1 − specificity (FPR)** · y-axis **sensitivity (TPR)** · ✓ measures **responsiveness** · area under curve (**C-statistic**): **0.5 = useless**, **>0.75 adequate**, higher = better



ROC curve

– – Active space – –

Tool	Purpose
ROC curve	**Responsiveness** / diagnostic ability
Funnel plot	**Publication bias** in meta-analysis
Forest plot	Relative treatment effects across studies (meta-analysis)
Kaplan–Meier	**Survivorship**
Cronbach α	**Internal consistency** of questionnaire items
Log rank	Compare survival curves
Multiple regression	Predict a variable from ≥2 others · controls confounders
Cohen's kappa	**Inter-observer** reliability
ICC	**Intra-observer** reliability

Incidence, prevalence & risk measures

Incidence ✓ **new cases** in a period per population at risk · long-term population/ cohort studies · e.g. 100 new cases among 400 at-risk workers (100 already affected excluded) = **25%**

Prevalence ✓ **total existing cases** at a time point · **cross-sectional** survey (census) · e.g. 200/500 = 40% · MRSA cases on 1 day = prevalence

Measure	Formula / source
Relative risk	[a/(a+b)] / [c/(c+d)] = incidence exposed / incidence unexposed · **cohort** studies · RR >1 → more outcome in exposed
Odds ratio	**(a × d)/(b × c)** · **case-control** (retrospective) + logistic regression · ≈ RR when outcome rare (<10%) · exaggerates risk when common
Absolute risk reduction	Risk difference between groups
Relative risk reduction	1 − RR
Attributable risk	Risk difference attributable to exposure
NNT	**1 / ARR** (NNS 1000 → ARR 0.1%)
Hazard ratio	Comparison of rates over time (survival curves)

Hep C example 75/595 transfused vs 16/712 not → RR = **incidence (75/595) ÷ incidence (16/712)**

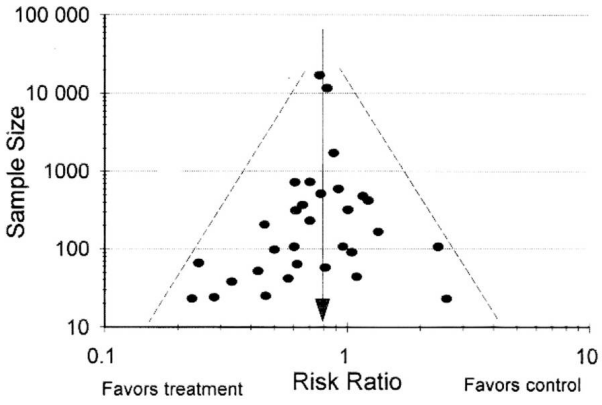
Odds ratio in forest plot OR <1 = ↓ odds in treatment · **CI includes 1 → not significant** · **narrower CI = more precise, less uncertainty** · OR size ≠ effect size/clinical importance · adjusted vs crude OR differ (confounding)

Study reporting cohort (e.g. BMI vs THA outcome in 500) → ✓ **relative risk** · case-control → odds ratio

Study design & meta-analysis

Design	Description
Case-control	Start with **outcome** (e.g. tibial nonunion) vs controls without → look back for risk factors → **OR**
Cohort	Start with **exposure** (e.g. daily vit D) → follow for outcome (hip fracture) → **RR** · prospective or retrospective
Cross-sectional	Single time point → **prevalence**
RCT	Random allocation, prospective
Case series	No comparison group
Meta-analysis	Combines studies · funnel plot for publication bias

Funnel plot effect size vs study size/precision · large studies less variable · ✓ **symmetric inverted funnel = no publication bias** · skewed = bias (negative studies unpublished)



Funnel plot

Statistical test selection (t-test, chi-square, ANOVA, non-parametric) (TO)

Parametric continuous, **normally distributed** · **non-parametric** – skewed continuous, **ordinal** or **categorical**

Comparison	Parametric	Non-parametric
2 groups, **paired**	**Paired t-test** (means in matched groups)	**Wilcoxon signed-rank** (Orthobullets) · OB1230/OB213156 call it Wilcoxon rank-sum – read the options
2 groups, **unpaired**	**Student / independent t-test**	**Mann-Whitney U** (= Wilcoxon rank sum)
≥3 groups	**ANOVA**	**Kruskal-Wallis** (Friedman for repeated)
Categorical, 2+ groups	**Chi-square**	Chi-square
Categorical, **small sample** (<5 per cell or total <~50)	**Fisher exact**	Fisher exact

– – Active space – –

ANOVA — one-way = **one independent variable** · two-way = **two independent variables** · ≥3 groups, normal data (e.g. BMI across pinning/hemi/THA; fastball velocity of 5 pitchers)

Choose a test

- Categorical / proportions → **CHI-SQUARE**
  - small cells (<5) → **FISHER EXACT**
- Continuous, normal → 2 groups T-TEST (paired if matched) · ≥3 groups → **ANOVA**
- Continuous skewed or ordinal → 2 groups MANN-WHITNEY U · ≥3 groups → **KRUSKAL-WALLIS**

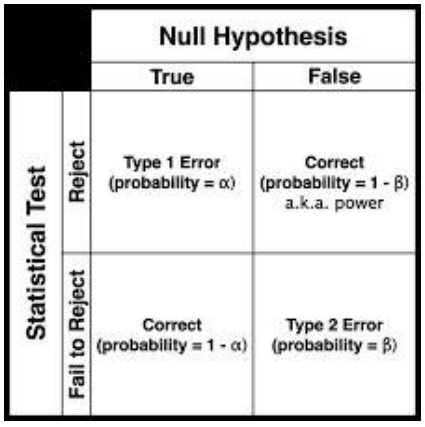
Ordinal grade (Herring) in two unpaired populations — ✓ **Mann-Whitney U (TO)**
✗ Wrong options: paired t · Wilcoxon signed-rank · Kruskal-Wallis

Examples — mean age 2 normal groups → **t-test** · CRPS rate 1% vs 9% → **chi-square** · single non-normal continuous variable → **Mann-Whitney U** · OR time robotic vs manual (normal) → **t-test** · ACL graft return-to-play scores (finite range) → chi-square per OB217862

5 Management

Type I & II errors & power

	H_0 true	H_0 false
Reject H_0	**Type I (α) error** – false positive	Correct – **power (1 − β)**
Fail to reject H_0	Correct (1 − α)	**Type II (β) error** – false negative



Type I and II errors

Type I (α) — ✓ **rejecting a true null** · α set at **0.05** (1/20) · ↓ by lowering α (0.01) or **Bonferroni** (but ↑ type II)

Type II (β) — ✓ **accepting a false null** · detecting no difference when one exists · commonest in **underpowered** studies · e.g. small trial negative, later 2000-patient trial positive · crossover with intent-to-treat (40% "no tourniquet" got tourniquet) · 2 groups of 5 with p = 0.079 → **may be type II**

Power ✓ **1 − β** · probability of detecting a true effect · usually **80%** (β 0.2) · depends on effect size, α, β, **n** · ✓ **↑ sample size → ↑ power, ↓ type II** · 84 needed per arm, β 0.2 → chance of false-negative = **0.2**
✗ Wrong options: prospective vs retrospective · case-control

Power analysis ✓ **determines minimum sample size** to show a clinically significant difference · ✓ **before the study** · elements: α, β, effect size (± variance), n · post-hoc power discouraged

Reduces type II	Doesn't
✓ **↑ sample size**	↓ α to 0.01 (↑ type II) · blinding (detection bias) · goniometer (measurement error) · randomisation (selection bias) · Bonferroni (↑ type II) · ↑ β

Effect size magnitude of difference in means between groups ÷ **pooled SD** · continuous variables (RRR for dichotomous) · not shown by p-value

Other terms **Hawthorne effect** = behaviour change when observed · **confounder** = associated with both exposure + outcome · selection bias · measurement error · sampling error · **fragile p-values** with small samples
✗ Wrong options: "error"

P-values, confidence intervals, correlation & effect size

P-value probability, **if H_0 true**, of a result **as or more extreme** · p = 0.09 → difference would arise by chance **9 in 100** times · ✓ p-value = probability of type I error (OB217456)
✗ Wrong options: "groups statistically equal" if p > 0.05 · gives effect size · clinical significance

Bonferroni correction ✓ **post-hoc correction of p-values for multiple simultaneous tests** on one data set · **α ÷ number of comparisons** (10 tests → 0.005)

Confidence level	Mean ± SD
85%	1.44
90%	1.65
95%	**1.96** (OB218892 uses ±2 SD)
98%	2.32
99%	**2.58**

Confidence interval range likely to contain the true parameter · shows **precision** + magnitude · ✓ **larger sample → narrower CI** (smaller margin of error) · 99% CI wider than 95% · overlapping CIs do not prove no significance – need formal test · e.g. FNA 0.5 ± 0.2 → 95% range **0.1–0.9 cm³**

Correlation R^2 = 0.97, p = 0.03 → strongly correlated, unlikely chance, does not **validate accuracy** (both methods could be wrong) · Pearson = 2 normal continuous variables · **Bayesian** = prior → posterior probability

Reliability, MCID, fragility index & survival

Psychometrics **reliability** = ✓ **reproducibility** between repeated tests / observers · **validity** = measures what it intends · **responsiveness** = detects change (ROC)

Inter-observer **Cohen's kappa** · **intra-observer** – **ICC**

– – Active space – –

MCID	difference in outcome that is clinically relevant · few tools have accepted MCID · statistically significant 5-point ASES difference with MCID 10–15 → ✓ **may not be clinically meaningful** (both groups' 22/27-point gains exceed MCID)
Fragility index	✓ **number of outcome changes (event ↔ non-event) needed to flip significance** · dichotomous outcomes · larger = more robust
Survivorship	joint replacement success · all assumed operated simultaneously · **life table** (annual, changes once/yr) vs **product limit** (recalculated at each failure) · **Kaplan–Meier**: x = time, y = survival probability · higher curve at a time point = better survival

How it's asked

"Equation for specificity?"	→ **TN / (TN + FP)** ✗ Wrong options: TP/(TP + FN) (sensitivity)
"Positive predictive value?"	→ **TP / (TP + FP)**
"86/(86 + 4) = 95.5% (US for meniscus tears)?"	→ **specificity**
"Sensitivity of an assay?"	→ **TP / (TP + FN)**
"Tool to determine responsiveness?"	→ **ROC curve** ✗ Wrong options: funnel plot, Kaplan–Meier, Cronbach α, forest plot
"Test value to avoid treating people without RA?"	→ **specificity** ✗ Wrong options: sensitivity · NPV
"Probability of finding a true association?"	→ **power**
"Accepting the null when false?"	→ **type II error**
"Statistical power defined by?"	→ **1 − probability of type II error** ✗ Wrong options: 1 − α
"Ability to detect a true difference between interventions?"	→ **power** ✗ Wrong options: effect size · Hawthorne
"Capsule repair study negative, later large study positive?"	→ **type II error**
"Rejecting a true null hypothesis?"	→ **type I error**
"Increase study power by?"	→ **increasing sample size**
"No difference when one truly exists?"	→ **type II error**
"Power analysis determines what, when?"	→ **sample size, before the study**
"84 per arm, β 0.2, α 0.05, p = 0.28 — chance null accepted when false?"	→ **0.2** ✗ Wrong options: 0.8 = power · 0.05
"Tourniquet RCT, 40% crossover, ITT no difference?"	→ **type 2 (β) error**
"Purpose and timing of power analysis?"	→ **minimum patients for clinically significant difference, before study**

"Lateralised RSA study negative, later RCTs positive?"	→ **type II error**
"Reduce type II error?"	→ **increase sample size** ✗ Wrong options: α 0.01 · blinding · β 0.9
"6300-patient trial positive, 150-patient follow-up negative?"	→ **beta error**
"Calculate power?"	→ **1 − type II error**
"Two groups of 5, p = 0.079 – statement?"	→ **may have made a type II error** ✗ Wrong options: Bonferroni · underpowered ↑ type I
"Reduce type 2 error in TKA rehab RCT?"	→ **increase sample size** ✗ Wrong options: goniometer · blinding · Bonferroni
"Herring grade in rural vs urban groups?"	→ **Mann-Whitney U (TO)**
"Paired t-test used for?"	→ **difference between two means in matched groups**
"Chi-square unreliable with small numbers – use?"	→ **Fisher exact test**
"Mean age, two normal groups?"	→ **Student t-test**
"CRPS rates 1% vs 9%?"	→ **chi-square**
"Chi-square most appropriate for?"	→ **categorical data between groups** ✗ Wrong options: <5 per group (Fisher)
"Normal data, ≥3 independent groups?"	→ **ANOVA** ✗ Wrong options: Kaplan–Meier · log rank
"Single continuous non-normal variable?"	→ **Mann-Whitney U**
"Ideal ANOVA scenario?"	→ **mean BMI across pinning, hemi, THA (parametric)**
"Data for chi-squared test?"	→ **categorical data of two large groups** ✗ Wrong options: two small (Fisher)
"Mean OR time robotic vs manual TKA, normal?"	→ **Student t-test**
"Best ANOVA application?"	→ **fastball velocity of top five pitchers**
"One-way vs two-way ANOVA?"	→ **one vs two independent variables, ≥3 groups**
"Definition of incidence?"	→ **new cases in a specific time period**
"Relative risk of hep C with transfusion?"	→ **(75/595) ÷ (16/712)**
"500 workers, 100 symptomatic at start, 100 new in 5 yr – incidence?"	→ **25%** ✗ Wrong options: 20% · 40% (prevalence)
"500 THAs followed by BMI – report effect as?"	→ **relative risk** ✗ Wrong options: odds ratio

– – Active space – –

"Total MRSA cases on one day?"	**→ prevalence**
"Cross-sectional survey establishes?"	**→ prevalence**
"RR of dehiscence anterior vs posterior (W, X, Y, Z table)?"	**→ W/(W + X) ÷ Y/(Y + Z)** ✗ Wrong options: (W/Y)/(X/Z) (OR)
"Effect size definition?"	**→ estimated magnitude of difference in means**
"Range likely to include the unknown parameter?"	**→ confidence interval**
"Bonferroni correction?"	**→ post-hoc p-value correction for multiple tests**
"Women vs men steps, p = 0.09 – interpretation?"	**→ difference would occur by chance 9 in 100 times if no true difference** ✗ Wrong options: statistically equal
"R^2 0.97, p 0.03 between FE and bench model?"	**→ strongly correlated, not chance, but doesn't validate the model**
"Mean ± 2.58 SD?"	**→ 99% CI**
"Meaning of p = 0.03?"	**→ probability of type I error**
"1.2 ± 0.3 vs 0.5 ± 0.2 cm^3 – conclusion?"	**→ FNA yields 0.1–0.9 cm^3 95% of the time** ✗ Wrong options: significance from CIs alone
"Narrower CI in final study – cause?"	**→ larger sample size** ✗ Wrong options: 99% CI · more variability
"Forest plot of aspirin ORs – accurate?"	**→ DVT estimate less uncertain than hematoma (narrower CI)**
"Funnel plot purpose?"	**→ detect publication bias** ✗ Wrong options: forest plot · ROC
"Nonunion patients vs non-nonunion patients questionnaire?"	**→ case-control study**
"Daily vit D group vs none followed for hip fracture?"	**→ cohort study**
"Reliability refers to?"	**→ reproducibility between repeated tests or observers**
"Kaplan–Meier – accurate statement?"	**→ group with highest curve at 1 month more likely alive**
"Fragility index?"	**→ number of outcome changes needed to change trial significance**
"ASES +22 vs +27, p = 0.03, MCID 10–15?"	**→ 5-point difference may not be clinically meaningful**

STRUCTURAL PROPERTIES

– Basic Science · 101 / 108 –

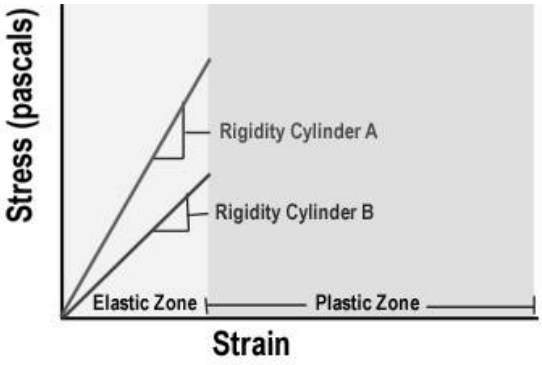
1 Basics

Implant behaviour — **structural properties** (shape, configuration — cylinder, rectangle) + **material properties** · structural ≠ strength characteristic

Bending rigidity (stiffness)

Definition — slope of **elastic range** on structural stress–strain curve · **THA stem stiffness** → proximal **stress shielding**

Shape	Bending rigidity ∝
Solid cylinder	**Radius⁴**
Hollow cylinder	**Radius³**
Rectangle (plate)	**Thickness³**



Bending rigidity ∝ radius⁴

Moments of inertia

Term	Meaning
Area moment of inertia (I)	Shape property predicting deflection, bending, stress · **∝ length⁴** · function of width, thickness, polar moment · = flexural stiffness
Polar moment of inertia (J)	Resistance to **torsion**
Moment of inertia	Resistance to change in rotation

Deflection — displacement under load · ∝ (force / elastic modulus) × area moment of inertia (as printed)

Bone anisotropy

Cortical bone — ✓ **anisotropic** (collagen fibre orientation) → mechanical properties **vary with orientation** · loaded along **long axis → highest stiffness + tensile strength** vs oblique/transverse
✗ Wrong options: hydroxyapatite/water content differences

– – Active space – –

How it's asked

"Why orient cortical bone tensile specimens relative to femoral shaft?"	**→ mechanical properties vary with orientation (anisotropy)** ✗ Wrong options: hydroxyapatite structure · water content · easier cutting

SUBSCAPULARIS

– Basic Science · 102 / 108 –

1 Basics

Anatomy

	Subscapularis
Origin	**Subscapular fossa**
Insertion	**Lesser tuberosity**
Action	**Internal (medial) rotation** + adduction · holds humeral head in glenoid
Innervation	**Upper + lower subscapular nerves (C5–C7)** – from ✓ **posterior cord**
Arterial supply	Subscapular artery



Subscapularis

Brachial plexus level	Branches
Posterior cord	✓ **Upper + lower subscapular** · thoracodorsal
Lateral cord	Lateral pectoral
Superior (upper) trunk	**Suprascapular** · nerve to subclavius
Middle / inferior trunks	No branches

3 Presentation

Mechanism **forced external rotation** injury → subscapularis tear

Tests **lift-off** (hand behind back, IR) → ✓ **subscapularis highest % maximal contraction** · **belly-press**
✗ Wrong options: pec major, lat dorsi, supraspinatus

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Sagittal oblique MRI (coracoid level)	✓ **subscapularis = anterior** · supraspinatus superior · infraspinatus, teres minor, teres major posterior · assess **atrophy + fatty infiltration** (repair prognosis)



Sagittal oblique MRI – subscapularis anterior

Arthroscopy (posterosuperior portal)	subscapularis tendon anteriorly · axial MRI fatty atrophy = chronic tear

How it's asked

"Muscle with highest activity in lift-off test?"	→ **subscapularis**
"Sagittal oblique MRI – anterior muscle (asterisk)?"	→ **subscapularis** ✗ Wrong options: supraspinatus · teres minor
"Forced ER injury – tendon's nerve branches from?"	→ **posterior cord** ✗ Wrong options: lateral cord · superior trunk

SUPERFICIAL PERONEAL NERVE

– Basic Science · 103 / 108 –

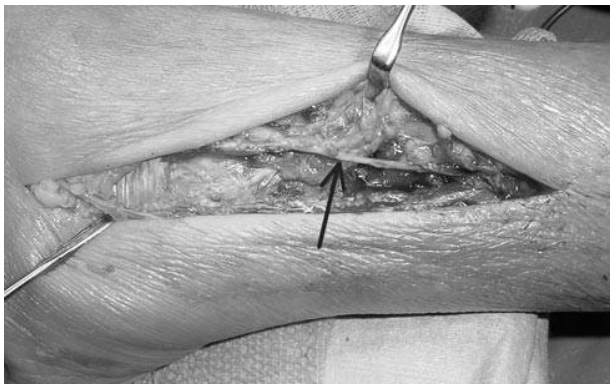
1 Basics

Anatomy & entrapment

Course branch of **common peroneal** after crossing fibular head → **lateral compartment** → ✓ **motor to peroneus longus + brevis** → pierces **crural fascia ~10–12 cm proximal to fibular tip** → **intermediate + medial dorsal cutaneous nerves** (dorsum of foot + toes)

At risk **lateral approach to ankle** (e.g. bimalleolar ORIF)

Nerve	Supplies
Superficial peroneal	**Lateral compartment** (PL, PB) · dorsum of foot except 1st web
Deep peroneal	**Anterior compartment** · **EDB, EHB** · **1st dorsal web space**
Sural	**Lateral foot** sensation



Superficial peroneal nerve at lateral ankle approach

3 Presentation

Entrapment runner · lateral distal leg pain + **dorsal foot dysaesthesia** · worse with **active/passive plantar flexion + inversion** (traction at fascial exit) · tender fullness (fascial hernia) **~10 cm above lateral malleolus** · **Tinel +** · no weakness · normal X-ray/MRI

5 Management

Entrapment ✓ **fascial release + neurolysis**
✗ Wrong options: **closure of fascial defect** (worsens, risk exertional compartment syndrome) · four-compartment fasciotomy (only for established compartment syndrome) · biopsy · epidural steroid

– – Active space – –

How it's asked

"Nerve at lateral ankle approach — true?"	→ **proximally innervates lateral compartment** ✗ Wrong options: anterior compartment · EDB · 1st web · purely sensory lateral foot
"Runner, dysaesthesia dorsum foot with PF + inversion, Tinel 10 cm above malleolus?"	→ **fascial release + neurolysis of SPN** ✗ Wrong options: fascial closure · fasciotomy

SYNOVIUM & SYNOVIAL FLUID

– Basic Science · 104 / 108 –

1 Basics

Synovium

Function nutrient exchange between blood and joint fluid

Structure vascularised, porous connective tissue · ✓ **no basement membrane**

Synovial cell	Origin / features	Function
Type A	**Macrophage-derived** · non-fixed · antigen-presenting · superficial	✓ **Phagocytosis**
Type B	**Fibroblast-like** · rich RER, dendritic processes · deeper	✓ **Produce synovial fluid** – hyaluronic acid, fibronectin, collagen
Type C	Intermediate · unknown origin/function	? multipotent precursor of A or B

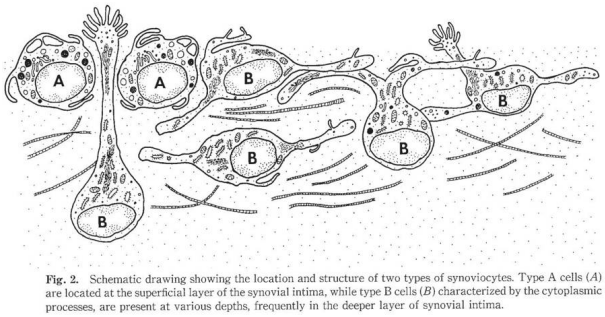


Fig. 2. Schematic drawing showing the location and structure of two types of synoviocytes. Type A cells (*A*) are located at the superficial layer of the synovial intima, while type B cells (*B*) characterized by the cytoplasmic processes, are present at various depths, frequently in the deeper layer of synovial intima.

Synovial cell types

Synovial cells & lubrication

Synovial fluid **ultrafiltrate of plasma** regulated by synovium · **~2 mL** in healthy knee · lubricates + **nourishes cartilage by diffusion**

Component	Role
Hyaluronan	Synthesis needs **uridine diphosphoglucose dehydrogenase**
✓ **Lubricin**	Glycoprotein (HA-binding) · **↓ coefficient of friction** · made by **superficial-zone chondrocytes** · deficiency → early arthritis
Urokinase-type plasminogen activator	Activates plasmin → **prevents intra-articular clot** (fibrinolysis)
Collagenases · prostaglandins	–
Proteolipids	Boundary lubrication

– – Active space – –

✗ Wrong option	Actual role
Decorin	Commonest tendon proteoglycan · regulates collagen fibril diameter
Scleraxis	Tendon/ligament transcription factor · inhibits osteoblastogenesis
Aggrecan	Cartilage swelling pressure
Cathepsin	Osteoclast protease at ruffled border

Biomechanics ✓ **non-Newtonian** · viscosity not constant, not linear · **viscosity ↑ as shear rate ↓**

Lipohaemarthrosis (supine) layer 1 **fat** · layer 2 **serum/synovial fluid** (✓ **uPA → inhibits clot**) · layer 3 **RBCs** (don't form fibrin clot)

4 Diagnosis & Classification

Synovial fluid analysis in OA (TO)

Parameter (TO)	Normal	OA	Gout	RA	Pyogenic	TB
Appearance	Straw/clear yellow	**Clear yellow**	Yellow–turbid milky	Yellow, cloudy	Purulent	Yellow, turbid
Viscosity	Normal	✓ **Normal**	↓	↓	↓	↓
WBC/mm³	≤200	✓ **≤2000**	2000–50,000	2000–50,000	**>50,000**	10,000–20,000
PMN	<20%	✓ **<20%**	60–70%	50–60%	**90%**	60%
Crystals	–	–	**Urate**	–	–	–
Glucose	Slightly < plasma	–	↓	↓	↓↓	↓↓

OA fluid ✓ **normal viscosity, <20% neutrophils**
✗ Wrong options: increased viscosity · WBC >2000 · decreased viscosity **(TO)**

How it's asked

"Cell producing synovial fluid?"	→ **type B synovial cell** ✗ Wrong options: type A (phagocytic) · fibrochondrocyte
"Synovial component most reducing friction?"	→ **lubricin** ✗ Wrong options: decorin, scleraxis, aggrecan, cathepsin
"Lipohaemarthrosis layers – true?"	→ **layer 2 contains uPA, inhibits intra-articular clot**
"OA synovial fluid – true statement?"	→ **normal viscosity with <20% neutrophils (TO)**

TENDONS

– Basic Science · 105 / 108 –

1 Basics

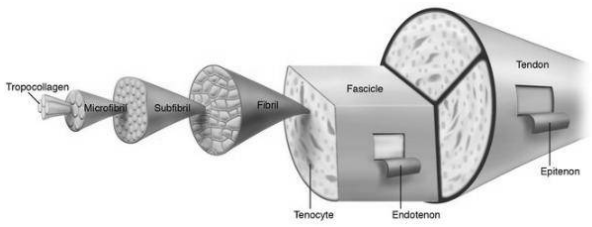
Function transfer muscle force to bone · orient along stress

	Paratenon-covered	Sheathed
Examples	**Patellar, Achilles**	**Hand flexors**
Blood supply	**Rich** → heal better	Avascular areas · **nutrition by diffusion**
Injury	Trauma · fail at **musculotendinous** or **tendon–bone junction**	**Laceration** · risk of **adhesions**

Composition (collagen, decorin, aggrecan, enthesis)

Structure fascicles separated by **endotenon**, surrounded by **epitenon** · hierarchical (microfibrils → fibrils → fascicles) · mostly **water**

Component	% dry weight / role
Type I collagen	**85%** (Orthobullets, OB213905) · **95%** (OB8280) – read the options
Type III collagen	**0–5%** (<5%) · also V, VI small
Proteoglycans	0–5% · attract water
✓ **Decorin**	**Most abundant tendon proteoglycan** (small leucine-rich PG) · ✓ **regulates collagen fibril diameter / size** · cross-links fibres, transfers load
✓ **Aggrecan**	Proteoglycan in **areas of compression**
Tenascin-C	Glycoprotein ↑ in tendinopathy
Elastin	Tissue recoil after stretch
Fibronectin	Cell adhesion, migration, differentiation
Fibrillin	Elastic fibre formation



Hierarchical tendon structure

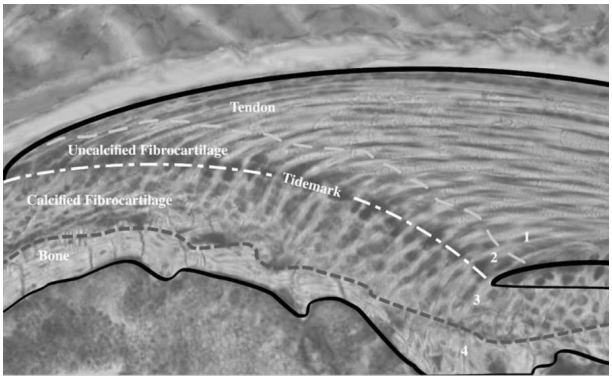
Cells **fibroblasts (tenocytes)**: spindle-shaped, parallel rows along load · make type I (85%) + type III (5%) collagen · drive healing

Blood supply musculotendinous junction · osseotendinous junction · surrounding tissue (paratenon, mesotenon, **vincula**)

– – Active space – –

Enthesis

Insertion zones (↑ modulus)	Contents
Tendon	Type I + III collagen, elastin, PGs, fibroblasts
Uncalcified fibrocartilage	Aggrecan, types I, II, III, fibrochondrocytes
– **Tidemark** –	Straight basophilic line = soft–hard mechanical boundary
Calcified fibrocartilage	**Type II**, aggrecan, types I + **X**, fibrochondrocytes · interlocks with bone
Bone	Osteocytes, osteoblasts, osteoclasts, type I, apatite



Four zones of the direct insertion

	Fibrous (indirect)	Fibrocartilaginous (direct)
Site	Metaphysis / diaphysis	Epiphysis / apophysis
Structure	Perforating mineralised (Sharpey) fibres	**4 zones**
Insertion angle	Changes little	**Changes greatly → overuse injury**
Examples	**Deltoid**, adductor magnus–linea aspera, pronator teres	**Rotator cuff, Achilles**

Nerve endings

✓ **enthesis = free nerve endings (nociception)** · myotendinous junction = **Golgi organs** (steady state), **Ruffini** (prolonged, sensitive), **Pacini** (fast) · **Meissner** = skin light touch

Vascular supply – vincula

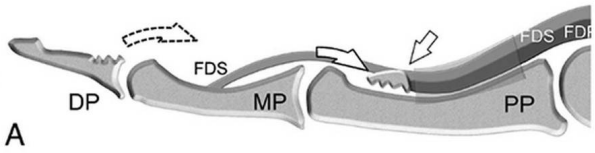
Sheathed tendons

vincula = **main blood supply** to bulk of FDS/FDP · FDS + FDP each have **vincula brevia + longa** (2 each, Orthobullets) · "5 vincula: 2 short, 3 long" (OB5552)

Vinculum	Origin → attachment
VBS (brevia superficialis)	PIPJ / adjacent proximal phalanx
VLS (longa superficialis)	Proximal P1 / MCPJ → 2 slips
✓ **VBP** (brevia profundus)	**DIPJ / distal middle phalanx → FDP at A5**
VLP (longa profundus)	Distal P1 / PIP volar plate → through VBS → FDP

Jersey finger (Leddy–Packer II, fragment at A2)

✓ **VBP disrupted** · VLP intact · FDS vincula unaffected
✗ Wrong options: volar plate



Vincula of FDS and FDP

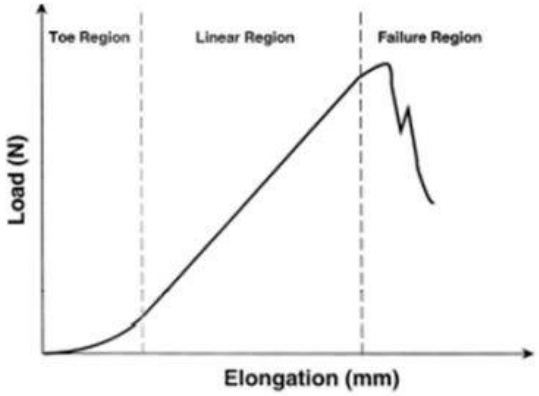
2 Mechanism

Mechanical behaviour (toe region, elastin)

Properties **more collagen, less viscoelastic than ligament** · viscoelastic, non-linear · **strain-rate dependent** · ✓ **strong in tension (5–10% strain vs bone 1–4%)** · buckle in compression · **creep + stress relaxation**

Modifiers exercise ↑ · **immobilisation ↓** · ↑ to maturity then ↓ · laser/heat → shrinkage · hydration, temperature, pH → **test under physiological conditions**

Load–elongation region	Explanation
✓ **Toe** (low load, non-linear)	**Crimped fibres straighten** · crimped elastin fibrils (OB5510) · tendon toe smaller than ligament (more parallel)
Linear (elastic)	Intermediate load · slope = **Young's modulus**
Yield point	Elastic → plastic
Failure	High load



Load–elongation curve

Soft tissue vs metal ✓ **toe region unique to soft tissue**
✗ Wrong options: modulus, plastic region, strain energy density, yield point (shared)

Mechanotransduction ✓ **deflection of primary cilium**
✗ Wrong options: apoptosis · cell elongation · pressurisation (older theories)

Training (8 wk half-marathon) ✓ **net ↑ type I collagen** (↑ turnover; ↓ cross-links with endurance)
✗ Wrong options: known effect on fibril diameter / CSA

– – Active space – –

5 Management

Healing & adaptation

Stage	Events	Time
Haemostasis	**Platelets** → coagulation · fibrin clot + fibronectin → chemotaxis	5–15 min
Inflammation	✓ **Neutrophils + M1 macrophages** · chemotaxis of fibroblasts · fibroblasts make **type III collagen**	**1–7 days**
Proliferation	Disorganised collagen + angiogenesis · GFs: PDGF, IGF, **BMP-12/13**, TGF-β · M2 shift · ↑ type I + neovascularisation	7–21 days
Remodelling	**Type III → type I** · fibrosis, ↓ cellularity	Up to **18 months**

Repair strength ✓ **weakest in inflammatory phase** · **7–10 days** (Orthobullets) · **5–21 days** (OB8271) · most strength by **21–28 days** · max at **6 months** · ✓ **final only ~2/3 normal**

Early mobilisation earlier ROM but ↓ repair strength · ✓ **flexor tendons: prevents adhesions**

Tendon vs Ligament

	Tendon	Ligament
Collagen.....101	More, more parallel	Less, less organised
Viscoelasticity	Less	More
Toe region	Smaller	Larger

How it's asked

"Decorin controls?"	→ **tendon collagen fibre size** ✗ Wrong options: myelin · axon branching
"Tendon matrix ratio?"	→ **95% type I, <5% type III**
"Found in areas of tendon compression?"	→ **aggrecan** ✗ Wrong options: decorin
"Enthesis contains?"	→ **free nerve endings** ✗ Wrong options: Golgi, Pacini, Ruffini (myotendinous) · Meissner (skin)
"Function of most abundant tendon proteoglycan?"	→ **regulates collagen fibril diameter** ✗ Wrong options: compression (aggrecan)
"Region X on load-elongation curve?"	→ **toe region with crimped fibres**
"Stress–strain portion due to elastin structure?"	→ **A – toe region** ✗ Wrong options: B elastic · E modulus
"Unique to ligament vs titanium curve?"	→ **toe region**
"Tendon repair weakest in which phase?"	→ **inflammatory**

"8 wk marathon training — Achilles change?"	→ **net increase type I collagen** ✗ Wrong options: ↑ fibril diameter · ↑ CSA
"How fibroblasts sense strain?"	→ **primary cilium deflection**
"Inflammatory phase hallmark (week 1)?"	→ **neutrophils + M1 macrophages** ✗ Wrong options: M2 (remodelling) · type I deposition
"Jersey finger with fragment at A2 — vascular structure injured?"	→ **vincula brevia to FDP**

– – Active space – –

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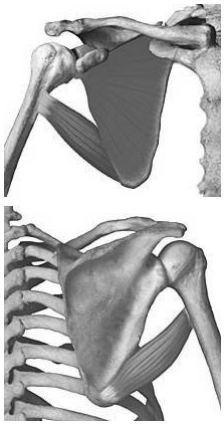
TERES MAJOR

– Basic Science · 106 / 108 –

1 Basics

Anatomy

	Teres major
Origin	Dorsal **inferior angle of scapula**
Insertion	✓ **Medial lip of intertubercular groove**
Action	**Adducts + internally rotates** arm
Innervation	✓ **Lower subscapular nerve** – **C6, C7** (Orthobullets) · **C5, C6** (OB) · **thoracodorsal in 20%**
Arterial supply	Subscapular + circumflex scapular arteries

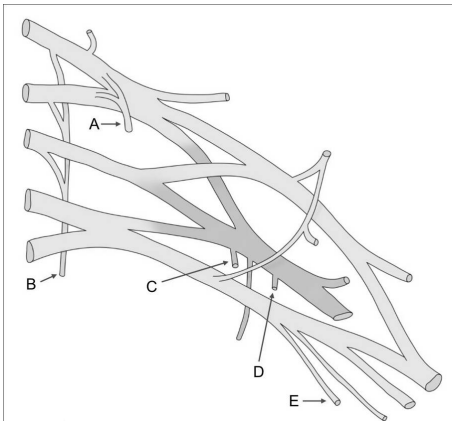


Teres major

Innervation

Lower subscapular nerve — **posterior cord** (with upper subscapular + thoracodorsal) · also supplies **lower subscapularis** · relevant to tendon transfers for massive cuff tears

Brachial plexus label (OB210271)	Nerve
A	Nerve to subclavius
B	Long thoracic
C	Upper subscapular
✓ **D**	**Lower subscapular**
E	Medial cutaneous nerve of arm



Brachial plexus — lower subscapular nerve

How it's asked

"IR muscle inserting on medial lip of bicipital groove — nerve?"	→ **lower subscapular nerve** ✗ Wrong options: upper subscapular · long thoracic

– – Active space – –

THROMBOEMBOLISM (PE & DVT) PROPHYLAXIS

– Basic Science · 107 / 108 –

1 Basics

VTE DVT + PE · US incidence: DVT **200,000/yr**, PE **50,000/yr**

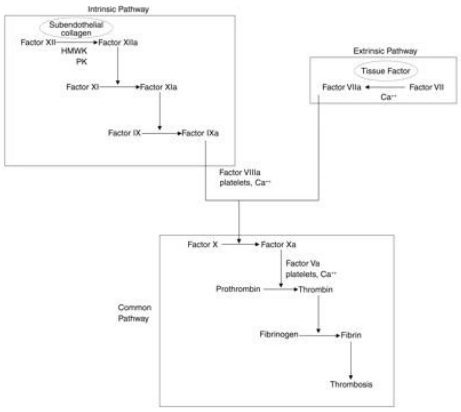
Pathophysiology (Virchow triad, thromboplastin, fibrin, embolisation)

Virchow's triad **venous stasis** · **hypercoagulability** · **endothelial (intimal) injury**
✗ Wrong options: thrombocytopenia · platelet dysfunction

Clot formation stasis → fibrin formation → clot retraction → propagation · DVT starts in **venous valve cusps**

Surgical trigger ✓ **thromboplastin (tissue factor, factor III)** released during dissection → **extrinsic pathway** (VII → X → prothrombin → thrombin → fibrinogen → fibrin)

Thrombus **fibrin + RBCs**, relatively **few platelets** (platelets key in **arterial** thrombi) · **fibrin = end product of coagulation cascade** (thrombin cleaves fibrinogen) · collagen starts intrinsic arm · fibrinolytic system breaks down fibrin



Coagulation cascade

Cement pressurisation of femoral canal ✓ **greatest V/Q mismatch risk in THA** (MMA + fat emboli)
✗ Wrong options: reaming, modular stem, SCD, posterior approach

Thrombophilias (factor V Leiden, protein C/S)

Inherited	Note
MTHFR C677T/TT	✓ **Highest risk** (✗ Wrong options: "no impact")
Factor V Leiden	Gln-for-Arg substitution at protein C cleavage site → **factor Va resists inactivation by activated protein C** → ↑ thrombin · >50% get DVT with one extra risk (fracture, TJA) · (✗ Wrong options: platelet / endothelial / hepatic disorder · mandatory Greenfield filter)
Antithrombin III deficiency	ATIII normally inhibits thrombin + Xa
Protein C / S deficiency	Protein S = cofactor to activated protein C; inactivates Va + VIIIa · ✓ **deficiency → VTE** (✗ Wrong options: fibrinogen, VII, VIII, prothrombin deficiency)
Activated protein C resistance · ↑ factor VIII · hyperhomocysteinaemia · **prothrombin G20210A**	–

✗ Wrong option	Actually
Familial dysfibrinogenaemia	Defective fibrin clot
Plasminogen deficiency	Can't lyse clots
Glanzmann thrombasthenia	Platelet defect → bleeding
von Willebrand disease	Bleeding disorder
Antiphospholipid syndrome	Autoimmune, antibodies to protein C/S → clotting

Risk factors (TO)

Acquired
malignancy (up to 20% of new VTE) · **hormones**: recombinant **EPO** (✓ thrombotic event before spinal fusion), **HRT**, OCP, late pregnancy · antiphospholipid (lupus anticoagulant, anticardiolipin) · prior VTE · **obesity** · CHF · MI · varicose veins · smoking · ↑ viscosity · thoracic outlet (UE DVT) · **type of surgery (TJA)** · **age** · family history
✗ Wrong options: **hypercholesterolaemia**
✗ Wrong options: protective

Anaesthesia
✓ **general > spinal/epidural** (conversion spinal → GA ↑ risk) · AAOS: neuraxial anaesthesia not shown to ↓ VTE (↓ blood loss)

Warfarin paradox
rapid INR rise on unopposed warfarin → early **protein C fall** (t½ 6–8 h) before factor II (48–120 h) → transient hypercoagulability

TXA
✓ **no ↑ DVT** (oral or IV)

Paediatric
rising (obesity, OCP, smoking) · ✓ **biggest risk = prior VTE** (+ thrombophilia, strong family history)
✗ Wrong options: nephrotic, obesity, OCP, SCI (lesser)

Caprini index
✓ **DVT risk assessment in trauma** · low/moderate/high/highest · **major LE fracture = highest** · **score ≥3 → prophylaxis (TO)**

– – Active space – –

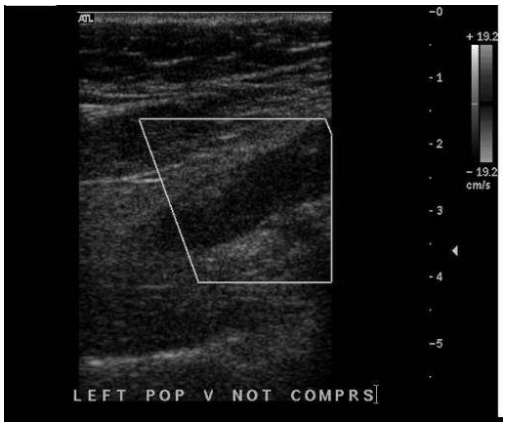
3 Presentation

DVT	calf pain · palpable cord · pitting swelling · **50% with classic signs have no DVT** · **50% venogram-positive have normal exam** · ~20% clinically silent after TJA
PE	most **asymptomatic** · pleuritic pain · dyspnoea · tachypnoea · saddle embolus → death · post-vertebroplasty PE may be delayed (PMMA, fat, marrow)
Timing	most symptomatic VTE **after discharge, 2–6 wk** · TKA: ↑ DVT but ↓ symptomatic PE vs THA · no prophylaxis TKA → symptomatic PE ~8% · DVT ~60% without chemoprophylaxis ✗ Wrong options: within 2 wk
Classification	acute <28 days · chronic >28 days · recurrent

4 Diagnosis & Classification

Key radiology findings

DVT	**duplex compression US** ("non-compressible vein", ~95% sens/spec) = **gold standard** (Orthobullets) · ✓ **venography = gold standard** (TO) – read the options · US best **non-invasive** test (97% proximal, ~75% calf) · symptomatic calf pain 2 wk post-TKA → ✓ **venous US** next **(TO)** ✗ Wrong options: MRI · CTA · aspiration
Screening tests (TO)	**D-dimer**: sensitive, not specific (inflammation, malignancy, post-op) – rules out · **digital photoplethysmography** ~100% sensitive **(TO)**
Routine post-op duplex screening	✓ **strong AAOS recommendation against** in asymptomatic elective arthroplasty, even with prior DVT / estrogen (cost ~$35,000 per extra DVT)



Duplex US – non-compressible vein

PE	✓ **CT pulmonary angiography = gold standard** · **V/Q scan** if **contrast allergy** · CXR usually normal early (oligaemia, prominent hilum) → late wedge/plate atelectasis



CTPA – saddle embolus

Diagnostic criteria – PE

Finding	PE
PaO_2 < 80 mmHg	✓
$PaCO_2$ < 35 mmHg (hypocapnia)	✓ · (✗ Wrong options: **$PaCO_2$ > 35**)
A–a gradient > 20 mmHg	✓
Tachypnoea with SpO_2 99%	✓ (pulse oximetry unreliable – hyperventilation)
Tachycardia + new **RBBB**	✓
ECG	**Sinus tachycardia (commonest**, 8–69%) · RAD · RBBB · **S1Q3T3 (McGinn-White)** 10–20%
Acid–base	**Respiratory alkalosis** (✗ Wrong options: hyperchloraemic metabolic acidosis)

Wells criteria clinical probability for DVT and PE **(TO)**

5 Management

Prophylaxis guidelines (AAOS, ACCP, mechanical, combined) (TO)

Risk stratification (ACCP 2008) ✓ **THA, TKA, hip fracture, major trauma, SCI = HIGH risk (TO)**
✗ Wrong options: "THR moderate risk"

AAOS approach ✓ **pre-op assessment of VTE + bleeding risk** · weigh **bleeding vs PE**

AAOS elevated risk of major bleeding	AAOS elevated risk of PE
Bleeding disorder (haemophilia, vWD) · **recent GI bleed** · **recent haemorrhagic stroke** · active liver disease	**Hypercoagulable state** (e.g. ✓ **protein C deficiency**) · **previous documented PE**

– – Active space – –

Operative criteria – who gets prophylaxis

Procedure	Prophylaxis
THA / TKA	✓ **All patients** – mechanical + pharmacological (moderate strength) · no optimal agent specified
Upper extremity · arthroscopy · isolated fracture knee and below (standard risk)	Not recommended
Foot & ankle	Not shown to ↓ VTE (20,043 patients)
Spine	No consensus · early mobilisation + IPC · chemo for long surgery, multilevel thoracolumbar, anterior TL · risk **epidural haematoma**
Shoulder arthroplasty	Mobilisation, mechanical, regional · LMWH only if ↑ risk
Trauma	✓ **Combined mechanical + chemical → ↓ DVT AND PE** · standard: IPC + enoxaparin
Liver failure / bleeding risk (INR 1.36)	✓ **Mechanical only** (✗ Wrong options: warfarin · none)
Contraindication to anticoagulation	IVC filter considered (✗ Wrong options: routine)

AAOS guideline points
✓ **mechanical compressive devices post-op** · aspirin post-op · ✓ **stop antiplatelets pre-op** · (✗ Wrong options: stockings alone sufficient · routine duplex · specific INR (2011)) · regional anaesthesia suggested · **early mobilisation (consensus)** · AAOS/ASH 2019: IPC > stockings, no IVC filter, aspirin or anticoagulant, DOAC > LMWH (conditional)
✗ Wrong options: start aspirin 1 wk pre-op

Shown to ↓ VTE after TKA
early mobilisation · aspirin · coumadin · pneumatic boots
✗ Wrong options: **neuraxial anaesthesia**

IPC mechanism
✓ **↑ venous blood flow/return** · ↑ endothelial **fibrinolysis** · ↓ venous **compliance** + stasis
✗ Wrong options: ↓ bleeding time · endothelial injury

2012 ACCP convergence with AAOS
✓ **↑ aspirin monotherapy prescribing** · ACCP agents ≥ **10–14 days**: ASA, LMWH, fondaparinux, apixaban, dabigatran, rivaroxaban, LDUH, VKA · dual mechanical + chemo inpatient

Compliance almost all receive prophylaxis but often not ACCP-compliant · LMWH compliant more than warfarin (THA 63 vs 33%) · VTE ↑ costs in- and post-discharge · most acute peri-op pulmonary events = **fat embolism / cardiac**

Agent (TO)	Dose / start	Duration	Notes / reversal
Aspirin	81 mg BID (favoured) or 160–325 mg · within 12–24 h	TKA 10–14 d · THA 28–35 d	AAOS 2007 + ACCP 2012 ✓ · **SIGN: not as sole agent** · platelets (no antidote)
LMWH (enoxaparin)	30 mg BID (US) / 40 mg OD · 12–24 h	10–14 d · THA up to 35 d	"Gold standard" (TO) · no aPTT monitoring · ↓ HIT, ↓ osteoporosis vs UFH
UFH	–	–	Monitor **aPTT** · replaced by LMWH
Warfarin (VKA)	INR **2–3, target 2.5** (ACCP) · **AAOS 2007: INR ≤ 2**	Up to 35 d	Vit K, PCC, FFP
Fondaparinux	**2.5 mg OD**	–	Anti-Xa · ✓ **no routine monitoring**
Rivaroxaban	**10 mg OD** · 6–10 h (TO) / 12–24 h (TO EPCAT table)	THA 35 d · TKA 14 d	Oral Xa · not for **hip fracture** · RECORD 1–4 · **andexanet alfa** · 20 mg = treatment dose
Apixaban	**2.5 mg BID** · 12–24 h	THA 35 d (32–38) · TKA 12 d (10–14)	Oral Xa · not for hip fracture · ADVANCE · **least bleeding** · **andexanet alfa** (✗ Wrong options: protamine)
Dabigatran	**220 mg OD** (half dose 1–4 h post-op; 110 mg if >75 or renal)	THA 28–35 d · TKA 10 d	**Direct thrombin (IIa) inhibitor** · RE-MODEL, RE-NOVATE · ✓ **idarucizumab**

Step-down (EPCAT II) (TO) ✓ **rivaroxaban 10 mg × 5 days → aspirin 81 mg** for 30 more days (THA) / 9 days (TKA) · non-inferior, less bleeding **(TO)**

Switching rules (TO) do not **overlap LMWH + DOAC** · give first DOAC dose **when next LMWH dose due** (24 h after 40 mg OD; 12 h after 30 mg BID) · LMWH → aspirin day 4–6 · do not step up aspirin → rivaroxaban 20 mg · CrCl <30 → adjust/avoid · epidural: hold LMWH 12 h before catheter removal, next dose ≥4 h after **(TO)**

Elective THA / TKA

- All → **MECHANICAL (IPC) + EARLY MOBILISATION**
- Standard risk → **ASPIRIN or ANTICOAGULANT (individualised)**
- ↑ PE risk (prior PE, hypercoagulable) → **PHARMACOLOGICAL ± MECHANICAL**
- ↑ Bleeding risk (bleeding disorder, liver disease, recent GI bleed / haemorrhagic stroke) → **MECHANICAL ONLY**
  - anticoagulation contraindicated + high PE risk → **CONSIDER IVC FILTER**

Treatment of VTE

Isolated calf thrombus <5 cm rarely needs treatment → **serial US**

Anticoagulation **~3 months after DVT** · **~12 months after PE**

– – Active space – –

Other	**early mobilisation** (dislodgement risk < further clot risk) · graduated compression hose → may prevent post-thrombotic syndrome · thrombolysis / thrombectomy controversial
Suspected intra-op PE in trauma	damage control: convert definitive (IMN) to provisional (ex-fix), get helical CT

6 Complications

DVT	PE · **post-thrombotic syndrome 43% at 2 yr** (30% mild, 10% moderate, 3% severe) · **recurrence 25%**
PE	sudden death · obstructive shock · PEA · arrhythmias · pulmonary hypertension · cor pulmonale · hypoxaemia · R→L shunt
Mortality	**6%** DVT and **12%** PE within 1 month · ~10% of PE die in first hour, 30% later from recurrence
Bleeding on dabigatran	✓ **idarucizumab (TO)**

DOAC comparison (TO)

	Rivaroxaban	Apixaban	Dabigatran
Target	Xa	Xa	**IIa (thrombin)**
Dose	10 mg OD	**2.5 mg BID**	**220 mg OD**
Reversal	Andexanet alfa	Andexanet alfa	**Idarucizumab**
Hip fracture	No	No	–

How it's asked

"THA step – greatest V/Q mismatch risk?"	→ **cement pressurisation of femoral canal**
"NOT part of Virchow's triad?"	→ **neither thrombocytopenia nor platelet dysfunction**
"Molecule released in dissection causing DVT?"	→ **thromboplastin** ✗ Wrong options: prothrombin · factor XI
"What is fibrin?"	→ **product of the coagulation cascade** ✗ Wrong options: fibrinolytic product · platelet cytoskeleton
"Variant cofactor resistant to activated protein C?"	→ **factor V Leiden** ✗ Wrong options: ATIII · protein S
"Variant cofactor can't be inactivated → ↑ thrombin?"	→ **factor V Leiden thrombophilia** ✗ Wrong options: Glanzmann · vWD
"Leiden thrombophilia best described?"	→ **abnormal factor V resists aPC inactivation → ↑ clotting**

"Deficiency raising VTE risk?"	→ **protein S** ✗ Wrong options: fibrinogen, VII, VIII, prothrombin
"Caprini index used for?"	→ **DVT risk in trauma (TO)**
"Pre-op EPO before scoliosis fusion — risk?"	→ **thrombotic event**
"THA patient — risk factors (BMI 34, age 70, cholesterol, son FVL)?"	→ **type of surgery, age, BMI** ✗ Wrong options: hypercholesterolaemia
"True about VTE risk?"	→ **conversion spinal → general ↑ risk** ✗ Wrong options: MTHFR no impact · TXA ↑ · estrogen protective
"Biggest paediatric VTE risk factor?"	→ **history of VTE**
"NOT true about VTE prophylaxis?"	→ **THR patients are moderate risk** (they're high) **(TO)**
"Step-down protocol after THA (EPCAT II)?"	→ **rivaroxaban 10 mg × 5 d → aspirin 81 mg × 30 d (TO)**
"GI bleed post-TKA on direct thrombin inhibitor — dose + reversal?"	→ **220 mg OD; idarucizumab (TO)**
"Twice-daily oral Xa inhibitor after THA — dose, start, reversal?"	→ **apixaban 2.5 mg BID, 12–24 h, andexanet alfa (TO)**
"Pneumatic compression devices associated with?"	→ **↑ venous blood flow** ✗ Wrong options: ↑ compliance · ↓ bleeding time
"AAOS recommendation after TKA?"	→ **mechanical compressive devices post-op** ✗ Wrong options: pre-op aspirin · screening duplex · INR 1.5–2.5
"Elevated bleeding risk EXCEPT?"	→ **protein C deficiency** (PE risk)
"AAOS TJA prophylaxis guideline includes?"	→ **pre-op assessment of VTE and bleeding risk** ✗ Wrong options: GA · routine IVC filter · INR 2–3
"Liver failure, INR 1.36, TKA — prophylaxis?"	→ **mechanical prophylaxis in hospital**
"Practice change after 2012 ACCP convergence?"	→ **↑ aspirin monotherapy**
"True regarding VTE after hip/knee surgery?"	→ **almost all get prophylaxis, often not ACCP-compliant** ✗ Wrong options: most within 2 wk
"NOT shown to ↓ VTE after TKA?"	→ **neuraxial anaesthesia**
"Combined mechanical + chemical in trauma?"	→ **lower DVT and PE**
"Post-op duplex screening after THA?"	→ **strongly recommended against**
"Gold standard for DVT post-TKR?"	→ **venography (TO)**
"Calf pain + low fever 2 wk post-TKA — next?"	→ **venous ultrasonography**

– – Active space – –

"Indication for helical chest CT EXCEPT?"	→ **$PaCO_2$ > 35 mmHg**
"Dyspnoea, tachycardia 3 days post-TKA – likely finding?"	→ **S1Q3T3 on ECG**
"Collapse with JVD days after repeat vertebral augmentation?"	→ **pulmonary embolism**

ULNAR NERVE

– Basic Science · 108 / 108 –

1 Basics

Anatomy

Course — **medial cord** → anteromedial arm → **pierces medial intermuscular septum** (upper arm) → posteromedial → **cubital tunnel** → motor branches **distal** to tunnel (FCU, FDP ring + small) → hand

No elbow function — ✓ transection above elbow → **no elbow limitation** (elbow movers = radial, musculocutaneous, median)

Motor innervation (interossei, adductor pollicis, palmaris brevis) (TO)

Region	Ulnar supply (TO)
Forearm ("1½ flexors")	**FCU** + **medial half of FDP** (ring, small)
Hand – thenar ("1½ muscles")	✓ **Adductor pollicis** + **deep (medial) head of FPB**
Hand – other	✓ **All interossei (palmar + dorsal)** · **medial 2 lumbricals** · hypothenars
Superficial branch	✓ **Palmaris brevis** (only intrinsic)
Deep branch	Hypothenars (FDM brevis, ADM, ODM) · medial lumbricals · adductor pollicis · interossei · FPB deep head

Median supply (TO)	Radial supply (TO)
Forearm: pronator teres, PL, FCR, FDS · **AIN**: FPL, pronator quadratus, lateral FDP	Before spiral groove: triceps long + medial · in groove: lateral head, anconeus
Hand: APB, opponens pollicis, superficial FPB, **lateral 2 lumbricals**	Below groove: BR, ECRL (brachialis sensory only) · **PIN**: supinator + APL, EPB, ECRB, EIP, EDC, EDM, ECU

3 Presentation

Cubital tunnel syndrome — hand clumsiness · intrinsic wasting · ✓ **Froment sign** · ↓ sensation ulnar small finger · **Tinel at medial elbow**



Cubital tunnel – intrinsic wasting

– – Active space – –

Sensory & motor distribution after injury

Martin-Gruber anastomosis	median → ulnar **motor** connections in forearm · distal ulnar transection with **intact finger abduction/adduction** · sensation: ✓ **dorsal 1st web + index tip intact, small finger tip lost** (sensory loss still ulnar)

How it's asked

"Ulnar nerve transected excising medial intermuscular septum — elbow limitation?"	→ **none**
"Distal ulnar transection, intact abduction/adduction — sensation?"	→ **dorsal 1st web + index tip intact, small finger tip lost**
"Nerve to adductor pollicis?"	→ **ulnar (TO)** ✗ Wrong options: median · both
"Nerve to palmar interossei?"	→ **ulnar (TO)**
"Intrinsic supplied by superficial branch of ulnar nerve?"	→ **palmaris brevis (TO)** ✗ Wrong options: adductor pollicis, FDM brevis, ADM

BRAIN ANATOMY

– Basic Science · TO notes –

1 Basics

General

Weight	**1300–1400 g** · **~10 glial cells per neuron** (TO)
Neurons vs glia	neurons **cannot multiply**, glia can → **gliomas** among commonest brain tumours · lost neurons after head injury **not replaced** (TO)
Coverings	dura · arachnoid · pia (as for the cord) (TO)
Parts	cerebrum (2 hemispheres) · cerebellum · brainstem (midbrain, pons, medulla) · diencephalon (thalamus, hypothalamus, epithalamus + pineal) (TO)
Contralateral control	motor fibres **decussate in the medulla** (TO)

Ventricles & CSF pathway

CSF pathway (TO)	lateral ventricle → **foramen of Monro** → 3rd ventricle → **aqueduct of Sylvius** → 4th ventricle → **foramina of Luschka + Magendie** → subarachnoid space + spinal central canal → absorbed by **arachnoid villi / granulations**
Third ventricle	between the thalami, above hypothalamus · **fourth ventricle** – between pons / medulla and cerebellum (TO)
Obstruction anywhere	**hydrocephalus** (TO)

![Ventricles and CSF pathway](../../TO notes/attachments/TOn-notes-anatomy-of-the-brain-aug-2026-p02-f004.jpg)

Cerebrum

Cortex	superficial **grey matter**; white matter deep · grey grows faster → **gyri, sulci, fissures** (TO)
Hemisphere connection	**corpus callosum** · separated by **longitudinal fissure** (TO)

Lobe	Position	Main functions (TO)
Frontal (largest)	Anterior; **central sulcus** separates from parietal	**Motor cortex**, personality, **Broca** (motor speech), planning, emotion
Parietal	Top of head	**Sensory cortex** (touch, pressure, temperature, pain), shapes on skin, language interpretation
Occipital	Back	**Visual cortex**
Temporal	Sides, above ears	**Auditory cortex**, hearing + **smell**; posterior part → speech comprehension

![Lobes of the cerebrum](../../TO notes/attachments/TOn-notes-anatomy-of-the-brain-aug-2026-p02-f005.jpg)

– – Active space – –

Cerebellum

Size 2nd largest; posterior cranial fossa; **~10% of brain weight but ~50% of neurons** (TO)

Parts **vermis** (posture, sitting upright) · anterior + posterior lobes (subconscious movement control) · **flocculonodular lobe** (equilibrium, balance) · folia · white matter = **arbor vitae** (TO)

Lesion **staccato ("scanning") speech** · incoordination, imbalance (TO)

Peduncle	Connects cerebellum to (TO)
Superior	**Midbrain**
Middle	**Pons**
Inferior	**Medulla**

Brainstem

Part	Key facts (TO)
Midbrain (~2.5 cm)	Tectum: **superior colliculi = vision**, **inferior colliculi = hearing** · tegmentum (eye movements) · **substantia nigra** (dopamine; lesion = **Parkinsonism**) · crus cerebri (corticospinal + corticonuclear) · arousal, sleep–wake, temperature
Pons	Links cerebellum to brain · respiration, swallowing, **pontine micturition centre**, hearing, balance, eye movement
Medulla (~3 cm)	Pons → foramen magnum · **motor decussation** · cardiovascular, respiratory rhythm, vasomotor centres · swallow, cough, sneeze, vomit · lower cranial nerve nuclei

Brainstem lesion drowsiness / **coma** (arousal centres) (TO)

Diencephalon

Thalamus ~**80%** of diencephalon · **gateway to cortex**: relays nearly all sensory input (TO)

Hypothalamus autonomic, temperature, water balance, food intake, **circadian rhythm**, hypothalamo-pituitary axis, HR + BP (with medulla) (TO)

Pineal **melatonin in darkness** → sleep · melatonin used for disturbed sleep after head injury (TO)

Thalamic nucleus	Relays (TO)
Medial geniculate	**Hearing**
Lateral geniculate	**Vision**
Ventral posterior	Taste + body sensation (touch, pain, cold)

Circumventricular organs

Definition periventricular structures **without a blood–brain barrier** ("windows of the brain"), sensory + secretory (TO)

"Medimix SOAP" (TO) **M**edian eminence · **S**ubfornical organ · **S**ubcommissural organ · **O**VLT · **A**rea postrema · **P**ineal · **P**osterior pituitary

2 Development

Embryology

Neural ectoderm (TO) **neural tube** (brain + cord) · **neural crest** (most of PNS) · **placodes** (sensory ganglia, anterior pituitary, inner ear)

Phase (TO)	Event	Defects
Dorsal induction (neurulation)	Neural plate **~4.5–5 wk** → groove → folds fuse → tube; secondary neurulation → distal cord	Primary: **anencephaly, cephalocele, Chiari** · secondary: **spinal dysraphism** (spina bifida, meningomyelocele)
Ventral induction	Rostral expansion → brain vesicles (proximal **2/3 brain**, caudal **1/3 cord**; lumen = ventricles)	**Holoprosencephaly**, hydrocephalus, **aqueductal stenosis**, agenesis of corpus callosum, **Dandy-Walker**
Proliferation, migration, histogenesis	Neurons multiply + migrate to cortex	**Lissencephaly**, schizencephaly, polymicrogyria, heterotopia, micro / megalencephaly, focal cortical dysplasia
Myelination	Myelin sheaths form	–

Primary vesicle	Secondary vesicle	Adult derivative (TO)
Prosencephalon	Telencephalon · diencephalon	Cerebral hemispheres · thalamus, hypothalamus, epithalamus
Mesencephalon	Mesencephalon	Midbrain
Rhombencephalon	**Metencephalon** · **myelencephalon**	**Pons + cerebellum** · **medulla**

![Primary and secondary brain vesicles](../../TO notes/attachments/TOn-notes-anatomy-of-the-brain-aug-2026-p05-f013.jpg)

3 Surface Anatomy

Cortical surface anatomy

Central sulcus joins Sylvian fissure only **~2%** (separated by **subcentral gyrus** in 98%) (TO)

Parietal lobules superior + inferior, split by **intraparietal sulcus** · **supramarginal gyrus (area 40)** caps Sylvian fissure · **angular gyrus (area 39)** caps superior temporal sulcus (TO)

Pars opercularis, triangularis, orbitalis **inferior FRONTAL gyrus** (TO)
✗ Wrong options: inferior temporal gyrus

– – Active space – –

Brodmann areas

Area	Function (TO)
4 (precentral)	**Primary motor**
6	Premotor / supplementary motor
8	**Frontal eye field**
3, 1, 2 (postcentral)	**Primary somatosensory**
44, 45 (inferior frontal)	**Broca** – **non-fluent**, comprehension preserved
22, 39, 40	**Wernicke** – **fluent**, comprehension lost
41, 42 (Heschl)	**Primary auditory**
17	**Primary visual**

Craniometric points

Pterion frontal + parietal + temporal + **greater wing of sphenoid** · 2 fingerbreadths above zygomatic arch, thumb's breadth behind frontal process of zygoma · **thinnest skull** · overlies **Sylvian fissure** (TO)

Pterion fracture **middle meningeal artery → extradural haematoma** (TO)

Others asterion · lambda · **bregma (coronal + sagittal sutures)** · stephanion · glabella · opisthion · inion (TO)

Superior sagittal sinus usually **slightly right** of the sagittal suture (TO)

![Craniometric points](../../TO notes/attachments/TOn-notes-anatomy-of-the-brain-aug-2026-p06-f016.jpg)

Homunculus

Large areas hand (thumb), face, lips, tongue, eyes · small: trunk, legs (TO)

Layout **leg + genitals medial** (interhemispheric) · **face + hand lateral** → explains deficits by arterial territory (TO)

X vs Y

	Broca	Wernicke
Areas	**44, 45**	**22, 39, 40**
Lobe	Inferior frontal	Posterior superior temporal
Speech	**Non-fluent**	**Fluent**
Comprehension	**Preserved**	**Lost**

	Medial geniculate	Lateral geniculate
Relays	**Hearing**	**Vision**
Midbrain partner	Inferior colliculus	Superior colliculus

How it's asked

"Cortical surface anatomy — false statement?"	→ **"pars opercularis, triangularis, orbitalis lie in inferior temporal gyrus"** (TO)
"Central sulcus joins Sylvian fissure in?"	→ **~2%** (TO)
"Cerebellar peduncles connect to?"	→ **superior–midbrain · middle–pons · inferior–medulla** (TO)
"Metencephalon gives rise to?"	→ **pons + cerebellum** (TO)
"Circumventricular organs?"	→ **"Medimix SOAP"** (TO)
"Pterion fracture → ?"	→ **middle meningeal artery → extradural haematoma** (TO)
"Substantia nigra lesion?"	→ **Parkinsonism** (TO)